

MOTIVIC HOMOTOPY THEORY

Notes

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PREFACE

These notes grew out of my participation in Marc Hoyois’s Oberseminar on motivic cohomology at the University of Regensburg (SoSe 2026), combined with my own study of the subject. The primary references are the work of Bachmann–Hoyois [BH21] on norms, and the recent paper [BEM25] of Bachmann, Elmanto, and Morrow (cited as BEM throughout). Their scope is motivic homotopy theory: the goal is to give a guided route from motivic spaces to stable motivic homotopy theory, its six operations, slice and homotopy filtrations, framed models, and algebraic cobordism, with conventions, prerequisites, and pointers to the primary sources made explicit.

ORGANIZATION

Chapter 1 constructs the unstable motivic homotopy category $H(S)$ via sifted cocompletion, Nisnevich sheafification, and $\mathbb{A}^1$ -localization, develops the pointed variant $H_*(S)$, and establishes the bigraded sphere structure together with the Tate object $\mathbb{T}_S$.

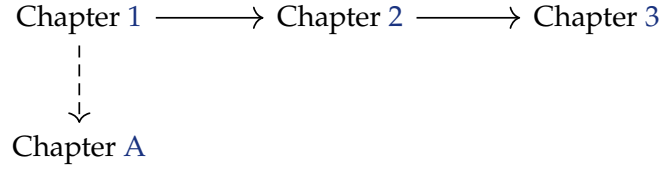
Chapter 2 formally inverts $\mathbb{T}_S$ to obtain the stable category $SH(S)$, briefly introduces the Annala–Iwasa generalization MotSp , and equips SH with a six-functor formalism via the span-category framework of Mann–Scholze and Cnossen–Lenz–Linskens. The chapter culminates in the Drew–Gallauer universal property, characterizing SH as the universal $\mathbb{A}^1$ -invariant six-functor formalism on schemes.

Chapter 3 develops the comparisons of $SH(S)$ with classical structures: the BEM comparison pinning down how $SH(S)$ differs from its naive stabilization $\text{Shv}_{\text{Nis}, \mathbb{A}^1}(\text{Sm}_S, \text{Sp})$ through the inversion of the Tate object (giving the motivic Eilenberg–MacLane functor and the construction of the motivic K -theory spectrum KGL); the comparison between Voevodsky’s slice filtration and Postnikov-type filtrations, culminating in the slice conjecture and $\mathbb{A}^1$ -motivic cohomology; the comparison between motivic and topological orientations (Chern classes and formal group laws); and the motivic recognition principle of [Elm+21], identifying very effective motivic spectra over a perfect field with grouplike framed motivic spaces. The chapter closes with the zeroth slice of the motivic sphere: the slice theorem $s^0(\mathbb{1}_k) \simeq H\mathbb{Z}_k$ of Levine and Voevodsky, approached through the birational geometry of framed Hilbert schemes and through Levine’s homotopy coniveau tower. The chapter culminates in the universal Thom spectrum MGL (Section 3.5): its construction via the motivic J-homomorphism following Bachmann–Hoyois, the universality theorem of Panin–Pimenov–Röndigs, the Hopkins–Morel–Hoyois equivalence $MGL/(a_1, a_2, \dots)[1/c] \simeq H\mathbb{Z}[1/c]$ with its consequences for the slices of MGL and of the sphere, the motivic Conner–Floyd theorem recovering KGL , and the framed model of MGL via finite syntomic schemes.

These three chapters are the complete scope of the notes. Motivic cohomology appears only where it is intrinsic to motivic homotopy theory — notably as the zeroth slice of KGL and of the sphere, and as the graded theory attached to the slice filtration on KH — rather than as the subject of a separate sequel. The appendix Chapter A supplies the construction of norms on motivic spectra, building on the sifted-cocompletion technique of Chapter 1.

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Solid arrows indicate logical dependence; dashed arrows indicate supporting material.



STATUS

This version completes the planned exposition within the scope stated above. Completeness here means that every announced chapter and argument is present; it is not a claim of encyclopedic coverage, and errors may remain. During the writing of this document, the author has been struggling with serious mental health difficulties as well as severe dry eye syndrome, which has made it necessary to rely on Claude Code as a writing assistant. AI-assisted writing carries the risk of inaccurate or fabricated citations and of subtle mathematical errors; to mitigate this, the text has undergone repeated systematic verification passes in which citations and citation-backed claims were checked against the cited sources. Feedback from readers is at all times warmly welcomed.

CONVENTIONS

- **Higher categories.** Throughout these notes, we work in the framework of ∞ -categories. The word “category” means ∞ -category unless otherwise specified; we write “ordinary category” or “1-category” for the classical notion. We use the word *anima* for spaces/ ∞ -groupoids and write $\mathbf{An}$ for the category of anima. For objects X, Y of a category C , we write $\mathrm{Map}_C(X, Y)$ for the mapping anima; when C is stable, $\mathrm{map}_C(X, Y)$ denotes the mapping spectrum.
- **Everything is derived.** All functors are implicitly derived, and we do not decorate them: $\Gamma(X, -)$ and $\Gamma_\tau(-, \mathcal{F})$ denote derived global sections (classically written $R\Gamma$), $\otimes$ denotes the derived tensor product, and pullbacks and pushforwards of sheaves are the derived functors. When an underived classical construction is meant, we say so explicitly.
- **Schemes.** All schemes are assumed to be quasi-compact and quasi-separated (qcqs) unless otherwise stated, and Sm_S denotes the category of smooth S -schemes of finite presentation.
- **Twists and shifts.** For spectra and sheaves of spectra, $[n]$ denotes the n -fold suspension. Our $\mathbb{T}_S$ -graded conventions are *shift-free*: the j -th level of a motivic spectrum is $E(j) = \omega^\infty(E \otimes \mathbb{T}_S^{\otimes j})$, and the $\mathbb{A}^1$ -motivic cohomology spectrum $\mathbb{Z}(j)^\mathbb{A}$ carries no Tate desuspension. This differs from the conventions of BEM [BEM25] by a shift of $[2j]$ in weight j ; see Warning 3.1.5 and Remark 3.2.43 and the dictionary below.

NOTATION

The table below collects the most frequently used symbols, with pointers to their definitions.

Symbol	Meaning	Defined in
Sm_S	smooth finitely presented S -schemes	Chapter 1
$H(S), H_*(S)$	(pointed) motivic spaces	Definitions 1.3.1 and 1.4.1
$\mathbb{T}_S = \mathbb{P}^1/\infty$	Tate object; $\mathbb{T}_S \simeq S^{2,1} = S^1 \wedge \mathbb{G}_m$	Definition 1.4.3 and Proposition 1.4.6
$S^{p,q}$	bigraded spheres; stably $S^{i,j} = \mathbb{T}_S^{\otimes j}[i - 2j]$	Definition 1.4.5 and Section 2.1.4
$\mathrm{SH}(S)$	stable motivic homotopy category $H_*(S)[\mathbb{T}_S^{-1}]$	Definition 2.1.5
$M_S, \widetilde{M}_S$	(un)reduced motivic spectrum; $\sigma^\infty \dashv \omega^\infty$	Section 2.1.4
$\mathrm{MotSp}(S)$	Annala–Hoyois–Iwasa motivic spectra	Definition 2.1.27
$\mathrm{Shv}_{\mathrm{Nis}, \mathbb{A}^1}$	$\mathbb{A}^1$ -invariant Nisnevich sheaves of spectra	Section 3.1
$(\mathrm{Sm}_S, \mathrm{Sp})$	j -th level $\omega^\infty(E \otimes \mathbb{T}_S^{\otimes j})$	Warning 3.1.5
$E(j)$	bigraded homotopy sheaves (a simplicial, b Tate)	Definition 3.1.25
$\pi_{a,b}(E)$	Tate-graded homotopy sheaves; $\pi_i(E)_j = \pi_{i-j, -j}(E)$	Definition 3.1.26
$\mathrm{KGL}_S, \mathrm{kg}l_S$	motivic K -theory spectrum, its effective cover	Definition 3.1.11 and Construction 3.2.8
$\mathrm{SH}^{\mathrm{eff}}, \mathrm{SH}^{\mathrm{veff}}$	effective, very effective motivic spectra	Definitions 3.2.5 and 3.2.13
$\mathrm{Fil}_{\mathrm{slice}}^j, s^j$	slice filtration and slices	Construction 3.2.8
$\mathbb{Z}(j)_{\mathbb{A}^1},$		
$\mathrm{Fil}_{\mathbb{A}^1}^* \mathrm{KH}$	$\mathbb{A}^1$ -motivic cohomology and filtration	Definition 3.2.42
$H\mathbb{Z}_S = s^0 \mathrm{KGL}_S$	motivic Eilenberg–MacLane spectrum	Remark 3.2.44
$\mathrm{DM}(X)$	$H\mathbb{Z}$ -modules in $\mathrm{SH}(X)$	Definition 3.2.68
$\mathrm{Corr}^{\mathrm{fr}}(\mathrm{Sm}_S),$		
γ	framed correspondences	Definition 3.3.7
$H^{\mathrm{fr}}(S), \mathrm{SH}^{\mathrm{fr}}(S)$	framed motivic spaces, spectra	Definition 3.3.10
$\mathrm{Hilb}^{\mathrm{fr}}$	framed Hilbert scheme	Construction 3.4.5
L_{bir}^d	d -birational localization	Definition 3.4.1

The following dictionary translates between our conventions and those of BEM [BEM25]. Here $E \in \mathrm{SH}(S)$, and $\pi_{a,b}$ denotes the bigraded homotopy sheaves of Definition 3.1.25.

	this text	BEM [BEM25]
bigraded spheres		$S^{i,j} = \mathbb{T}_S^{\otimes j}[i - 2j]$ (identical)
j -th level	$E(j) = \omega^\infty(E \otimes \mathbb{T}_S^{\otimes j})$	$E(j)_{\text{BEM}} = E(j)[-2j]$
homotopy of levels	$\pi_i E(j) = \pi_{i-j, -j}(E)$	$\pi_i E(j)_{\text{BEM}} = \pi_{i+j, -j}(E)$; equivalently $\pi_{i+2j} E(j)$
	$\mathbb{Z}(j)^\mathbb{A} =$	$\mathbb{Z}(j)_{\text{BEM}}^\mathbb{A} =$
motivic cohomology	$\text{map}(\mathbb{1}, s^j \text{KGL})$	$\mathbb{Z}(j)^\mathbb{A}[-2j]$
cohomology groups	$\pi_{2j-p} \mathbb{Z}(j)^\mathbb{A}(X)$	$H^p(X, \mathbb{Z}(j)) = \pi_{-p} \mathbb{Z}(j)_{\text{BEM}}^\mathbb{A}(X)$
$\mathbb{A}^1$ -motivic filtration		$\text{Fil}_\mathbb{A}^\star \text{KH}$ (identical)

In short: in weight j , our objects are BEM's shifted by $[2j]$, and formulas migrate by absorbing the corresponding Tate shifts — for instance, BEM's $\text{gr}_\mathbb{A}^j \text{KH} \simeq \mathbb{Z}(j)_{\text{BEM}}^\mathbb{A}[2j]$ reads $\text{gr}_\mathbb{A}^j \text{KH} \simeq \mathbb{Z}(j)^\mathbb{A}$ here. See Warning 3.1.5 and Remark 3.2.43 for further discussion.

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UNSTABLE MOTIVIC HOMOTOPY THEORY

The starting point of motivic homotopy theory is a striking analogy between algebraic geometry and topology. In topology, for any paracompact Hausdorff space X , the projection $X \times I \rightarrow X$ (with $I = [0, 1]$) induces a bijection

$$\pi_0 \mathrm{Vect}_r^{\mathrm{top}}(X) \xrightarrow{\sim} \pi_0 \mathrm{Vect}_r^{\mathrm{top}}(X \times I).$$

Vector bundles, and many other invariants, are insensitive to the cylinder direction. The classical Quillen–Suslin theorem [Qui76; Sus76] provides the algebraic counterpart: every finitely generated projective module over $k[x_1, \dots, x_n]$ is free; equivalently, the pullback

$$\pi_0 \mathrm{Vect}_r(\mathrm{Spec}(k)) \xrightarrow{\sim} \pi_0 \mathrm{Vect}_r(\mathbb{A}_k^n)$$

is a bijection. Under this analogy, the affine line $\mathbb{A}^1$ plays the role of the interval I : both serve as “cylinder objects” in their respective categories, suggesting a homotopy theory for algebraic geometry in which the projection $\mathbb{A}^1 \times X \rightarrow X$ is declared a homotopy equivalence.

Supporting this intuition, many cohomology theories on smooth schemes are $\mathbb{A}^1$ -invariant. For a smooth scheme X over a field k :

- **Chow groups:** $\mathrm{CH}^*(X) \xrightarrow{\sim} \mathrm{CH}^*(X \times \mathbb{A}^1)$;
- **Grothendieck group:** $K_0(X) \xrightarrow{\sim} K_0(X \times \mathbb{A}^1)$;
- **Étale cohomology** (for ℓ invertible in k): $H_{\mathrm{\acute{e}t}}^*(X, \mu_\ell) \xrightarrow{\sim} H_{\mathrm{\acute{e}t}}^*(X \times \mathbb{A}^1, \mu_\ell)$.

Remark 1.0.1 ($\mathbb{A}^1$ -invariance fails on non-affine schemes). The functor $\pi_0 \mathrm{Vect}_r : \mathrm{Sm}_k^{\mathrm{op}} \rightarrow \mathrm{Set}$ is *not* $\mathbb{A}^1$ -invariant in general. Take the rank-2 bundle E on $\mathbb{P}^1 \times \mathbb{A}^1$ obtained by gluing the trivial bundle on the standard charts $U_0 = \mathrm{Spec}(k[x]) \times \mathbb{A}^1$ and $U_1 = \mathrm{Spec}(k[x^{-1}]) \times \mathbb{A}^1$ via the transition matrix

$$g_{01}(x, t) = \begin{pmatrix} x & t \\ 0 & x^{-1} \end{pmatrix} \in \mathrm{GL}_2(k[x, x^{-1}, t])$$

(here a section of E is a pair of sections (s_0, s_1) of the trivial bundles over the two charts with $s_1 = g_{01} s_0$ on the overlap, so that the transition function x defines $\mathcal{O}(-1)$). At $t = 0$, the matrix is diagonal, giving $E|_{\mathbb{P}^1 \times \{0\}} \simeq \mathcal{O}(1) \oplus \mathcal{O}(-1)$. At $t = 1$, the matrix defines a non-trivial extension

$$0 \longrightarrow \mathcal{O}(-1) \longrightarrow E|_{\mathbb{P}^1 \times \{1\}} \longrightarrow \mathcal{O}(1) \longrightarrow 0$$

whose class in $\mathrm{Ext}^1(\mathcal{O}(1), \mathcal{O}(-1)) = H^1(\mathbb{P}^1, \mathcal{O}(-2)) \cong k$ is non-zero; by Grothendieck’s splitting theorem, this extension is isomorphic as a vector bundle to $\mathcal{O} \oplus \mathcal{O}$. Since $E|_{\mathbb{P}^1 \times \{0\}} \neq E|_{\mathbb{P}^1 \times \{1\}}$ on $\mathbb{P}^1$, the bundle E cannot lie in the image of $\pi_0 \mathrm{Vect}_2(\mathbb{P}^1) \rightarrow \pi_0 \mathrm{Vect}_2(\mathbb{P}^1 \times \mathbb{A}^1)$.

Nevertheless—and this is the deep theorem of Morel [Mor12], simplified by Asok–Hoyois–Wendt [AHW17]—if we forcibly $\mathbb{A}^1$ -localize Vect_r , the resulting functor still computes the correct values on *smooth affine* schemes. This affine representability theorem is one of the cornerstone applications of motivic homotopy theory.

Overview of the construction. Fix a quasi-compact quasi-separated scheme S , and let Sm_S denote the category of smooth S -schemes of finite presentation. Our goal is to construct the **motivic homotopy category** $\mathrm{H}(S)$: an enlargement of Sm_S in which the projections $\mathbb{A}^1 \times_S X \rightarrow X$ become equivalences. The construction proceeds in three universal steps, each with a clear geometric meaning:

- (i) **Sifted cocompletion.** Embed Sm_S into $\mathcal{P}_\Sigma(\mathrm{Sm}_S)$, the category of presheaves of anima that convert finite coproducts to finite products.
- (ii) **Nisnevich descent.** Force a chosen class of étale-like covers (the *Nisnevich covers*) to become equivalences, yielding $\mathrm{Shv}_{\mathrm{Nis}}(\mathrm{Sm}_S)$.
- (iii) **$\mathbb{A}^1$ -localization.** Force the projections $\mathbb{A}^1 \times X \rightarrow X$ to become equivalences, yielding the motivic homotopy category $\mathrm{H}(S)$.

Remark 1.0.2 (Why $\mathcal{P}_\Sigma$ rather than PShv ?). One could equally well start from the full presheaf category $\mathrm{PShv}(\mathrm{Sm}_S)$. The advantage of $\mathcal{P}_\Sigma(\mathrm{Sm}_S)$ is threefold:

- (i) Since Sm_S is extensive, $\mathcal{P}_\Sigma(\mathrm{Sm}_S) \simeq \mathrm{Shv}_\sqcup(\mathrm{Sm}_S)$ [BH21, Lemma 2.4], which is already a topos for the coproduct topology.
- (ii) The Yoneda embedding $\mathrm{Sm}_S \hookrightarrow \mathcal{P}_\Sigma(\mathrm{Sm}_S)$ preserves finite coproducts, and every object is a sifted colimit of representables.
- (iii) The universal property of $\mathcal{P}_\Sigma$ as the free sifted cocompletion is essential to the construction of norm functors in Appendix A; see [BH21, §3].

Since finite coproduct decompositions are themselves Nisnevich covers, we have natural inclusions

$$\mathrm{Shv}_{\mathrm{Nis}}(\mathrm{Sm}_S) \subset \mathcal{P}_\Sigma(\mathrm{Sm}_S) \subset \mathrm{PShv}(\mathrm{Sm}_S),$$

and we write $L_{\mathrm{Nis}} : \mathcal{P}_\Sigma(\mathrm{Sm}_S) \rightarrow \mathrm{Shv}_{\mathrm{Nis}}(\mathrm{Sm}_S)$ for the Nisnevich sheafification functor.

1.1. THE NISNEVICH TOPOLOGY

The Nisnevich topology, also called the completely decomposed topology, sits strictly between Zariski and étale. In motivic homotopy theory, its traditional motivation is summarized by two complementary optimality slogans: it is fine enough for purity and still coarse enough for algebraic K-theory to be representable (Remark 1.1.1). These slogans compare the standard topologies used here; they are not meant as an abstract uniqueness theorem among all Grothendieck topologies.

1.1.1. Three topologies

For S qcqs, there are three commonly used Grothendieck topologies on Sm_S :

$$\mathrm{Zar} \subset \mathrm{Nis} \subset \mathrm{ét}.$$

- **Zariski:** covers are jointly surjective open immersions.
- **Étale:** covers are jointly surjective étale morphisms.
- **Nisnevich:** covers are jointly surjective étale morphisms such that every point of the base admits a preimage inducing an isomorphism of residue fields.

The functor-of-points perspective makes the Nisnevich condition transparent. For a smooth k -scheme X , an étale cover $\{U_i \rightarrow X\}$ need not satisfy $\coprod_i U_i(k) \twoheadrightarrow X(k)$: the preimage of a

k -rational point might only exist over a finite separable extension L/k . A Nisnevich cover excludes this phenomenon at every point: for every field K , every K -point of X lifts to some U_i . In fact, joint surjectivity on K -points for all fields K characterizes Nisnevich covers among étale families [BH21, Proposition A.2(a)].

Remark 1.1.1 (Why Nisnevich?). Among the standard Zariski–Nisnevich–étale choices, the role of the Nisnevich topology is motivated by two complementary considerations:

- **Purity.** The purity theorem asserts that for a closed immersion $Z \hookrightarrow X$ in Sm_S with normal bundle ν_Z , there is an $\mathbb{A}^1$ -local equivalence $X/(X \setminus Z) \simeq \mathrm{Th}(\nu_Z)$. Zariski descent alone is too coarse for the usual proof.
- **Representability of K-theory.** Non-connective K-theory $K: \mathrm{Sm}_S^{\mathrm{op}} \rightarrow \mathrm{Sp}$ satisfies Nisnevich descent but not étale descent in general.

1.1.2. Nisnevich covers

Definition 1.1.2 (Nisnevich cover). A family of étale morphisms $(f_i: U_i \rightarrow X)_{i \in I}$ in Sm_S is a **Nisnevich cover** if for every $x \in X$, there exists $i \in I$ and $u \in U_i$ with $f_i(u) = x$ such that the induced extension $\kappa(x) \xrightarrow{\sim} \kappa(u)$ of residue fields is an isomorphism.

Remark 1.1.3. The term “completely decomposed” originates in arithmetic: a prime ideal $\mathfrak{p}$ of $\mathcal{O}_K$ is “completely decomposed” in an étale cover precisely when its residue field does not extend.

Theorem 1.1.4 (Equivalent characterizations; cf. [BH21, Lemma A.1, Proposition A.2]). Let X be qcqs and $(f_i: U_i \rightarrow X)_{i \in I}$ a family of étale morphisms. The following are equivalent:

- (**Rational points**) For every $x \in X$, there exists i and $u \in U_i$ with $f_i(u) = x$ and $\kappa(x) \xrightarrow{\sim} \kappa(u)$.
- (**Finite stratification**) There is a chain of finitely presented closed subschemes

$$\emptyset = Z_0 \subset Z_1 \subset \cdots \subset Z_n = X$$

such that some f_i admits a section over each stratum $Y_j = Z_j \setminus Z_{j-1}$.

- (**Refinement by Nisnevich squares**) There is a stratification as above and a finitely presented étale family $(g_j: V_j \rightarrow X)_{j=1}^n$ refining (f_i) , with each g_j restricting to an isomorphism over Y_j .

Proof. The implications (iii) $\Rightarrow$ (ii) $\Rightarrow$ (i) are immediate. We prove (i) $\Rightarrow$ (iii).

Replacing each U_i by an affine open cover, we may assume all U_i are quasi-compact and separated, hence all f_i are of finite presentation. Consider the collection of closed subschemes $Y \subseteq X$ for which (iii) fails; we show this collection is empty by Zorn’s lemma.

Let $(Y_\alpha)_{\alpha \in A}$ be a descending chain and $Y = \bigcap_{\alpha} Y_\alpha$. If (iii) holds for Y , then by finite presentation it holds for some Y_α (limits of finitely presented data along cofiltered systems). Hence if (iii) fails, there exists a minimal closed $Y \subseteq X$ for which it fails. Replacing X by Y , we may assume (iii) fails for X but holds for all proper closed subschemes.

Clearly $X \neq \emptyset$. Take a maximal point $x \in X$ (a generic point of an irreducible component, so that $\mathcal{O}_{X,x}$ is zero-dimensional). Since $\mathcal{O}_{X,x}^{\mathrm{red}} = \kappa(x)$ is a field, the local ring $\mathcal{O}_{X,x}$ is henselian. By (i), for some i there is a morphism $\mathrm{Spec}(\kappa(x)) \rightarrow U_i$ lifting $\mathrm{Spec}(\kappa(x)) \rightarrow X$; by [Gro67, Théorème 18.5.17], this section extends to a section $\mathrm{Spec}(\mathcal{O}_{X,x}) \rightarrow U_i$. Passing to a limit, we obtain a quasi-compact open neighborhood $V \subseteq X$ of x and a section $s: V \rightarrow U_i$.

Take a finitely presented closed $Z \subset X$ with $V = X \setminus Z$. By minimality, (iii) holds for Z : there is a stratification $\emptyset = Z_0 \subset \cdots \subset Z_n = Z$ and a refining family $(h_j: W_j \rightarrow Z)$ with

the required properties. For each j , the refinement map $W_j \rightarrow U_{i_j} \times_X Z$ restricts over Y_j , where h_j is an isomorphism, to a section $t_j: Y_j \rightarrow U_{i_j} \times_X Y_j$ of the projection. Sections of étale morphisms are open immersions, so $t_j(Y_j)$ is open in $U_{i_j} \times_X Y_j$ and hence in the closed subscheme $U_{i_j} \times_X Z_j \subseteq U_{i_j}$, since $Y_j \subseteq Z_j$ is open. Choose a quasi-compact open $V_j \subseteq U_{i_j}$ with $V_j \cap (U_{i_j} \times_X Z_j) = t_j(Y_j)$; then $g_j := f_{i_j}|_{V_j}: V_j \rightarrow X$ is finitely presented étale and restricts to an isomorphism over Y_j . Extending the chain to $\emptyset = Z_0 \subset \cdots \subset Z_n \subset Z_{n+1} = X$ with $V_{n+1} = V$ and $g_{n+1} = f_i \circ s$ realizes (iii) for X , contradicting minimality. $\square$

For a qcqs scheme X , the **small Nisnevich topos** X_{Nis} of X is the topos of Nisnevich sheaves of anima on the category Et_X^{fp} of finitely presented étale X -schemes.

Proposition 1.1.5 ([BH21, A.3]). *Let X be a qcqs scheme.*

(i) *For every $Y \in \text{Et}_X^{\text{fp}}$ and every point $y \in Y$, the functor*

$$F \longmapsto F(\text{Spec } \mathcal{O}_{Y,y}^h),$$

where $\mathcal{O}_{Y,y}^h$ is the henselization of Y at y , is the inverse image functor of a point of the hyper-completion $X_{\text{Nis}}^\wedge$. These points form a conservative family.

(ii) *X_{Nis} is coherent in the sense of [Lur18, Definition A.2.1.6].*

1.1.3. Nisnevich squares

The key feature of the Nisnevich topology in practice is that it is generated by a particularly simple class of covers.

Definition 1.1.6 (Nisnevich square). A pullback square in Sm_S

$$\begin{array}{ccc} U \times_X V & \longrightarrow & V \\ \downarrow & \lrcorner & \downarrow p \\ U & \xhookrightarrow{j} & X \end{array}$$

is a **Nisnevich square** if

- j is an open immersion;
- p is étale;
- p induces an isomorphism $p^{-1}(X \setminus U)_{\text{red}} \xrightarrow{\sim} (X \setminus U)_{\text{red}}$.

From the functor-of-points perspective, a Nisnevich square decomposes the field-valued points of X into two classes: those landing in the open U , handled by j ; and those in the closed complement $X \setminus U$, covered by p without any residue field extension. More generally, an R -point of X with R henselian local either factors through U or sends the closed point into $X \setminus U$, in which case it lifts uniquely to V — compare the conservative family of points in Proposition 1.1.5.

Lemma 1.1.7. *The family $\{j: U \hookrightarrow X, p: V \rightarrow X\}$ is a Nisnevich cover of X .*

Proof. For $x \in U$, j has x in its image with trivial residue extension. For $x \in X \setminus U$, the assumption gives $v \in p^{-1}(x)$ with $\kappa(v) \simeq \kappa(x)$. $\square$

The deeper result is that Nisnevich squares *generate* the Nisnevich topology:

Proposition 1.1.8 (Brown–Gersten). *Let $\mathcal{D}$ be an ∞ -category admitting small limits. A $\mathcal{D}$ -valued presheaf $\mathcal{F}$ on Sm_S is a Nisnevich sheaf if and only if $\mathcal{F}(\emptyset)$ is terminal and $\mathcal{F}$ sends every Nisnevich square to a pullback square:*

$$\mathcal{F}(X) \xrightarrow{\sim} \mathcal{F}(V) \times_{\mathcal{F}(U \times_X V)} \mathcal{F}(U).$$

Proof. By [BH21, Proposition A.2(d)], the Nisnevich topology on qcqs schemes — and hence its restriction to Sm_S — is generated by the empty cover and distinguished Nisnevich squares. Voevodsky’s descent criterion for this cd-structure [AHW17, Theorem 3.2.5] identifies descent with the conditions that $\mathcal{F}(\emptyset)$ be terminal and that $\mathcal{F}$ send every distinguished square to a pullback. The criterion is limit-theoretic and therefore applies to presheaves valued in any ∞ -category $\mathcal{D}$ admitting small limits. $\square$

1.1.4. Examples

Example 1.1.9 (Zariski covers are Nisnevich). Open immersions induce identity maps on residue fields, so Zariski covers are automatically Nisnevich. The standard cover $\{\mathbb{A}^1, \mathbb{A}^1\} \rightarrow \mathbb{P}^1$ corresponds to the Nisnevich square

$$\begin{array}{ccc} \mathbb{G}_m & \longrightarrow & \mathbb{A}^1 \\ \downarrow & & \downarrow \\ \mathbb{A}^1 & \longrightarrow & \mathbb{P}^1. \end{array}$$

Example 1.1.10 (The m -th power map). Let $m \geq 1$, let k be a field with $\mathrm{char}(k) \nmid m$, and let $a \in k^\times$. Consider

$$\begin{aligned} f: (\mathbb{A}^1 \setminus \{a\}) \sqcup (\mathbb{A}^1 \setminus \{0\}) &\longrightarrow \mathbb{A}^1, \\ \text{first component: } x &\longmapsto x, \\ \text{second component: } y &\longmapsto y^m. \end{aligned}$$

The first component is an open immersion; the second is étale. This f is a Nisnevich cover if and only if a has an m -th root in k .

Example 1.1.11 (Non-connective algebraic K-theory is a Nisnevich sheaf). By Brown–Gersten it suffices to check Nisnevich excision (the empty case is trivial). For a Nisnevich square

$$\begin{array}{ccc} W & \longrightarrow & V \\ \downarrow & & \downarrow \\ U & \xrightarrow{j} & Y, \end{array}$$

Perfect complexes form an fpqc stack, hence in particular a Nisnevich sheaf of stable categories. Applying this descent statement to the distinguished square gives a pullback square

$$\begin{array}{ccc} \mathrm{Perf}(W) & \longleftarrow & \mathrm{Perf}(V) \\ \uparrow & & \uparrow \\ \mathrm{Perf}(U) & \longleftarrow & \mathrm{Perf}(Y). \end{array}$$

Setting $Z = Y \setminus U = V \setminus W$, let $\mathrm{Perf}_Z(Y) := \mathrm{fib}(\mathrm{Perf}(Y) \rightarrow \mathrm{Perf}(U))$, and define $\mathrm{Perf}_Z(V)$ similarly. Nisnevich excision identifies these two supported categories. The resulting rows

are Karoubi sequences — fiber sequences of stable categories whose quotient becomes an equivalence after idempotent completion. Since non-connective K-theory sends Karoubi sequences to fiber sequences of spectra [TT90; BGT13], the K-theory square is a pullback. (For the connective cover k this fails in general: connective truncation does not preserve the localization fiber sequence when boundary maps meet negative K-groups.)

1.1.5. Milnor excision

Closely related to Nisnevich descent, though in general a logically independent condition, is the property of *Milnor excision*, formulated in terms of ring-theoretic squares.

Definition 1.1.12 (Milnor square). A **Milnor square** (or *excision square*) is a commutative square of rings

$$\begin{array}{ccc} A & \longrightarrow & A' \\ \downarrow & & \downarrow \\ A/I & \longrightarrow & A'/I' \end{array}$$

that is both cartesian and cocartesian as a square of A -modules; equivalently, the induced map of non-unital A -algebras $I \rightarrow I'$ is an isomorphism [BEM25, §2.2.4].

Example 1.1.13 ((Henselian) valuation squares). For a valuation ring V and prime ideal $\mathfrak{p} \subset V$, the square

$$\begin{array}{ccc} V & \longrightarrow & V_{\mathfrak{p}} \\ \downarrow & & \downarrow \\ V/\mathfrak{p} & \longrightarrow & \kappa(\mathfrak{p}) \end{array}$$

is a Milnor square; see [BEM25, §2.2.4] and the references therein. When V is moreover henselian (as a local ring), we call it a **henselian valuation square**; henselianness ensures that $(V, \mathfrak{p})$ and $(V_{\mathfrak{p}}, \mathfrak{p}V_{\mathfrak{p}})$ are henselian pairs, and it is these squares that enter the excision criterion below.

A Milnor square is not a Nisnevich square: the map $\mathrm{Spec}(A/I) \rightarrow \mathrm{Spec}(A)$ is a closed immersion rather than an étale morphism. A functor $\mathcal{F}$ valued in an ∞ -category with finite limits is said to satisfy **Milnor excision** for a given Milnor square if it sends the square to a cartesian square. This is in general incomparable with Nisnevich descent: a Milnor square involves a closed immersion rather than an étale morphism, so neither condition formally implies the other, and a Nisnevich sheaf need not be Milnor excisive. The genuine comparison instead runs through the finer cdh topology. More precisely, if B has finite valuative dimension and $\mathcal{F}: (\mathrm{Sch}_B^{\mathrm{qcqs}})^{\mathrm{op}} \rightarrow \mathrm{Sp}$ is a finitary cdh sheaf, then it suffices for Milnor excision to check the square of Example 1.1.13 for every finite-rank henselian valuation ring V equipped with a B -algebra structure and every prime ideal $\mathfrak{p} \subset V$; see [EHIK21, Theorem 3.3.4 and Remark 3.3.3] and [BEM25, §2.2.4].

1.2. $\mathbb{A}^1$ -LOCALIZATION

Write $\mathrm{PShv}_{\mathbb{A}^1}(\mathrm{Sm}_S) \subset \mathrm{PShv}(\mathrm{Sm}_S)$ for the full subcategory of $\mathbb{A}^1$ -**invariant** presheaves: those $\mathcal{F}$ with $\mathcal{F}(U) \xrightarrow{\sim} \mathcal{F}(U \times_S \mathbb{A}^1)$ for all $U \in \mathrm{Sm}_S$. The inclusion admits a left adjoint

$$L_{\mathbb{A}^1}: \mathrm{PShv}(\mathrm{Sm}_S) \longrightarrow \mathrm{PShv}_{\mathbb{A}^1}(\mathrm{Sm}_S),$$

called the $\mathbb{A}^1$ -**localization functor**.

Definition 1.2.1 ($\mathbb{A}^1$ -local equivalence and $\mathbb{A}^1$ -homotopy). Let $f: \mathcal{F} \rightarrow \mathcal{G}$ be a morphism in $\mathrm{PShv}(\mathrm{Sm}_S)$.

- f is an $\mathbb{A}^1$ -**local equivalence** if $L_{\mathbb{A}^1}(f)$ is an isomorphism.
- An **elementary $\mathbb{A}^1$ -homotopy** between maps $f, g: \mathcal{F} \rightarrow \mathcal{G}$ is a map $H: \mathcal{F} \times \mathbb{A}^1 \rightarrow \mathcal{G}$ with $H \circ i_0 \simeq f$ and $H \circ i_1 \simeq g$, where $i_0, i_1: \mathcal{F} \rightarrow \mathcal{F} \times \mathbb{A}^1$ are the sections $i_s = (\mathrm{id}_{\mathcal{F}}, s)$ induced by the points $s \in \{0, 1\} \subset \mathbb{A}^1$. Two maps are $\mathbb{A}^1$ -**homotopic** if they are connected by a finite zigzag of elementary $\mathbb{A}^1$ -homotopies; the zigzag is genuinely needed, as two elementary $\mathbb{A}^1$ -homotopies cannot in general be concatenated algebraically (cf. [Hoy, Definition 2.1]).
- f is an $\mathbb{A}^1$ -**homotopy equivalence** if there exists g with fg and gf each $\mathbb{A}^1$ -homotopic to the appropriate identity.

Proposition 1.2.2. Every $\mathbb{A}^1$ -homotopy equivalence is an $\mathbb{A}^1$ -local equivalence. In particular, the projection $p: \mathcal{F} \times \mathbb{A}^1 \rightarrow \mathcal{F}$ is an $\mathbb{A}^1$ -local equivalence for every $\mathcal{F}$.

Proof. We first show $p: \mathcal{F} \times \mathbb{A}^1 \rightarrow \mathcal{F}$ is an $\mathbb{A}^1$ -homotopy equivalence for every presheaf $\mathcal{F}$. The zero section $s: \mathcal{F} \rightarrow \mathcal{F} \times \mathbb{A}^1$ satisfies $p \circ s = \mathrm{id}_{\mathcal{F}}$. The reverse composite $s \circ p$ is $\mathbb{A}^1$ -homotopic to $\mathrm{id}_{\mathcal{F} \times \mathbb{A}^1}$ via

$$(\mathcal{F} \times \mathbb{A}^1) \times \mathbb{A}^1 \xrightarrow{\mathrm{id}_{\mathcal{F}} \times \mu} \mathcal{F} \times \mathbb{A}^1,$$

where $\mu: \mathbb{A}^1 \times \mathbb{A}^1 \rightarrow \mathbb{A}^1$ is the ring multiplication: at $t = 1$ this gives $\mathrm{id}_{\mathcal{F} \times \mathbb{A}^1}$, at $t = 0$ it gives $s \circ p$. (Both the section and the homotopy are induced by maps of representables, so no descent to representables is needed.)

Now suppose $a, b: \mathcal{F} \rightarrow \mathcal{G}$ are $\mathbb{A}^1$ -homotopic; by induction along the zigzag we may assume they are related by an elementary homotopy H . We claim $L_{\mathbb{A}^1}(p)$ is an isomorphism for every $\mathcal{F}$: for representable $\mathcal{F} = U$ this holds by the very definition of $\mathbb{A}^1$ -invariance ($\mathrm{Map}(U, \mathcal{Z}) \simeq \mathrm{Map}(U \times \mathbb{A}^1, \mathcal{Z})$ for all $\mathbb{A}^1$ -invariant $\mathcal{Z}$), and in general because $\mathcal{F} \times \mathbb{A}^1 \simeq \mathrm{colim}_i (U_i \times \mathbb{A}^1)$ for any presentation $\mathcal{F} = \mathrm{colim}_i U_i$ by universality of colimits, and $L_{\mathbb{A}^1}$ preserves colimits. Since $p \circ i_0 = p \circ i_1 = \mathrm{id}$, applying $L_{\mathbb{A}^1}$ gives $L_{\mathbb{A}^1}(i_0) = L_{\mathbb{A}^1}(i_1)$. Hence

$$L_{\mathbb{A}^1}(a) = L_{\mathbb{A}^1}(H) \circ L_{\mathbb{A}^1}(i_0) = L_{\mathbb{A}^1}(H) \circ L_{\mathbb{A}^1}(i_1) = L_{\mathbb{A}^1}(b).$$

The result follows from the definition of homotopy equivalence. $\square$

1.2.1. Interval objects

So far we have two ways of inverting $\mathbb{A}^1$ on $\mathrm{PShv}(\mathrm{Sm}_S)$: localize at the $\mathbb{A}^1$ -local equivalences, or localize at the (a priori smaller) class of $\mathbb{A}^1$ -homotopy equivalences. A third construction, the singular complex built from the algebraic simplices $\Delta_S^\bullet$, will appear in Theorem 1.2.11 below. Do all three constructions produce the same category? Following the exposition of [Hoy], the answer is yes, and this is a formal consequence of $\mathbb{A}^1$ being an **interval object**.

Definition 1.2.3 (Bipointed object). Let C be a category with finite products. A **bipointed object** is an object $I \in C$ with two global sections $0, 1: * \rightarrow I$.

Definition 1.2.4 (Three homotopy categories). Let C have finite products and I be a bipointed object.

- (i) **I -localization** (nullification): formally invert all projections $X \times I \rightarrow X$:

$$C/I := C[\{X \times I \rightarrow X\}^{-1}].$$

(ii) **Localization at I -homotopy equivalences:**

$$C[\mathrm{hoeq}_I^{-1}].$$

(iii) **I -homotopy category**, via simplicial enrichment:

$$\mathrm{Ho}(C, I), \quad \mathrm{Map}_{\mathrm{Ho}(C, I)}(X, Y) = \mathrm{colim}_{[n] \in \Delta^{\mathrm{op}}} \mathrm{Map}_C(X \times I^n, Y).$$

Here $[n] \mapsto I^n$ carries a canonical *cosimplicial* structure, which is genuinely non-obvious (the powers of a bipointed object naturally form a cubical object): the coface $d^i: I^{n-1} \rightarrow I^n$ prepends the section 1 (for $i = 0$), duplicates the i -th coordinate via the diagonal (for $0 < i < n$), or appends the section 0 (for $i = n$), and the codegeneracies omit coordinates. Equivalently, $I^\bullet$ is the right Kan extension of the bipointed object along $\Delta_{\leq 1} \hookrightarrow \Delta$ [Hoy, Constructions 1.3–1.4].

In general, there are canonical functors

$$C[\mathrm{hoeq}_I^{-1}] \longrightarrow \mathrm{Ho}(C, I) \longrightarrow C/I,$$

neither of which is generally an equivalence.

Definition 1.2.5 (Interval object). A bipointed object $I \in C$ is an **interval object** if

- (i) $X \times I$ exists for all $X \in C$;
- (ii) I is I -contractible: $I \rightarrow *$ is an I -homotopy equivalence.

Example 1.2.6. • The unit interval $[0, 1] \in \mathrm{Top}$ with multiplication $\mu(x, y) = xy$.

- The affine line $\mathbb{A}^1 \in \mathrm{Sm}_S$ with ring multiplication.
- The real line $\mathbb{R}$ and complex plane $\mathbb{C}$ in Top , both with multiplication.

The notion of interval object is due to Voevodsky, and the following unification theorem to Morel–Voevodsky; we follow the ∞ -categorical formulation of [Hoy], cf. [MV99, §2.3].

Theorem 1.2.7 (Morel–Voevodsky; see [Hoy, Propositions 4.3 and 4.6]). *For any category C and interval object $I \in C$,*

$$C[\mathrm{hoeq}_I^{-1}] \xrightarrow{\sim} \mathrm{Ho}(C, I) \xrightarrow{\sim} C/I.$$

Proof sketch. The key observation is that I -contractibility makes every projection $\mathrm{pr}_1: X \times I \rightarrow X$ an I -homotopy equivalence: the zero section $(\mathrm{id}_X, 0)$ is its I -homotopy inverse, and the required zigzag of I -homotopies is obtained from one between id_I and the constant map at 0 by taking the product with id_X . Conversely, after the projections have been inverted, I -homotopic maps become equal, so every I -homotopy equivalence becomes invertible. This identifies the first and third localizations.

The comparison with the middle term uses the full cosimplicial coherence. After passing to presheaves, the I -contractibility of Δ_I^n makes

$$F \longrightarrow \underline{\mathrm{Map}}(\Delta_I^n, F)$$

an I -homotopy equivalence for every n . Consequently the unit

$$F \longrightarrow \mathrm{colim}_{[n] \in \Delta^{\mathrm{op}}} \underline{\mathrm{Map}}(\Delta_I^n, F)$$

is an I -equivalence; it need not itself be an I -homotopy equivalence. Lemma 4.5 of [Hoy] shows that the target is I -invariant, and [Hoy, Proposition 4.6] therefore identifies it with $L_I F$. This yields $\mathrm{Ho}(C, I) \simeq C/I$. $\square$

Corollary 1.2.8. *The affine line $\mathbb{A}^1$ with sections 0, 1 and ring multiplication is an interval object in Sm_S , in $\mathcal{P}_\Sigma(\mathrm{Sm}_S)$, and in $\mathrm{PShv}(\mathrm{Sm}_S)$. Hence the three a priori distinct constructions of §1.2 produce the same category, on $\mathrm{PShv}(\mathrm{Sm}_S)$ and likewise on $\mathcal{P}_\Sigma(\mathrm{Sm}_S)$:*

- (i) $\mathbb{A}^1$ -localization (inverting $\mathcal{F} \times \mathbb{A}^1 \rightarrow \mathcal{F}$);
- (ii) localization at $\mathbb{A}^1$ -homotopy equivalences;
- (iii) the singular complex $L_{\mathbb{A}^1}$.

Remark 1.2.9. The multiplication $\mu(x, y) = xy$ on $\mathbb{A}^1$ provides the $\mathbb{A}^1$ -homotopy from the constant map at 0 (at $y = 0$) to $\mathrm{id}_{\mathbb{A}^1}$ (at $y = 1$). The two sections of the bipointed structure play distinct roles: 1 is the multiplicative *unit*, at which the homotopy realizes the identity, and 0 is the *absorbing element*, at which it collapses to the constant map. Both are needed to exhibit $\mathbb{A}^1$ as an interval object.

1.2.2. Explicit formula for $L_{\mathbb{A}^1}$

Corollary 1.2.8 identifies $\mathbb{A}^1$ -localization abstractly with the $\mathbb{A}^1$ -homotopy category

$$\mathrm{Ho}(\mathrm{PShv}(\mathrm{Sm}_S), \mathbb{A}^1).$$

Its mapping anima are computed by

$$\mathrm{colim}_{\Delta^{\mathrm{op}}} \mathrm{Map}(- \times \mathbb{A}^\bullet, -).$$

We now convert this into an explicit “singular complex” formula, modeled on its topological namesake.

Definition 1.2.10 (Algebraic simplex). The standard n -simplex over S is

$$\Delta_S^n := \mathrm{Spec}(\mathcal{O}_S[T_0, \dots, T_n]/(T_0 + \dots + T_n - 1)) \subset \mathbb{A}_S^{n+1}.$$

Face and degeneracy maps are defined as in topology, yielding a cosimplicial scheme $\Delta_S^\bullet: \Delta \rightarrow \mathrm{Sch}_S$; this cosimplicial scheme goes back to Reotor [Rec71, Remark 2.5]. Note $\Delta_S^n \simeq \mathbb{A}_S^n$ as S -schemes via the cumulative coordinates $x_j = \sum_{i \geq j} T_i$; these isomorphisms even assemble into a canonical isomorphism of cosimplicial schemes between $\Delta_S^\bullet$ and the cosimplicial extension $[n] \mapsto (\mathbb{A}_S^1)^n$ of the interval $\mathbb{A}^1$ used in Corollary 1.2.8 — indeed there is a *unique* cosimplicial object of the form $[n] \mapsto \mathbb{A}^n$ in the category of affine spaces [Hoy, Example 4.9]. Beware that the face maps are *not* the coordinate inclusions $\mathbb{A}_S^{n-1} \subset \mathbb{A}_S^n$.

Theorem 1.2.11 (Explicit formula for $\mathbb{A}^1$ -localization; cf. [MV99, §2.3], [Hoy, Proposition 4.6]). For $\mathcal{F} \in \mathrm{PShv}(\mathrm{Sm}_S)$ and $X \in \mathrm{Sm}_S$,

$$(L_{\mathbb{A}^1} \mathcal{F})(X) \simeq \mathrm{colim}_{[n] \in \Delta^{\mathrm{op}}} \mathcal{F}(X \times \Delta_S^n).$$

Proof. Write $\mathcal{F}'(X) := \mathrm{colim}_{[n] \in \Delta^{\mathrm{op}}} \mathcal{F}(X \times \Delta_S^n)$.

$\mathbb{A}^1$ -invariance. The projection $p: X \times \mathbb{A}^1 \rightarrow X$ and zero section s induce maps on the cosimplicial objects $X \times \Delta_S^\bullet$ and $X \times \mathbb{A}^1 \times \Delta_S^\bullet$. We have $s^* \circ p^* = \mathrm{id}$, and the multiplication $\mu: \mathbb{A}^1 \times \mathbb{A}^1 \rightarrow \mathbb{A}^1$ defines an $\mathbb{A}^1$ -homotopy $p^* s^* \simeq \mathrm{id}$. Since $\Delta_S^1 \simeq \mathbb{A}_S^1$, this $\mathbb{A}^1$ -homotopy translates into a simplicial homotopy on $\mathcal{F}(X \times \Delta_S^\bullet)$ between $p^* s^*$ and id , which becomes an equivalence after $\mathrm{colim}_{\Delta^{\mathrm{op}}}$. Hence $p^*: \mathcal{F}'(X) \xrightarrow{\sim} \mathcal{F}'(X \times \mathbb{A}^1)$.

Universal property. Each unit map $\mathcal{F} \rightarrow \mathcal{F}(- \times \Delta_S^n)$ is an $\mathbb{A}^1$ -homotopy equivalence: $\Delta_S^n \simeq \mathbb{A}_S^n$ is $\mathbb{A}^1$ -contractible, and cotensoring its contraction yields the elementary $\mathbb{A}^1$ -homotopy

from the composite with the vertex evaluation $\mathcal{F}(- \times \Delta_S^n) \rightarrow \mathcal{F}$ to the identity. By Proposition 1.2.2, each unit is therefore an $\mathbb{A}^1$ -local equivalence. Since $\mathbb{A}^1$ -local equivalences are closed under colimits and Δ^{op} is weakly contractible, the canonical map $\mathcal{F} \simeq \text{colim}_{\Delta^{\text{op}}} \mathcal{F} \rightarrow \mathcal{F}'$ is an $\mathbb{A}^1$ -local equivalence. Combined with the $\mathbb{A}^1$ -invariance of $\mathcal{F}'$ established above, this gives $\text{Map}(\mathcal{F}', \mathcal{G}) \xrightarrow{\sim} \text{Map}(\mathcal{F}, \mathcal{G})$ for every $\mathbb{A}^1$ -invariant $\mathcal{G}$, i.e. $\mathcal{F}' \simeq L_{\mathbb{A}^1} \mathcal{F}$ (cf. [Hoy, Proposition 4.6]). $\square$

Remark 1.2.12 (Geometric meaning). Theorem 1.2.11 is the algebro-geometric analogue of the topological singular complex $\text{Sing}_\bullet$. The simplicial anima $\mathcal{F}(X \times \Delta_S^\bullet)$ encodes algebraic homotopy information: 0-simplices are sections, 1-simplices parametrize $\mathbb{A}^1$ -paths, and higher simplices record the coherences among those paths. Its geometric realization therefore forms the $\mathbb{A}^1$ -local replacement while retaining the higher homotopy data; it is not merely a quotient of the set of sections.

Caveat: $L_{\mathbb{A}^1}$ does not generally preserve Nisnevich sheaves, so the motivic localization L_{mot} is *not* simply $L_{\mathbb{A}^1} \circ L_{\text{Nis}}$; see Remark 1.3.3.

1.3. MOTIVIC SPACES

We now assemble the construction. Let S be qcqs. Since Sm_S is extensive, $\mathcal{P}_\Sigma(\text{Sm}_S) \simeq \text{Shv}_\square(\text{Sm}_S)$, and finite coproduct decompositions are Nisnevich covers, so $\text{Shv}_{\text{Nis}}(\text{Sm}_S) \subset \mathcal{P}_\Sigma(\text{Sm}_S)$.

Definition 1.3.1 (Motivic spaces). The category of **motivic spaces** over S is

$$H(S) := \text{Shv}_{\text{Nis}}(\text{Sm}_S) \cap \text{PShv}_{\mathbb{A}^1}(\text{Sm}_S) \subset \text{PShv}(\text{Sm}_S),$$

consisting of presheaves of anima satisfying both Nisnevich descent and $\mathbb{A}^1$ -invariance. The corresponding localization functor is

$$L_{\text{mot}} : \text{PShv}(\text{Sm}_S) \longrightarrow H(S).$$

A morphism in $\text{PShv}(\text{Sm}_S)$ is a **motivic equivalence** if L_{mot} sends it to an equivalence.

Fact 1.3.2. (i) L_{Nis} is left exact. Both $L_{\mathbb{A}^1}$ and L_{mot} preserve finite products [Hoy14, Proposition C.6].

(ii) Motivic equivalences form the smallest strongly saturated class in $\text{PShv}(\text{Sm}_S)$ containing both Nisnevich-local and $\mathbb{A}^1$ -local equivalences.

Remark 1.3.3 (Non-commutativity of localizations). In general, neither $L_{\text{mot}} = L_{\mathbb{A}^1} \circ L_{\text{Nis}}$ nor $L_{\text{mot}} = L_{\text{Nis}} \circ L_{\mathbb{A}^1}$ holds: $L_{\mathbb{A}^1}$ can destroy Nisnevich descent, while L_{Nis} can destroy $\mathbb{A}^1$ -invariance. Nevertheless, $H(S)$ is defined as an intersection, and is therefore independent of the order of localization. Concretely, the motivic localization is computed by ping-ponging:

$$L_{\text{mot}} \simeq (L_{\text{Nis}} \circ L_{\mathbb{A}^1})^{\circ \mathbb{N}},$$

i.e., by alternating the two localizations countably many times and passing to the colimit (cf. [AE16, Theorem 4.27]).

In the spectra-valued setting, however, $L_{\mathbb{A}^1}$ preserves Nisnevich descent: since Sp is stable, the pullback square in Proposition 1.1.8 is also a pushout, and the Brown–Gersten condition self-dualizes. One thus obtains a functor

$$L_{\mathbb{A}^1} \circ L_{\text{Nis}} : \text{PShv}(\text{Sm}_S, \text{Sp}) \rightarrow \text{Shv}_{\text{Nis}, \mathbb{A}^1}(\text{Sm}_S, \text{Sp}).$$

We record a first consequence of $\mathbb{A}^1$ -invariance that will be used repeatedly: invariance under locally trivial bundles with affine-space fibers.

Corollary 1.3.4 ($\mathbb{A}^1$ -bundle invariance). *Let $p : E \rightarrow B$ be a Zariski-locally trivial $\mathbb{A}^1$ -bundle in Sm_S (for instance, the total space of a line bundle). Then $L_{\mathrm{mot}}(p)$ is an equivalence in $H(S)$.*

Proof. Let $U_\bullet$ be the Čech nerve of a trivializing Zariski cover of B , so that $B \simeq \mathrm{colim}_{\Delta^{\mathrm{op}}} U_\bullet$ in $\mathrm{Shv}_{\mathrm{Nis}}(\mathrm{Sm}_S)$, and likewise $E \simeq \mathrm{colim}_{\Delta^{\mathrm{op}}} p^{-1}(U_\bullet)$. Levelwise, $p^{-1}(U_\bullet) \rightarrow U_\bullet$ is of the form $U \times \mathbb{A}^1 \rightarrow U$, a motivic equivalence by Proposition 1.2.2; since L_{mot} preserves colimits, p is a motivic equivalence. $\square$

1.3.1. Motivic spaces and K-theory

Convention 1.3.5 (Connective vs. non-connective K-theory). Throughout these notes we distinguish two versions of algebraic K-theory by case:

- $K : \mathrm{Sch}^{\mathrm{qcqs}, \mathrm{op}} \rightarrow \mathrm{Sp}$ denotes **non-connective** algebraic K-theory (Bass–Thomason–Trobeaugh); we always use uppercase K for this version.
- $k : \mathrm{Sch}^{\mathrm{qcqs}, \mathrm{op}} \rightarrow \mathrm{Sp}$ denotes **connective** algebraic K-theory; we always use lowercase k for this version. For every qcqs X one has $k(X) \simeq \tau_{\geq 0}K(X)$: the Bass–Thomason–Trobeaugh delooping modifies K only in negative degrees. On regular schemes one moreover has $k(X) \simeq K(X)$ (Proposition 1.3.11 below).

Write $\mathrm{KH} = L_{\mathbb{A}^1}K$ for homotopy K-theory; we do not use the notation kh . In the stable chapter, KGL denotes the Bott-periodic motivic spectrum representing KH , whereas $\mathrm{kgl} := \mathrm{Fil}_{\mathrm{slice}}^0 \mathrm{KGL}$ denotes its effective cover. Thus the pair $\mathrm{KGL}/\mathrm{kgl}$ is not obtained merely by applying the uppercase/lowercase convention to K/k . We also silently use the equivalence $\mathrm{Sp}_{\geq 0} \simeq \mathrm{CMon}^{\mathrm{gp}}(\mathrm{An})$: when a connective spectrum-valued presheaf such as k occurs in the space-valued category $H(S)$, the same symbol denotes its grouplike $\mathbb{E}_\infty$ -space $\Omega^\infty k$, or its underlying motivic space when the group structure is not at issue.

For $r \geq 0$ let $\mathrm{BGL}_r \in \mathrm{Shv}_{\mathrm{Nis}}(\mathrm{Sm}_S)$ denote the Nisnevich classifying stack of rank- r vector bundles: the Nisnevich sheafification of the objectwise classifying space of the group sheaf GL_r . Since vector bundles are Zariski-locally trivial, $\pi_0 \mathrm{BGL}_r(U) \cong \pi_0 \mathrm{Vect}_r(U)$ for $U \in \mathrm{Sm}_S$. Recall from Remark 1.0.1 that $\pi_0 \mathrm{Vect}_r$ is not $\mathbb{A}^1$ -invariant; nevertheless, after motivic localization BGL_r still classifies rank- r vector bundles on smooth *affine* schemes (Remark 1.3.7 below). Stabilizing with respect to rank via the direct-sum maps $E \mapsto E \oplus \mathcal{O}$ yields

$$\mathrm{BGL}_\infty := \mathrm{colim}_n \mathrm{BGL}_n \in \mathrm{PShv}(\mathrm{Sm}_S),$$

the colimit taken at the presheaf level (it is only in the theorem below that we localize), which classifies stable vector bundles up to addition of trivial summands; the $\mathbb{Z}$ factor in what follows records the virtual rank. This is the motivic analogue of the classical topological identification $\mathbb{Z} \times \mathrm{BU} \simeq \Omega^\infty \mathrm{ku}$ with the zeroth space of the connective complex K-theory spectrum ku , and the special case $r = 1$ gives $\mathrm{BGL}_1 = \mathrm{BG}_m$, classifying line bundles.

Concretely, when the base S is regular Noetherian, every $X \in \mathrm{Sm}_S$ is regular, so $k \simeq K$ on Sm_S (Proposition 1.3.11 below); in particular k satisfies Nisnevich descent (Example 1.1.11) and is $\mathbb{A}^1$ -invariant (Remark 1.3.10), hence defines an object of $H(S)$.

Theorem 1.3.6 (Representability of K-theory; Morel–Voevodsky [MV99, §4.3]). *For S regular Noetherian of finite Krull dimension,*

$$L_{\mathrm{mot}}(\mathbb{Z} \times \mathrm{BGL}_\infty) \simeq k.$$

In particular, $[U, \mathbb{Z} \times \mathrm{BGL}_\infty]_{\mathrm{H}(S)} \simeq K_0(U)$ for every $U \in \mathrm{Sm}_S$ (affineness is needed only for the fixed-rank statements of Remark 1.3.7).

Remark 1.3.7 (Affine representability). Although Vect_r is not $\mathbb{A}^1$ -invariant on all smooth schemes (Remark 1.0.1), motivic localization still yields the correct values on smooth affine schemes: Asok–Hoyois–Wendt [AHW17] proved that if X is a presheaf on Sm_S such that $\pi_0(X)$ is $\mathbb{A}^1$ -invariant on affine schemes and X satisfies affine Nisnevich excision, then $\pi_0(X)(U) \xrightarrow{\sim} [U, X]_{\mathrm{H}(S)}$ for all smooth affine U . Taking $X = \mathrm{BGL}_r$ gives $\pi_0 \mathrm{Vect}_r(U) \simeq [U, \mathrm{BGL}_r]_{\mathrm{H}(S)}$ for affine U — provided $\pi_0 \mathrm{Vect}_r$ is $\mathbb{A}^1$ -invariant on smooth affine S -schemes, a Bass–Quillen-type hypothesis made explicit in [AHW17, Theorem 5.2.3]. By the theorems of Lindel and Popescu it holds in all ranks when S is ind-smooth over a Dedekind ring with perfect residue fields, e.g. $S = \mathrm{Spec} k$ or $\mathrm{Spec} \mathbb{Z}$ [AHW17, Theorem 5.2.1]; for $r = 1$ it holds over any regular base [AHW17, Remark 5.2.4]; for general regular S and $r \geq 2$ it is the open Bass–Quillen conjecture.

Definition 1.3.8 (Higher K-groups). For $X \in \mathrm{Sm}_S$ and $n \in \mathbb{Z}$, the n -th algebraic K-group of X is

$$K_n(X) := \pi_n K(X).$$

In particular, $K_0(X)$ is the Grothendieck group of the stable ∞ -category $\mathrm{Perf}(X)$ of perfect complexes on X — the free abelian group on equivalence classes modulo $[B] = [A] + [C]$ for each cofiber sequence $A \rightarrow B \rightarrow C$ — and $K_1(X)$ contains $\mathcal{O}_X(X)^\times = \mathrm{GL}_1(X)$ as a direct summand, split off by the determinant $\det: K_1(X) \rightarrow \mathcal{O}_X(X)^\times$.

Example 1.3.9 (K-theory of a field). For k a field, $K_0(k) = \mathbb{Z}$ and $K_1(k) = k^\times$. Matsumoto’s theorem identifies $K_2(k)$ with the Milnor K-group $K_2^M(k)$, namely the quotient of $k^\times \otimes_{\mathbb{Z}} k^\times$ by the Steinberg relation $\{a, 1 - a\} = 0$ for $a \neq 0, 1$. Closed-form descriptions of $K_n(k)$ for $n \geq 3$ are generally unavailable.

Remark 1.3.10 (The role of regularity). Nisnevich descent of K-theory holds without regularity (Example 1.1.11); $\mathbb{A}^1$ -invariance does not. For regular Noetherian X , Quillen proved $K_n(X) \xrightarrow{\sim} K_n(X \times \mathbb{A}^1)$ for all $n \geq 0$. For singular schemes this fails: with $R = k[\varepsilon]/(\varepsilon^2)$,

$$K_1(R[t]) \neq K_1(R),$$

since $(1 + \varepsilon t) \in R[t]^\times$ contributes a class in $K_1(R[t])$ not coming from $K_1(R)$.

To restore $\mathbb{A}^1$ -invariance on non-regular schemes, Weibel introduced **homotopy K-theory**

$$\mathrm{KH}(X) := \mathrm{colim}_{[n] \in \Delta^{\mathrm{op}}} K(X \times \Delta^n) \in \mathrm{Sp},$$

where the colimit is taken in spectra. This is the analogous $\mathbb{A}^1$ -localization in presheaves of spectra on $\mathrm{Sch}^{\mathrm{qcqs}}$; after restriction to Sm_S , it agrees with the localization used above.

Proposition 1.3.11 (Vanishing of negative K-theory on regular schemes). For X regular Noetherian, $K_n(X) = 0$ for all $n < 0$. In particular, $k(X) \simeq K(X)$ on regular schemes.

Proof. $\mathrm{Perf}(X) \simeq D^b(\mathrm{Coh}(X))$ by regularity, and the latter carries a bounded t -structure with Noetherian heart; the claim follows from Antieau–Gepner–Heller [AGH19, Theorem 1.2]. $\square$

Remark 1.3.12 (Bass splitting and the Bott class). Classically, Bass’s fundamental theorem yields a natural splitting

$$K_n(X \times \mathbb{G}_m) \simeq K_n(X) \oplus K_{n-1}(X)$$

for regular X . The reduced Bass summand determines the Bott class

$$\beta \in \widetilde{K}_1(\mathbb{G}_m, 1),$$

or equivalently, using $\mathbb{T}_S \simeq \Sigma(\mathbb{G}_m, 1)$, a morphism $\beta: \mathbb{T}_S \rightarrow \mathrm{KH}$. Under the model $\mathbb{T}_S \simeq \mathbb{P}_S^1/\infty$, the same class is represented by the reduced projective-bundle class

$$1 - [\mathcal{O}(-1)] \in \widetilde{K}_0(\mathbb{P}_S^1, \infty),$$

with the sign convention used in the stable chapter. The motivic spectrum KGL_S represents KH , and multiplication by β induces the Bott equivalence

$$\mathbb{T}_S \otimes \mathrm{KGL}_S \xrightarrow{\simeq} \mathrm{KGL}_S.$$

Thus Bass delooping and the projective-bundle formula give compatible models of the same Bott class, rather than separate $(1, 1)$ - and simplicial summands; see Definition 3.1.11.

1.4. POINTED MOTIVIC SPACES

1.4.1. Basic constructions

Definition 1.4.1 (Pointed motivic spaces). The category $H_*(S)$ of **pointed motivic spaces** is the category of pointed objects in $H(S)$.

The functor $(-)_+: H(S) \rightarrow H_*(S)$ adds a disjoint basepoint, $X_+ = X \sqcup *$, and is left adjoint to the forgetful functor.

The **smash product** on $H_*(S)$ is the pushout

$$X \wedge Y := \mathrm{cofib}(X \vee Y \rightarrow X \times Y)$$

(where $X \vee Y := X \sqcup_* Y$ is the wedge), making $H_*(S)$ symmetric monoidal with unit S_+ [Lur17, Proposition 4.8.2.11].

We adopt the basepoint conventions:

- $\mathbb{A}^1$ is pointed at 1;
- $\mathbb{G}_m$ is pointed at 1.

The sphere computations below all rest on the following bridge from Nisnevich descent to pushouts of representables, which is the space-valued counterpart of the self-dualization noted in Remark 1.3.3 for spectra.

Lemma 1.4.2 (Nisnevich squares are motivic pushouts). *The motivic localization of a Nisnevich square in Sm_S is a pushout square in $H(S)$. If moreover the square is pointed compatibly (e.g. by a section of the initial corner $W = U \times_X V$), it is a pushout square in $H_*(S)$.*

Proof. For every $\mathcal{F} \in \mathrm{Shv}_{\mathrm{Nis}}(\mathrm{Sm}_S)$, Proposition 1.1.8 identifies $\mathrm{Map}(X, \mathcal{F}) \simeq \mathcal{F}(X)$ with $\mathcal{F}(U) \times_{\mathcal{F}(W)} \mathcal{F}(V) \simeq \mathrm{Map}(U \sqcup_W V, \mathcal{F})$, where the pushout is computed in presheaves; so the comparison map $U \sqcup_W V \rightarrow X$ is a Nisnevich-local, hence motivic, equivalence. Since L_{mot} is a left adjoint it preserves the pushout, and pushouts of pointed objects are computed in the underlying category because the indexing span is weakly contractible. $\square$

For $\mathbb{P}_S^1$, rather than fixing a basepoint we work throughout with the following object.

Definition 1.4.3 (Tate object). The **Tate object** is

$$\mathbb{T}_S := \operatorname{cofib}(\infty_+ : S_+ \longrightarrow (\mathbb{P}_S^1)_+) \in H_*(S),$$

i.e. $\mathbb{P}_S^1$ with the section $\infty = [1 : 0]$ collapsed to the basepoint. We use $\mathbb{T}_S$ throughout in place of $(\mathbb{P}_S^1, \infty)$, both in this chapter and in the stable chapter.

Definition 1.4.4 (Suspension and loops). For $X \in H_*(S)$,

- the **suspension** $\Sigma X := S^1 \wedge X$, where $S^1 = [1]/\partial[1]$ is the simplicial circle (as a constant sheaf);
- the **loop space** $\Omega X := \operatorname{fib}(* \rightarrow X)$.

These form an adjunction $\Sigma \dashv \Omega$.

1.4.2. Bigraded spheres

The computations of this subsection are due to Morel–Voevodsky [MV99, §3.2]. In $H_*(S)$ there are two essentially different “circles”.

Definition 1.4.5 (Motivic spheres). • The **simplicial circle** is $S^{1,0} := S^1$ (constant sheaf).
 • The **Tate circle** is $S^{1,1} := \mathbb{G}_m$ (pointed at 1).
 • The **motivic (p, q) -sphere**, for $p \geq q \geq 0$, is

$$S^{p,q} := (S^{1,0})^{\wedge(p-q)} \wedge (S^{1,1})^{\wedge q}.$$

The key computation:

Proposition 1.4.6 (Tate object as a motivic sphere). In $H_*(S)$, $\mathbb{T}_S \simeq S^{2,1} = S^1 \wedge \mathbb{G}_m$.

Proof. Consider the standard Nisnevich square for $\mathbb{P}^1$:

$$\begin{array}{ccc} \mathbb{G}_m & \longrightarrow & \mathbb{A}^1 \\ \downarrow & & \downarrow \\ \mathbb{A}^1 & \longrightarrow & \mathbb{P}^1. \end{array}$$

Point the two affine charts and their intersection at 1; the induced basepoint of $\mathbb{P}^1$ is then $[1 : 1]$. By Lemma 1.4.2, the displayed square is a pushout in $H_*(S)$. Both copies of $(\mathbb{A}^1, 1)$ are pointed-contractible, so this pushout identifies $(\mathbb{P}^1, [1 : 1])$ with the suspension of $(\mathbb{G}_m, 1)$. Finally, the algebraic path

$$\mathbb{A}_S^1 \longrightarrow \mathbb{P}_S^1, \quad t \longmapsto [1 : t],$$

joins $\infty = [1 : 0]$ to $[1 : 1]$ and hence identifies the pointed objects $(\mathbb{P}^1, \infty) = \mathbb{P}^1/\infty$ and $(\mathbb{P}^1, [1 : 1])$ in $H_*(S)$. Therefore

$$\mathbb{T}_S \simeq * \sqcup_{\mathbb{G}_m} * \simeq \Sigma \mathbb{G}_m = S^1 \wedge \mathbb{G}_m = S^{2,1}. \quad \square$$

Remark 1.4.7. This makes precise the observation that $\mathbb{G}_m$ realizes a kind of “algebraic circle”: under Betti realization $\mathbb{G}_m(\mathbb{C}) = \mathbb{C}^\times$ is homotopy equivalent to the topological S^1 . The motivic sphere $\mathbb{G}_m$ plays the role of $S^{1,1}$, while the simplicial circle S^1 is $S^{1,0}$. Their smash product $\mathbb{T}_S \simeq S^{2,1}$ is the natural “algebraic 2-sphere”.

This bigraded structure is what makes motivic homotopy theory richer than classical homotopy theory: the weight direction q encodes Tate twists specific to algebraic geometry.

Proposition 1.4.8 (Punctured affine spaces). *For $n \geq 1$, $\mathbb{A}^n \setminus \{0\} \simeq S^{2n-1,n}$ in $H_*(S)$, pointed at $(1, \dots, 1)$. (The explicit $\mathbb{A}^1$ -homotopy $t \mapsto (1, t, \dots, t)$ connects this section to $(1, 0, \dots, 0)$, so the latter gives an equivalent pointed object.)*

Proof. By induction on n . The case $n = 1$ is $\mathbb{A}^1 \setminus \{0\} = \mathbb{G}_m \simeq S^{1,1}$. For $n \geq 2$, the Nisnevich square

$$\begin{array}{ccc} (\mathbb{A}^{n-1} \setminus \{0\}) \times \mathbb{G}_m & \longrightarrow & \mathbb{A}^{n-1} \times \mathbb{G}_m \\ \downarrow & & \downarrow \\ (\mathbb{A}^{n-1} \setminus \{0\}) \times \mathbb{A}^1 & \longrightarrow & \mathbb{A}^n \setminus \{0\} \end{array}$$

is a pushout in $H_*(S)$ by Lemma 1.4.2, pointed at $((1, \dots, 1), 1)$. Since $\mathbb{A}^{n-1} \simeq *$ and $\mathbb{A}^1 \simeq *$, the top-right and bottom-left corners contract onto $\mathbb{G}_m$ and $\mathbb{A}^{n-1} \setminus \{0\}$ respectively, so the square exhibits $\mathbb{A}^n \setminus \{0\}$ as the pushout of the two projections

$$\mathbb{A}^{n-1} \setminus \{0\} \longleftarrow (\mathbb{A}^{n-1} \setminus \{0\}) \times \mathbb{G}_m \longrightarrow \mathbb{G}_m,$$

that is, as the join $(\mathbb{A}^{n-1} \setminus \{0\}) * \mathbb{G}_m$. For pointed motivic spaces X, Y there is a natural equivalence $X * Y \simeq \Sigma(X \wedge Y)$, exactly as in topology [MV99, §3.2]. Hence, by the induction hypothesis,

$$\mathbb{A}^n \setminus \{0\} \simeq \Sigma((\mathbb{A}^{n-1} \setminus \{0\}) \wedge \mathbb{G}_m) \simeq S^{1,0} \wedge S^{2n-3,n-1} \wedge S^{1,1} = S^{2n-1,n}. \quad \square$$

Corollary 1.4.9 (Thom spaces). *In $H_*(S)$,*

$$\mathbb{A}^n / (\mathbb{A}^n \setminus \{0\}) \simeq S^{2n,n} \quad \text{and} \quad \mathbb{P}^n / \mathbb{P}^{n-1} \simeq S^{2n,n}.$$

Proof. The cofiber sequence $\mathbb{A}^n \setminus \{0\} \rightarrow \mathbb{A}^n \rightarrow \mathbb{A}^n / (\mathbb{A}^n \setminus \{0\})$ together with $\mathbb{A}^n \simeq *$ gives

$$\mathbb{A}^n / (\mathbb{A}^n \setminus \{0\}) \simeq \Sigma(\mathbb{A}^n \setminus \{0\}) \simeq \Sigma S^{2n-1,n} = S^{2n,n}.$$

For the second, write $\mathbb{P}^n = \mathbb{A}^n \cup (\mathbb{P}^n \setminus \{0\})$ via the standard chart $\mathbb{A}^n \subset \mathbb{P}^n$ and the open complement of the origin of that chart; their intersection is $\mathbb{A}^n \setminus \{0\}$. The resulting Zariski (hence Nisnevich) square

$$\begin{array}{ccc} \mathbb{A}^n \setminus \{0\} & \longrightarrow & \mathbb{A}^n \\ \downarrow & & \downarrow \\ \mathbb{P}^n \setminus \{0\} & \longrightarrow & \mathbb{P}^n \end{array}$$

is a pushout in $H_*(S)$ by Lemma 1.4.2, pointed at $(1, \dots, 1)$. The radial projection from $0 \in \mathbb{A}^n$ realizes $\mathbb{P}^n \setminus \{0\}$ as the total space of a line bundle over the hyperplane at infinity $\mathbb{P}^{n-1}$, hence $\mathbb{P}^n \setminus \{0\} \simeq \mathbb{P}^{n-1}$ in $H_*(S)$ by Corollary 1.3.4. Collapsing $\mathbb{P}^{n-1} \subset \mathbb{P}^n$ in the pushout therefore identifies $\mathbb{P}^n / \mathbb{P}^{n-1}$ with $\mathbb{A}^n / (\mathbb{A}^n \setminus \{0\}) \simeq S^{2n,n}$. $\square$

STABLE MOTIVIC HOMOTOPY THEORY I — CONSTRUCTION AND SIX FUNCTORS

Grothendieck envisioned, long before stable categories were available, that every cohomology theory on schemes should factor through a single universal “category of motives” $\text{Mot}(k)$. In the modern dictionary “studying cohomology” is increasingly understood as “studying six-functor formalisms”, and Grothendieck’s program becomes the search for a universal such formalism. The motivic stable homotopy category $\text{SH}(S)$ of Morel–Voevodsky is the answer for $\mathbb{A}^1$ -invariant cohomology theories, and constructing it together with its six-functor structure is the goal of this chapter.

The starting point is the unstable category $H_*(S)$ from the previous chapter — pointed objects in $H(S) = \text{Shv}_{\text{Nis}}(\text{Sm}_S) \cap \text{PShv}_{\mathbb{A}^1}(\text{Sm}_S)$ — which is already rich enough to host a great deal of algebraic geometry. What it lacks is the formal machinery of a stable homotopy theory: fiber and cofiber sequences that coincide, the suspension–loop equivalence $\Sigma \simeq \Omega^{-1}$, and the six-functor formalism that underlies modern thinking about cohomology. All of these become available after one further universal step: formally inverting the Tate object $\mathbb{T}_S \simeq S^1 \wedge \mathbb{G}_m$ under the smash product. The result is $\text{SH}(S)$, the **motivic stable homotopy category**, the central object of this chapter.

Drew–Gallauer [DG22] supplied a precise form of Grothendieck’s search: there is a universal presentable six-functor formalism on finite-type schemes which is cohomologically smooth, satisfies excision, and kills the $\mathbb{A}^1$ -homology. This universal object is precisely the motivic formalism SH of Morel–Voevodsky. Their proof is constructive and recovers the construction we shall give: the passage to $\text{Shv}_{\text{Nis}}(\text{Sm}_-, \text{Sp})$ is forced by excision, the further $\mathbb{A}^1$ -localization by the triviality of $\mathbb{A}^1$ -homology, and the $\otimes$ -inversion of $\mathbb{T}_S$ by cohomological smoothness (the invertibility of the dualizing object of $\mathbb{A}^1$).

It bears emphasizing that SH is not the final word: the axioms above — especially $\mathbb{A}^1$ -invariance — exclude important arithmetic-geometric cohomology theories. Prismatic cohomology does not factor through SH , and more generally p -adic cohomology theories in characteristic p require fundamentally different frameworks. Recently, Annala–Iwasa [AI25] have proposed a non- $\mathbb{A}^1$ -invariant refinement called *motivic spectra* MotSp ; the full subcategory of Nisnevich-local, $\mathbb{A}^1$ -invariant objects is SH , or equivalently $\text{SH} \simeq L_{\text{Nis}, \mathbb{A}^1} \text{MotSp}$. This broader framework recovers algebraic K -theory more directly via a Conner–Floyd-style isomorphism [AH24a] and is the home of Atiyah duality for arbitrary qcqs derived schemes [AH24b]. We should therefore think of $\text{SH}(S)$ as the universal coefficient system for $\mathbb{A}^1$ -invariant cohomology, not as the definitive answer to Grothendieck’s question; the broader MotSp is briefly introduced in Section 2.1.5, but SH remains the protagonist of these notes.

The chapter is organized as follows.

- Section 2.1 constructs $\text{SH}(S)$ as the formal inversion of $\mathbb{T}_S$ in $H_*(S)$, developing the symmetric-spectrum and telescope models on which the construction rests. We close the section with the Annala–Iwasa generalization MotSp and its relation to SH .
- Section 2.2 sets up the language of six-functor formalisms via the span-category framework of Mann–Scholze, in the 2-categorical form due to Crossen–Lenz–Linskens.

- Section 2.3 applies this framework to SH via the cdh topology, equipping it with its full six-functor structure (cdh descent as a corollary) and developing the relative duality theory of cohomologically smooth, étale, and proper morphisms à la Heyer–Mann.
- Section 2.4 establishes the Drew–Gallauer universal property — SH is initial among presentable six-functor formalisms which are cohomologically smooth, satisfy excision, and kill $\mathbb{A}^1$ -homology.

The next chapter (Chapter 3) takes up the comparisons of $\mathrm{SH}(S)$ with classical structures: with the naive stabilization $\mathrm{Shv}_{\mathrm{Nis}, \mathbb{A}^1}(\mathrm{Sm}_S, \mathrm{Sp})$, with Postnikov-type filtrations, and with the topological theory of orientations and formal group laws.

2.1. FORMAL INVERSION

This section gives a careful construction of the stable motivic category $\mathrm{SH}(S)$, together with the technology—symmetric sequences, free algebras, Bousfield localization—needed both for this construction and for the construction of specific motivic spectra in subsequent sections.

The goal of this section is to invert the Tate object $\mathbb{T}_S$ of Definition 1.4.3 under the smash product, producing a presentably symmetric monoidal category $\mathrm{SH}(S)$ that extends the construction chain

$$\mathrm{Sm}_S \xrightarrow{\cdot} \mathrm{PShv}(\mathrm{Sm}_S) \xrightarrow{L_{\mathrm{mot}}} \mathrm{H}(S) \xrightarrow{(-)_+} \mathrm{H}_*(S) \xrightarrow{\sigma_{\mathbb{T}_S}^\infty} \mathrm{SH}(S),$$

from Chapter 1 by one further universal step.

2.1.1. The universal $\otimes$ -inversion problem

Let $C \in \mathrm{CAlg}(\mathrm{Pr}^L)$ be a presentably symmetric monoidal category and let $T \in C$. We seek the initial presentably symmetric monoidal C -algebra in which the image of T becomes $\otimes$ -invertible: that is, the initial object of $\mathrm{CAlg}(\mathrm{Pr}^L)_C /$ whose structure functor sends T to an invertible object.

Definition 2.1.1 ([AI25, §1.1]). Let $\mathcal{D}$ be a category tensored over a symmetric monoidal category C . Given a Bousfield localization $L: \mathcal{D} \rightarrow \mathcal{D}'$, we say L is **smashing** if the composition $\mathcal{D} \xrightarrow{L} \mathcal{D}' \hookrightarrow \mathcal{D}$ is of the form $A \otimes -$ for some $A \in \mathrm{CAlg}(C)$.

Remark 2.1.2. The name comes from the classical case $C = \mathcal{D} = \mathrm{Sp}$, where such a localization is given by smashing with the spectrum A . (We do not require $\mathcal{D}$ to be presentable here: the definition will be applied below to $\mathcal{D} = \mathrm{Mod}_C(\mathrm{Pr}^L)$ itself, tensored over itself.) A smashing localization is in particular compatible with the C -module structure: its local equivalences are closed under $A \otimes -$ for all $A \in C$. The converse fails: a Bousfield localization with $\otimes$ -closed local equivalences need not be smashing. For instance, for $n \geq 1$ the localization $L_{K(n)}: \mathrm{Sp} \rightarrow \mathrm{Sp}$ has $\otimes$ -closed local equivalences by the Künneth isomorphism for Morava K -theory, but it is not of the form $A \otimes -$: smashing localizations preserve colimits, while $L_{K(n)}$ famously does not. The restriction $n \geq 1$ matters here: $K(0) \simeq \mathrm{H}\mathbb{Q}$ gives rationalization, which is smashing.

Proposition 2.1.3 ([Rob15, Proposition 2.9], [AI25, p. 1.2.2]). *There exists a smashing Bousfield localization*

$$(-)[T^{-1}]: \mathrm{Mod}_C(\mathrm{Pr}^L) \rightarrow \mathrm{Mod}_C(\mathrm{Pr}^L)$$

whose essential image is spanned by those C -modules on which T acts as an equivalence, i.e. those $\mathcal{D}$ for which $T \otimes -: \mathcal{D} \rightarrow \mathcal{D}$ is an equivalence.

For a C -module $\mathcal{D}$, we refer to $\mathcal{D}[T^{-1}]$ as the **formal inversion** of T in $\mathcal{D}$.

Remark 2.1.4. Since the localization $(-)[T^{-1}]$ is smashing, the unit map $u: C \rightarrow C[T^{-1}]$ exhibits $C[T^{-1}]$ as an idempotent in $\text{Mod}_C(\text{Pr}^L)$, and thus u induces a symmetric monoidal structure on $C[T^{-1}]$, which in turn induces an equivalence

$$\text{Mod}_{C[T^{-1}]}(\text{Pr}^L) \xrightarrow{\sim} \text{Mod}_C(\text{Pr}^L)[T^{-1}].$$

The idempotency also resolves the universal problem posed at the start of this subsection: for every $\mathcal{E} \in \text{CAlg}(\text{Pr}^L)_C$, the space of C -algebra maps $C[T^{-1}] \rightarrow \mathcal{E}$ is contractible when the image of T in $\mathcal{E}$ is $\otimes$ -invertible and empty otherwise [Lur17, §4.8.2], so $C[T^{-1}]$ is precisely the initial C -algebra inverting T (cf. [Rob15, §2]).

This motivates the following abstract definition.

Definition 2.1.5. The **motivic stable homotopy category** is the formal inversion of $\mathbb{T}_S$ in $H_*(S)$:

$$\text{SH}(S) := H_*(S)[\mathbb{T}_S^{-1}].$$

This definition is too abstract to work with geometrically; we need a concrete model of the localization $(-)[\mathbb{T}_S^{-1}]$.

$$\begin{array}{ccc} & & C[T^{-1}] \\ & \nearrow \text{formal inversion} & \downarrow \simeq \\ C & \xrightarrow{\sigma^\infty} \text{Sp}_T(C) & \\ & \searrow N_T & \downarrow \simeq \text{when } T \text{ is symmetric} \\ & & \text{Tel}_T(C) \end{array}$$

Here σ^∞ is the free T -spectrum functor — the free lax- T -spectrum functor F_T of Section 2.1.3 followed by localization into T -spectra — and N_T is the telescope construction; the two agree when T is symmetric, see Proposition 2.1.17.

2.1.2. Symmetric sequences and free algebras

Definition 2.1.6 (Symmetric sequences). Let $\mathbb{F} = \text{Fin}^\simeq$ denote the groupoid of finite sets and bijections. For $C \in \text{CAlg}(\text{Pr}^L)$, the category of **symmetric sequences** in C is

$$\text{SSeq}(C) := \text{Fun}(\mathbb{F}, C),$$

equipped with the Day convolution symmetric monoidal structure induced by the disjoint-union monoidal structure on $\mathbb{F}$; see [AI25, §A.4] for the Day convolution.

In fact, an object in $\text{SSeq}(C)$ is given by a sequence $E_\bullet = (E_0, E_1, \dots)$ in C with a Σ_n -action on E_n . For $X_\bullet, Y_\bullet \in \text{SSeq}(C)$, we have a formula

$$(X_\bullet \otimes Y_\bullet)_n = \bigoplus_{p+q=n} \Sigma_{p+q} \otimes_{\Sigma_p \times \Sigma_q} (X_p \otimes Y_q).$$

From this perspective, one obtains several canonical functors:

- $U: \mathbf{SSeq}(C) \rightarrow C$, carrying $X_\bullet$ to X_0 .
- $F: C \rightarrow \mathbf{SSeq}(C)$, carrying C to $(C, \emptyset, \emptyset, \dots)$, and left adjoint to U .
- A shift functor $s_-: \mathbf{SSeq}(C) \rightarrow \mathbf{SSeq}(C)$, given by

$$(s_-Y)_\bullet := (Y_1, Y_2, \dots),$$

where Σ_n acts on $(s_-Y)_n = Y_{n+1}$ by restriction along the inclusion $\Sigma_n \subset \Sigma_{n+1}$.

- s_- admits a left adjoint s_+ , given by

$$(s_+Y)_\bullet := (\emptyset, Y_0, \dots, \Sigma_n \otimes_{\Sigma_{n-1}} Y_{n-1}, \dots).$$

- For $n \geq 0$, the level- n evaluation functor $U_n := U \circ s_-^n: \mathbf{SSeq}(C) \rightarrow C$, carrying $X_\bullet$ to X_n (so $U_0 = U$).

Remark 2.1.7 (Notation). The notation s_+ (left adjoint, “shift up”) and s_- (right adjoint, “shift down”) is borrowed from [AI25, p. 1.3.1]. We will see in Lemma 2.1.12 that, after passing to $\mathbf{Sp}_T(C)$, the localized s_- becomes equivalent to $T \otimes -$ and s_+ becomes equivalent to $\underline{\mathrm{Map}}(T, -)$. Despite this, we will retain the traditional notation σ^∞ and ω^∞ for the resulting suspension spectrum / underlying object adjunction, in line with the analogy with classical spectra; see the discussion after Definition 2.1.11.

Definition 2.1.8 (Symmetric algebra). For $X \in C$, write $\mathrm{Sym}(X) \in \mathbf{CAlg}(C)$ for the **free commutative algebra** generated by X , i.e. the image of X under the left adjoint of the forgetful functor $\mathbf{CAlg}(C) \rightarrow C$. Since the tensor product of C commutes with colimits in each variable, its underlying object is computed by the familiar formula

$$\mathrm{Sym}(X) \simeq \bigoplus_{n \geq 0} (X^{\otimes n})_{h\Sigma_n}$$

[Lur17, §3.1.3].

2.1.3. Lax T -spectra, T -spectra, and T -telescopes

For $T \in C$, consider the symmetric sequence

$$T_{\mathrm{sym}} := s_+F(T) = (\emptyset, T, \emptyset, \emptyset, \dots) \in \mathbf{SSeq}(C).$$

We may form its symmetric algebra $\mathrm{Sym}(T_{\mathrm{sym}}) \in \mathbf{CAlg}(\mathbf{SSeq}(C))$.

Definition 2.1.9 (Lax T -spectra). The category of **lax T -spectra** in C is

$$\mathbf{Sp}_T^{\mathrm{lax}}(C) := \mathrm{Mod}_{\mathrm{Sym}(T_{\mathrm{sym}})}(\mathbf{SSeq}(C)) \in \mathbf{CAlg}(\mathbf{Pr}^L)_C/.$$

The adjunction (F, U) , together with the free-module functor $\mathrm{Sym}(T_{\mathrm{sym}}) \otimes -$, induces an adjunction

$$C \begin{array}{c} \xrightarrow{F_T} \\ \perp \\ \xleftarrow{U} \end{array} \mathbf{Sp}_T^{\mathrm{lax}}(C)$$

We may therefore use U to study the structure of $\mathbf{Sp}_T^{\mathrm{lax}}(C)$.

Lemma 2.1.10 ([AI25, p. 1.3.6]). For each $n \geq 0$, $U_n(\mathrm{Sym}(T_{\mathrm{sym}})) \simeq T^{\otimes n}$.

Proof. Since $T_{\text{sym}} = s_+ F(T)$ is concentrated at level 1, the Day convolution formula forces $T_{\text{sym}}^{\otimes m}$ to be concentrated at level m ; hence only the summand $\text{Sym}^n(T_{\text{sym}})$ of $\text{Sym}(T_{\text{sym}})$ contributes at level n , giving

$$U_n(\text{Sym}(T_{\text{sym}})) \simeq U_n(\text{Sym}^n(s_+ F(T))) \simeq (\Sigma_n \otimes T^{\otimes n})_{h\Sigma_n}.$$

The Σ_n -action on $\Sigma_n \otimes T^{\otimes n}$ is diagonal, so the homotopy orbit is equivalent to $T^{\otimes n}$. $\square$

This lemma shows that lax T -spectra “see” $T^{\otimes n}$ at level n ; consequently, by Bousfield localizing at the relevant bonding maps, we will obtain a concrete model of T -spectra from $\text{Sp}_T^{\text{lax}}(C)$.

The bonding maps arise as follows. For a lax T -spectrum E , the natural equivalences

$$T \otimes s_+ E \simeq s_+(T \otimes E) \simeq T_{\text{sym}} \otimes E,$$

together with the module multiplication $T_{\text{sym}} \otimes E \rightarrow E$, supply a map $\sigma_E : s_+(T \otimes E) \rightarrow E$ of lax T -spectra; we write

$$\sigma_E^\# : E \longrightarrow \underline{\text{Map}}(T, s_- E)$$

for its adjoint under $s_+ \dashv s_-$ and $T \otimes - \dashv \underline{\text{Map}}(T, -)$, and call it the **structure map** of E [AI25, p. 1.3.7].

Definition 2.1.11 (T -spectra). A **T -spectrum** in C is a lax T -spectrum E whose structure map $\sigma_E^\#$ is an equivalence. The full subcategory $\text{Sp}_T(C) \subset \text{Sp}_T^{\text{lax}}(C)$ spanned by the T -spectra is an accessible localization

$$L_{\text{Sp}} : \text{Sp}_T^{\text{lax}}(C) \rightarrow \text{Sp}_T(C)$$

[AI25, p. 1.3.9].

The underlying symmetric sequence of a T -spectrum is thus a sequence $E_\bullet = (E_0, E_1, \dots)$ in C with Σ_n -action on E_n , together with bonding equivalences

$$E_n \xrightarrow{\sim} \underline{\text{Map}}(T, E_{n+1}) \quad (n \geq 0).$$

By [Lur17, Proposition 4.1.7.4], $\text{Sp}_T(C)$ admits a unique presentably symmetric monoidal structure for which L_{Sp} is symmetric monoidal.

Lemma 2.1.12 ([AI25, p. 1.3.13]). *On $\text{Sp}_T(C)$, there are natural equivalences*

$$s_- \simeq T \otimes -, \quad L_{\text{Sp}} s_+ \simeq \underline{\text{Map}}(T, -),$$

and T acts as an equivalence on $\text{Sp}_T(C)$. In particular, $L_{\text{Sp}} s_+$ and s_- are mutually inverse equivalences of $\text{Sp}_T(C)$.

The adjunction $F_T \dashv U$ can be derived to the following adjunction:

$$C \begin{array}{c} \xrightarrow{\sigma^\infty} \\ \perp \\ \xleftarrow{\omega^\infty} \end{array} \text{Sp}_T(C)$$

where $\omega^\infty E = E_0$ on the underlying symmetric sequence. The notation is consistent with the analogy with classical spectra: σ^∞ sends an object of C to its T -suspension spectrum, while ω^∞ extracts the underlying 0-th level. Since T becomes invertible in $\text{Sp}_T(C)$, this adjunction extends naturally: for each $j \in \mathbb{Z}$ one has an adjunction $\sigma^{\infty-j} \dashv \omega^{\infty-j}$, where

$$\sigma^{\infty-j} := T^{\otimes(-j)} \otimes \sigma^\infty(-), \quad \omega^{\infty-j} := \omega^\infty(T^{\otimes j} \otimes -).$$

By Lemma 2.1.12, for $j \geq 0$ the object $\omega^{\infty-j}E$ is the j -th level E_j of the underlying symmetric sequence, while for $j < 0$ it is $\underline{\text{Map}}(T^{\otimes(-j)}, E_0)$; thus $\omega^{\infty-j}$ deserves the name “ j -th level” for all $j \in \mathbb{Z}$.

Corollary 2.1.13 (Levelwise description of T -spectra). *Every $E \in \text{Sp}_T(C)$ determines, functorially:*

- a sequence of objects $(E(j))_{j \in \mathbb{Z}}$ in C , with $E(j) := \omega^{\infty-j}E$;
- equivalences $E(j) \xrightarrow{\sim} \underline{\text{Map}}(T, E(j+1))$ for every $j \in \mathbb{Z}$.

If E is a commutative algebra, the levels inherit compatible graded multiplications $E(i) \otimes E(j) \rightarrow E(i+j)$.

Proof. The first two clauses follow from Lemma 2.1.12; the multiplicative statement is dealt with in Section 3.1.1. $\square$

Warning 2.1.14. This data does *not* recover E in general. A $\mathbb{Z}$ -sequence with bonding equivalences is precisely an object of the limit $\lim(\cdots \rightarrow C \xrightarrow{\underline{\text{Map}}(T, -)} C) \simeq \text{Tel}_T(C)$, so the assertion that such data determines an object of $\text{Sp}_T(C)$ is exactly the telescope comparison below (Proposition 2.1.17), which requires T to be symmetric. Likewise the graded multiplications do not assemble into a graded-commutative structure in general: the $\mathbb{E}_2$ -obstruction is the swap on $T^{\otimes 2}$ (cf. Section 3.1.2).

Proposition 2.1.15 (Formal inversion and T -spectra, [AI25, p. 1.3.14]). *There is a categorical equivalence*

$$C[T^{-1}] \simeq \text{Sp}_T(C).$$

Recall that in stable homotopy theory, stabilization is realized as a colimit. We now apply the same idea to give a more direct construction of $\text{SH}(S)$: namely, by writing $C[T^{-1}]$ as a colimit along $T \otimes -$ in $\text{Mod}_C(\text{Pr}^L)$. This construction, however, requires an additional symmetry condition on T ; see Proposition 2.1.17.

Definition 2.1.16. The category of T -telescopes in C is given by the following colimit in $\text{Mod}_C(\text{Pr}^L)$:

$$\text{Tel}_T(C) := \text{colim}(C \xrightarrow{T \otimes -} C \xrightarrow{T \otimes -} \cdots).$$

Proposition 2.1.17 (Telescope comparison, [AI25, p. 1.6.3]). *If T is **symmetric** in the sense that the cyclic permutation*

$$T^{\otimes 3} \longrightarrow T^{\otimes 3}, \quad a \otimes b \otimes c \longmapsto b \otimes c \otimes a,$$

is homotopic to the identity in C , then there is an equivalence

$$C[T^{-1}] \simeq \text{Sp}_T(C) \simeq \text{Tel}_T(C)$$

in $\text{Mod}_C(\text{Pr}^L)$.

2.1.4. Specialization to motivic homotopy theory

We now apply the preceding machinery to motivic homotopy theory. The input is the pointed motivic homotopy category $H_*(S)$ from Chapter 1, equipped with the smash product, and the Tate object $\mathbb{T}_S$.

Proposition 2.1.18 ($\mathbb{T}_S$ is symmetric). *The cyclic permutation $\sigma: \mathbb{T}_S^{\wedge 3} \rightarrow \mathbb{T}_S^{\wedge 3}$ is $\mathbb{A}^1$ -homotopic to the identity in $H_*(S)$.*

Proof. By Proposition 1.4.6 we have $\mathbb{T}_S \simeq \mathbb{A}^1/(\mathbb{A}^1 \setminus 0)$ in $H_*(S)$: both sides are identified with $\Sigma \mathbb{G}_m$, using $\mathbb{A}^1 \simeq *$ for the right-hand side. Since smash products commute with cofibers and the union of the opens $\{x_i \neq 0\}$ inside $\mathbb{A}^n$ is $\mathbb{A}^n \setminus 0$, smashing n copies gives a canonical equivalence

$$\mathbb{T}_S^{\wedge n} \simeq \mathbb{A}^n/(\mathbb{A}^n \setminus 0)$$

under which permuting the smash factors corresponds to permuting the coordinates; more generally, every $M \in \mathrm{GL}_n(\mathbb{Z})$ acts on the right-hand side after applying the canonical homomorphism $\mathrm{GL}_n(\mathbb{Z}) \rightarrow \mathrm{GL}_n(\Gamma(S, \mathcal{O}_S))$ and using the resulting linear action on $\mathbb{A}^n$.

Step 1: elementary matrices act trivially. For an elementary matrix $E_{ij}(\lambda)$, the assignment $(t, x) \mapsto E_{ij}(t\lambda) \cdot x$ is a morphism $\mathbb{A}^1 \times \mathbb{A}^n \rightarrow \mathbb{A}^n$ — here t enters as a *coefficient*, so the formula is polynomial in t — which is a linear automorphism over each point of $\mathbb{A}^1$; in particular it fixes the origin and preserves $\mathbb{A}^1 \times (\mathbb{A}^n \setminus 0)$, hence descends to a pointed $\mathbb{A}^1$ -homotopy of self-maps of $\mathbb{A}^n/(\mathbb{A}^n \setminus 0)$ from the identity to the action of $E_{ij}(\lambda)$. Consequently, matrices differing by a product of elementary matrices act $\mathbb{A}^1$ -homotopically equally.

Step 2: the swap is a $\langle -1 \rangle$ -twist. For $n = 2$, the matrix identities

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = E_{21}(-1) E_{12}(1) E_{21}(-1),$$

combined with Step 1 show that the transposition τ of the two smash factors of $\mathbb{T}_S^{\wedge 2}$ is $\mathbb{A}^1$ -homotopic to the action of $\mathrm{diag}(-1, 1)$, that is, to $\langle -1 \rangle \wedge \mathrm{id}$, where $\langle -1 \rangle$ denotes the automorphism of $\mathbb{T}_S = \mathbb{A}^1/(\mathbb{A}^1 \setminus 0)$ induced by $-1 \in \mathrm{GL}_1$.

Step 3: the 3-cycle. The cyclic permutation σ is the composite of two adjacent transpositions, so by Step 2 it is $\mathbb{A}^1$ -homotopic to the action of $\mathrm{diag}(-1, -1, 1)$. This matrix has determinant 1, and $\mathrm{SL}_3(\mathbb{Z})$ is generated by elementary matrices since $\mathbb{Z}$ is Euclidean; Step 1 therefore gives $\sigma \simeq \mathrm{id}$ on $\mathbb{T}_S^{\wedge 3}$. $\square$

Corollary 2.1.19 (The swap is a $\langle -1 \rangle$ -twist). *The transposition of the two factors of $\mathbb{T}_S^{\wedge 2}$ is $\mathbb{A}^1$ -homotopic to $\langle -1 \rangle \wedge \mathrm{id}$ in $H_*(S)$ (Step 2 of the preceding proof). In particular, while all even permutations of smash factors are $\mathbb{A}^1$ -homotopic to the identity, the swap itself is in general not: its class survives stabilization as the motivic swap class $\langle -1 \rangle$ on $\mathbb{T}_S^{\otimes 2}$, the $\mathbb{E}_2$ -obstruction discussed in Section 3.1.2.*

Remark 2.1.20 (Why the argument is linear). The proof lives on $\mathbb{A}^3$, where the entries of $E_{ij}(t\lambda)$ are polynomial in t . No analogous homotopy exists on the factor $\mathbb{G}_m^{\wedge 3}$ alone: there an elementary matrix would have to act by the exponent action, e.g. $(s_1, s_2, s_3) \mapsto (s_1 s_2^t, s_2, s_3)$, which is not algebraic in t — indeed tori are $\mathbb{A}^1$ -rigid (for reduced R one has $R[t]^\times = R^\times$, so pointed $\mathbb{A}^1$ -families of maps between tori are constant), and a general elementary matrix does not even preserve the fat wedge, hence induces no self-map of $\mathbb{G}_m^{\wedge 3}$.

Corollary 2.1.21 (Telescope model of $\mathrm{SH}(S)$). *Applying Proposition 2.1.17 to $(H_*(S), \mathbb{T}_S)$ yields*

$$\mathrm{SH}(S) \simeq \mathrm{colim} \left(H_*(S) \xrightarrow{-\wedge \mathbb{T}_S} H_*(S) \xrightarrow{-\wedge \mathbb{T}_S} \dots \right)$$

in $\mathrm{Mod}_{H_*(S)}(\mathrm{Pr}^L)$.

Terminology 2.1.22. Objects of $\mathrm{SH}(S)$ are called ($\mathbb{A}^1$ -invariant) **motivic spectra**.

Proposition 2.1.23 (Functoriality of SH, [Rob15, §2]; see also [BH21, §4.1]). *The assignments $S \mapsto \mathrm{SH}(S)$ and $(f: S' \rightarrow S) \mapsto (f^*: \mathrm{SH}(S) \rightarrow \mathrm{SH}(S'))$, where f^* is the symmetric monoidal pullback, define a functor*

$$\mathrm{SH}: (\mathrm{Sch}^{\mathrm{qcqs}})^{\mathrm{op}} \longrightarrow \mathrm{CAlg}(\mathrm{Pr}_{\mathrm{st}}^L).$$

Proof sketch. The coherence of the assignments follows from the universal property of the construction (Proposition 2.1.3 applied over each base, cf. the initiality in the remark above). For stability: $\mathbb{T}_S \simeq S^1 \wedge \mathbb{G}_m$, so a smash factor of the invertible $\mathbb{T}_S$ is invertible; in particular the simplicial suspension $S^1 \wedge -$ is an equivalence, and a presentable pointed category with invertible suspension is stable. $\square$

Putting together all the constructions from Chapter 1 and this section, the motivic spectrum functor $M_S: \mathrm{Sm}_S \rightarrow \mathrm{SH}(S)$ factors as the composite

$$\mathrm{Sm}_S \xrightarrow{\mathfrak{z}} \mathrm{PShv}(\mathrm{Sm}_S) \xrightarrow{L_{\mathrm{mot}}} H(S) \xrightarrow{(-)_+} H_*(S) \xrightarrow{\sigma_{\mathbb{T}_S}^\infty} \mathrm{SH}(S),$$

where each step realizes a universal property:

- $\mathfrak{z}$: free cocompletion;
- L_{mot} : free imposition of Nisnevich descent and $\mathbb{A}^1$ -invariance;
- $(-)_+$: free adjunction of a basepoint;
- $\sigma_{\mathbb{T}_S}^\infty$: free $\otimes$ -inversion of $\mathbb{T}_S$ (Definition 2.1.5).

Concretely, $M_S(X) = \sigma_{\mathbb{T}_S}^\infty(L_{\mathrm{mot}}(X)_+)$ for $X \in \mathrm{Sm}_S$. The subscript $\mathbb{T}_S$ distinguishes this one-step passage from $H_*(S)$ to $\mathrm{SH}(S)$ from the generic formal-inversion functor σ^∞ of Section 2.1.3 and from the simplicial-suspension functor Σ^∞ introduced below.

Remark 2.1.24. Since every step can be upgraded to a symmetric monoidal functor, we have $M_S(S) \simeq \mathbb{1}_S \in \mathrm{SH}(S)$ and $M_S(X \times_S Y) \simeq M_S(X) \otimes M_S(Y)$.

One also has the following properties:

- the map $M_S(\mathbb{A}^1) \rightarrow M_S(S)$ induced by $\mathbb{A}^1 \rightarrow S$ is an equivalence;
- the Tate object $\mathbb{T}_S$ (called the *Tate motive* in [BEM25]) is the cofiber of $M_S(S) \xrightarrow{\infty} M_S(\mathbb{P}_S^1)$, and is invertible by construction;
- M_S carries Nisnevich distinguished squares in Sm_S to cartesian squares in $\mathrm{SH}(S)$.

In fact, $\mathrm{SH}(S)$, together with the symmetric monoidal functor $M_S: \mathrm{Sm}_S \rightarrow \mathrm{SH}(S)$, is initial among presentably symmetric monoidal stable categories $\mathcal{D}$ equipped with a symmetric monoidal functor $\mathrm{Sm}_S \rightarrow \mathcal{D}$ satisfying the properties above [Rob15, Corollary 2.39].

Recall from Definition 1.4.5 that there are two types of “circles” in $H_*(S)$. After stabilization the simplicial circle becomes the categorical suspension $[1]$, computed by the pushout

$$\begin{array}{ccc} E & \longrightarrow & 0 \\ \downarrow & \lrcorner & \downarrow \\ 0 & \longrightarrow & E[1]. \end{array}$$

Combined with the unstable identity $\mathbb{T}_S \simeq S^1 \wedge \mathbb{G}_m$ in $H_*(S)$, one finds

$$S^{i,j} \simeq \mathbb{T}_S^{\otimes j}[i - 2j] \quad (i \geq j \geq 0),$$

where the left-hand side is the bigraded sphere of Definition 1.4.5. For arbitrary $i, j \in \mathbb{Z}$ we take this formula as the *definition* of the bigraded sphere $S^{i,j} \in \mathrm{SH}(S)$ (cf. [BEM25, §3.1]), and for any $E \in \mathrm{SH}(S)$ we write $\Sigma^{i,j}E := E \otimes S^{i,j}$.

Instead of inverting $\mathbb{T}_S$ in a single step, one may first stabilize $H_*(S)$ with respect to the simplicial circle S^1 . This produces the category of $\mathbb{A}^1$ -invariant Nisnevich sheaves of spectra

$$\mathrm{Sp}(H_*(S)) \simeq \mathrm{Shv}_{\mathrm{Nis}, \mathbb{A}^1}(\mathrm{Sm}_S, \mathrm{Sp})$$

(indeed, $\mathbb{A}^1$ -invariant objects are closed under limits in $\mathrm{Shv}_{\mathrm{Nis}}(\mathrm{Sm}_S, \mathrm{Spc}_*)$, so a spectrum object of $H_*(S)$ is the same as an Ω -spectrum object of $\mathrm{Shv}_{\mathrm{Nis}}(\mathrm{Sm}_S, \mathrm{Spc}_*)$ with $\mathbb{A}^1$ -invariant levels, i.e. an $\mathbb{A}^1$ -invariant Nisnevich sheaf of spectra), together with the simplicial suspension functor $\Sigma^\infty: H_*(S) \rightarrow \mathrm{Shv}_{\mathrm{Nis}, \mathbb{A}^1}(\mathrm{Sm}_S, \mathrm{Sp})$. Since S^1 is already invertible in $\mathrm{Shv}_{\mathrm{Nis}, \mathbb{A}^1}(\mathrm{Sm}_S, \mathrm{Sp})$, inverting $\mathbb{T}_S$ there is equivalent to inverting $\mathbb{G}_m$; write

$$\sigma^\infty: \mathrm{Shv}_{\mathrm{Nis}, \mathbb{A}^1}(\mathrm{Sm}_S, \mathrm{Sp}) \longrightarrow \mathrm{SH}(S)$$

for the resulting formal-inversion functor, and ω^∞ for its right adjoint. We then have the factorization

$$\sigma_{\mathbb{T}_S}^\infty \simeq \sigma^\infty \circ \Sigma^\infty: H_*(S) \longrightarrow \mathrm{SH}(S),$$

and the identification $\mathrm{SH}(S) \simeq \mathrm{Shv}_{\mathrm{Nis}, \mathbb{A}^1}(\mathrm{Sm}_S, \mathrm{Sp})[\mathbb{G}_m^{-1}]$ [ABH26, p. 2.2.3]. The lax symmetric monoidal right adjoint

$$\omega^\infty: \mathrm{SH}(S) \longrightarrow \mathrm{Shv}_{\mathrm{Nis}, \mathbb{A}^1}(\mathrm{Sm}_S, \mathrm{Sp})$$

is explicitly described by $\omega^\infty(E) = \mathrm{map}_{\mathrm{SH}(S)}(M_S(-), E)$.

Remark 2.1.25. Objects of $\mathrm{Shv}_{\mathrm{Nis}, \mathbb{A}^1}(\mathrm{Sm}_S, \mathrm{Sp})$ are also known as **motivic S^1 -spectra**, and this category is sometimes denoted $\mathrm{SH}^{S^1}(S)$; we will not use this alternative notation.

2.1.5. The Annala–Iwasa motivic spectra category MotSp

The construction of $\mathrm{SH}(S)$ given above incorporates $\mathbb{A}^1$ -invariance from the start: we $\otimes$ -invert $\mathbb{T}_S$ *within* the $\mathbb{A}^1$ -localized world $H_*(S)$. A more recent perspective due to Annala–Iwasa [AI25] reverses this priority: one performs the $\mathbb{P}^1$ -inversion in a much larger, non- $\mathbb{A}^1$ -invariant base, and recovers SH at the end by imposing both Nisnevich descent and $\mathbb{A}^1$ -invariance. The resulting non- $\mathbb{A}^1$ -invariant framework is the natural home of recent advances such as the algebraic Conner–Floyd theorem [AHI24a] and motivic Atiyah duality [AHI24b] (the latter in the Nisnevich variant; see the remark below), and we record the definition of MotSp here for use later in these notes.

The base category is the elementary-blowup-excisive Zariski sheaf category. Recall that an **abstract blowup square** (Definition 2.3.1) consists of a closed immersion $Z \hookrightarrow X$ together with a proper map $\tilde{X} \rightarrow X$ that is an isomorphism over $X \setminus Z$, and that an **elementary blowup square** in Sm_S is the blowup square of an *elementary* closed immersion $i: Z \hookrightarrow X$ — one that, Zariski-locally on X , is the inclusion of the zero section of $\mathbb{A}_Z^n \sqcup Y$ for some $n \geq 0$ and some Y [AHI24a, Definition 2.1]. The basic example is the square

$$\begin{array}{ccc} \mathbb{P}_X^{n-1} & \longrightarrow & \mathrm{Bl}_{\{0\}}(\mathbb{A}_X^n) \\ \downarrow & & \downarrow \\ \{0\}_X & \longrightarrow & \mathbb{A}_X^n \end{array}$$

for $X \in \mathrm{Sm}_S$ and $n \geq 1$.

Definition 2.1.26 (Elementary blowup excision). Following [AHI24a, Definition 2.1], a presheaf $\mathcal{F} : (\mathrm{Sm}_S)^{\mathrm{op}} \rightarrow \mathrm{Sp}$ satisfies **elementary blowup excision (ebu)** if it sends every elementary blowup square in Sm_S to a pullback square. For Zariski sheaves this agrees with the original formulation [AI25, Definition 3.3.1], which imposes excision only for the standard squares displayed above. Excision for blowups along *arbitrary* closed immersions of smooth S -schemes — all of which are regularly immersed — is the a priori stronger *smooth blowup excision*; the two notions agree for Nisnevich sheaves of spectra [AHI24a, Proposition 2.2], but not obviously for the Zariski sheaves used here. We write

$$\mathcal{P}_{\mathrm{Zar},\mathrm{ebu}}(\mathrm{Sm}_S, \mathrm{Sp}) \subseteq \mathrm{PShv}(\mathrm{Sm}_S, \mathrm{Sp})$$

for the full subcategory of Zariski sheaves of spectra satisfying elementary blowup excision.

Definition 2.1.27 (Annala–Hoyois–Iwasa motivic spectra, [AI25, §2.2], [AHI24a, §§2 and 4]). The **(non- $\mathbb{A}^1$ -invariant) motivic spectra over S** are

$$\mathrm{MotSp}(S) := \mathcal{P}_{\mathrm{Zar},\mathrm{ebu}}(\mathrm{Sm}_S, \mathrm{Sp})[(\mathbb{P}_S^1, \infty)^{-1}],$$

the formal inversion of the pointed projective line $(\mathbb{P}_S^1, \infty)$ in Zariski sheaves of spectra satisfying elementary blowup excision. (A spectrum-valued sheaf is already canonically pointed, so no extra subscript $*$ is needed.) Annala–Iwasa originally defined MotSp in [AI25, §2.2] via Zariski sheaves of anima alone; the additional ebu enforcement, which is the form we use, is the variant adopted in [AHI24a, §§2 and 4], where the category is denoted MS_S .

Remark 2.1.28 (Three differences from $\mathrm{SH}(S)$). Comparing Definition 2.1.27 with the construction of $\mathrm{SH}(S)$ above, the differences are:

- *Topology*. The base topology is Zariski, not Nisnevich. (The sequel [AHI24b] works with the Nisnevich variant, which Annala–Iwasa now regard as the fundamental setting [AI25, §0.5]; the results of [AI25; AHI24a] hold in the Zariski setting used here.)
- *Excision*. A weak excision condition, ebu, is imposed by hand; nothing of the sort appears in the construction of $\mathrm{SH}(S)$, where excision comes for free ($\mathbb{A}^1$ -invariant Nisnevich sheaves satisfy even smooth blowup excision, cf. Proposition 2.1.29). Ebu is strictly weaker than the abstract-blowup excision underlying the cdh topology (Definition 2.3.1): genuine K-theory satisfies ebu but not abstract blowup excision.
- *No $\mathbb{A}^1$ -invariance*. Crucially, $\mathrm{MotSp}(S)$ is *not* required to be $\mathbb{A}^1$ -invariant. This is the main novelty.

Proposition 2.1.29 (SH inside MotSp , [AHI24a, §4]; cf. [AI25, §2.2]). *The full subcategory of $\mathrm{MotSp}(S)$ spanned by the objects that are $\mathbb{A}^1$ -invariant and satisfy Nisnevich descent is the Morel–Voevodsky category $\mathrm{SH}(S)$; here $\mathrm{SH}(S) \subseteq \mathrm{MotSp}(S)$ because objects of $\mathrm{SH}(S)$ satisfy even smooth blowup excision [MV99, §3, Remark 2.30]. We write*

$$L_{\mathrm{Nis},\mathbb{A}^1} : \mathrm{MotSp}(S) \longrightarrow \mathrm{SH}(S)$$

for the localization onto this full subcategory, which enforces both conditions. Equivalently,

$$\mathrm{SH}(S) \simeq \mathrm{Shv}_{\mathrm{Nis},\mathbb{A}^1}(\mathrm{Sm}_S, \mathrm{Sp})[(\mathbb{P}_S^1, \infty)^{-1}] \simeq L_{\mathrm{Nis},\mathbb{A}^1} \mathrm{MotSp}(S).$$

It does not seem to be known whether $\mathbb{A}^1$ -invariance alone forces Nisnevich descent in $\mathrm{MotSp}(S)$: [AI25, §2.2] accordingly present $\mathrm{SH}(S)$ as the localization of the $\mathbb{P}^1$ -stable Zariski theory at Nisnevich descent *together with* $\mathbb{A}^1$ -invariance, and [AHI24a, §4] expect the “true”

category of motivic spectra to be a further localization of $\mathrm{MotSp}(S)$ enforcing Nisnevich descent.

The advantage of MotSp over SH is that genuine algebraic K -theory $K(-)$, which is *not* $\mathbb{A}^1$ -invariant on singular schemes, lives natively in MotSp rather than only in its $\mathbb{A}^1$ -localized homotopy variant KH . The Annala–Hoyois–Iwasa Conner–Floyd theorem [AHI24a, Theorem 1.2] states that for every qcqs derived scheme X there is an isomorphism of bigraded rings

$$\mathrm{MGL}^{**}(X) \otimes_L \mathbb{Z}[\beta^{\pm 1}] \simeq K^{**}(X),$$

where L is the Lazard ring, acting on $\mathbb{Z}[\beta^{\pm 1}]$ through the multiplicative formal group law $x + y - \beta xy$, and $K^{p,q}(X) = K_{2q-p}(X)$. Here $\otimes_L$ is the ordinary tensor product of bigraded rings, not the derived tensor product of the standing convention. The isomorphism sees the full K -theory rather than KH precisely because the universal oriented motivic ring spectrum MGL [AHI24a, Theorem 1.1] is now constructed inside MotSp .

Warning 2.1.30 (Scope of these notes). Outside this subsection, the main structural development works with SH , since the BEM comparison, the slice conjecture, and the homotopy t -structure assume $\mathbb{A}^1$ -invariance; MotSp reappears only in later comparison remarks. Refining these structures to MotSp is an active area of research; we point the reader to [AI25; AHI24b; AHI24a] for the current state of the art.

2.2. SIX-FUNCTOR FORMALISMS

Having constructed SH as a functor $(\mathrm{Sch}^{\mathrm{qcqs}})^{\mathrm{op}} \rightarrow \mathrm{CAlg}(\mathrm{Pr}_{\mathrm{st}}^L)$, we now extend it to a full six-functor formalism. The present section sets up the abstract framework: general definition (Section 2.2.1), recollement and base change (Section 2.2.2), and the span-category encoding via the Cnossen–Lenz–Linskens universal property (Section 2.2.3). The application of this framework to SH , including cdh descent and the Heyer–Mann theory of cohomologically smooth, étale, and proper morphisms, is the subject of Section 2.3; the universal property of SH in its precise Drew–Gallauer form is taken up in Section 2.4.

2.2.1. Six-functor formalisms

A modern viewpoint, originating in Grothendieck’s school, organizes “cohomology with coefficients” on a class of geometric objects $\mathcal{C}$ as a single coherent package: an assignment $X \mapsto D(X)$ of a category of “coefficient systems” to each object, together with six functors relating these categories under morphisms.

An informal definition. A **six-functor formalism** on $\mathcal{C}$ assigns to each $X \in \mathcal{C}$ a presentable symmetric monoidal stable category $D(X)$, and to each morphism $f: Y \rightarrow X$ in $\mathcal{C}$ four functors

$$f^* \dashv f_*: D(Y) \rightleftarrows D(X), \quad f_! \dashv f^!: D(X) \rightleftarrows D(Y),$$

together with the symmetric monoidal product $\otimes$ and internal hom $\underline{\mathrm{Map}}$ on each $D(X)$, such that:

- f^* and $f_!$ are colimit-preserving and symmetric monoidal (resp. projection-formula compatible);

- base change: for every pullback square in $\mathcal{C}$,

$$\begin{array}{ccc} Y' & \xrightarrow{g'} & Y \\ f' \downarrow & \lrcorner & \downarrow f \\ X' & \xrightarrow{g} & X \end{array}$$

the canonical maps $g^* f_! \rightarrow f'_! g'^*$ and $f^! g_* \rightarrow g'_* f'^!$ are equivalences;

- projection formula: $f_!(A \otimes f^* B) \simeq f_!(A) \otimes B$ for $A \in \mathcal{D}(Y)$, $B \in \mathcal{D}(X)$.

Coherent versions of these data are encoded by a lax symmetric monoidal functor out of a span category — and, in the Cnossen–Lenz–Linskens refinement we will use, by a 2-functor out of a span 2-category; this span-category formulation is the precise way of encoding all the higher coherences and is the subject of Section 2.2.3.

The six functors carry distinct geometric content:

- f^* is *pullback*, f_* the *direct image*, $\otimes$ the *symmetric monoidal product* encoding cup products, and $\underline{\mathrm{Map}}$ the right adjoint to $\otimes$.
- $f_!$ is the *exceptional / proper-supported pushforward*: it agrees with f_* when f is proper and with $f_\#$ (extension by zero) when f is an open immersion.
- $f^!$ is the *exceptional pullback*, encoding orientation and duality data; in many examples, for f smooth of relative dimension d , $f^! \simeq f^* \otimes \omega_f$ with ω_f the relative dualizing complex.

The three operations f_* , $f^!$, $\underline{\mathrm{Map}}$ are *right adjoints* of f^* , $f_!$, $\otimes$ respectively, hence determined by the others. Modern formulations therefore typically specify only the three left functors $(f^*, f_!, \otimes)$ (a so-called **three-functor formalism**) and recover the rest by adjunction.

Three representative examples are:

- *Locally compact Hausdorff topological spaces* (Heyer–Mann [HM24]; see also [Sch26, Lecture VII]): the coefficient category is the derived category $\mathcal{D}(X, \Lambda)$ of sheaves of Λ -modules, for Λ an $\mathbb{E}_\infty$ -ring ($\Lambda = \mathbb{Z}$ in [Sch26]); Heyer–Mann in fact construct the formalism on the larger category of *condensed anima*, which contains locally compact Hausdorff spaces as a full subcategory.
- *ℓ -adic étale cohomology of schemes* [Sch26, Appendix to Lecture VII].
- *The motivic formalism* SH [Sch26, Lecture XI] — the universal $\mathbb{A}^1$ -invariant one by Drew–Gallauer (Theorem 2.4.1).

For pedagogical introductions complementing the present chapter, we recommend Scholze’s lecture notes [Sch26], Gallauer’s mini-course [Gal21] (the source of much of the present subsection’s exposition), and the textbook treatment in Cisinski–Déglise [CD19]. The span-category encoding adopted here is closest to Mann [Man22] and Scholze [Sch26]; its 2-categorical refinement, which we follow, is due to Cnossen–Lenz–Linskens [CLL25]. We turn to these after recording the recollement and base-change conventions in the next subsection.

2.2.2. Recollement and base change

The Cnossen–Lenz–Linskens universal property in the next subsection will be phrased in terms of (I, P) -biadjointability, which packages three pieces of data: left base change for I -morphisms, right base change for P -morphisms, and a mixed Beck–Chevalley compatibility between the two (condition (3) of Theorem 2.2.4). The *localization* (or recollement) axiom is a logically independent geometric input, from which both descent and — for SH — the mixed compatibility itself will be extracted in Section 2.3. We record these notions here.

Definition 2.2.1 (Recollement, localization axiom; [Lur17, Definition A.8.1]). Let C be an ∞ -category which admits finite limits, and let $C_0, C_1 \subseteq C$ be full subcategories. We say C is a **recollement** of C_0 and C_1 if the following conditions are satisfied:

- (i) the full subcategories C_0 and C_1 are stable under equivalence;
- (ii) the inclusions $C_0 \hookrightarrow C$ and $C_1 \hookrightarrow C$ admit left adjoints L_0 and L_1 ;
- (iii) the functors L_0 and L_1 are left exact;
- (iv) L_1 carries every object of C_0 to the final object of C ;
- (v) if α is a morphism in C such that $L_0(\alpha)$ and $L_1(\alpha)$ are equivalences, then α is an equivalence.

A functor $D: (\mathrm{Sch}^{\mathrm{qcqs}})^{\mathrm{op}} \rightarrow \mathrm{Pr}_{\mathrm{st}}^L$ satisfies the **localization axiom** if for every $X \in \mathrm{Sch}^{\mathrm{qcqs}}$ and every closed immersion $i: Z \hookrightarrow X$ whose open complement $j: U \hookrightarrow X$ is quasi-compact (e.g. every finitely presented closed immersion), so that $U \in \mathrm{Sch}^{\mathrm{qcqs}}$: (i) the right adjoints i_* and j_* of i^* and j^* are fully faithful; and (ii) the category $D(X)$ is a recollement of their essential images $i_*D(Z)$ and $j_*D(U)$. By (i) the recollement localizations are identified as $L_0 \simeq i_*i^*$ and $L_1 \simeq j_*j^*$, so conditions (4) and (5) say precisely that $j^*i_* \simeq 0$ and that (i^*, j^*) is jointly conservative — the form in which the axiom enters Proposition 2.3.8. Equivalently, $D(Z) \xrightarrow{i_*} D(X) \xrightarrow{j^*} D(U)$ is a recollement in the classical sense, with adjoint strings $i^* \dashv i_* \dashv i^!$ and $j^! \dashv j^* \dashv j_*$.

Definition 2.2.2 (Base change properties). Let $D: (\mathrm{Sch}^{\mathrm{qcqs}})^{\mathrm{op}} \rightarrow \mathrm{Pr}_{\mathrm{st}}^L$ be a functor.

- A morphism $g: C \rightarrow D$ satisfies **left base change** if for every pullback square

$$\begin{array}{ccc} A & \xrightarrow{k} & B \\ h \downarrow & \lrcorner & \downarrow f \\ C & \xrightarrow{g} & D \end{array}$$

the functor $k^*: D(B) \rightarrow D(A)$ admits a left adjoint $k_{\#}$, and the canonical morphism $k_{\#}h^* \rightarrow f^*g_{\#}$ is an equivalence.

- A morphism $f: B \rightarrow D$ satisfies **right base change** if for every pullback square as above, $h^*: D(C) \rightarrow D(A)$ admits a right adjoint h_* , and the canonical morphism $g^*f_* \rightarrow h_*k^*$ is an equivalence.
- For wide subcategories I, P closed under base change, D is (I, P) -**biadjointable** if every $i \in I$ satisfies left base change, every $p \in P$ satisfies right base change, and for every pullback square as above with horizontal legs k, g in I and vertical legs h, f in P the mixed Beck–Chevalley map $g_{\#}h_* \rightarrow f_*k_{\#}$ is an equivalence [CLL25, §2].

The link between recollement, base change, and Nisnevich/cdh descent will be proved in Proposition 2.3.8 after we set up the cdh topology in Section 2.3.1.

2.2.3. Span categories and suitable decompositions

The economical encoding of a six-functor formalism, initiated by Liu–Zheng [LZ24] and brought to its present form by Mann [Man22] and Scholze [Sch26], packages all the operations $(f^*, f_!, \otimes)$ and their coherences into a single lax symmetric monoidal functor on a span category. Given a wide subcategory $E \subset C$ closed under composition and base change, the **span category** $\mathrm{Span}(C, E)$ has the same objects as C and morphisms $X \rightarrow Y$ given by spans

$$X \leftarrow Z \xrightarrow{e} Y, \quad e \in E,$$

composed by pullback; when C has finite products, $\text{Span}(C, E)$ inherits a symmetric monoidal structure induced by the finite products of C , with unit the terminal object. (This monoidal structure is *not* Cartesian: the tensor is induced by the product of C , while the categorical product of $\text{Span}(C, E)$ — which coincides with the coproduct, span categories being semiadditive — is instead the disjoint union, provided C is extensive and E contains the summand inclusions, as in all our examples.) A **three-functor formalism** on (C, E) is then a lax symmetric monoidal functor

$$D: \text{Span}(C, E) \longrightarrow \text{Cat},$$

encoding f^* (backward legs), $f_!$ (forward legs), and $\otimes$ (monoidal structure) in a single coherent package; the remaining operations f_* , $f^!$, $\underline{\text{Map}}$ are obtained as right adjoints whenever these exist — automatically so for formalisms valued in Pr_{st}^L with $f_!$ and $\otimes$ colimit-preserving, as in the informal definition above — and are therefore *properties* of D rather than additional data.

The challenge is that constructing a lax monoidal functor out of $\text{Span}(C, E)$ directly is difficult. In practice, one starts from a much simpler piece of data, namely a functor

$$D_0: C^{\text{op}} \longrightarrow \text{CAlg}(\text{Cat})$$

encoding only f^* and $\otimes$, and asks when it extends to $\text{Span}(C, E)$. The key input is a **suitable decomposition** of E in the sense of Mann [Man22, Definition A.5.9]: a pair of wide subcategories $I, P \subset E$ (thought of as “étale-like” and “proper-like” morphisms respectively) such that every $e \in E$ factors as $e = p \circ i$ with $i \in I$ and $p \in P$, subject to mild cancellation and truncation axioms. The extension theorem then reads: if $i^* = D_0(i)$ admits a left adjoint $i_{\#}$ and $p^* = D_0(p)$ admits a right adjoint p_* satisfying the expected base change and projection formulae—in particular the mixed Beck–Chevalley equivalence $i_{\#}p'_* \simeq p_*i'_{\#}$ —then D_0 admits an extension to a three-functor formalism D (essentially unique when all maps in I and P are truncated, by Kuijper and Dauser–Kuijper; uniqueness becomes automatic in the 2-categorical formulation below), in which

$$f_! = p_* i_{\#} \quad \text{for any factorization } f = p \circ i.$$

Recently, Cnossen–Lenz–Linskens [CLL25] gave a conceptually streamlined account of this extension theorem. They enhance $\text{Span}(C, E)$ to a 2-category of spans of spans — a mild generalization of a construction of Haugseng [Hau18] — whose 2-morphisms encode the adjunction data of $i_{\#}$ and p_* in a fully coherent way, and they prove that the inclusion $C^{\text{op}} \hookrightarrow \text{Span}_2(C, E)_{I, P}$ is a universal (I, P) -biadjointable functor. In this setup, Mann’s extension theorem becomes a formal corollary; as a byproduct, the unit and counit witnessing $i_{\#} \dashv i^*$ are part of the structure of D , rather than non-canonical choices recoverable only after the fact.

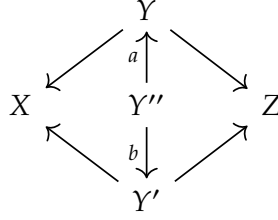
Construction 2.2.3 (Span 2-category; Haugseng [Hau18], Cnossen–Lenz–Linskens [CLL25]). The 2-category $\text{Span}_2(C, E)_{I, P}$ has:

- *objects*: objects in C ;
- *1-morphisms* $X \rightarrow Z$: spans

$$\begin{array}{ccc} & Y & \\ f \swarrow & & \searrow g \\ X & & Z \end{array}$$

with $g \in E$;

- 2-morphisms between two such spans: diagrams



with $a \in P$ and $b \in I$.

Theorem 2.2.4 ([CLL25, Theorem 5.17]). *Let $\mathcal{C}$ be a category with finite products and let $I, P \subseteq E \subseteq \mathcal{C}$ be wide subcategories closed under base change such that*

- (i) *I and P are left cancellable;*
- (ii) *every map in E factors as a map in I followed by a map in P ;*
- (iii) *every map in $I \cap P$ is truncated.*

Let $D_0: \mathcal{C}^{\text{op}} \rightarrow \text{CAlg}(\text{Cat})$ be a functor satisfying:

- (i) *for every $i: a \rightarrow b$ in I , the functor $i^*: D_0(b) \rightarrow D_0(a)$ admits a left adjoint $i_{\#}$ satisfying the Beck–Chevalley condition with respect to pullbacks in $\mathcal{C}$ and the projection formula $i_{\#}(- \otimes i^*(-)) \simeq i_{\#}(-) \otimes (-)$;*
- (ii) *for every $p: a \rightarrow b$ in P , the functor $p^*: D_0(b) \rightarrow D_0(a)$ admits a right adjoint p_* satisfying the dual Beck–Chevalley condition and dual projection formula;*
- (iii) *for every pullback square*

$$\begin{array}{ccc}
 a & \xrightarrow{i} & b \\
 p \downarrow & \lrcorner & \downarrow q \\
 c & \xrightarrow{j} & d
 \end{array}$$

in $\mathcal{C}$ with vertical maps in P and horizontal maps in I , the mixed Beck–Chevalley map $j_{\#}p_ \rightarrow q_*i_{\#}$ is an equivalence.*

Then the lax symmetric monoidal functor $(\mathcal{C}^{\times})^{\text{op}} \rightarrow \text{Cat}^{\times}$ corresponding to D_0 admits a unique extension to a lax symmetric monoidal 2-functor

$$D: \text{Span}_2(\mathcal{C}, E)_{I,P}^{\otimes} \longrightarrow \text{CAT}^{\times},$$

where CAT denotes the $(\infty, 2)$ -category of ∞ -categories, so that the extension retains the non-invertible 2-cells encoding the adjunction data. Moreover, for any $e \in E$ with factorization $e = p \circ i$ ($i \in I$, $p \in P$), the exceptional pushforward is $e_! \simeq p_ i_{\#}$.*

The above is the Mann-style extension formulation; equivalently, the inclusion $\mathcal{C}^{\text{op}} \hookrightarrow \text{Span}_2(\mathcal{C}, E)_{I,P}$ is the universal (I, P) -biadjointable functor (in the sense of Definition 2.2.2).

Sketch of proof. The argument proceeds in three steps.

Step 1. One verifies directly that the inclusion $\mathcal{C}^{\text{op}} \hookrightarrow \text{Span}_2(\mathcal{C}, E)_{I,P}$ is (I, P) -biadjointable, so that it is a candidate for the universal functor.

Step 2. Formal 2-categorical arguments produce an abstract initial (I, P) -biadjointable 2-category $\mathbb{B}(\mathcal{C}; I, P)$ together with a comparison functor

$$\mathbb{B}(\mathcal{C}; I, P) \longrightarrow \text{Span}_2(\mathcal{C}, E)_{I,P}.$$

This functor is essentially surjective on objects and 1-morphisms, so one reduces to checking that it induces equivalences on Hom-categories.

Step 3. The Hom-category comparison reduces in turn to identifying, for each $a \in C$, the free (I, P) -biadjointable functor generated by the identity of a , which is computed explicitly as

$$Y \longmapsto \text{Span}_{P,I}(C/a \times_C E/Y),$$

the category of spans $a \leftarrow u \rightarrow Y$ whose forward leg lies in E , with backward and forward legs of the span structure of type P and I respectively [CLL25, Theorem 3.7]. This identification completes the proof. $\square$

Thus, to construct a six-functor formalism it suffices to verify the biadjointability conditions of Theorem 2.2.4: left base change and the projection formula for I , right base change and the dual projection formula for P , and the mixed Beck–Chevalley equivalence. All coherences come for free from the universal property.

This is the framework in which we place SH. Taking $C = \text{Sch}^{\text{qcqs}}$, $E =$ morphisms locally of finite presentation, $I =$ étale morphisms, $P =$ finitely presented proper morphisms, the six-functor formalism on SH is obtained by verifying that the pullback functors f^* and the smash product satisfy precisely the (I, P) -biadjointability hypotheses; the deep geometric input is smooth base change and proper base change for SH, due to Morel–Voevodsky [MV99] and Ayoub [Ayo07] / Cisinski–Déglise [CD19]. We carry out the verification in Section 2.3.2 below.

2.3. CDH DESCENT AND THE SIX-FUNCTOR FORMALISM ON SH

We now apply the framework of Section 2.2 to the motivic stable category SH. The key technical input is the cdh topology, which couples the Nisnevich topology with abstract blowup squares; we set it up in Section 2.3.1, verify the localization axiom and base-change properties for SH to assemble the resulting six-functor formalism in Section 2.3.2, and then develop the relative duality theory of cohomologically smooth, étale, and proper morphisms in Section 2.3.3.

2.3.1. cdh topology

Abstract blowup squares

Definition 2.3.1 (Abstract blowup square). A commutative diagram in Sch^{qcqs}

$$\begin{array}{ccc} E & \longrightarrow & \tilde{X} \\ q \downarrow & \lrcorner & \downarrow p \\ Z & \xrightarrow{i} & X \end{array}$$

is an **abstract blowup square** if

- i is a finitely presented closed immersion;
- p is proper and finitely presented;
- p induces an isomorphism $p^{-1}(X \setminus Z) \xrightarrow{\sim} X \setminus Z$;
- the square is cartesian: $E = \tilde{X} \times_X Z$.

Remark 2.3.2 (Comparison with Nisnevich squares). Abstract blowup squares are structurally parallel to Nisnevich squares:

	Nisnevich square	Abstract blowup square
Decomposition of X	$U \xrightarrow{\text{open}} X$	$Z \xrightarrow{\text{closed}} X$
Covering morphism	$V \rightarrow X$ étale	$\tilde{X} \rightarrow X$ proper
Isomorphism over	$X \setminus U$	$X \setminus Z$
Typical source	étale neighborhoods	Blowups, normalizations

Both decompose X into a known part and a part needing coverage, then handle the latter via the appropriate class of morphisms.

Example 2.3.3 (cdh cover of the nodal curve). Let k be a field with $\text{char}(k) \neq 2$, and let $N = \text{Spec } k[x, y]/(y^2 - x^2(x + 1))$ be the plane nodal cubic, with node at the origin:

$$\begin{array}{ccc} E & \xrightarrow{\quad} & \tilde{N} \simeq \mathbb{A}^1 \\ \downarrow & \lrcorner & \downarrow p \\ Z = \{0\} & \hookrightarrow & N, \end{array}$$

where p is the normalization $t \mapsto (t^2 - 1, t^3 - t)$ (finite, hence proper) and $E = \text{Spec}(k[t]/(t^2 - 1)) \simeq \text{Spec}(k) \sqcup \text{Spec}(k)$, the two preimages $t = \pm 1$ of the node, encodes the two tangent directions at the node. For a cdh sheaf $\mathcal{F}$,

$$\mathcal{F}(N) \simeq \mathcal{F}(\mathbb{A}^1) \times_{\mathcal{F}(\{t=1\} \sqcup \{t=-1\})} \mathcal{F}(\{0\}).$$

Thus invariants of the singular curve N are completely determined by their values on the smooth model $\mathbb{A}^1$, the singular point $\{0\}$, and the gluing data over E .

Definition of the cdh topology

Definition 2.3.4 (cdh topology). For X qcqs, the **cdh topology** on $\text{Sch}_X^{\text{qcqs}}$ is the Grothendieck topology generated by

- (i) Nisnevich squares (Definition 1.1.6), and
- (ii) abstract blowup squares (Definition 2.3.1).

A presheaf $\mathcal{F}$ on $\text{Sch}_X^{\text{qcqs}}$ is a **cdh sheaf** if and only if $\mathcal{F}(\emptyset)$ is terminal and $\mathcal{F}$ sends both types of squares to pullback squares—equivalently, $\mathcal{F}$ is a Nisnevich sheaf (cf. Proposition 1.1.8, which includes the empty-scheme condition) satisfying excision for abstract blowup squares.

Cohomological dimension of the Nisnevich and cdh topoi

We pause to record two structural results — bounds on the homotopy dimension of the Nisnevich and cdh topoi — that are not needed for the construction of the six-functor formalism on SH, but enter crucially in later applications such as the homotopy t -structure of Section 3.1.7 and the comparison with cdh-motivic cohomology of Section 3.2.9. Recall that a topos X has **homotopy dimension** $\leq d$ if every $(d - 1)$ -connected sheaf on X admits a global section.

Theorem 2.3.5 (Clausen–Mathew, [CM21]). *For any qcqs scheme X of finite Krull dimension,*

$$\dim \mathrm{Shv}_{\mathrm{Nis}}(\mathrm{Et}_X) \leq \mathrm{Kr}.\dim(X),$$

where the left-hand side is the homotopy dimension of the Nisnevich topos on the small étale site of X .

This generalizes the classical vanishing of Nisnevich cohomology above the Krull dimension (Grothendieck, Kato–Saito) by controlling not only cohomology but the truncation behavior of arbitrary sheaves of anima: on a finite-dimensional scheme, every Nisnevich sheaf of spectra is hypercomplete, and every Postnikov tower converges. The bound is sharp and reflects a structural advantage of the Nisnevich topology over its neighbors: the Zariski topos has the same dimensional bound but is too coarse to see Galois descent, while the étale topos has no analogous finite bound (its cohomological dimension can be infinite, e.g. for fields of large absolute Galois group). Geometrically, X has no non-trivial Nisnevich coverings iff X is henselian local, so Nisnevich stalks are computed at henselian local rings $\mathcal{O}_{X,x}^h$.

The cdh topos admits a parallel bound in terms of the *valuative dimension*: $\mathrm{v}.\dim(X) \leq d$ iff for every affine open $\mathrm{Spec}(A) \subseteq X$, every prime $\mathfrak{p} \in A$, and every strict chain $A/\mathfrak{p} \subseteq V_0 \subsetneq V_1 \subsetneq \cdots \subsetneq V_n \subseteq \mathrm{Frac}(A/\mathfrak{p})$ of valuation subrings of $\mathrm{Frac}(A/\mathfrak{p})$, one has $n \leq d$. One always has $\mathrm{v}.\dim(X) \geq \mathrm{Kr}.\dim(X)$, with equality when X is Noetherian.

Theorem 2.3.6 (Elmanto–Hoyois–Iwasa–Kelly, [EHIK21]). *For any qcqs scheme X of finite valuative dimension,*

$$\dim \mathrm{Shv}_{\mathrm{cdh}}(\mathrm{Sch}_X^{\mathrm{fp}}) \leq \mathrm{v}.\dim(X).$$

Geometrically, X has no non-trivial cdh coverings iff X is the spectrum of a henselian valuation ring, so cdh stalks are computed at henselian valuation rings. The valuative dimension is the right invariant because its growth under affine-space products is controlled ($\mathrm{v}.\dim(X \times \mathbb{A}^n) = \mathrm{v}.\dim(X) + n$) and it is well-behaved on non-Noetherian schemes, where Krull dimension fails. Theorem 2.3.6 is the technical input behind several recent advances in motivic cohomology over singular and non-Noetherian bases, including the Elmanto–Morrow construction of motivic cohomology of equicharacteristic schemes [EM23] and its extension to arbitrary qcqs schemes, with the comparison to prismatic cohomology, in [BEM25].

Remark 2.3.7 (Finitarity). The Nisnevich and cdh topologies are both *finitary*: every covering family admits a finite subcover. As a consequence, sheafification commutes with filtered colimits, the topoi $\mathrm{Shv}_{\mathrm{Nis}}(\mathrm{Sm}_S)$ and $\mathrm{Shv}_{\mathrm{cdh}}(\mathrm{Sch}_S^{\mathrm{fp}})$ are compactly generated by the sheafifications of representables, and hypercompletion is automatic on schemes of finite Krull dimension (for the Nisnevich topology), respectively finite valuative dimension (for the cdh topology), by Theorems 2.3.5 and 2.3.6.

2.3.2. The six-functor formalism for SH

We now apply the framework of Section 2.2.3 to SH. The geometric input is the localization axiom together with base change for étale and proper morphisms; from these the descent properties of SH (Nisnevich and cdh) follow by the general fact below.

Proposition 2.3.8 (Localization plus base change implies descent). *Let $D: (\mathrm{Sch}^{\mathrm{qcqs}})^{\mathrm{op}} \rightarrow \mathrm{Pr}_{\mathrm{st}}^L$ satisfy the localization axiom (Definition 2.2.1).*

- (i) *If every étale morphism satisfies left base change (Definition 2.2.2), then D satisfies Nisnevich descent.*

- (ii) If every finitely presented proper morphism satisfies right base change, then D satisfies abstract blowup descent.

Proof. Note first that the localization axiom applied to the closed immersion $\emptyset \hookrightarrow X$ (with open complement id_X) forces $D(\emptyset) \simeq 0$: condition (4) of Definition 2.2.1 makes every object of $i_*D(\emptyset)$ zero, and i_* is fully faithful. So the empty-cover condition in the sheaf criterion of Proposition 1.1.8 (cf. Definition 2.3.4) holds automatically.

We treat abstract blowup descent; the Nisnevich case is similar (and dual). Consider an abstract blowup square

$$\begin{array}{ccc} E & \xrightarrow{k} & \tilde{X} \\ q \downarrow & \lrcorner & \downarrow p \\ Z & \xrightarrow{i} & X. \end{array}$$

By assumption, p^* and q^* admit right adjoints p_* and q_* . We must show that the comparison functor

$$(p^*, i^*): D(X) \longrightarrow D(\tilde{X}) \times_{D(E)} D(Z)$$

is an equivalence. Its right adjoint G sends a pair (A, B) with common image $C \in D(E)$ to the fiber product

$$G(A, B) = p_*A \times_{(iq)_*C} i_*B.$$

Conservativity of (p^, i^*) .* By the localization axiom, (i^*, j^*) is jointly conservative, where $j: U = X \setminus Z \hookrightarrow X$. Since p restricts to an isomorphism over U , we have $j^* \simeq h^*p^*$ (with $h: U \hookrightarrow \tilde{X}$ the open complement of k), so (p^*, i^*) is also jointly conservative.

Counit is an equivalence. It remains to show that for every (A, B) , the counit $(p^*, i^*) \circ G(A, B) \rightarrow (A, B)$ is an equivalence. We verify the two components separately.

(i^ -component.)* Since i is a closed immersion, i_* is fully faithful by the localization axiom, so $i^*i_* \simeq \text{id}$. Combined with right base change for p along i , which gives $i^*p_*A \simeq q_*k^*A \simeq q_*C$, the fiber product becomes

$$i^*(p_*A \times_{(iq)_*C} i_*B) \simeq q_*C \times_{q_*C} B \simeq B.$$

(p^ -component.)* We must show $p^*(p_*A \times_{(iq)_*C} i_*B) \xrightarrow{\sim} A$. By the localization axiom, (k^*, h^*) is jointly conservative on $D(\tilde{X})$, so it suffices to check after applying each.

Applying k^* : using $k^*p^* = q^*i^*$ and the previous case,

$$k^*p^*(p_*A \times_{(iq)_*C} i_*B) \simeq q^*B \simeq C \simeq k^*A.$$

Applying h^* : since i_*B and $(iq)_*C$ are supported on Z , applying $j^* = h^*p^*$ kills them, leaving j^*p_*A . We must show $j^*p_*A \simeq h^*A$. Since p restricts to an isomorphism over U , the square

$$\begin{array}{ccc} U & \xrightarrow{h} & \tilde{X} \\ p|_U \downarrow & \lrcorner & \downarrow p \\ U & \xrightarrow{j} & X \end{array}$$

is cartesian with $p|_U$ an isomorphism. Right base change for the proper morphism p along j (Definition 2.2.2, the hypothesis of part (2)) therefore gives $j^*p_*A \simeq (p|_U)_*h^*A \simeq h^*A$, as required. $\square$

We now state the localization axiom for SH.

Theorem 2.3.9 (Morel–Voevodsky, [MV99]; for the general case, [Hoy17]). *The stable motivic functor $\mathrm{SH}: (\mathrm{Sch}^{\mathrm{qcqs}})^{\mathrm{op}} \rightarrow \mathrm{Pr}_{\mathrm{st}}^L$ satisfies the localization axiom. (Morel–Voevodsky proved this over a Noetherian base of finite Krull dimension; the extension to qcqs schemes, by Noetherian approximation, is due to Hoyois [Hoy17], in line with Remark 2.3.11 below.)*

Theorem 2.3.10 (Existence: Ayoub [Ayo07], Cisinski–Déglise [CD19]; universal property: Cnossen–Lenz–Linskens [CLL25]). *There is an essentially unique lax symmetric monoidal extension making the following diagram commute:*

$$\begin{array}{ccc} (\mathrm{Sch}^{\mathrm{qcqs}})^{\mathrm{op}} & \xrightarrow{\mathrm{SH}} & \mathrm{Pr}_{\mathrm{st}}^L \\ \downarrow & \nearrow \text{dashed} & \\ \mathrm{Span}_2(\mathrm{Sch}^{\mathrm{qcqs}}, E)_{I,P}^{\otimes} & & \end{array}$$

with $E =$ morphisms locally of finite presentation, $I =$ étale morphisms, and $P =$ finitely presented proper morphisms. (Here the top arrow is the underlying functor of SH , and the dashed extension is a lax symmetric monoidal 2-functor into $(\mathrm{Pr}_{\mathrm{st}}^L)^{\otimes}$; its restriction along the vertical inclusion recovers SH valued in $\mathrm{CAlg}(\mathrm{Pr}_{\mathrm{st}}^L)$, since the backward-leg functors f^* are symmetric monoidal, but the extension itself is not CAlg -valued: the forward legs $f_!$ are only laxly compatible with $\otimes$, via the projection formula.) Explicitly, $\mathrm{Span}_2(\mathrm{Sch}^{\mathrm{qcqs}}, E)_{I,P}$ has:

- objects: quasi-compact quasi-separated schemes;
- 1-morphisms $X \rightarrow Z$: spans

$$\begin{array}{ccc} & Y & \\ f \swarrow & & \searrow g \\ X & & Z \end{array}$$

with g locally of finite presentation;

- 2-morphisms between two such spans: diagrams

$$\begin{array}{ccccc} & & Y & & \\ & \swarrow & \uparrow a & \searrow & \\ X & & Y'' & & Z \\ & \swarrow & \downarrow b & \searrow & \\ & & Y' & & \end{array}$$

with a finitely presented proper and b étale.

In this extension, the **exceptional pushforward** along $f \in E$ is given by $f_! = p_* i_{\#}$ for any factorization $f = p \circ i$ with i étale and p finitely presented proper.

Remark 2.3.11 (Structure of the proof). The proof decomposes into a formal step and a deep geometric step.

The *formal step* is the universal property of $\mathrm{Span}_2(-, E)_{I,P}^{\otimes}$ (Theorem 2.2.4): given any lax symmetric monoidal functor $D_0: \mathcal{C}^{\mathrm{op}} \rightarrow \mathrm{CAlg}(\mathrm{Cat})$ which is (I, P) -biadjointable and satisfies the projection formula, there is an essentially unique extension to $\mathrm{Span}_2(\mathcal{C}, E)_{I,P}^{\otimes}$. This step is purely 2-categorical and contains no algebro-geometric content.

The *geometric step* is to verify the input hypothesis, namely that SH is (I, P) -biadjointable. This breaks further into three ingredients:

- **Étale base change and the projection formula for $i_{\#}$:** these follow essentially from the construction of SH. More generally, for any smooth $i: T \rightarrow S$ the functor $i_{\#}$ on $\mathrm{PShv}(\mathrm{Sm}_-)$ is the left Kan extension along the forgetful functor $\mathrm{Sm}_T \rightarrow \mathrm{Sm}_S$, $(Y \rightarrow T) \mapsto (Y \rightarrow T \rightarrow S)$ — equivalently, its right adjoint i^* is restriction along this functor — and is compatible with base change at the presheaf level; the case required by the theorem is the specialization to i étale.
- **Localization:** the recollement (i^*, j^*) for closed immersions i with open complement j , i.e. Theorem 2.3.9.
- **Proper base change and the projection formula for p_* :** this is the deep input. Ayoub [Ayo07] proved proper base change for projective morphisms; Cisinski–Déglise [CD19] extended this to arbitrary proper morphisms using Chow’s lemma and cdh descent, by induction on the dimension, in their setting of Noetherian finite-dimensional schemes; the further extension to arbitrary qcqs bases proceeds by Noetherian approximation [Hoy17].

Once all three ingredients are in hand, the full six-functor package on SH follows by invoking the universal property (Theorem 2.2.4). The mixed Beck–Chevalley equivalence $i_{\#}p'_* \simeq p_*i'_{\#}$ is not an additional input: for a pullback square one of whose edges is étale, it is deduced by combining all three ingredients above — étale base change, proper base change, and the joint conservativity supplied by localization (one checks the map after restriction to an open subscheme and to its closed complement).

By Proposition 2.3.8 together with Theorem 2.3.9 and the étale and proper base change ingredients of Remark 2.3.11, we obtain:

Corollary 2.3.12. *The functor SH is a cdh sheaf.*

2.3.3. Cohomological smoothness, étaleness, and properness

Three classes of morphisms organize the entire content of any six-functor formalism: the cohomologically smooth, étale, and proper ones. The Drew–Gallauer characterization (Theorem 2.4.1 below) is phrased in their language, and they appear throughout applications. We give the definitions and basic properties here, following Heyer–Mann [HM24].

Throughout this subsection, (C, E) denotes a **geometric setup**: a category C together with a class E of morphisms containing all isomorphisms, closed under composition and pullback along arbitrary morphisms, and closed under diagonals ($f \in E \Rightarrow \Delta_f \in E$) — equivalently, E is right cancellative; equivalently, the wide subcategory $C_E \subseteq C$ admits and the inclusion preserves fiber products [HM24, Definition 2.1.1, Lemma 2.1.5]. The diagonal condition, not needed in Section 2.2.3, is what makes the identity kernels $\Delta_{f!}(\mathbb{1})$ below well defined; it holds for $E =$ morphisms locally of finite presentation, the class used for SH in Theorem 2.3.10. As before, $D: \mathrm{Span}(C, E) \rightarrow \mathrm{Cat}$ is a three-functor formalism (or a six-functor formalism, when stated), and we write Cat_2 for the 2-category of categories.

Cohomological smoothness

The motivating example is Poincaré duality.

Theorem 2.3.13 (Poincaré duality, classical). *Let $f: X \rightarrow Y$ be a proper map of topological manifolds of relative dimension d (i.e. a topological submersion: locally on X of the form $V \times \mathbb{R}^d \rightarrow V$ for $V \subseteq Y$ open). Then $f_*: D(X, \mathbb{Z}) \rightarrow D(Y, \mathbb{Z})$ admits a right adjoint $f^!$, and there is a sheaf*

$\omega_f := f^!(\mathbb{1}_Y) \in D(X, \mathbb{Z})$, locally isomorphic to $\mathbb{Z}[d]$, with

$$f^! \simeq f^* \otimes \omega_f.$$

The sheaf ω_f is the **dualizing complex** of f .

A morphism in E which satisfies the analogue of Poincaré duality in D will be called **cohomologically smooth** (or **D-smooth**). Two requirements are immediate:

- There is an isomorphism $f^! \simeq f^* \otimes \omega_f$ for some sheaf ω_f ;
- $\omega_f := f^!(\mathbb{1}_Y)$ is invertible.

But this is not yet enough. Consider a pullback square

$$\begin{array}{ccc} X' & \xrightarrow{f'} & Y' \\ g' \downarrow & \lrcorner & \downarrow g \\ X & \xrightarrow{f} & Y. \end{array}$$

We expect f' to be D-smooth as well, and we expect the dualizing complexes to be compatible with restriction:

$$g'^* \omega_f \simeq \omega_{f'}.$$

This compatibility is not automatic from pure category theory; it must be imposed.

Definition 2.3.14 (Cohomologically smooth morphism). A morphism $f: X \rightarrow Y$ in E is **cohomologically smooth** (or **D-smooth**) if it satisfies:

- (i) The functor $f_!$ admits a right adjoint $f^!$, and the natural transformation

$$f^!(\mathbb{1}_Y) \otimes f^*(-) \rightarrow f^!(-)$$

is an equivalence; we write $\omega_f := f^!(\mathbb{1}_Y)$ and call it the **dualizing complex** of f .

- (ii) The dualizing complex ω_f is $\otimes$ -invertible in $D(X)$.

- (iii) For every morphism $g: Y' \rightarrow Y$ in C and pullback square as above, the morphism f' also satisfies ((i))–((ii)), and the canonical map $g'^* f^!(\mathbb{1}_Y) \rightarrow f'^!(\mathbb{1}_{Y'})$ is an equivalence.

Remark 2.3.15. Cohomological smoothness in the sense of Definition 2.3.14 is weaker than the geometric notion of a smooth morphism. For instance, in the étale six-functor formalism $X \mapsto \mathrm{Shv}(X_{\text{ét}}, \widehat{D}_{\text{pf}}(\mathbb{Z}))$ of [Sch26] (coefficients in profinitely complete, equivalently torsion, complexes), every universal homeomorphism between schemes (a morphism inducing a homeomorphism of underlying topological spaces after arbitrary base change) is cohomologically smooth: étale topology is insensitive to such maps, so they look like isomorphisms at the level of D , and isomorphisms are trivially D-smooth.

The kernel 2-category and Fourier–Mukai transforms

Verifying Definition 2.3.14 directly is awkward because of the base-change clause ((iii)). The next definitions package D-smoothness into a single 2-categorical statement, after which the base-change clause is automatic.

Fix $Y \in C$, and let $C_E \subseteq C$ be the wide subcategory whose morphisms are those in E . The slice $(C_E)_{/Y}$, paired with the class of *all* morphisms, is itself a geometric setup, and D restricts to a functor

$$D_Y: \mathrm{Span}((C_E)_{/Y}, \mathrm{all})^{\otimes} \rightarrow \mathrm{Span}(C, E)^{\otimes} \xrightarrow{D} \mathrm{Cat}^{\times}.$$

Definition 2.3.16 (Fourier–Mukai transform). Given $X_1, X_2 \in (C_E)_{/Y}$ and $K \in D(X_1 \times_Y X_2)$, the **Fourier–Mukai transform with kernel K** is the functor

$$F_K: D(X_1) \rightarrow D(X_2), \quad A \mapsto (\pi_2)_!(\pi_1^* A \otimes K).$$

A span $X_1 \xleftarrow{h} Z \xrightarrow{k} X_2$ over Y induces a Fourier–Mukai transform whose kernel is $K = (h, k)_! \mathbb{1}_Z$.

Remark 2.3.17. The terminology is borrowed from the classical analytic transform $T: C^\infty(X) \rightarrow C^\infty(Y)$, $f \mapsto \int_X K(x, y) f(x) dx$, determined by its kernel K . Definition 2.3.16 categorifies this construction: $\pi_1^* A$ pulls A back to a “function of two variables”, $\otimes K$ multiplies pointwise by the kernel, and $(\pi_2)_!$ is integration along the first variable.

Specializing to $f: X \rightarrow Y$ in E :

- Setting $X_1 = X$, $X_2 = Y$, and $K_1 = \mathbb{1}_X \in D(X \times_Y Y) = D(X)$, the Fourier–Mukai transform is $A \mapsto f_! A$.
- Setting $X_1 = Y$, $X_2 = X$, and $K_2 = \omega_f = f^!(\mathbb{1}_Y) \in D(Y \times_Y X) = D(X)$, the transform is $B \mapsto f^* B \otimes \omega_f$.

Idea. The desired identity $f^! \simeq f^* \otimes \omega_f$ should arise by exhibiting the kernels K_1 and K_2 as an adjoint pair in a 2-category whose objects are schemes over Y and whose 1-morphisms are kernels. Since 2-functors preserve adjunctions, the resulting adjunction is automatically compatible with arbitrary base change $g: Y' \rightarrow Y$, taking care of clause ((iii)) of Definition 2.3.14 for free.

The invertibility of ω_f is a separate, relatively independent condition; this is what motivates the notion of suaveness below.

Definition 2.3.18 (2-category of kernels). Let $Y \in C$. The **2-category of kernels** $\mathbb{K}_{D,Y}$ is the 2-category whose:

- *Objects* are objects of $(C_E)_{/Y}$, i.e., morphisms $X \rightarrow Y$ in E .
- *1-morphism category* from X_1 to X_2 is $\text{Fun}_Y(X_1, X_2) := D(X_1 \times_Y X_2)$.
- *Composition* of $M \in \text{Fun}_Y(X_1, X_2)$ and $N \in \text{Fun}_Y(X_2, X_3)$ is

$$N \circ M := (\pi_{13})_!(\pi_{12}^* M \otimes \pi_{23}^* N) \in D(X_1 \times_Y X_3),$$

where the π_{ij} are the projections from $X_1 \times_Y X_2 \times_Y X_3$.

The identity 1-morphism on X is $\Delta_{f!}(\mathbb{1}_X) \in D(X \times_Y X)$, where $\Delta_f: X \rightarrow X \times_Y X$ is the relative diagonal.

Conceptually, $\mathbb{K}_{D,Y}$ is obtained from $\text{Span}((C_E)_{/Y}, \text{all})$, which is closed (being a span category over a category with finite limits) and hence self-enriched, by base-changing the self-enrichment along D_Y ; see [Aok26, Definition 2.4].

Remark 2.3.19 (The presentable 2-category of kernels). When D takes values in Pr , Aoki [Aok26, Definition 2.5] gives a closely related variant that bypasses enriched category theory entirely. Regarding D_Y as a commutative algebra in $\text{Fun}(\text{Span}((C_E)_{/Y}, \text{all}), \text{Pr})$ via Day convolution, the **presentable 2-category of kernels** is

$$\mathbb{K}_{D,Y}^{\text{pres}} := \text{Mod}_{D_Y}(\text{Fun}(\text{Span}((C_E)_{/Y}, \text{all}), \text{Pr})).$$

This is the Pr -valued presheaf cocompletion of the enriched kernel 2-category from Definition 2.3.18: by [Aok26, Theorem 2.6],

$$\mathbb{K}_{D,Y}^{\text{pres}} \simeq \text{PShv}^{\text{Pr}}(\mathbb{K}_{D,Y}^{\text{op}}),$$

so $\mathbb{K}_{D,Y}$ embeds fully faithfully into $\mathbb{K}_{D,Y}^{\text{pres}}$ via the enriched Yoneda embedding. The two formulations are interchangeable for the developments below: properties such as “being a left adjoint” are detected on the Yoneda image, so the kernel-level adjunctions defining suaveness, primness, and smoothness in Definitions 2.3.14 and 2.3.21 can equivalently be formulated in either $\mathbb{K}_{D,Y}$ or $\mathbb{K}_{D,Y}^{\text{pres}}$.

There is a canonical 2-functor

$$\Psi_{D,Y} = \text{Fun}_Y(Y, -) : \mathbb{K}_{D,Y} \longrightarrow \text{Cat}_2,$$

sending an object X to $D(X)$ and a 1-morphism $M \in \text{Fun}_Y(X_1, X_2)$ to the Fourier–Mukai transform $(\pi_2)_!(\pi_1^*(-) \otimes M)$.

For the reader’s convenience, we record three facts about adjunctions in 2-categories that we will use freely (proofs are standard, see [HM24, Appendix D]):

- Any 2-functor $F : \mathbb{C} \rightarrow \mathbb{D}$ preserves adjunctions; in fact, it preserves the entire mate correspondence [HM24, Lemma D.2.7].
- Adjunctions are unique up to a contractible space of choices: any two right adjoints to a given 1-morphism are canonically equivalent.
- A pair (f, g) with candidate unit η and counit ε such that the triangle composites are isomorphisms (not necessarily identities) can be corrected to a strict adjunction by modifying η .

Theorem 2.3.20 (Smoothness via kernels; cf. [HM24, Definition 4.5.1, Remark 4.5.2]). *A morphism $f : X \rightarrow Y$ in E is D-smooth if and only if*

- The 1-morphism $\mathbb{1}_X \in \text{Fun}_Y(X, Y) = D(X)$ is a left adjoint in $\mathbb{K}_{D,Y}$. Denote its right adjoint by $\omega_f \in \text{Fun}_Y(Y, X) = D(X)$.*
- ω_f is $\otimes$ -invertible in $D(X)$.*

When these hold, $\omega_f \simeq f^!(\mathbb{1}_Y)$.

Proof sketch. ($\Rightarrow$) Suppose f is D-smooth. Apply clause ((iii)) of Definition 2.3.14 to the self-base-change $g = f$: in the resulting square $X' = X \times_Y X$, $f' = \pi_2$, $g' = \pi_1$, it says that the canonical map $\pi_1^* f^!(\mathbb{1}_Y) \rightarrow \pi_2^!(\mathbb{1}_X)$ is an equivalence. By the concrete criterion Proposition 2.3.27(1) below applied to $A = \mathbb{1}_X$ (= [HM24, Lemma 4.5.4]), this exhibits $\mathbb{1}_X$ as a left adjoint in $\mathbb{K}_{D,Y}$ with right adjoint $\omega_f = f^!(\mathbb{1}_Y)$; invertibility is clause ((ii)). (Clauses ((i))–((ii)) alone would not suffice: a functor-level adjunction does not lift along $\Psi_{D,Y}$, which is not 2-fully faithful — this is precisely why clause ((iii)) is imposed.)

($\Leftarrow$) Apply $\Psi_{D,Y}$ to the adjunction $\mathbb{1}_X \dashv \omega_f$. Since 2-functors preserve adjunctions, the resulting functors $\Psi_{D,Y}(\mathbb{1}_X) \simeq f_!(-)$ and $\Psi_{D,Y}(\omega_f) \simeq f^*(-) \otimes \omega_f$ form an adjunction in Cat_2 . By uniqueness of right adjoints, $f^*(-) \otimes \omega_f \simeq f^!(-)$, which is clause ((i)) of Definition 2.3.14; setting $- = \mathbb{1}_Y$ gives $\omega_f \simeq f^!(\mathbb{1}_Y)$. The base-change clause ((iii)) follows from the fact that $S \mapsto \mathbb{K}_{D,S}$ assembles into a lax symmetric monoidal functor $\text{Span}(C, E)^{\otimes} \rightarrow 2\text{Cat}^{\times}$ into the category of 2-categories [HM24, Theorem 4.2.4], so the adjunction $\mathbb{1}_X \dashv \omega_f$ is preserved by base change. $\square$

Suaveness and primness

Theorem 2.3.20 decomposes cohomological smoothness as “left-adjointness in $\mathbb{K}_{D,Y}$ ” plus “invertibility of the dualizing complex”. Dropping the invertibility yields broader classes of objects.

Definition 2.3.21 (Suave and prim objects). Let $f: X \rightarrow Y$ be in E and $A \in D(X)$.

- (i) A is f -**suave** if, viewed as a 1-morphism $X \rightarrow Y$ in $\mathbb{K}_{D,Y}$, it is a left adjoint. Its right adjoint is the f -**suave dual** $SD_f(A) \in D(X)$. Write $\text{Suave}_f(X) \subseteq D(X)$ for the full subcategory of f -suave objects.
- (ii) A is f -**prim** if, viewed as a 1-morphism $Y \rightarrow X$ in $\mathbb{K}_{D,Y}$, it is a left adjoint. Its right adjoint is the f -**prim dual** $PD_f(A) \in D(X)$. Write $\text{Prim}_f(X) \subseteq D(X)$ for the full subcategory of f -prim objects.

Remark 2.3.22 (Arrow convention for prim duality). We have used the canonical self-duality $\mathbb{K}_{D,Y} \simeq \mathbb{K}_{D,Y}^{\text{op}}$ to phrase both clauses of Definition 2.3.21 as left-adjoint conditions. In the convention of Heyer–Mann, the same prim object A is instead displayed as a right adjoint $A: X \rightarrow Y$, with $PD_f(A): Y \rightarrow X$ its left adjoint. Reversing 1-morphisms gives exactly our displayed adjunction $A: Y \rightarrow X \dashv PD_f(A): X \rightarrow Y$; only the arrow convention has changed.

Remark 2.3.23 (Origin of the terminology). The terms are due to Scholze (suave) and Hansen (prim); applications typically require D to be a six-functor formalism, although the definition makes sense for three-functor formalisms.

- **Suave** (*relaxed smoothness*): when $A = \mathbb{1}_X$ is f -suave, applying $\Psi_{D,Y}$ to the adjunction $\mathbb{1}_X \dashv SD_f(\mathbb{1}_X)$ gives $f^*(-) \otimes SD_f(\mathbb{1}_X) \simeq f^!(-)$. More generally, every f -suave A produces a relation between f^* and $f^!$.
- **Prim** (*primitive properness*): when $A = \mathbb{1}_X$ is f -prim, the dual statement gives $f_!(PD_f(\mathbb{1}_X) \otimes -) \simeq f_*(-)$. Primness is the phenomenon that f_* is recovered from $f_!$ by twisting; this is strictly weaker than the proper-base-change identity $f_! \simeq f_*$, so it is “primitive” in this sense.

Remark 2.3.24 (Stack-theoretic perspective). Aoki [Aok26, Definition 4.5] reformulates these notions at the stack level: a morphism $[X] \rightarrow [Y]$ in the presentable kernel 2-category $\mathbb{K}_{D,Y}^{\text{pres}}$ of Remark 2.3.19 is called *weakly suave* if it admits an $[X]$ -linear left adjoint, *prim* if it admits an $[X]$ -linear right adjoint, and *suave* if it is weakly suave and $[Y]$ is dualizable over $[X]$. Modulo passage between the enriched and presentable kernel 2-categories, Aoki’s *weakly suave* corresponds to D -suaveness of f , i.e. the case $A = \mathbb{1}_X$ of Definition 2.3.21 (cf. Definition 2.3.30 below); when D takes values in Pr^{st} , dualizability of $[Y]$ over $[X]$ is automatic, so *suave* and *weakly suave* agree in this case [Aok26, Example 4.8]. Neither notion includes invertibility of ω_f : cohomological smoothness corresponds instead to Aoki’s separate notion of *smooth*, namely suave with $f^!\mathbb{1}$ invertible [Aok26, Definition 4.29].

In a precise sense, A is f -suave (or f -prim) iff it is “finite-dimensional relative to f ”: it admits a reflexive duality.

Proposition 2.3.25 (Reflexive duality; cf. [HM24, §4.4]). Let D be a six-functor formalism on (C, E) and $f: X \rightarrow Y$ in E .

- (i) If $A \in D(X)$ is f -suave, then $SD_f(A)$ is also f -suave (with right adjoint A in $\mathbb{K}_{D,Y}$), and there is a natural isomorphism

$$SD_f(A) \otimes f^*(-) \simeq \underline{\text{Map}}(A, f^!(-)): D(Y) \rightarrow D(X).$$

Setting $- = \mathbb{1}_Y$ yields $SD_f(A) \simeq \underline{\text{Map}}(A, f^!(\mathbb{1}_Y))$, exhibiting $SD_f(A)$ as a relative Verdier dual of A . The bidual identity $SD_f(SD_f(A)) \simeq A$ then holds, and SD_f defines a contravariant equivalence

$$SD_f: \text{Suave}_f(X)^{\text{op}} \xrightarrow{\sim} \text{Suave}_f(X).$$

(ii) Dually, if $A \in \mathcal{D}(X)$ is f -prim, then $\mathrm{PD}_f(A)$ is also f -prim, with

$$f_!(\mathrm{PD}_f(A) \otimes -) \simeq f_* \underline{\mathrm{Map}}(A, -): \mathcal{D}(X) \rightarrow \mathcal{D}(Y),$$

and the bidual $\mathrm{PD}_f(\mathrm{PD}_f(A)) \simeq A$ holds. Consequently PD_f defines a contravariant equivalence

$$\mathrm{PD}_f: \mathrm{Prim}_f(X)^{\mathrm{op}} \xrightarrow{\sim} \mathrm{Prim}_f(X).$$

Proof sketch. The proofs are exercises in the 2-categorical adjunction calculus; we sketch the suave case. The kernel 2-category has a built-in self-duality $\mathbb{K}_{D,Y} \simeq \mathbb{K}_{D,Y}^{\mathrm{op}}$ obtained by reversing 1-morphisms, under which a left adjoint becomes a right adjoint and vice versa; the bidual identity follows. The Hom formula is obtained by applying $\Psi_{D,Y}$ to the adjunction $A \dashv \mathrm{SD}_f(A)$ and unfolding via $f_! \dashv f^!$ together with the definition of internal hom. $\square$

A more concrete characterization of suaveness and primness is available, useful in computations. It rests on the following standard 2-categorical lemma.

Lemma 2.3.26 (Adjoints in 2-categories, [HM24, Proposition D.2.8]). *Let $\mathbb{C}$ be a 2-category and $f: X \rightarrow Y$ a 1-morphism. The following are equivalent:*

- (i) f is a left adjoint.
- (ii) For each $Z \in \{X, Y\}$, the post-composition functor $f_*: \mathrm{Fun}_{\mathbb{C}}(Z, X) \rightarrow \mathrm{Fun}_{\mathbb{C}}(Z, Y)$, $h \mapsto f \circ h$, admits a right adjoint G_Z , and the canonical morphism $G_Y(\mathrm{id}_Y) \circ f \rightarrow G_X(f)$ becomes an isomorphism after applying $\mathrm{Hom}_{\mathrm{Fun}(X,X)}(\mathrm{id}_X, -)$.
- (iii) For each $Z \in \mathbb{C}$, f_* admits a right adjoint G_Z , and for every $h: Z \rightarrow Z'$ in $\mathbb{C}$, the canonical map $h^* G_{Z'} \rightarrow G_Z h^*$ is an equivalence of functors.

The right adjoint of f is then $G_Y(\mathrm{id}_Y): Y \rightarrow X$.

Proposition 2.3.27 (Concrete criterion for suave/prim objects, [HM24, Lemmas 4.4.5, 4.4.6]). *Let $\mathcal{D}$ be a six-functor formalism on $(\mathcal{C}, E)$ and $f: X \rightarrow Y$ in E . Write $\pi_i: X \times_Y X \rightarrow X$ for the projections.*

(i) $A \in \mathcal{D}(X)$ is f -suave iff the natural map

$$\pi_1^* \underline{\mathrm{Map}}(A, f^! \mathbb{1}_Y) \otimes \pi_2^* A \longrightarrow \underline{\mathrm{Map}}(\pi_1^* A, \pi_2^! A)$$

becomes an isomorphism after applying $\mathrm{Hom}_{\mathcal{D}(X \times_Y X)}(\Delta_{f!} \mathbb{1}_X, -) \simeq \mathrm{Hom}_{\mathcal{D}(X)}(\mathbb{1}_X, \Delta_f^!(-))$. In this case, for every $Z \in (\mathcal{C}_E)_Y$ and $M \in \mathcal{D}(Z)$,

$$\mathrm{pr}_X^* \mathrm{SD}_f(A) \otimes \mathrm{pr}_Z^* M \xrightarrow{\sim} \underline{\mathrm{Map}}(\mathrm{pr}_X^* A, \mathrm{pr}_Z^! M) \quad \text{in } \mathcal{D}(X \times_Y Z).$$

(ii) $A \in \mathcal{D}(X)$ is f -prim iff the natural map

$$f_!(\pi_{2*} \underline{\mathrm{Map}}(\pi_1^* A, \Delta_{f!} \mathbb{1}_X) \otimes A) \longrightarrow f_* \underline{\mathrm{Map}}(A, A)$$

becomes an isomorphism after applying $\mathrm{Hom}_{\mathcal{D}(Y)}(\mathbb{1}_Y, -)$. In this case,

$$\mathrm{PD}_f(A) \simeq \pi_{2*} \underline{\mathrm{Map}}(\pi_1^* A, \Delta_{f!} \mathbb{1}_X).$$

Moreover, for every $Z \in (\mathcal{C}_E)_Y$ and $M \in \mathcal{D}(X \times_Y Z)$,

$$\mathrm{pr}_{Z!}(\mathrm{pr}_X^* \mathrm{PD}_f(A) \otimes M) \xrightarrow{\sim} \mathrm{pr}_{Z*} \underline{\mathrm{Map}}(\mathrm{pr}_X^* A, M) \quad \text{in } \mathcal{D}(Z).$$

Proof. Apply Lemma 2.3.26 to the kernel 2-category $\mathbb{K}_{D,Y}$. $\square$

Corollary 2.3.28 (Suave–prim duality pairing). *Let $f : X \rightarrow Y$ in E , with $A \in D(X)$ f -suave and $B \in D(X)$ f -prim. Then there is a natural isomorphism*

$$f_* \underline{\mathrm{Map}}(B, \mathrm{SD}_f(A)) \simeq \left(f_* \underline{\mathrm{Map}}(\mathrm{PD}_f(B), A) \right)^\vee.$$

Proof. Direct from Proposition 2.3.25 and Proposition 2.3.27. $\square$

Proposition 2.3.29 (Closure properties, [HM24, Lemma 4.4.12, Corollary 4.4.13]). *Let $f : X \rightarrow Y$ be in E .*

- $\mathrm{Suave}_f(X)$ and $\mathrm{Prim}_f(X)$ are closed under retracts in $D(X)$.
- For any small $\mathcal{I}$ such that D is compatible with $\mathcal{I}$ -indexed colimits and $\mathcal{I}^{\mathrm{op}}$ -indexed limits, $\mathrm{Suave}_f(X)$ and $\mathrm{Prim}_f(X)$ are closed under such colimits and limits.

Here D is compatible with $\mathcal{I}$ -indexed colimits if every $D(X)$ admits them, $\otimes$ preserves them in each variable, and $f^*, f_!$ preserve them for all $f \in E$ [HM24, Definition 3.1.6]; stable below means that all $D(X)$ are stable and D is compatible with finite limits and colimits. In particular, when D is a stable three-functor formalism, $\mathrm{Suave}_f(X)$ and $\mathrm{Prim}_f(X)$ are closed under retracts, fibers, and cofibers, hence are thick subcategories of $D(X)$.

Suave morphisms and prim morphisms

Specializing Definition 2.3.21 to $A = \mathbb{1}_X$ yields classes of *morphisms*.

Definition 2.3.30 (Suave and prim morphisms). *Let $f : X \rightarrow Y$ be in E .*

- f is **D-suave** if $\mathbb{1}_X \in \mathrm{Suave}_f(X)$. We then write $\omega_f := \mathrm{SD}_f(\mathbb{1}_X) \in D(X)$ for the **dualizing complex** of f .
- f is **D-prim** if $\mathbb{1}_X \in \mathrm{Prim}_f(X)$. We then write $\delta_f := \mathrm{PD}_f(\mathbb{1}_X) \in D(X)$ for the **co-dualizing complex** of f .

Remark 2.3.31. Combining Theorem 2.3.20 with Definition 2.3.30 gives

$$f \text{ is D-smooth} \iff f \text{ is D-suave and } \omega_f \text{ is invertible.}$$

Proposition 2.3.25 and Proposition 2.3.27 provide explicit formulas:

$$\omega_f \simeq f^!(\mathbb{1}_Y) \quad \text{and} \quad \delta_f \simeq \pi_{2*} \Delta_{f!} \mathbb{1}_X,$$

where $\pi_2 : X \times_Y X \rightarrow X$ is the projection onto the second factor.

For D-suave f , the formula $\omega_f \otimes f^*(-) \xrightarrow{\sim} f^!(-)$ has a companion: applying Proposition 2.3.25(1) to the f -suave object $\omega_f = \mathrm{SD}_f(\mathbb{1}_X)$, whose suave dual is $\mathbb{1}_X$ by the bidual identity, gives $f^*(-) \xrightarrow{\sim} \underline{\mathrm{Map}}(\omega_f, f^!(-))$. Plugging in $\mathbb{1}_Y$ yields $\mathbb{1}_X \xrightarrow{\sim} \underline{\mathrm{Map}}(\omega_f, \omega_f)$, and for dualizable ω_f this gives $\mathbb{1}_X \simeq \omega_f^\vee \otimes \omega_f$. We record this as an observation.

Observation 2.3.32. For a D-suave morphism f , the dualizing complex ω_f is $\otimes$ -invertible if and only if it is dualizable. Consequently the verification of D-smoothness reduces to checking suaveness and dualizability of ω_f .

We close the subsubsection with the base-change identities for suave and prim morphisms; they will guide the definitions of cohomologically étale and proper morphisms in the next subsubsection.

Lemma 2.3.33 (Base change for suave and prim morphisms, [HM24, Lemma 4.5.13]). *Let D be a six-functor formalism on (C, E) and consider a pullback square*

$$\begin{array}{ccc} X' & \xrightarrow{f'} & Y' \\ g' \downarrow & \lrcorner & \downarrow g \\ X & \xrightarrow{f} & Y \end{array}$$

in C_E .

(i) *If f is D -suave, the natural transformations*

$$g'_* f'^* \simeq f^* g_*, \quad g'_! f'^! \simeq f^! g_!, \quad f'^* g^! \simeq g'^! f^*, \quad g'^* f^! \simeq f'^! g^*$$

are all isomorphisms.

(ii) *If f is D -prim, the natural transformations*

$$g^* f_* \simeq f'_* g'^*, \quad g^! f_! \simeq f'_! g'^!, \quad f_! g'_* \simeq g_* f'_!, \quad f_* g'_! \simeq g'_! f_*$$

are all isomorphisms.

Cohomological étaleness and properness

We now ask: under what conditions are there genuine isomorphisms $f^! \simeq f^*$ and $f_! \simeq f_*$, rather than mere twists by ω_f or δ_f ?

A candidate map $f^! \rightarrow f^*$ is constructed using the diagonal. Consider

$$\begin{array}{ccccc} X & & \xrightarrow{\text{id}_X} & & X \\ & \searrow \Delta_f & & \searrow \pi_2 & \\ & X \times_Y X & \xrightarrow{\pi_2} & X & \\ & \pi_1 \downarrow & \lrcorner & \downarrow f & \\ & X & \xrightarrow{f} & Y. & \end{array}$$

id_X (curved arrow from X to X)

We have $f \pi_1 = f \pi_2$ and $\pi_1 \Delta_f = \pi_2 \Delta_f = \text{id}_X$. Suppose inductively that Δ_f satisfies $\Delta_f^! \simeq \Delta_f^*$. Then

$$\begin{aligned} f^! &\simeq \text{id}_X^* f^! \simeq \Delta_f^* \pi_2^* f^! \xrightarrow{\sim} \Delta_f^! \pi_2^* f^! \\ &\rightarrow \Delta_f^! \pi_1^! f^* \simeq (\pi_1 \Delta_f)^! f^* \simeq f^*, \end{aligned}$$

where $\pi_2^* f^! \rightarrow \pi_1^! f^*$ is the exchange map induced (via right adjoints) from the base-change isomorphism $\pi_{1!} \pi_2^* \simeq f^* f_!$. By Lemma 2.3.33, when f is D -suave this exchange map is an isomorphism, and so $f^! \simeq f^*$.

A dual construction gives $f_! \rightarrow f_*$, an isomorphism when Δ_f satisfies $\Delta_f^! \simeq \Delta_{f*}$ and f is D -prim. To make the recursion well-founded we impose a truncatedness hypothesis: the recursion terminates at -2 -truncated morphisms (i.e., isomorphisms), where the conclusion is trivial.

Definition 2.3.34 (Cohomologically étale and proper morphisms, [HM24, Definition 4.6.1]). Let D be a six-functor formalism on (C, E) , and let $f : X \rightarrow Y$ be a truncated morphism in E (i.e., n -truncated for some $n \in \mathbb{N}$).

- f is **D-étale** (or **cohomologically étale**) if f is D-suave and Δ_f is either D-étale or an isomorphism.
- f is **D-proper** (or **cohomologically proper**) if f is D-prim and Δ_f is either D-proper or an isomorphism.

These definitions are well-founded because n -truncated morphisms have $(n - 1)$ -truncated diagonals.

Proposition 2.3.35 (Characterization of étaleness and properness, [HM24, Lemma 4.6.4]). *Let D be a six-functor formalism on (C, E) and $f: X \rightarrow Y$ in E .*

(i) *Suppose Δ_f is D-étale. There is a natural map $f^! \rightarrow f^*$, and the following are equivalent:*

- (a) *f is D-étale;*
- (b) *the natural map $f^! \mathbb{1}_Y \rightarrow f^* \mathbb{1}_Y$ is an equivalence in $D(X)$;*
- (c) *the natural map $f^! \rightarrow f^*$ is an equivalence of functors.*

(ii) *Suppose Δ_f is D-proper. There is a natural map $f_! \rightarrow f_*$, and the following are equivalent:*

- (a') *f is D-proper;*
- (b') *the natural map $f_! \mathbb{1}_X \rightarrow f_* \mathbb{1}_X$ is an equivalence in $D(Y)$;*
- (c') *the natural map $f_! \rightarrow f_*$ is an equivalence of functors.*

Proof sketch. By duality it suffices to prove (b) $\Rightarrow$ (a) for the étale case. We must show that $\mathbb{1}_X$ is f -suave, i.e., via Proposition 2.3.27, that the criterion map $\pi_1^* f^! \mathbb{1}_Y \rightarrow \pi_2^! \mathbb{1}_X$ becomes an isomorphism after applying $\mathrm{Hom}_{D(X)}(\mathbb{1}_X, \Delta_f^!(-))$. Unwinding the construction of the natural map $f^! \mathbb{1}_Y \rightarrow f^* \mathbb{1}_Y$, it is obtained from the criterion map by applying $\Delta_f^!$ and using $\Delta_f^! \simeq \Delta_f^*$ (as Δ_f is D-étale) together with $\pi_i \Delta_f = \mathrm{id}_X$. Hence hypothesis (b) says exactly that the criterion map becomes an equivalence after applying $\mathrm{Hom}_{D(X)}(\mathbb{1}_X, \Delta_f^!(-))$, which is the criterion. $\square$

2.4. THE UNIVERSAL PROPERTY OF SH

With the six-functor formalism on SH now in hand, we return to the question advertised in the chapter introduction: in what precise sense is SH universal? The Drew–Gallauer theorem (Section 2.4.1) singles it out by three axioms; every cohomology theory of varieties satisfying those axioms then receives a unique *realization* of SH.

2.4.1. The Drew–Gallauer universal property

Theorem 2.4.1 (Drew–Gallauer [DG22]; cf. [Sch26, Theorem 11.1]). *Let k be a Noetherian base ring with $\mathrm{Spec}(k)$ of finite Krull dimension, and let C be the category of separated finite-type k -schemes, equipped with $E = \mathrm{Mor}(C)$, $I =$ open immersions, and $P =$ proper morphisms. Consider the category of presentable six-functor formalisms*

$$D: \mathrm{Span}_2(C, E)_{I, P}^{\otimes} \longrightarrow \mathrm{Pr}_{\mathrm{st}}^L$$

(this is the Drew–Gallauer setup, independently of the larger qcqs setup of Theorem 2.3.10: every morphism of C is separated and of finite type, and the required (I, P) -factorization is Nagata’s compactification theorem) in which open immersions are cohomologically étale and proper maps are cohomologically proper (Definition 2.3.34), and which satisfy:

- (i) **Cohomological smoothness:** every smooth morphism is cohomologically smooth in $\mathcal{D}$ (Definition 2.3.14);
- (ii) **Excision:** for any closed immersion $i: Z \hookrightarrow X$ with open complement $j: U \hookrightarrow X$, the sequence

$$j_! \mathbb{1} \longrightarrow \mathbb{1} \longrightarrow i_* \mathbb{1}$$

is a fiber sequence in $\mathcal{D}(X)$;

- (iii) **$\mathbb{A}^1$ -homology trivial:** for $\pi: \mathbb{A}^1 \rightarrow \mathrm{Spec}(k)$, the canonical map $\mathbb{1} \rightarrow \pi_! \pi^! \mathbb{1}$ is an equivalence.¹

This category has an initial object, and that initial object is the motivic six-functor formalism SH of Morel–Voevodsky.

We have stated the theorem in the reformulation of [Sch26, Theorem 11.1]. The original result of [DG22] asserts that SH is initial among cocomplete *coefficient systems*: there, the axioms posit left adjoints $f_{\sharp}$ along smooth f (with base change and projection formula) and invertibility of $p_{\sharp} s_* \mathbb{1}$, in place of conditions phrased through cohomological smoothness, étaleness, and properness; see [Sch26, Remark 11.2] for the translation.

The proof of [DG22] is constructive, and recovers the construction of SH given in Section 2.1: enforcing the existence of left adjoints $f_{\sharp}$ to smooth pullbacks gives the stable presheaf category $\mathrm{PShv}(\mathrm{Sm}_X; \mathrm{Sp})$; excision implies Nisnevich descent and forces passage to $\mathrm{Shv}_{\mathrm{Nis}}(\mathrm{Sm}_X; \mathrm{Sp})$; triviality of $\mathbb{A}^1$ -homology forces $\mathbb{A}^1$ -localization, yielding $\mathrm{Shv}_{\mathrm{Nis}, \mathbb{A}^1}(\mathrm{Sm}_X; \mathrm{Sp})$; finally, $\otimes$ -inverting the Tate object yields SH . We refer the reader to [Sch26, §11] for a clean exposition of the proof sketch.

2.4.2. Realizations

The payoff of initiality is that realization functors out of SH become essentially free once the three axioms of Theorem 2.4.1 have been verified. Many standard topological and prime-to-the-residue-characteristic arithmetic six-functor formalisms satisfy them, and initiality then produces a unique morphism of six-functor formalisms out of SH [Sch26, §11]. Such a morphism intertwines f^* , $f_!$ and $\otimes$ by specified coherent equivalences (and, in the coefficient-system formulation of [DG22], also $f_{\sharp}$ for f smooth); the remaining operations f_* , $f^!$, $\underline{\mathrm{Hom}}$ are intertwined laxly, via the induced mate maps. The two basic examples are the following.

Example 2.4.2 (Betti realization). Take $k = \mathbb{C}$, and let

$$\mathrm{D}_B(X) := \mathrm{Shv}(X(\mathbb{C}); \mathrm{Sp})$$

be the category of sheaves of spectra on the complex-analytic space $X(\mathbb{C})$. Analytification preserves fibre products and carries open immersions, proper morphisms, and smooth morphisms to open embeddings, proper maps, and submersions; so the six-functor formalism on locally compact Hausdorff spaces of Section 2.2.1 (with coefficients $\Lambda = \mathbb{S}$) restricts to a presentable six-functor formalism D_B on separated finite-type $\mathbb{C}$ -schemes. The three axioms are classical topology: smooth morphisms are cohomologically smooth because submersions are, by local Poincaré duality; excision is the recollement for the closed decomposition $X(\mathbb{C}) = Z(\mathbb{C}) \sqcup U(\mathbb{C})$; and $\mathbb{A}^1$ -homology triviality holds because $\mathbb{A}^1(\mathbb{C}) = \mathbb{C} \cong \mathbb{R}^2$, where $\pi^! \mathbb{1} \simeq \mathbb{1}[2]$ and $\Gamma_c(\mathbb{R}^2, \mathbb{S}) \simeq \mathbb{S}[-2]$ combine to give $\pi_! \pi^! \mathbb{1} \simeq \mathbb{1}$. Theorem 2.4.1 therefore yields a unique morphism of six-functor formalisms

$$\mathrm{Re}_B: \mathrm{SH}(X) \longrightarrow \mathrm{Shv}(X(\mathbb{C}); \mathrm{Sp}),$$

¹The map is the composite $\mathbb{1} \simeq \pi_! s_! s^! \pi^! \mathbb{1} \rightarrow \pi_! \pi^! \mathbb{1}$ induced by the counit $s_! s^! \rightarrow \mathrm{id}$ for the zero section $s: \mathrm{Spec}(k) \rightarrow \mathbb{A}^1$, using $\pi s = \mathrm{id}$; this is the “adjunction map” of [Sch26, Theorem 11.1].

the **Betti realization**, first constructed by Ayoub [Ayo10]. Compatibility with $f_{\#}$ pins down its values: for $Y \rightarrow X$ smooth, $\mathrm{Re}_B(M_X(Y))$ is the fibrewise suspension spectrum of $Y(\mathbb{C})$ over $X(\mathbb{C})$ — over $X = \mathrm{Spec}(\mathbb{C})$, simply $\Sigma_+^\infty Y(\mathbb{C})$. In particular $\mathbb{T}_{\mathbb{C}} \mapsto \mathbb{P}^1(\mathbb{C})/\infty \simeq S^2$, so the bigraded spheres collapse as $S^{p,q} \mapsto S^p$: the weight q is invisible to topology, and only the comparison with structures remembering the weight (Hodge theory, Galois actions) can detect it.

Example 2.4.3 (Étale realization). Let k be Noetherian of finite Krull dimension and let n be invertible in k . Let

$$\mathrm{D}_{\mathrm{\acute{e}t}}(X; \mathbb{Z}/n) := \mathrm{Shv}_{\mathrm{\acute{e}t}}(X; \mathrm{D}(\mathbb{Z}/n))$$

be the (hypercomplete) étale derived category of $\mathbb{Z}/n$ -modules. The six operations here are classical (SGA 4): cohomological smoothness of smooth morphisms is étale Poincaré duality, in the form $\pi^! \simeq \pi^*(d)[2d]$ for π smooth of relative dimension d ; excision is again the closed/open recollement; and $\mathbb{A}^1$ -homology triviality reduces, by proper base change, to the computation $\Gamma_c(\mathbb{A}_{\mathbb{S}}^1, \mathbb{Z}/n(1)) \simeq \mathbb{Z}/n[-2]$ on geometric fibres. The resulting **étale realization**

$$\mathrm{Re}_{\mathrm{\acute{e}t}}: \mathrm{SH}(X) \longrightarrow \mathrm{D}_{\mathrm{\acute{e}t}}(X; \mathbb{Z}/n)$$

sends $\mathbb{T}_X$ to the Tate twist $\mathbb{Z}/n(1)[2]$, so here the weight *is* remembered, by the twist; see Cisinski–Déglise [CD16] for a comprehensive treatment of étale motives and the comparison with the étale derived category. The hypothesis that n be invertible is essential. With p -torsion coefficients in characteristic p , étale cohomology is not $\mathbb{A}^1$ -invariant: Artin–Schreier theory already gives

$$H_{\mathrm{\acute{e}t}}^1(\mathbb{A}_{\mathbb{F}_p}^1, \mathbb{Z}/p) \neq 0.$$

Thus axiom (3) fails and no Drew–Gallauer realization — equivalently, no morphism of six-functor formalisms of the preceding kind — can exist with these coefficients. This failure marks a genuine boundary of the $\mathbb{A}^1$ -invariant realization formalism; its p -adic replacements lie outside the scope of these notes.

Remark 2.4.4 (Scope). Not every six-functor formalism receives a realization this way: the axioms of Theorem 2.4.1 require the adjoints to point in the directions fixed by the span category, and for some formalisms — algebraic $\mathcal{D}$ -modules being the standard example — the natural adjunctions go the other way [Sch26, §11, footnote]. For a panorama of further realizations from the coefficient-system viewpoint, including the Hodge realization, see [Gal21].

STABLE MOTIVIC HOMOTOPY THEORY II — COMPARISONS

The previous chapter constructed $\mathrm{SH}(S)$ and characterized it abstractly via the Drew–Gallauer theorem. The present chapter turns from intrinsic structure to *comparisons*: how does $\mathrm{SH}(S)$ relate to other categories arising in algebraic geometry, and how do its internal structures echo the classical machinery of algebraic topology?

A useful compass is the analogy with sheaf theory on a topological space:

Sheaves on a space X		Motivic world over S
$\mathrm{Shv}(X, \mathrm{An})$	$\longleftrightarrow$	$\mathrm{H}(S)$
$\mathrm{Shv}(X, \mathrm{An}_*)$	$\longleftrightarrow$	$\mathrm{H}_*(S)$
$\mathrm{Shv}(X, \mathrm{Sp})$	$\longleftrightarrow$	$\mathrm{SH}(S)$

The bottom row is imperfect. The naive analogue of $\mathrm{Shv}(X, \mathrm{Sp})$ on the motivic side is

$$\mathrm{Shv}_{\mathrm{Nis}, \mathbb{A}^1}(\mathrm{Sm}_S, \mathrm{Sp}),$$

and $\mathrm{SH}(S)$ differs from it precisely because we have inverted the Tate object rather than the simplicial circle alone. Pinning down the relationship between the two is the first comparison of the chapter.

The chapter is organized around four nested comparisons:

- **Categorical.** The BEM equivalence (Proposition 3.1.8) identifies graded $\mathbb{A}^1$ -invariant Nisnevich sheaves of spectra satisfying the $\mathbb{P}^1$ -bundle formula with $\mathcal{P}'_S$ -modules in $\mathrm{SH}(S)$. The resulting motivic Eilenberg–MacLane functor H lifts those variants of classical theories such as de Rham, Betti, and étale cohomology that satisfy these hypotheses to motivic ring spectra; we apply it to construct the motivic K -theory spectrum KGL .
- **Filtration.** The homotopy t -structure on $\mathrm{SH}(S)$ and Voevodsky’s slice filtration are the algebraic-geometric counterparts of the Postnikov t -structure and the chromatic filtrations of stable homotopy theory. The dictionary, mediated by $\sigma^\infty \dashv \omega^\infty$ and Morel’s stable $\mathbb{A}^1$ -connectivity theorem, culminates in Voevodsky’s slice conjecture: the slice filtration of KGL is stable under arbitrary base change.
- **Orientation.** A motivic ring spectrum $E \in \mathrm{CAlg}(\mathrm{hSH}(S))$ is *oriented* when the basepoint $\mathbb{T}_S \rightarrow \widetilde{M}_S(\Omega^\infty \mathcal{P}ic)$ admits a retraction after tensoring with E , mirroring the classical complex orientation $\mathrm{MU}(1) \simeq \Sigma^\infty \mathbb{P}\mathbb{C}\mathbb{P}^\infty[-2] \rightarrow E$. From this single retraction unfold Chern classes, the projective bundle formula, formal group laws, and Quillen universality, with HZ (additive), KGL (multiplicative), and MGL (universal) populating the chromatic picture.

- **Recognition.** Over a perfect field k , the very effective category $\mathrm{SH}^{\mathrm{veff}}(k)$ admits an unstable description analogous to $\mathrm{Sp}_{\geq 0} \simeq \mathrm{CMon}^{\mathrm{gp}}(\mathrm{An})$, with finite sets replaced by *framed correspondences* (after Voevodsky and Garkusha–Panin, in the form of [Elm+21]). The recognition principle of Section 3.3 is the motivic Barratt–Priddy–Quillen theorem, and yields framed-theoretic models for $\mathbb{1}_k$, $H\mathbb{Z}_k$, and kgl_k .

The four comparisons interlock: H builds KGL , whose slice filtration produces $\mathbb{A}^1$ -motivic cohomology and ultimately the slice conjecture, and whose canonical orientation drives the projective bundle formula; the framed presentation supplies an alternative model for both the sphere and the effective cover of K -theory, culminating in the zeroth-slice theorem $s^0(\mathbb{1}_k) \simeq H\mathbb{Z}_k$ of Section 3.4. The closing section (Section 3.5) binds every thread to a single organizing Thom spectrum, MGL : its orientation exists over any base and has the stated universal property over a field; the Hopkins–Morel–Hoyois theorem identifies its additive quotient with motivic cohomology under the stated characteristic hypotheses, Conner–Floyd recovers K -theory by Landweber base change, the slices of the sphere carry Adams–Novikov-type MGL -cooperation data, and finite syntomic schemes give its framed model.

3.1. RELATING $\mathrm{SH}(S)$ AND $\mathrm{Shv}_{\mathrm{Nis}, \mathbb{A}^1}(\mathrm{Sm}_S, \mathrm{Sp})$

A motivic spectrum $E \in \mathrm{SH}(S)$ is determined by its underlying graded-sheaf data $(E(j))_{j \in \mathbb{Z}}$ with bonding equivalences $E(j) \simeq \mathrm{Map}(\mathbb{T}_S, E(j+1))$ — this is the telescope description of Corollary 2.1.21, available because $\mathbb{T}_S$ is symmetric (cf. Warning 2.1.14). On the graded-sheaf side, suitably chosen $\mathbb{A}^1$ -invariant variants of de Rham, Betti, and étale cohomology come equipped with additional structure: multiplicative products, first Chern classes, and the projective bundle formula. The main result of the section (Proposition 3.1.8) is a symmetric monoidal equivalence

$$\mathrm{GrShv}_{\mathrm{Nis}, \mathbb{A}^1}(\mathrm{Sm}_S, \mathrm{Sp})_{\mathrm{c}, \mathrm{pbf}} \simeq \mathrm{Mod}_{\mathcal{P}'_S}(\mathrm{SH}(S))$$

identifying such graded data with modules over a commutative algebra $\mathcal{P}'_S \in \mathrm{CAlg}(\mathrm{SH}(S))$; composition with the forgetful functor yields the **motivic Eilenberg–MacLane functor** H , which lifts classical cohomology theories to motivic ring spectra.

Our path is indirect. The tautological adjunction $\sigma^{\infty, \mathrm{gr}} \dashv \omega^{\infty, \mathrm{gr}}$ fails to be $\mathbb{E}_2$ -monoidal because the switch map on $\mathbb{T}_S \otimes \mathbb{T}_S$ need not be homotopic to the identity in $\mathrm{SH}(S)$ (it acts as $\langle -1 \rangle$, which equals id only when -1 is a square); passing to Bott-periodic modules builds in the missing symmetry, and the equivalence drops out.

Notation 3.1.1 (Tensor symbols). Throughout this section the symbol $\otimes$ denotes the ambient monoidal product, whose meaning (in $\mathrm{SH}(S)$, $\mathrm{GrSH}(S)$, $\mathrm{GrShv}_{\mathrm{Nis}, \mathbb{A}^1}(\mathrm{Sm}_S, \mathrm{Sp})$, SSeq , etc.) is determined by typing information; $\wedge$ remains reserved for pure $H_*(S)$ -level expressions such as $\mathbb{T}_S \simeq S^1 \wedge \mathbb{G}_m$.

Notation 3.1.2 (Reduced motivic spectrum functor $\widetilde{M}_S$). Throughout this chapter we make repeated use of the **reduced motivic spectrum functor**, the dashed arrow in the commuta-

tive diagram [BEM25, §2.2]

$$\begin{array}{ccc}
 \mathrm{Sm}_S & \xrightarrow{M_S} & \mathrm{SH}(S) \\
 \downarrow \mathbb{A}^1 & & \uparrow \sigma^\infty \\
 \mathrm{PShv}(\mathrm{Sm}_S) & & \\
 \downarrow (-)_+ & \nearrow \tilde{M}_S & \\
 \mathrm{PShv}(\mathrm{Sm}_S)_* & & \\
 \downarrow \Sigma^\infty & & \\
 \mathrm{PShv}(\mathrm{Sm}_S, \mathrm{Sp}) & \xrightarrow{L_{\mathrm{Nis}, \mathbb{A}^1}} & \mathrm{Shv}_{\mathrm{Nis}, \mathbb{A}^1}(\mathrm{Sm}_S, \mathrm{Sp})
 \end{array}$$

extending the motivic spectrum functor M_S of Section 2.1.4 from Sm_S to pointed presheaves. In the language of Section 2.1.4 the composite is

$$\tilde{M}_S = \sigma_{\mathbb{T}_S}^\infty \circ L_{\mathrm{mot}} : \mathrm{PShv}(\mathrm{Sm}_S)_* \longrightarrow \mathrm{SH}(S),$$

and on a pointed presheaf $(F, x) \in \mathrm{PShv}(\mathrm{Sm}_S)_*$,

$$\tilde{M}_S(F, x) \simeq \mathrm{cofib}(M_S(\mathrm{pt}) \xrightarrow{x_*} M_S(F)).$$

The triangular commutativity $M_S \simeq \tilde{M}_S \circ (-)_+$ on Sm_S records that adjoining and then forgetting a disjoint basepoint recovers the original motive.

Two pointed objects whose reduced motives appear repeatedly:

- The projective line pointed at infinity, $(\mathbb{P}_S^1, \infty)$, has reduced motive $\tilde{M}_S(\mathbb{P}_S^1, \infty) \simeq \mathbb{T}_S$, recovering the Tate object.
- The motivic Picard stack $\Omega^\infty \mathcal{P}ic \simeq \mathrm{B}\mathbb{G}_m$, pointed at the trivial line bundle, has reduced motive $\tilde{M}_S(\Omega^\infty \mathcal{P}ic) \in \mathrm{SH}(S)$; this is the natural source of universal first Chern class data in Section 3.2.4.

3.1.1. Graded algebras and the projective bundle formula

Construction 3.1.3. Equip $\mathbb{Z}^\delta$ with its discrete groupoid structure, so that $\mathrm{Fun}(\mathbb{Z}^\delta, C)$ acquires a symmetric monoidal structure via Day convolution with respect to the addition operation on $\mathbb{Z}$. Consider the square of commutative monoids in $\mathbf{An}$:

$$\begin{array}{ccc}
 \mathbb{F} & \longrightarrow & \mathbb{N} \\
 \downarrow & & \downarrow \\
 \mathbb{S} & \longrightarrow & \mathbb{Z}
 \end{array}$$

where the vertical maps are group completions and the horizontal maps are induced by π_0 . Restriction along $\mathbb{F} \rightarrow \mathbb{N} \rightarrow \mathbb{Z}$ yields a lax symmetric monoidal forgetful functor

$$\mathrm{res} : \mathrm{Gr}(C) = \mathrm{Fun}(\mathbb{Z}^\delta, C) \longrightarrow \mathrm{SSeq}(C).$$

Let $E^\star \in \text{CAlg}(\text{Gr}(C))$ be a $\mathbb{Z}$ -graded commutative algebra in C , and let $c: T \rightarrow E^1$ be a morphism in C (a **Bott element**). By the universal property of Sym , the element c extends uniquely to a morphism

$$\text{Sym}(T_{\text{sym}}) \longrightarrow \text{res}(E)$$

of commutative algebras in $\text{SSeq}(C)$, making $\text{res}(E)$ canonically into a commutative algebra in $\text{Sp}_T^{\text{lax}}(C)$. Moreover, if the resulting bonding maps

$$E^n \otimes T \xrightarrow{1 \otimes c} E^n \otimes E^1 \longrightarrow E^{n+1}$$

have adjoints that are equivalences, then (E, c) lifts to a commutative algebra in $\text{Sp}_T(C)$ with $\omega^{\infty-n}(E, c) \simeq E^n$ for all $n \in \mathbb{Z}$.

The motivic case

From this point on, abbreviate

$$C_S := \text{Shv}_{\text{Nis}, \mathbb{A}^1}(\text{Sm}_S, \text{Sp}).$$

We specialize to $C = C_S$ and $T = \Sigma^\infty \mathbb{T}_S$, where $\mathbb{T}_S$ is the Tate object of Definition 1.4.3. Here $\Sigma^\infty: H_*(S) \rightarrow C_S$ is the suspension functor, and a Bott element is a morphism $\Sigma^\infty \mathbb{T}_S \rightarrow E^1$.

Construction 3.1.4 (The projective-bundle algebra). Let $\Sigma^\infty \mathbb{T}_S \langle 1 \rangle \in \text{Gr}(C_S)$ denote the graded sheaf concentrated in degree 1 with value $\Sigma^\infty \mathbb{T}_S$, and set

$$Q_S := \text{Sym}(\Sigma^\infty \mathbb{T}_S \langle 1 \rangle) \in \text{CAlg}(\text{Gr}(C_S)).$$

One checks that $\sigma^\infty(Q_S) \simeq \text{Sym}(\mathbb{T}_S \langle 1 \rangle) \in \text{Gr}(\text{SH}(S))$, where on the right $\mathbb{T}_S$ stands for the image of $\Sigma^\infty \mathbb{T}_S$ under $\sigma^\infty: \text{Shv}_{\text{Nis}, \mathbb{A}^1}(\text{Sm}_S, \text{Sp}) \rightarrow \text{SH}(S)$, consistent with the notation of Section 2.1.

We therefore consider the category of Q_S -modules, for which we introduce the abbreviation

$$\text{GrShv}_{\text{Nis}, \mathbb{A}^1}(\text{Sm}_S, \text{Sp})_c := \text{Mod}_{Q_S}(\text{GrShv}_{\text{Nis}, \mathbb{A}^1}(\text{Sm}_S, \text{Sp})).$$

Given $E \in \text{CAlg}(\text{GrShv}_{\text{Nis}, \mathbb{A}^1}(\text{Sm}_S, \text{Sp})_c)$, the bonding maps of Construction 3.1.3 take the form

$$b^n: E^n \otimes \Sigma^\infty \mathbb{T}_S \longrightarrow E^{n+1}, \quad n \in \mathbb{Z}.$$

Passing to adjoints, these become

$$b^{n, \dagger}: E^n \longrightarrow \Omega_{\Sigma^\infty \mathbb{T}_S} E^{n+1} \simeq \text{fib}(E^{n+1}(\mathbb{P}_-^1) \xrightarrow{\infty^*} E^{n+1}),$$

where ∞^* denotes restriction along the inclusion of the point at infinity. The requirement that every $b^{n, \dagger}$ is an equivalence is precisely the $\mathbb{P}^1$ **bundle formula** (pbf).

We write

$$\text{GrShv}_{\text{Nis}, \mathbb{A}^1}(\text{Sm}_S, \text{Sp})_{c, \text{pbf}}$$

for the full subcategory of Q_S -modules satisfying the $\mathbb{P}^1$ bundle formula.

The passage from such data to a genuine motivic spectrum is encoded in the functor H announced at the start of the section; we construct it in Section 3.1.4, after the Bott-periodic groundwork of Section 3.1.3 is in place.

3.1.2. The $\sigma^{\infty, \text{gr}} \dashv \omega^{\infty, \text{gr}}$ adjunction and periodization

To construct the Eilenberg–MacLane functor H , we approach the problem from the reverse direction: from a motivic spectrum $E \in \text{SH}(S)$, extract a graded sheaf and ask when this graded sheaf lies in $\text{GrShv}_{\text{Nis}, \mathbb{A}^1}(\text{Sm}_S, \text{Sp})_{\text{c, pbf}}$.

By Corollary 2.1.13, for each $j \in \mathbb{Z}$, we can extract an $\mathbb{A}^1$ -invariant Nisnevich sheaf from E via the formula

$$E(j) = \text{map}_{\text{SH}(S)}(M_S(-), E \otimes \mathbb{T}_S^{\otimes j}).$$

Warning 3.1.5. Our notation differs from [BEM25, §3.2], where the j -th level is

$$E(j)_{\text{BEM}} := \text{map}_{\text{SH}(S)}(M_S(-), E \otimes \mathbb{T}_S^{\otimes j}[-2j]).$$

In view of $\mathbb{T}_S \simeq S^1 \wedge \mathbb{G}_m$, this is equivalently

$$\text{map}_{\text{SH}(S)}(M_S(-), E \otimes \mathbb{G}_m^{\otimes j}[-j]).$$

Thus BEM carries an additional $[-2j]$ desuspension relative to our $E(j)$. Our $\mathbb{T}_S$ -graded version is closer to the T -spectra of Section 2.1.3, where $\mathbb{T}_S$ is the invertible object.

Assembling these levels and packaging the bonding data, one obtains an adjunction

$$\text{GrShv}_{\text{Nis}, \mathbb{A}^1}(\text{Sm}_S, \text{Sp}) \begin{array}{c} \xrightarrow{\sigma^{\infty, \text{gr}}} \\ \perp \\ \xleftarrow{\omega^{\infty, \text{gr}}} \end{array} \text{SH}(S)$$

with

$$\sigma^{\infty, \text{gr}}(E^\star) = \bigoplus_{j \in \mathbb{Z}} \sigma^\infty(E^j) \otimes \mathbb{T}_S^{\otimes(-j)}, \quad \omega^{\infty, \text{gr}}(E)^j \simeq E(j) \simeq \omega^\infty(E \otimes \mathbb{T}_S^{\otimes j}).$$

Mind the twist $\mathbb{T}_S^{\otimes(-j)}$ on the left adjoint.¹ The right adjoint $\omega^{\infty, \text{gr}}$ does not commute with pullback functors, and it is also poorly multiplicative. To diagnose the obstruction: under the identification $\text{Gr}(C) \simeq \text{PShv}(\mathbb{Z}^\delta) \otimes C$ [Lur17, Proposition 4.8.1.17], an $\mathbb{E}_n$ -monoidal structure on $\sigma^{\infty, \text{gr}}$ corresponds to an $\mathbb{E}_n$ -monoidal map $\mathbb{Z} \rightarrow \mathcal{P}ic(\text{SH}(S))$ sending $1 \mapsto \mathbb{T}_S^{\otimes(-1)}$. For $n = 1$, the free $\mathbb{E}_1$ -monoid $\mathbb{Z}$ on an invertible generator delivers such a map canonically, and $\sigma^{\infty, \text{gr}}$ becomes monoidal — so $\omega^{\infty, \text{gr}}$ is lax monoidal and induces a functor

$$\text{Alg}(\text{GrShv}_{\text{Nis}, \mathbb{A}^1}(\text{Sm}_S, \text{Sp})) \longleftarrow \text{Alg}(\text{SH}(S)).$$

For $n = 2$, however, the lift requires the swap on $\mathbb{T}_S \otimes \mathbb{T}_S$ to be homotopic to the identity in $\text{SH}(S)$ — but the swap acts over a field $S = \text{Spec}(k)$ as the class $\langle -1 \rangle \in \pi_0(\mathbb{1}_k) \cong \text{GW}(k)$ (combining the degree- (-1) transposition on the simplicial factor $(S^1)^{\wedge 2}$ with the $\mathbb{G}_m$ -swap contribution; see the footnote to [BEM25, Remark 3.10]), which equals id if and only if -1 is a square in k , so the desired $\mathbb{E}_2$ -monoidal lift does not exist in general.

To bypass this obstruction we work with extra data on the $\text{SH}(S)$ side.

Definition 3.1.6. For $E \in \text{CAlg}(\text{SH}(S))$, a **periodization** of E is a compatible commutative algebra structure on $E \otimes \mathbb{T}_S^{\otimes \star} \in \text{GrSH}(S)$. We write $\text{CAlg}(\text{SH}(S))_{\text{per}}$ for the category of pairs $(E, \text{periodization})$, with morphisms respecting the periodization.

¹For the right adjoint to have j -th component $\omega^\infty(E \otimes \mathbb{T}_S^{\otimes j})$, the left adjoint must twist by $\mathbb{T}_S^{\otimes(-j)}$, as one checks on $C = \text{Sp}$, $T = S^2$: the functor $A^\star \mapsto \bigoplus_j A^j[-2j]$ has right adjoint with j -th component $F \mapsto F[2j]$. The formula for $\sigma^{\infty, \text{gr}}$ in [BEM25, §3.2] is stated with the twist $\mathbb{T}^{\otimes j}$ and is incompatible with the $\omega^{\infty, \text{gr}}$ -formula given there, which is the load-bearing one.

By the resulting lax symmetric monoidality of ω^∞ , the right adjoint upgrades on this enriched source to

$$\mathrm{CAlg}(\mathrm{GrShv}_{\mathrm{Nis}, \mathbb{A}^1}(\mathrm{Sm}_S, \mathrm{Sp})) \longleftarrow \mathrm{CAlg}(\mathrm{SH}(S))_{\mathrm{per}}.$$

The category $\mathrm{CAlg}(\mathrm{SH}(S))_{\mathrm{per}}$ is still defined extrinsically (in terms of $\mathrm{GrSH}(S)$); the next subsection reinterprets it as the category of commutative algebras over a single commutative ring in $\mathrm{SH}(S)$ via the Bott-periodic-module construction.

3.1.3. Bott-periodic modules and the BEM equivalence

Definition 3.1.7 (Bott-periodic modules). Let $E \in \mathrm{CAlg}(C)$ and let $\beta: T \rightarrow E$ be a morphism in C (the **Bott element**). The category of β -**periodic** E -**modules** is the monoidal Bousfield localization

$$P_\beta \mathrm{Mod}_E(C) \subset \mathrm{Mod}_E(C)$$

obtained by inverting the map $\beta: T \otimes E \rightarrow E$.

Mirroring Construction 3.1.4, consider $\mathcal{P}_S := \mathrm{Sym}(\mathbb{T}_S\langle 1 \rangle) \in \mathrm{CAlg}(\mathrm{GrSH}(S))$ with Bott element $\beta: \mathcal{P}_S \otimes \mathbb{T}_S\langle 1 \rangle \rightarrow \mathcal{P}_S$. By [BH21, Lemma 12.1], the Bott-periodization $P_\beta \mathcal{P}_S$ is computed by the telescope

$$\mathrm{colim} \left(\mathcal{P}_S \xrightarrow{\beta} \mathcal{P}_S \otimes \mathbb{T}_S^{\otimes -1} \langle -1 \rangle \xrightarrow{\beta} \mathcal{P}_S \otimes \mathbb{T}_S^{\otimes -2} \langle -2 \rangle \xrightarrow{\beta} \dots \right).$$

Set $\mathcal{P}'_S := (P_\beta \mathcal{P}_S)^0$, the degree-zero component; for any $\mathcal{P}'_S$ -algebra E , one has $E \otimes \mathbb{T}_S^{\otimes \star} \simeq E \otimes_{\mathcal{P}'_S} P_\beta \mathcal{P}_S$ in $\mathrm{GrSH}(S)$.

This converts a periodization into a $\mathcal{P}'_S$ -algebra structure and back. A map $\mathcal{P}'_S \rightarrow E$ in $\mathrm{CAlg}(\mathrm{SH}(S))$ produces $E \otimes_{\mathcal{P}'_S} P_\beta \mathcal{P}_S$, the periodization of E . Conversely, a periodization yields $\mathbb{T}_S \rightarrow E \otimes \mathbb{T}_S$ from the unit, which induces $\mathcal{P}_S \rightarrow E \otimes \mathbb{T}_S^{\otimes \star}$ and (since the target is Bott-periodic) factors as $P_\beta \mathcal{P}_S \rightarrow E \otimes \mathbb{T}_S^{\otimes \star}$; passing to the degree-zero component gives $\mathcal{P}'_S \rightarrow E$. We obtain an equivalence

$$\mathrm{CAlg}(\mathrm{SH}(S))_{\mathrm{per}} \simeq \mathrm{CAlg}(\mathrm{SH}(S))_{\mathcal{P}'_S} = \mathrm{CAlg}(\mathrm{Mod}_{\mathcal{P}'_S}(\mathrm{SH}(S))),$$

identifying periodized motivic ring spectra with commutative algebras in $\mathrm{Mod}_{\mathcal{P}'_S}(\mathrm{SH}(S))$ — an intrinsic interpretation, as promised.

Proposition 3.1.8 ([BEM25, Proposition 3.15]). *There is a monoidal equivalence $\omega_{\otimes}^{\infty, \mathrm{gr}}$ filling the following commutative square of monoidal categories and lax monoidal functors:*

$$\begin{array}{ccc} \mathrm{GrShv}_{\mathrm{Nis}, \mathbb{A}^1}(\mathrm{Sm}_S, \mathrm{Sp})_{\mathrm{c}, \mathrm{pbf}} & \xleftarrow{\omega_{\otimes}^{\infty, \mathrm{gr}}} & \mathrm{Mod}_{\mathcal{P}'_S}(\mathrm{SH}(S)) \\ \downarrow & & \downarrow \\ \mathrm{GrShv}_{\mathrm{Nis}, \mathbb{A}^1}(\mathrm{Sm}_S, \mathrm{Sp}) & \xleftarrow{\omega_{\otimes}^{\infty, \mathrm{gr}}} & \mathrm{SH}(S). \end{array}$$

Moreover, $\omega_{\otimes}^{\infty, \mathrm{gr}}$ upgrades to a symmetric monoidal equivalence. Consequently, the motivic Eilenberg–MacLane functor H is the inverse of $\omega_{\otimes}^{\infty, \mathrm{gr}}$ followed by the forgetful functor $\mathrm{Mod}_{\mathcal{P}'_S}(\mathrm{SH}(S)) \rightarrow \mathrm{SH}(S)$.

Proof. Set $v: \mathcal{Q}_S \otimes \Sigma^\infty \mathbb{T}_S\langle 1 \rangle \rightarrow \mathcal{Q}_S$; by construction

$$\mathrm{GrShv}_{\mathrm{Nis}, \mathbb{A}^1}(\mathrm{Sm}_S, \mathrm{Sp})_{\mathrm{c}, \mathrm{pbf}} = P_v \mathrm{GrShv}_{\mathrm{Nis}, \mathbb{A}^1}(\mathrm{Sm}_S, \mathrm{Sp})_{\mathrm{c}}.$$

Since Q_S is the unit, [Hoy20, Proposition 3.1] identifies this v -localization with the formal inversion (Proposition 2.1.3) of $Q_S \otimes \Sigma^\infty \mathbb{T}_S \langle 1 \rangle$. Using $\sigma^\infty(Q_S) = \mathcal{P}_S$ and [BH21, Lemma 12.1],

$$\mathrm{GrShv}_{\mathrm{Nis}, \mathbb{A}^1}(\mathrm{Sm}_S, \mathrm{Sp})_{c, \mathrm{pbf}} \simeq P_\beta \mathrm{Mod}_{\mathcal{P}_S}(\mathrm{GrSH}(S)) \simeq \mathrm{Mod}_{P_\beta \mathcal{P}_S}(\mathrm{GrSH}(S)).$$

The discussion preceding the proposition supplies the final equivalence $\mathrm{Mod}_{P_\beta \mathcal{P}_S}(\mathrm{GrSH}(S)) \simeq \mathrm{Mod}_{\mathcal{P}'_S}(\mathrm{SH}(S))$. $\square$

3.1.4. The motivic Eilenberg–MacLane functor H

With Proposition 3.1.8 in hand, we can finally define H as the inverse of the equivalence $\omega_{\otimes}^{\infty, \mathrm{gr}}$, followed by the forgetful functor $\mathrm{Mod}_{\mathcal{P}'_S}(\mathrm{SH}(S)) \rightarrow \mathrm{SH}(S)$.

Definition 3.1.9. The procedure above defines the **motivic Eilenberg–MacLane functor**

$$H: \mathrm{GrShv}_{\mathrm{Nis}, \mathbb{A}^1}(\mathrm{Sm}_S, \mathrm{Sp})_{c, \mathrm{pbf}} \rightarrow \mathrm{SH}(S).$$

Those variants of algebraic de Rham, Betti, and étale cohomology that are $\mathbb{A}^1$ -invariant and satisfy the projective bundle formula furnish objects of $\mathrm{CAlg}(\mathrm{GrShv}_{\mathrm{Nis}, \mathbb{A}^1}(\mathrm{Sm}_S, \mathrm{Sp})_{c, \mathrm{pbf}})$ after a standard shearing of the Tate grading. For example, the étale theory belongs here with torsion coefficients invertible on S . If $F_{\mathrm{old}}(j)$ is written in the traditional cohomological convention, throughout this paragraph we set

$$F(j) := F_{\mathrm{old}}(j)[2j].$$

Thus the traditional Bott/Chern map $\mathbb{T}_S \rightarrow F_{\mathrm{old}}(1)[2]$ becomes a degree-zero map $\mathbb{T}_S \rightarrow F(1)$ in our shift-free convention. With this reindexing, such a theory comes equipped with:

- a family of $\mathbb{A}^1$ -invariant Nisnevich $D(\mathbb{Z})$ -valued sheaves $F(j)$ together with multiplications $F(i) \otimes F(j) \rightarrow F(i+j)$;
- a first Chern class map $c_1: \Gamma_{\mathrm{Nis}}(-, \mathbb{G}_m)[1] \rightarrow F(1)$;
- the projective bundle formula: for every $X \in \mathrm{Sm}_S$, $j \in \mathbb{Z}$ and $n \geq 1$,

$$\sum_{i=0}^n c_1(\mathcal{O}(1))^i: \bigoplus_{i=0}^n F(j-i)(X) \xrightarrow{\sim} F(j)(\mathbb{P}_X^n),$$

which for $n = 1$ recovers the $\mathbb{P}^1$ -bundle formula of Construction 3.1.4.

The multiplications assemble into a commutative algebra structure in $\mathrm{GrShv}_{\mathrm{Nis}, \mathbb{A}^1}(\mathrm{Sm}_S, D(\mathbb{Z}))$; forgetting the $\mathbb{Z}$ -linear structure produces $F(\star) \in \mathrm{GrShv}_{\mathrm{Nis}, \mathbb{A}^1}(\mathrm{Sm}_S, \mathrm{Sp})$. Restricting the first Chern class to $\mathcal{O}(1) \in \mathrm{Pic}(\mathbb{P}_S^1)$ yields a morphism

$$\Sigma^\infty \mathbb{T}_S \simeq \Sigma^\infty(\mathbb{P}_S^1, \infty) \xrightarrow{\Sigma^\infty[\mathcal{O}(1)]} \Sigma^\infty \Omega^\infty \mathcal{P}ic \xrightarrow{\varepsilon} \mathcal{P}ic \simeq \tau^{\leq 1} \Gamma_{\mathrm{Zar}}(-, \mathbb{G}_m)[1] \longrightarrow F(1)$$

(the classifying map $[\mathcal{O}(1)]$ is pointed, since $\mathcal{O}(1)$ restricts trivially to ∞), where ε is the counit of $\Sigma^\infty \dashv \Omega^\infty$ at the level of presheaves of spectra, upgrading $F(\star)$ to an object of $\mathrm{GrShv}_{\mathrm{Nis}, \mathbb{A}^1}(\mathrm{Sm}_S, \mathrm{Sp})_{c, \mathrm{pbf}}$, which corresponds via H to a motivic spectrum.

3.1.5. Promoting Bott-periodic algebras to T -spectra

As a final application of the Bott-periodic framework, we record a general mechanism for promoting a Bott-periodic commutative algebra to a T -spectrum. This will be the key tool in the construction of motivic K-theory below.

Proposition 3.1.10 (Promotion to T -spectra, [Hoy20, Proposition 3.1]). *Let $E \in \mathbf{CAlg}(C)$ and $\beta: T \rightarrow E$ be a Bott element. Then the suspension functor induces an equivalence*

$$P_\beta \mathbf{Mod}_E(C) \simeq P_\beta \mathbf{Mod}_{\sigma^\infty(E)}(\mathbf{Sp}_T(C)).$$

*Consequently, if (E, β) is moreover **Bott-periodic** — the adjoint $E \rightarrow \mathbf{Map}(T, E)$ of the multiplication by β is an equivalence — then E admits an essentially unique promotion to a commutative algebra*

$$\underline{E} \in \mathbf{CAlg}(\mathbf{Sp}_T(C)) \quad \text{with } \omega^{\infty-j} \underline{E} \simeq E \text{ for all } j \in \mathbb{Z}.$$

3.1.6. Example: the motivic K-theory spectrum $\mathbf{KGL}$

We now apply the Bott-periodic construction (Proposition 3.1.10) to build the motivic K-theory spectrum $\mathbf{KGL}_S \in \mathbf{CAlg}(\mathbf{SH}(S))$. Throughout, we follow the case convention of Convention 1.3.5: uppercase for non-connective, lowercase for connective; in particular, the effective cover of $\mathbf{KGL}$ constructed below (Definition 3.1.11) is identified with $\mathbf{kgl}$, not $\mathbf{KGL}$.

Define homotopy K-theory

$$\mathbf{KH}(X) := L_{\mathbb{A}^1} K(X) = \operatorname{colim}_{[n] \in \Delta^{\text{op}}} K(X \times \Delta^n).$$

Restricted to $\mathbf{Sm}_S$, $\mathbf{KH}$ defines an object of $\mathbf{CAlg}(\mathbf{Shv}_{\mathbf{Nis}, \mathbb{A}^1}(\mathbf{Sm}_S, \mathbf{Sp}))$.

Functor-of-points recollection of $\mathcal{O}(1)$. The Serre twist $\mathcal{O}(1) \in \mathbf{Pic}(\mathbb{P}_S^1)$ is determined by its functor of points: for $x: \operatorname{Spec}(R) \rightarrow \mathbb{P}_S^1$ corresponding to a surjection $R^2 \twoheadrightarrow L$ onto an invertible R -module L , the pullback $x^* \mathcal{O}(1)$ is L . Equivalently, $\mathcal{O}(1)$ classifies a map $\mathbb{P}_S^1 \rightarrow \Omega^\infty \mathcal{P}ic$ into the motivic Picard stack (Notation 3.1.2). Its dual $\mathcal{O}(-1) = \mathcal{O}(1)^\vee$ is the *tautological* line bundle, appearing as the kernel of $\mathcal{O}^{\oplus 2} \twoheadrightarrow \mathcal{O}(1)$; both are distinguished classes in $K_0(\mathbb{P}^1)$.

The Bott element. By the projective bundle formula, $K_0(\mathbb{P}_S^1) \simeq K_0(S) \oplus K_0(S) \cdot \beta$; in its reduced summand we single out

$$\beta := 1 - [\mathcal{O}(-1)] \in \operatorname{fib}(K(\mathbb{P}^1) \rightarrow K(\infty)),$$

the **Bott element**. The sign and the choice of $\mathcal{O}(-1)$ are not cosmetic: this formula is fixed by compatibility with the canonical orientation map $1 - (-)^\vee: \Omega^\infty \mathcal{P}ic \rightarrow \Omega^\infty k$ of [BEM25, Definition 4.4] (written there as K^{cn} in our notation k), whose composition with the classifying map $[\mathcal{O}(1)]: \mathbb{P}^1 \rightarrow \Omega^\infty \mathcal{P}ic$ assigns to the Serre twist its first Chern class $1 - [\mathcal{O}(1)^\vee] = 1 - [\mathcal{O}(-1)] = \beta$. We will see this explicitly in Example 3.2.31; for now it suffices that under the motivic identification $\mathbb{T}_S \simeq \mathbb{P}_S^1 / \{\infty\}$, the class β lifts canonically to a morphism

$$\beta: \mathbb{T}_S \longrightarrow \mathbf{KH}$$

in $\mathbf{Shv}_{\mathbf{Nis}, \mathbb{A}^1}(\mathbf{Sm}_S, \mathbf{Sp})$.

Why single out this element. Two facts make β the right choice. First, iterated multiplication by β realizes the Bott equivalence $\mathbf{KH} \simeq \Omega_{\mathbb{T}_S} \mathbf{KH}$, which is what promotes $\mathbf{KH}$ to a $\mathbb{T}_S$ -spectrum. Second, β is the restriction to the $\mathbb{T}_S$ -cell of the canonical $\mathbf{KGL}$ -orientation, hence the universal first Chern class of $\mathcal{O}(1)$ in $\mathbf{KH}$ -theory; this is developed in Example 3.2.31.

Definition 3.1.11 ($\mathbf{KGL}$). The pair $(\mathbf{KH}, \beta)$ is Bott-periodic, and Proposition 3.1.10 promotes it to a commutative algebra

$$\mathbf{KGL}_S \in \mathbf{CAlg}(\mathbf{SH}(S))$$

with $\omega^{\infty-j} \mathbf{KGL}_S \simeq \mathbf{KH}$ for all $j \in \mathbb{Z}$, reflecting the Bott periodicity $\mathbb{T}_S \otimes \mathbf{KGL}_S \simeq \mathbf{KGL}_S$.

Proposition 3.1.12 (Base change for KGL). *For any morphism of schemes $f : Y \rightarrow X$, the natural map*

$$f^* \mathrm{KGL}_X \xrightarrow{\sim} \mathrm{KGL}_Y$$

is an equivalence in $\mathrm{SH}(Y)$.

To show this, we need the following proposition.

Proposition 3.1.13 (Bhatt–Lurie [Elm+20, Appendix A]). *The connective K-theory functor*

$$k : \mathrm{CRing}_R \longrightarrow \mathrm{Sp}$$

is left Kan extended from its restriction to the subcategory SmRing_R of smooth R -algebras.

Proof of Proposition 3.1.13. It suffices to show that the rank- n vector bundle stack

$$\mathrm{Vect}_n : \mathrm{CRing}_R \longrightarrow \mathrm{An}$$

is left Kan extended from SmRing_R . A presheaf is left Kan extended from a subcategory iff it is a colimit of representables from that subcategory. Consider the chain of morphisms of stacks

$$\bigsqcup_k U_{n,k} \longrightarrow \bigsqcup_k \mathrm{Gr}_n(\mathbb{A}^k) \longrightarrow \mathrm{Vect}_n,$$

where $\mathrm{Gr}_n(\mathbb{A}^k)$ is the Grassmannian of rank- n quotients of $\mathcal{O}^k$ (classifying rank- n vector bundles equipped with a surjection from $\mathcal{O}^k$), and $U_{n,k} \subset \mathrm{Gr}_n(\mathbb{A}^k)$ is the standard smooth affine open chart. On affine schemes the second map is surjective (any rank- n vector bundle on an affine scheme is a quotient of some trivial bundle), and the first map is an open cover by smooth affines; the composite is therefore an effective epimorphism, and Vect_n is the colimit of its Čech nerve, consisting of disjoint unions of smooth affines. $\square$

Proof of Proposition 3.1.12. By Proposition 3.1.13, k is left Kan extended from smooth affine algebras; the same holds for KH after $\mathbb{A}^1$ -localization. Left Kan extension is automatically compatible with base change, so $f^* \mathrm{KH}_X \simeq \mathrm{KH}_Y$. Since the Bott element is also natural, the Bott-periodic construction (Proposition 3.1.10) gives $f^* \mathrm{KGL}_X \simeq \mathrm{KGL}_Y$. $\square$

Corollary 3.1.14 (Cisinski). *The presheaf KH is a cdh sheaf on $\mathrm{Sch}^{\mathrm{qcqs}}$.*

Proof. By Corollary 2.3.12, SH is a cdh sheaf on $\mathrm{Sch}^{\mathrm{qcqs}}$. Combined with Proposition 3.1.12, the assignment $X \mapsto \mathrm{KGL}_X \in \mathrm{SH}(X)$ is a cdh-local section of the cocartesian fibration $\int \mathrm{SH}$. Since the unit $\mathbb{1}$ is preserved by pullback, the functor $\mathrm{Map}_{\mathrm{SH}(-)}(\mathbb{1}, -)$ applied to this section preserves cdh descent. Combined with the identification $\omega^\infty \mathrm{KGL}_X|_{\mathrm{Sm}_X} \simeq \mathrm{KH}|_{\mathrm{Sm}_X}$ (Definition 3.1.11), we obtain

$$\mathrm{KH}(X) \simeq \mathrm{Map}_{\mathrm{SH}(X)}(\mathbb{1}, \mathrm{KGL}_X),$$

which inherits cdh descent. $\square$

3.1.7. The homotopy t -structure

We end this section with the homotopy t -structure on $\mathrm{SH}(S)$ and compare it, via the $\sigma^\infty + \omega^\infty$ adjunction, with the Postnikov t -structure on $\mathrm{Shv}_{\mathrm{Nis}, \mathbb{A}^1}(\mathrm{Sm}_S, \mathrm{Sp})$. We follow [BH21, Appendix B].

Notation 3.1.15 (Bigrading and tensor product). Recall the motivic bigraded sphere $S^{p,q}$ of Definition 1.4.5, with $\mathbb{T}_S \simeq S^{2,1}$. For the t -structure discussion in this section it is convenient to parameterize bidegrees by pairs (a, b) with a the simplicial degree and b the Tate degree:

$$(S^1)^{\otimes a} \otimes \mathbb{G}_m^{\otimes b} \simeq S^{a+b, b}.$$

Unless stated otherwise, $\otimes$ denotes the monoidal product of $\mathrm{SH}(S)$ and $\wedge$ that of $\mathrm{H}_*(S)$.

Notation 3.1.16. Let $I \subset \mathbb{Z} \times \mathbb{Z}$ be a subset, where $(i, j) \in I$ represents the bidegree $(a, b) = (\text{simplicial}, \text{Tate})$. We denote by $\mathrm{SH}(S)_{\geq I}$ the full subcategory of $\mathrm{SH}(S)$ generated under colimits and extensions by

$$\{S^{i+j, j} \otimes M_S(X) \mid X \in \mathrm{Sm}_S, (i, j) \in I\}.$$

Example 3.1.17. When $I = \emptyset$, we have $\mathrm{SH}(S)_{\geq \emptyset} = \{0\}$.

Since $\mathrm{SH}(S)$ is presentable stable and the generating family above is small, [Lur17, Proposition 1.4.4.11] implies that $\mathrm{SH}(S)_{\geq I}$ is the connective part of an accessible t -structure on $\mathrm{SH}(S)$. We denote the corresponding coconnective part by $\mathrm{SH}(S)_{< I}$, consisting of all $E \in \mathrm{SH}(S)$ such that $\mathrm{Map}(F, E) \simeq *$ for every $F \in \mathrm{SH}(S)_{\geq I}$. We denote by $\tau_{\geq I}$ the right adjoint to $\mathrm{SH}(S)_{\geq I} \hookrightarrow \mathrm{SH}(S)$ and $\tau_{< I}$ the left adjoint to $\mathrm{SH}(S)_{< I} \hookrightarrow \mathrm{SH}(S)$.

Different choices of I produce different t -structures adapted to different purposes. Among them, one is distinguished by its role in Morel's work [Mor05] on the stable $\mathbb{A}^1$ -connectivity theorems.

Definition 3.1.18 (Homotopy t -structure). The **homotopy t -structure** on $\mathrm{SH}(S)$ is the t -structure determined by $I = \{0\} \times \mathbb{Z}$. We write $\mathrm{SH}(S)_{\geq 0} := \mathrm{SH}(S)_{\geq \{0\} \times \mathbb{Z}}$, and more generally $\mathrm{SH}(S)_{\geq d} := \mathrm{SH}(S)_{\geq \{d\} \times \mathbb{Z}}$ for $d \in \mathbb{Z}$. We denote by $\tau_{\geq d}$ the right adjoint to $\mathrm{SH}(S)_{\geq d} \hookrightarrow \mathrm{SH}(S)$ and by $\tau_{< d}$ the left adjoint to $\mathrm{SH}(S)_{< d} \hookrightarrow \mathrm{SH}(S)$.

Unpacking the notation, $\mathrm{SH}(S)_{\geq d}$ is the full subcategory generated under colimits and extensions by the objects $\Sigma^{d+j, j} M_S(X)$ with $X \in \mathrm{Sm}_S$ and $j \in \mathbb{Z}$.

Next, we record some functoriality properties of $\mathrm{SH}(S)_{\geq I}$. For every morphism $f: S' \rightarrow S$, the functor $f^*: \mathrm{SH}(S) \rightarrow \mathrm{SH}(S')$ sends the generators $\Sigma^{i+j, j} M_S(X)$ to $\Sigma^{i+j, j} M_{S'}(X \times_S S')$, which are again generators; hence f^* is right t -exact, and consequently its right adjoint f_* is left t -exact. The following lemmas improve on these observations.

Lemma 3.1.19 ([BH21, B.1]). *If $f: S' \rightarrow S$ is a pro-smooth morphism, then $f^*: \mathrm{SH}(S) \rightarrow \mathrm{SH}(S')$ is t -exact.*

Proof. Right t -exactness was noted above, so we must show that f^* preserves the coconnective part $\mathrm{SH}(S)_{< I}$.

We first treat the smooth case. If f is smooth, then f^* admits a left adjoint $f_{\#}: \mathrm{SH}(S') \rightarrow \mathrm{SH}(S)$ sending $\Sigma^{i+j, j} M_{S'}(X') \mapsto \Sigma^{i+j, j} M_S(X')$ (where X' is regarded as smooth over S via f). Thus $f_{\#}$ is right t -exact, and consequently f^* is left t -exact. Combined with right t -exactness, f^* is t -exact in the smooth case.

Now suppose $f: S' \rightarrow S$ is pro-smooth, with $S' = \lim_{\alpha \in A} S_{\alpha}$ a cofiltered limit of smooth S -schemes $f_{\alpha}: S_{\alpha} \rightarrow S$ with affine transition maps. Let $E \in \mathrm{SH}(S)_{< I}$; we show $f^*E \in \mathrm{SH}(S')_{< I}$, equivalently $\mathrm{Map}(F, f^*E) \simeq *$ for every $F = \Sigma^{i+j, j} M_{S'}(X')$ with $X' \in \mathrm{Sm}_{S'}$ and $(i, j) \in I$.

By [Gro67, 8.8.2 and 17.7.8], there exists an index $\alpha_0 \in A$ and a smooth S_{α_0} -scheme X_{α_0} with $X' \simeq S' \times_{S_{\alpha_0}} X_{\alpha_0}$. By continuity of $\mathrm{SH}(-)$ [CD19, Proposition 4.3.4],

$$\mathrm{Map}_{\mathrm{SH}(S')}(\Sigma^{i+j, j} M_{S'}(X'), f^*E) \simeq \operatorname{colim}_{\alpha \geq \alpha_0} \mathrm{Map}_{\mathrm{SH}(S_{\alpha})}(\Sigma^{i+j, j} M_{S_{\alpha}}(X_{\alpha}), f_{\alpha}^*E),$$

where $X_\alpha := S_\alpha \times_{S_{\alpha_0}} X_{\alpha_0}$. Since each f_α is smooth, the already established smooth case gives $f_\alpha^* E \in \mathrm{SH}(S_\alpha)_{<I}$, so every term in the colimit is contractible. $\square$

Lemma 3.1.20 ([BH21, B.2]). *If $\iota: Z \rightarrow S$ is a closed immersion, then $\iota_*: \mathrm{SH}(Z) \rightarrow \mathrm{SH}(S)$ is t -exact.*

Proof. Left t -exactness follows from the general fact noted above. For right t -exactness, since ι_* preserves colimits, the full subcategory $\{E \in \mathrm{SH}(Z) \mid \iota_* E \in \mathrm{SH}(S)_{\geq I}\}$ is closed under colimits and extensions. It therefore suffices to show $\iota_*(\Sigma^{a+j,j} M_Z(Y)) \in \mathrm{SH}(S)_{\geq I}$ for all $Y \in \mathrm{Sm}_Z$ and $(a, j) \in I$. By [Gro67, p. 18.1.1], locally on Y we can write $Y \simeq X \times_S Z$ for some $X \in \mathrm{Sm}_S$, so it suffices to show that $\iota_* \iota^*$ preserves $\mathrm{SH}(S)_{\geq I}$.

Consider the localization exact sequence

$$j_! j^* E \longrightarrow E \longrightarrow \iota_* \iota^* E,$$

where $j: U \hookrightarrow S$ is the open complement of Z . Since j is an open immersion, $j_! \simeq j_\#$ is right t -exact (as in the smooth case of Lemma 3.1.19), and j^* is right t -exact generally, so $j_! j^*$ is right t -exact. If $E \in \mathrm{SH}(S)_{\geq I}$, then $j_! j^* E \in \mathrm{SH}(S)_{\geq I}$, and so by the exact sequence $\iota_* \iota^* E \in \mathrm{SH}(S)_{\geq I}$. $\square$

By the lemmas above, one can derive the following property, which says that motivic spectra are detected by their pullbacks to points.

Proposition 3.1.21 ([BH21, B.3]). *Let S be a qcqs scheme locally of finite Krull dimension and let $E \in \mathrm{SH}(S)$. Then $E \in \mathrm{SH}(S)_{\geq I}$ if and only if $s^* E \in \mathrm{SH}(\kappa(s))_{\geq I}$ for every point $s \in S$, where $s: \mathrm{Spec} \kappa(s) \rightarrow S$ is the canonical map.*

Proof. Necessity is clear: $s^*: \mathrm{SH}(S) \rightarrow \mathrm{SH}(\kappa(s))$ is right t -exact.

For sufficiency, we proceed by induction on $d := \dim S$. Assume $s^* E \in \mathrm{SH}(\kappa(s))_{\geq I}$ for every $s \in S$; we show $\tau_{<I} E \simeq *$.

By Proposition 1.1.5 and Theorem 2.3.5, $\tau_{<I} E \simeq *$ can be checked on the henselizations $S_s^h := \mathrm{Spec} \mathcal{O}_{S,s}^h$ for $s \in S$. Since $S_s^h \rightarrow S$ is pro-smooth, Lemma 3.1.19 gives $(\tau_{<I} E)|_{S_s^h} \simeq \tau_{<I}(E|_{S_s^h})$, so we may replace S by S_s^h and assume S is local Henselian with closed point $\iota: \mathrm{Spec} \kappa(s) \hookrightarrow S$ and open complement $j: U \hookrightarrow S$, where $\dim U < \dim S$.

Consider the localization cofiber sequence

$$j_! j^* E \longrightarrow E \longrightarrow \iota_* \iota^* E.$$

For any $u \in U$, $(j^* E)_u \simeq E_{j(u)} \in \mathrm{SH}(\kappa(u))_{\geq I}$ by assumption, so by the induction hypothesis $j^* E \in \mathrm{SH}(U)_{\geq I}$, and hence $j_! j^* E \in \mathrm{SH}(S)_{\geq I}$ (since j is an open immersion, $j_! \simeq j_\#$ is right t -exact, as in the proof of Lemma 3.1.20). Also, $\iota^* E = s^* E \in \mathrm{SH}(\kappa(s))_{\geq I}$ by assumption, and Lemma 3.1.20 gives $\iota_* \iota^* E \in \mathrm{SH}(S)_{\geq I}$. Both outer terms of the cofiber sequence lie in $\mathrm{SH}(S)_{\geq I}$, hence so does E . $\square$

Postnikov t -structure and Homotopy t -structure

Recall the adjunction

$$\sigma^\infty: \mathrm{Shv}_{\mathrm{Nis}, \mathbb{A}^1}(\mathrm{Sm}_S, \mathrm{Sp}) \rightleftarrows \mathrm{SH}(S): \omega^\infty$$

introduced in Section 2.1.4. To match the homotopy t -structure on $\mathrm{SH}(S)$ with connectivity on the unstable side, we first isolate a natural connective class. It is the non-negative part of a genuine Postnikov t -structure whenever the base satisfies the $\mathbb{A}^1$ -connectivity property introduced below.

Definition 3.1.22 (Postnikov connectivity). The **Postnikov-connective objects** of C_S form the full subcategory

$$(C_S)_{\geq 0} := \{ F \mid \pi_i(F) = 0 \text{ as Nisnevich sheaves for all } i < 0 \},$$

where $\pi_i(F)$ denotes the Nisnevich sheafification of the presheaf $X \mapsto \pi_i F(X)$.

Remark 3.1.23. For an arbitrary base this definition specifies only a connective class; ordinary Postnikov truncation need not preserve $\mathbb{A}^1$ -local objects. If S satisfies the $\mathbb{A}^1$ -connectivity property of Definition 3.1.28, the class above is the non-negative part of an accessible t -structure, which we call the **Postnikov t -structure**. Over more general bases, Theorem 3.1.30 supplies only a shifted connectivity estimate, not the missing truncation functors.

This connective class pairs well with σ^∞ :

Proposition 3.1.24. *The functor $\sigma^\infty: C_S \rightarrow \mathrm{SH}(S)$ carries Postnikov-connective objects to connective objects for the homotopy t -structure. When S satisfies the $\mathbb{A}^1$ -connectivity property, this says that σ^∞ is right t -exact.*

Proof. Let $F \in (C_S)_{\geq 0}$. We must show $\sigma^\infty F \in \mathrm{SH}(S)_{\geq 0}$, equivalently $\mathrm{Map}_{\mathrm{SH}(S)}(\sigma^\infty F, E) \simeq *$ for every $E \in \mathrm{SH}(S)_{< 0}$. By adjunction this is $\mathrm{Map}(F, \omega^\infty E) \simeq *$.

It therefore suffices to show that $\omega^\infty E$ lies in $(C_S)_{< 0}$, i.e. has $\pi_i = 0$ as a Nisnevich sheaf for $i \geq 0$. By definition of $\mathrm{SH}(S)_{< 0}$ (the right orthogonal of $\mathrm{SH}(S)_{\geq \{0\} \times \mathbb{Z}}$), we have

$$\mathrm{Map}_{\mathrm{SH}(S)}(\Sigma^{j,j} M_S(X), E) \simeq * \quad \text{for all } X \in \mathrm{Sm}_S, j \in \mathbb{Z}.$$

Taking $j = 0$, we get $\mathrm{Map}(M_S(X), E) \simeq *$, i.e. $\pi_i \omega^\infty E(X) = 0$ for all $i \geq 0$ and $X \in \mathrm{Sm}_S$. The presheaves $\pi_i \omega^\infty E$ thus vanish for $i \geq 0$, and so do their Nisnevich sheafifications. $\square$

The homotopy t -structure on $\mathrm{SH}(S)$ also admits a description directly in terms of the bi-graded homotopy sheaves of E .

Definition 3.1.25 (Bigraded homotopy sheaves). For $E \in \mathrm{SH}(S)$ and integers $a, b \in \mathbb{Z}$, the (a, b) -**th homotopy sheaf** of E is the Nisnevich sheaf $\pi_{a,b}(E)$ associated with the presheaf

$$X \mapsto \pi_0 \mathrm{Map}_{\mathrm{SH}(S)}((S^1)^{\otimes a} \otimes \mathbb{G}_m^{\otimes b} \otimes M_S(X), E).$$

By [Mor03, Proposition 5.1.14], the family $\{\pi_{a,b}\}_{a,b \in \mathbb{Z}}$ is jointly conservative on $\mathrm{SH}(S)$: a morphism in $\mathrm{SH}(S)$ is an equivalence if and only if it induces isomorphisms on all bigraded homotopy sheaves.

To relate the bigraded picture to the Postnikov one, we use the T -spectrum structure on $\mathrm{SH}(S)$ with $T = \mathbb{T}_S = \Sigma^\infty(\mathbb{P}_S^1, \infty)$. Recall from Corollary 2.1.13 that a motivic spectrum E has a family of “levels”

$$E(j) := \omega^{\infty-j} E \simeq \omega^\infty(E \otimes \mathbb{T}_S^{\otimes j}) \in \mathrm{Shv}_{\mathrm{Nis}, \mathbb{A}^1}(\mathrm{Sm}_S, \mathrm{Sp}),$$

one for each $j \in \mathbb{Z}$.

Definition 3.1.26 (Tate-graded homotopy sheaves). For $E \in \mathrm{SH}(S)$ and $i, j \in \mathbb{Z}$, set

$$\pi_i(E)_j := \pi_i E(j) = \pi_i \omega^\infty(E \otimes \mathbb{T}_S^{\otimes j}),$$

the i -th (classical) Nisnevich homotopy sheaf of the j -th level $E(j)$ of E .

Remark 3.1.27. Unwinding the definition,

$$\pi_i(E)_j = \pi_{i-j, -j}(E),$$

since $\mathbb{T}_S = S^1 \wedge \mathbb{G}_m$. The convention of [BEM25, §3.2] (cf. Warning 3.1.5) instead defines $E(j) = \omega^\infty(E \otimes \mathbb{T}_S^{\otimes j}[-2j])$, with an extra $[-2j]$ desuspension; the corresponding formula is then $\pi_i(E)_j = \pi_{i+j, -j}(E)$. Our choice aligns with the general T -spectrum formalism of Section 2.1.3, where $\mathbb{T}_S$ is the invertible object.

The conservativity of $\{\pi_{a,b}\}$ on $\mathrm{SH}(S)$ (Proposition above) translates, via the substitution $(a, b) \leftrightarrow (i, j)$ of Definition 3.1.26, into the conservativity of $\{\pi_i(E)_j\}_{i,j}$: a morphism in $\mathrm{SH}(S)$ is an equivalence iff it induces isomorphisms on $\pi_i(-)_j$ for all i, j . This is the precise sense in which the Postnikov t -structure on each Tate level $E(j)$ controls E .

The two t -structures — the homotopy t -structure on $\mathrm{SH}(S)$ and the Postnikov t -structure on each level — agree, in a precise sense, as soon as the base S satisfies Morel’s $\mathbb{A}^1$ -connectivity property.

Morel originally isolated a fundamental property of the base scheme S which governs the interaction between $\mathbb{A}^1$ -localization and Nisnevich connectivity.

Definition 3.1.28 ($\mathbb{A}^1$ -connectivity property, [Mor05, Definition 1]). A scheme S is said to satisfy the $\mathbb{A}^1$ -connectivity property if the $\mathbb{A}^1$ -localization functor

$$L_{\mathbb{A}^1} : \mathrm{Shv}_{\mathrm{Nis}}(\mathrm{Sm}_S, \mathrm{Sp}) \longrightarrow \mathrm{Shv}_{\mathrm{Nis}, \mathbb{A}^1}(\mathrm{Sm}_S, \mathrm{Sp})$$

preserves Postnikov-connective objects. Equivalently, it preserves the connective (≥ 0) objects of the ambient Nisnevich sheaf category.

Theorem 3.1.29 (Stable $\mathbb{A}^1$ -connectivity, [Mor05, Theorem 6.1.8]). Every field k satisfies the $\mathbb{A}^1$ -connectivity property. That is, for $S = \mathrm{Spec} k$ and every $F \in \mathrm{Shv}_{\mathrm{Nis}}(\mathrm{Sm}_S, \mathrm{Sp})_{\geq n}$, one has $L_{\mathbb{A}^1} F \in \mathrm{Shv}_{\mathrm{Nis}, \mathbb{A}^1}(\mathrm{Sm}_S, \mathrm{Sp})_{\geq n}$.

Morel conjectured that the $\mathbb{A}^1$ -connectivity property holds more generally; it is known to fail for some schemes, but the following *shifted* version recovers a uniform statement for all qcqs schemes of finite valuative dimension.

Theorem 3.1.30 (Shifted stable $\mathbb{A}^1$ -connectivity, [BK25]). Let S be a qcqs scheme with $d := \mathrm{v.dim}(S) < \infty$. For every integer n and every $F \in \mathrm{Shv}_{\mathrm{Nis}}(\mathrm{Sm}_S, \mathrm{Sp})_{\geq n}$, the $\mathbb{A}^1$ -localization satisfies

$$L_{\mathbb{A}^1} F \in \mathrm{Shv}_{\mathrm{Nis}, \mathbb{A}^1}(\mathrm{Sm}_S, \mathrm{Sp})_{\geq n-d}.$$

In particular, $L_{\mathbb{A}^1}$ loses at most d degrees of connectivity, and recovers Theorem 3.1.29 when $d = 0$.

It follows from a theorem of Hoyois that, when S satisfies the $\mathbb{A}^1$ -connectivity property, the homotopy t -structure on $\mathrm{SH}(S)$ is detected by the Tate-graded homotopy sheaves:

Proposition 3.1.31 ([Hoy15, Theorem 2.3]). Let S be a scheme satisfying the $\mathbb{A}^1$ -connectivity property, and let $E \in \mathrm{SH}(S)$. Then:

- (i) $E \in \mathrm{SH}(S)_{\geq d}$ if and only if $\pi_i(E)_j = 0$ for all $i - j < d$.
- (ii) $E \in \mathrm{SH}(S)_{\leq d}$ if and only if $\pi_i(E)_j = 0$ for all $i - j > d$.

Equivalently, substituting $(a, b) = (i - j, -j)$: $E \in \mathrm{SH}(S)_{\geq d}$ iff $\pi_{a,b}(E) = 0$ for all $a < d$, and similarly for the coconnective part.

Combining Proposition 3.1.24 with Theorem 3.1.30, one sees that the composite

$$\sigma^\infty \circ L_{\mathbb{A}^1} : \mathrm{Shv}_{\mathrm{Nis}}(\mathrm{Sm}_S, \mathrm{Sp}) \longrightarrow \mathrm{SH}(S)$$

sends the n -connective part of the Postnikov t -structure into the $(n - d)$ -connective part of the homotopy t -structure, and is right t -exact over a base satisfying the $\mathbb{A}^1$ -connectivity property.

3.2. SLICE FILTRATION AND ORIENTATION

This section develops two interlocking pieces of the comparison program. Voevodsky's *slice filtration* on $\mathrm{SH}(S)$ (Construction 3.2.8) decomposes a motivic spectrum by its $\mathbb{T}$ -cellular complexity, mirroring the way the Postnikov tower decomposes a classical spectrum by its homotopical complexity. Applied to KGL , the graded pieces define $\mathbb{A}^1$ -motivic cohomology (Definition 3.2.42), and the BEM comparison theorem (Theorem 3.2.53) identifies it with the cdh -motivic cohomology built from K -theory directly, culminating in Voevodsky's slice conjecture (Theorem 3.2.64). *Orientation* (Definition 3.2.24) is the structural input that makes the slice machinery computable on projective bundles — it is the motivic avatar of complex orientation, and the source of the projective bundle formula, Chern classes, and formal group laws.

3.2.1. Recollection of filtrations

Definition 3.2.1. Let C be a category. The category of **(decreasing) filtrations** $\mathrm{Fil}(C)$ is defined to be

$$\mathrm{Fil}(C) := \mathrm{Fun}(\mathbb{Z}^{\mathrm{op}}, C).$$

The category of **increasing filtrations** is defined similarly, with $\mathbb{Z}^{\mathrm{op}}$ replaced by $\mathbb{Z}$.

Intuitively, a filtration $F^\star$ looks like a sequence

$$\dots \longrightarrow F^{s+1} \longrightarrow F^s \longrightarrow F^{s-1} \longrightarrow \dots$$

The constant-filtration functor $C \rightarrow \mathrm{Fil}(C)$ is fully faithful and exhibits C as a full subcategory of $\mathrm{Fil}(C)$.

Definition 3.2.2. Let C be a category. A **filtration on an object** $X \in C$ is a filtration $F^\star$ together with a map $F^\star \rightarrow X$ of filtrations (regarding X as a constant filtration).

More generally, let $\mathbb{Z}^{\mathrm{op}} \cup \{-\infty\}$ denote the poset obtained from $\mathbb{Z}^{\mathrm{op}}$ by adjoining an object $-\infty$ together with a unique morphism $n \rightarrow -\infty$ for every $n \in \mathbb{Z}$. Then the **category of pairs** $(F^\star \rightarrow X)$ **of filtrations on objects of C** is the functor category $\mathrm{Fun}(\mathbb{Z}^{\mathrm{op}} \cup \{-\infty\}, C)$.

The picture to keep in mind is the diagram

$$\begin{array}{ccccccc} \dots & \longrightarrow & F^{s+1} & \longrightarrow & F^s & \longrightarrow & F^{s-1} \longrightarrow \dots \\ & & & \searrow & \downarrow & \swarrow & \\ & & & & X. & & \end{array}$$

Definition 3.2.3. Let C be a category.

- A filtration $F^\star$ on X is called **exhaustive** if the canonical map $\mathrm{colim}_{s \in \mathbb{Z}^{\mathrm{op}}} F^s \rightarrow X$ is an equivalence.

- A filtration $F^\bullet$ is called **complete** if $F^\infty := \lim_s F^s \simeq 0$.

Definition 3.2.4. Let C be a category with a terminal object $*$ that admits cofibers, and let $F^\bullet \in \text{Fil}(C)$ be a filtration. For $i \in \mathbb{Z}$, the i -th associated graded piece is

$$\text{gr}_F^i = \text{cofib}(F^{i+1} \rightarrow F^i).$$

The associated graded functor is denoted $\text{gr}^\bullet : \text{Fil}(C) \rightarrow \text{Gr}(C)$.

3.2.2. Effective motivic spectra and the slice filtration

In motivic homotopy theory, a filtration of particular interest is the slice filtration.

Definition 3.2.5. The category of **effective motivic spectra**

$$\text{SH}^{\text{eff}}(S) \subseteq \text{SH}(S)$$

is the smallest full subcategory of $\text{SH}(S)$ closed under colimits and extensions containing $M_S(X)[i]$ for all $X \in \text{Sm}_S$ and $i \in \mathbb{Z}$. More generally, for any $j \in \mathbb{Z}$, the category of j -**effective motivic spectra** is

$$\text{SH}^{\text{eff}}(S)(j) := \mathbb{T}_S^{\otimes j} \otimes \text{SH}^{\text{eff}}(S).$$

By construction, $\text{SH}^{\text{eff}}(S)$ coincides with $\text{SH}(S)_{\geq \mathbb{Z} \times \{0\}}$ in the sense of Notation 3.1.16, and more generally $\text{SH}^{\text{eff}}(S)(j)$ coincides with $\text{SH}(S)_{\geq \mathbb{Z} \times \{j\}}$. The subcategory $\text{SH}^{\text{eff}}(S)$ is closed under the tensor product of $\text{SH}(S)$, and hence inherits a presentably symmetric monoidal structure.

Proposition 3.2.6. The inclusion functor $\iota^j : \text{SH}^{\text{eff}}(S)(j) \hookrightarrow \text{SH}(S)$ admits a right adjoint r^j .

Proof. By construction, $\text{SH}^{\text{eff}}(S)(j)$ is a full subcategory of $\text{SH}(S)$ closed under colimits, hence ι^j preserves colimits.

Moreover, $\text{SH}^{\text{eff}}(S)(j) = \text{SH}(S)_{\geq \mathbb{Z} \times \{j\}}$ is the connective part of an accessible t -structure by [Lur17, Proposition 1.4.4.11], exactly as in Notation 3.1.16; in particular it is presentable. The adjoint functor theorem [Lur09, Corollary 5.5.2.9] then gives a right adjoint r^j . $\square$

Remark 3.2.7 (Compact generation). For qcqs S , the category $\text{SH}(S)$ is compactly generated, with compact generators the objects $\mathbb{T}_S^{\otimes j} \otimes M_S(X)[i]$ for $X \in \text{Sm}_S$ affine and $i, j \in \mathbb{Z}$: the category $\text{Shv}_{\text{Nis}, \mathbb{A}^1}(\text{Sm}_S, \text{Sp})$ is compactly generated because the Nisnevich topology is finitary (Remark 2.3.7) and $\mathbb{A}^1$ -invariance is a condition closed under filtered colimits, and the formal $\mathbb{T}$ -inversion preserves compact generation since $\Sigma^\infty \mathbb{T}_S$ is compact (see the proof of [BH21, Lemma 4.1]). The same applies to $\text{SH}^{\text{eff}}(S)(j)$ with its generators. In particular, the right adjoints r^j preserve filtered colimits, and mapping spectra out of the generators commute with filtered colimits — facts used silently in exhaustiveness and base-change arguments below.

Construction 3.2.8 (Slice filtration). By Proposition 3.2.6, there is a sequence of colimit-preserving inclusions

$$\cdots \subset \text{SH}^{\text{eff}}(S)(j+1) \subset \text{SH}^{\text{eff}}(S)(j) \subset \text{SH}^{\text{eff}}(S)(j-1) \subset \cdots \subset \text{SH}(S),$$

giving rise, for every motivic spectrum $E \in \text{SH}(S)$, to a functorial filtration

$$\cdots \rightarrow \iota^{j+1} \circ r^{j+1} E \rightarrow \iota^j \circ r^j E \rightarrow \iota^{j-1} \circ r^{j-1} E \rightarrow \cdots \rightarrow E.$$

We denote this filtration by $\mathrm{Fil}_{\mathrm{slice}}^{\star}(E)$ and refer to it as the **slice filtration** of E . We will call $\mathrm{Fil}_{\mathrm{slice}}^0(E)$ the **effective cover** of E .

The slice filtration satisfies the following properties:

- The formation $E \mapsto \mathrm{Fil}_{\mathrm{slice}}^{\star}(E)$ is functorial in E and hence provides a functor $\mathrm{SH}(S) \rightarrow \mathrm{FilSH}(S)$.
- $\mathrm{Fil}_{\mathrm{slice}}^j(E) = \iota^j \circ r^j(E) \rightarrow E$ is universal for maps out of a j -effective motivic spectrum into E : any map $F \rightarrow E$ with $F \in \mathrm{SH}^{\mathrm{eff}}(S)(j)$ factors uniquely through $\mathrm{Fil}_{\mathrm{slice}}^j(E) \rightarrow E$.
- The graded piece

$$s^j(E) := \mathrm{cofib}(\mathrm{Fil}_{\mathrm{slice}}^{j+1}(E) \rightarrow \mathrm{Fil}_{\mathrm{slice}}^j(E)),$$

called the j -th **slice** of E , satisfies $\mathrm{SH}^{\mathrm{eff}}(S)(j+1) \perp s^j(E)$, i.e. $\mathrm{Map}_{\mathrm{SH}(S)}(F, s^j(E)) \simeq *$ for every $F \in \mathrm{SH}^{\mathrm{eff}}(S)(j+1)$.

- $\mathrm{Fil}_{\mathrm{slice}}^{\star}(E)$ is exhaustive (Definition 3.2.3): since every generator $M_S(X)$ of $\mathrm{SH}^{\mathrm{eff}}(S)$ lies in $\mathrm{SH}^{\mathrm{eff}}(S)(j)$ for j sufficiently small (in fact, for all $j \leq 0$), the functors $\iota^j r^j$ are eventually the identity on any fixed compact generator, so $\mathrm{colim}_{j \rightarrow -\infty} \iota^j r^j E \simeq E$.

In general, the slice filtration is not complete. The question of convergence is subtle, and addressing it will motivate the introduction of the *effective homotopy t -structure* and, in particular, of *very effective* motivic spectra, to which we now turn.

Applying the functor

$$\mathrm{SH}(S) \xrightarrow{\omega^{\infty}} \mathrm{Shv}_{\mathrm{Nis}, \mathbb{A}^1}(\mathrm{Sm}_S, \mathrm{Sp}) \xrightarrow{\Gamma} \mathrm{Sp}, \quad (3.2.1)$$

where Γ denotes the global sections, we equip $(\omega^{\infty}E)(S)$ with the filtration

$$\mathrm{Fil}_{\mathrm{slice}}^{\star}(\omega^{\infty}E)(S) := (\omega^{\infty} \mathrm{Fil}_{\mathrm{slice}}^{\star}(E))(S),$$

which, by a slight abuse of terminology, we will also refer to as the slice filtration on $(\omega^{\infty}E)(S)$. Since ω^{∞} is exact, the graded pieces of this filtration are $\mathrm{gr}_{\mathrm{slice}}^j(\omega^{\infty}E)(S) \simeq \omega^{\infty} s^j(E)(S) \in \mathrm{Sp}$ for $j \in \mathbb{Z}$.

Remark 3.2.9 (Multiplicativity). For any scheme S , the formation of the slice filtration defines a lax symmetric monoidal functor

$$\mathrm{Fil}_{\mathrm{slice}}^{\star} : \mathrm{SH}(S) \rightarrow \mathrm{FilSH}(S),$$

where the target is endowed with the Day convolution symmetric monoidal structure. Similarly, passing to graded pieces defines a lax symmetric monoidal functor

$$\mathrm{gr}_{\mathrm{slice}}^{\star} : \mathrm{SH}(S) \rightarrow \mathrm{GrSH}(S).$$

Furthermore, each of the functors in (3.2.1) is lax symmetric monoidal.

Remark 3.2.10 (Base change for the slice filtration). Let $f : S' \rightarrow S$ be a morphism of qcqs schemes. Since f^* preserves colimits and $f^*M_S(X) \simeq M_{S'}(X \times_S S')$ by base change, f^* sends $\mathrm{SH}^{\mathrm{eff}}(S)(j)$ into $\mathrm{SH}^{\mathrm{eff}}(S')(j)$ for every $j \in \mathbb{Z}$.

Consequently, the canonical map $f^* \mathrm{Fil}_{\mathrm{slice}}^{\star}(E) \rightarrow f^*E$ of filtered spectra factors uniquely, by the universal property of $\mathrm{Fil}_{\mathrm{slice}}^{\star}(f^*E)$, through a map

$$f^* \mathrm{Fil}_{\mathrm{slice}}^{\star}(E) \longrightarrow \mathrm{Fil}_{\mathrm{slice}}^{\star}(f^*E),$$

which in turn induces on graded pieces a map of graded spectra

$$f^* s^* E \longrightarrow s^* f^* E.$$

If f is smooth, then f^* admits a left adjoint $f_{\sharp}: \mathrm{SH}(S') \rightarrow \mathrm{SH}(S)$, which preserves j -effective spectra (it preserves colimits, and $f_{\sharp} M_{S'}(X') \simeq M_S(X')$ for $X' \in \mathrm{Sm}_{S'}$, regarded as smooth over S via f). By the adjunctions $f_{\sharp} \dashv f^*$ and $\iota^j \dashv r^j$, this gives a natural equivalence $f^* \circ r^j \simeq r^j \circ f^*$, so f^* commutes with the formation of the slice filtration and its slices.

By continuity of $\mathrm{SH}(-)$ ([CD19, Proposition 4.3.4]), the same conclusion holds more generally when f is a pro-smooth affine morphism.

We can also construct the unstable² analogue of the slice filtration. For any qcqs scheme S and $j \geq 0$, set

$$\mathrm{Shv}_{\mathrm{Nis}, \mathbb{A}^1}(\mathrm{Sm}_S, \mathrm{Sp})(j) \subset \mathrm{Shv}_{\mathrm{Nis}, \mathbb{A}^1}(\mathrm{Sm}_S, \mathrm{Sp})$$

be the subcategory generated under colimits by $\Sigma^{\infty} \mathbb{T}_S^{\otimes j} \otimes E$, for $E \in \mathrm{Shv}_{\mathrm{Nis}, \mathbb{A}^1}(\mathrm{Sm}_S, \mathrm{Sp})$.

Definition 3.2.11. We say an $\mathbb{A}^1$ -invariant Nisnevich sheaf $E \in \mathrm{Shv}_{\mathrm{Nis}, \mathbb{A}^1}(\mathrm{Sm}_S, \mathrm{Sp})$ is **unstably j -effective**, if $E \in \mathrm{Shv}_{\mathrm{Nis}, \mathbb{A}^1}(\mathrm{Sm}_S, \mathrm{Sp})(j)$.

By the same argument as in Proposition 3.2.6, the inclusion $\iota^j: \mathrm{Shv}_{\mathrm{Nis}, \mathbb{A}^1}(\mathrm{Sm}_S, \mathrm{Sp})(j) \hookrightarrow \mathrm{Shv}_{\mathrm{Nis}, \mathbb{A}^1}(\mathrm{Sm}_S, \mathrm{Sp})$ admits a right adjoint r^j . In the same way as for the slice filtration, this induces a functorial, exhaustive, multiplicative filtration for $E \in \mathrm{Shv}_{\mathrm{Nis}, \mathbb{A}^1}(\mathrm{Sm}_S, \mathrm{Sp})$, now $\mathbb{N}$ -indexed. We denote this filtration by

$$\cdots \rightarrow \mathrm{Fil}_{\mathrm{un-slice}}^{j+1}(E) \rightarrow \mathrm{Fil}_{\mathrm{un-slice}}^j(E) \rightarrow \mathrm{Fil}_{\mathrm{un-slice}}^{j-1}(E) \rightarrow \cdots \rightarrow \mathrm{Fil}_{\mathrm{un-slice}}^0(E) \simeq E$$

and write $s_u^j(E)$ for the j -th graded piece, which we call the **unstably slices of E** .

We now compare the unstable slice filtration to the slice filtration on $\mathrm{SH}(S)$. Observing that

$$\sigma^{\infty}(\mathrm{Shv}_{\mathrm{Nis}, \mathbb{A}^1}(\mathrm{Sm}_S, \mathrm{Sp})(j)) \subset \mathrm{SH}^{\mathrm{eff}}(S)(j),$$

we obtain Beck–Chevalley style transformations $\iota^j \circ \omega^{\infty} \rightarrow \omega^{\infty} \circ \iota^j$. They induce, for each $E \in \mathrm{SH}(S)$ and $j \geq 0$, natural maps

$$\mathrm{Fil}_{\mathrm{un-slice}}^j \omega^{\infty} E \rightarrow \omega^{\infty} \mathrm{Fil}_{\mathrm{slice}}^j(E) \quad \text{and} \quad s_u^j \omega^{\infty} E \rightarrow \omega^{\infty} s^j E.$$

Since all these functors and transformations are compatibly lax monoidal, the transformations above upgrade to lax monoidal transformations of $\mathbb{N}$ -indexed filtered and graded objects respectively:

$$\mathrm{Fil}_{\mathrm{un-slice}}^j \omega^{\infty} \rightarrow \omega^{\infty} \mathrm{Fil}_{\mathrm{slice}}^j \quad \text{and} \quad s_u^j \omega^{\infty} \rightarrow \omega^{\infty} s^j.$$

Effective homotopy t -structure and very effective spectra

The category $\mathrm{SH}^{\mathrm{eff}}(S)(j)$ inherits a natural t -structure from the homotopy t -structure on $\mathrm{SH}(S)$.

²“unstably” here means unstable under $\otimes \mathbb{T}_S$

Definition 3.2.12 (Effective homotopy t -structure). The **effective homotopy t -structure** on $\mathrm{SH}^{\mathrm{eff}}(S)(j)$ is the t -structure whose connective part is

$$\mathrm{SH}^{\mathrm{eff}}(S)(j)_{\geq d} := \mathrm{SH}^{\mathrm{eff}}(S)(j) \cap \mathrm{SH}(S)_{\geq d}, \quad d \in \mathbb{Z}.$$

Unwinding the definition via Notation 3.1.16: this intersection is generated under colimits and extensions by the single bidegree (d, j) — indeed $S^{d+j', j'} M_S(Y)$ for $j' \geq j$ is a summand of $S^{d+j, j} M_S(Y \times \mathbb{G}_m^{j'-j})$ — whence

$$\mathrm{SH}^{\mathrm{eff}}(S)(j)_{\geq d} = \mathrm{SH}(S)_{\geq \{(d, j)\}},$$

and [Lur17, Proposition 1.4.4.11] makes this the connective part of an accessible t -structure on $\mathrm{SH}^{\mathrm{eff}}(S)(j)$. (Beware that the coconnective part is the right orthogonal taken *inside* $\mathrm{SH}^{\mathrm{eff}}(S)(j)$; it contains, but a priori differs from, the intersection $\mathrm{SH}^{\mathrm{eff}}(S)(j) \cap \mathrm{SH}(S)_{\leq d}$. We shall only use the connective part.)

The connective part of the effective homotopy t -structure on $\mathrm{SH}^{\mathrm{eff}}(S)$ itself (i.e. $j = 0, d = 0$) is of independent interest.

Definition 3.2.13 (Very effective motivic spectra). The category of **very effective motivic spectra**

$$\mathrm{SH}^{\mathrm{veff}}(S) \subseteq \mathrm{SH}^{\mathrm{eff}}(S)$$

is the smallest full subcategory of $\mathrm{SH}(S)$ closed under colimits and extensions containing $M_S(X)$ for all $X \in \mathrm{Sm}_S$. More generally, for $j \in \mathbb{Z}$, set

$$\mathrm{SH}^{\mathrm{veff}}(S)(j) := \mathbb{T}_S^{\otimes j} \otimes \mathrm{SH}^{\mathrm{veff}}(S).$$

By construction, $\mathrm{SH}^{\mathrm{veff}}(S) = \mathrm{SH}(S)_{\geq \{(0, 0)\}} = \mathrm{SH}^{\mathrm{eff}}(S)_{\geq 0}$, so very effective spectra are precisely the connective objects in $\mathrm{SH}^{\mathrm{eff}}(S)$ for the effective homotopy t -structure.

Remark 3.2.14 (Shift of grading on twists). The tensor product with $\mathbb{T}_S^{\otimes j}$ shifts the bigrading: since $\mathbb{T}_S \simeq S^{2, 1}$, we have $\mathbb{T}_S^{\otimes j} \otimes M_S(X) = \Sigma^{2j, j} M_S(X)$, which lies in bidegree (j, j) in the (simplicial, Tate) coordinates of Notation 3.1.16. Consequently, under our inherited convention for the effective homotopy t -structure on $\mathrm{SH}^{\mathrm{eff}}(S)(j)$, the very effective part sits at $\geq j$ rather than ≥ 0 :

$$\mathrm{SH}^{\mathrm{veff}}(S)(j) = \mathrm{SH}(S)_{\geq \{(j, j)\}} = \mathrm{SH}^{\mathrm{eff}}(S)(j)_{\geq j}.$$

Some authors instead define the t -structure on $\mathrm{SH}^{\mathrm{eff}}(S)(j)$ by transferring along $\mathbb{T}_S^{\otimes j}$, under which convention very effective objects would sit at ≥ 0 . We do not adopt this convention, as the ambient homotopy t -structure on $\mathrm{SH}(S)$ is what we ultimately care about.

Proposition 3.2.15. *Let S be a Noetherian scheme of dimension $\leq d$.*

- (i) *Given $E \in \mathrm{SH}^{\mathrm{veff}}(S)$, then $\omega^\infty(E)$ is $(-d)$ -connective (Definition 3.1.22); more generally, $\omega^{\infty, \mathrm{gr}}(E)^j = E(j)$ is $(j - d)$ -connective for any $j \geq 0$.*
- (ii) *Suppose $d = 1$, and let $E \in \mathrm{SH}^{\mathrm{eff}}(S)$. If $\omega^\infty(E)$ is connective, then $E \in \mathrm{SH}^{\mathrm{veff}}(S)$.*

Proof. (i) For Noetherian S with $\mathrm{Kr.dim}(S) \leq d$ we have $\mathrm{v.dim}(S) = \mathrm{Kr.dim}(S) \leq d$ (see the discussion following Section 2.3.1), so Theorem 3.1.30 applies with this d . The claim then follows by combining it with Proposition 3.1.24, tracking connectivity through the $\sigma^\infty + \omega^\infty$ adjunction on each Tate level.

- (ii) The functor $\omega^\infty : \mathrm{SH}^{\mathrm{eff}}(S) \rightarrow \mathrm{Shv}_{\mathrm{Nis}, \mathbb{A}^1}(\mathrm{Sm}_S, \mathrm{Sp})$ is conservative and preserves colimits. Set $T := \sigma^\infty \circ \omega^\infty$, the comonad on $\mathrm{SH}^{\mathrm{eff}}(S)$ associated to the adjunction $\sigma^\infty \dashv \omega^\infty$. Its canonical resolution is the augmented simplicial object

$$T^{\circ \bullet + 1} E \longrightarrow E,$$

which is ω^∞ -split via the extra-degeneracy splitting of a comonad resolution (cf. the proof of [Lur17, Lemma 4.7.3.13]). Since ω^∞ is conservative and preserves colimits, applying it to this split simplicial object gives $\mathrm{colim}_{m \in \Delta^{\mathrm{op}}} \omega^\infty T^{\circ m + 1} E \xrightarrow{\sim} \omega^\infty E$, and conservativity then yields

$$\mathrm{colim}_{m \in \Delta^{\mathrm{op}}} T^{\circ m + 1} E \xrightarrow{\sim} E.$$

Now assume that $\omega^\infty(E)$ is connective, and set $E_\bullet := T^{\circ \bullet + 1} E$. By [Lur11, Remark 4.3.4],

$$E \simeq |E_\bullet| \simeq \mathrm{colim}_{m \rightarrow \infty} \left(\mathrm{colim}_{\Delta_{\leq m}^{\mathrm{op}}} E_\bullet|_{\Delta_{\leq m}^{\mathrm{op}}} \right),$$

and filtered colimits preserve connectivity, so it suffices to show that each finite colimit $\mathrm{colim}_{\Delta_{\leq m}^{\mathrm{op}}} E_\bullet|_{\Delta_{\leq m}^{\mathrm{op}}}$ is very effective. By *loc. cit.*, there is a cofiber sequence

$$\mathrm{colim}_{\Delta_{\leq (m-1)}^{\mathrm{op}}} E_\bullet|_{\Delta_{\leq (m-1)}^{\mathrm{op}}} \longrightarrow \mathrm{colim}_{\Delta_{\leq m}^{\mathrm{op}}} E_\bullet|_{\Delta_{\leq m}^{\mathrm{op}}} \longrightarrow M_m[m],$$

where $M_m[m]$ is a summand of E_m . Since $\mathrm{SH}^{\mathrm{veff}}(S)$ is closed under colimits and extensions, induction on m reduces the problem to showing that each $M_m[m]$ is very effective; since M_m is a summand of $E_m = T^{\circ m + 1} E$, it further reduces to proving

$$T^{\circ m + 1} E \in \mathrm{SH}(S)_{\geq -m} \quad \text{for every } m \geq 0. \quad (3.2.2)$$

Indeed, granting (3.2.2), we have $M_m \in \mathrm{SH}(S)_{\geq -m}$, hence $M_m[m] \in \mathrm{SH}(S)_{\geq 0}$; since $M_m[m]$ is also effective (as a summand of a shift of an effective spectrum), it is very effective. We prove (3.2.2) by induction on m . For $m = 0$: by hypothesis $\omega^\infty E \in \mathrm{Shv}_{\mathrm{Nis}, \mathbb{A}^1}(\mathrm{Sm}_S, \mathrm{Sp})_{\geq 0}$, so by Proposition 3.1.24 we have $TE = \sigma^\infty \omega^\infty E \in \mathrm{SH}(S)_{\geq 0}$. For the inductive step, assume $T^{\circ m} E \in \mathrm{SH}(S)_{\geq -(m-1)}$, i.e. $T^{\circ m} E[m-1] \in \mathrm{SH}^{\mathrm{veff}}(S)$. Applying (i) with $d = 1$, we obtain $\omega^\infty(T^{\circ m} E[m-1]) \in \mathrm{Shv}_{\mathrm{Nis}, \mathbb{A}^1}(\mathrm{Sm}_S, \mathrm{Sp})_{\geq -1}$, i.e. $\omega^\infty T^{\circ m} E \in \mathrm{Shv}_{\mathrm{Nis}, \mathbb{A}^1}(\mathrm{Sm}_S, \mathrm{Sp})_{\geq -m}$. By Proposition 3.1.24,

$$T^{\circ m + 1} E = \sigma^\infty \omega^\infty T^{\circ m} E \in \mathrm{SH}(S)_{\geq -m},$$

which is (3.2.2). □

Remark 3.2.16 (Motivic recognition principle). The category $\mathrm{SH}^{\mathrm{veff}}(S)$ admits a concrete unstable description, analogous to the classical identification $\mathrm{Sp}_{\geq 0} \simeq \mathrm{CMon}^{\mathrm{gp}}(\mathrm{An})$ between connective spectra and grouplike $\mathbb{E}_\infty$ -spaces. In the motivic setting, the role played classically by finite sets (via Σ_+^∞) is taken over by *framed correspondences* $\mathrm{Corr}^{\mathrm{fr}}(\mathrm{Sm}_S)$, leading to a symmetric monoidal equivalence $\mathrm{H}^{\mathrm{fr}}(k)^{\mathrm{gp}} \simeq \mathrm{SH}^{\mathrm{veff}}(k)$ over a perfect field k . We develop this circle of ideas in Section 3.3; the precise statement is Theorem 3.3.11.

3.2.3. Slice filtration for KGL

Recall from Definition 3.1.11 that the motivic K-theory spectrum KGL is constructed as the β -periodization of KH at $\beta: \mathbb{T}_S \xrightarrow{1-[O(-1)]} \text{KH}$. We now study its slice filtration.

Fix a qcqs scheme S , and denote the effective cover of KGL_S by

$$\text{kgl}_S := \text{Fil}_{\text{slice}}^0 \text{KGL}_S \in \text{CAlg}(\text{SH}(S)).$$

For any $E \in \text{SH}(S)$, the slice filtration satisfies the functorial identifications

$$\text{Fil}_{\text{slice}}^j(\mathbb{T}_S \otimes E) \simeq \mathbb{T}_S \otimes \text{Fil}_{\text{slice}}^{j-1}(E), \quad s^j(\mathbb{T}_S \otimes E) \simeq \mathbb{T}_S \otimes s^{j-1}(E).$$

Applied to $E = \text{KGL}_S$ and combined with Bott periodicity $\beta: \mathbb{T}_S \otimes \text{KGL}_S \xrightarrow{\sim} \text{KGL}_S$, this yields

$$\mathbb{T}_S \otimes \text{Fil}_{\text{slice}}^{j-1} \text{KGL}_S \xrightarrow{\sim} \text{Fil}_{\text{slice}}^j \text{KGL}_S, \quad \mathbb{T}_S \otimes s^{j-1}(\text{KGL}_S) \xrightarrow{\sim} s^j(\text{KGL}_S).$$

This observation leads to an alternative description of the slice filtration on KGL_S . The Bott element $\beta: \mathbb{T}_S \rightarrow \text{kgl}_S$ together with the ring structure on kgl_S defines

$$\beta: \mathbb{T}_S \otimes \text{kgl}_S \xrightarrow{\beta \otimes \text{can}} \text{kgl}_S \otimes \text{kgl}_S \xrightarrow{\text{mult}} \text{kgl}_S,$$

and by construction the following diagram commutes:

$$\begin{array}{ccc} \mathbb{T}_S \otimes \text{kgl}_S & \xrightarrow{\sim} & \text{Fil}_{\text{slice}}^1 \text{KGL}_S \\ & \searrow \beta & \swarrow \text{can} \\ & \text{kgl}_S & \end{array}$$

More generally, for each $j \geq 0$ there is a compatible commutative diagram in which $\mathbb{T}_S$ is replaced by $\mathbb{T}_S^{\otimes j}$ and β by its j -fold iterate $\beta^j: \mathbb{T}_S^{\otimes j} \otimes \text{kgl}_S \rightarrow \text{kgl}_S$. We thus obtain an equivalence between the slice filtration on KGL_S

$$\text{Fil}_{\text{slice}}^\star \text{KGL}_S = \cdots \longrightarrow \text{Fil}_{\text{slice}}^{j+1} \text{KGL}_S \longrightarrow \text{Fil}_{\text{slice}}^j \text{KGL}_S \longrightarrow \text{Fil}_{\text{slice}}^{j-1} \text{KGL}_S \longrightarrow \cdots$$

and the **Bott filtration**

$$\mathbb{T}_S^{\otimes \star} \otimes \text{kgl}_S = \cdots \longrightarrow \mathbb{T}_S^{\otimes j+1} \otimes \text{kgl}_S \longrightarrow \mathbb{T}_S^{\otimes j} \otimes \text{kgl}_S \longrightarrow \mathbb{T}_S^{\otimes j-1} \otimes \text{kgl}_S \longrightarrow \cdots$$

Passing to associated graded yields a description of the slices of KGL_S as Tate twists of the 0-th slice:

$$s^j \text{KGL}_S \simeq \mathbb{T}_S^{\otimes j} \otimes s^0 \text{KGL}_S \quad \text{for every } j \in \mathbb{Z}.$$

From this we deduce convergence of the slice filtration in low dimensions.

Lemma 3.2.17. *Let S be a regular Noetherian scheme of Krull dimension ≤ 1 . Then $\text{kgl}_S \in \text{SH}^{\text{veff}}(S)$, and, for every $j \geq 0$, $\omega^\infty \text{Fil}_{\text{slice}}^j \text{KGL}_S$ is $(j - \text{Kr.dim}(S))$ -connective as a Nisnevich sheaf of spectra.*

Proof. For the first claim, by Proposition 3.2.15(ii) it suffices to show that $\omega^\infty \mathrm{kgl}_S$ is connective. By construction $\mathrm{kgl}_S = \iota^0 r^0 \mathrm{KGL}_S$, and for every $X \in \mathrm{Sm}_S$ the spectrum $M_S(X) \in \mathrm{SH}^{\mathrm{eff}}(S)$ lies in the essential image of ι^0 . Combining the adjunction $\iota^0 \dashv r^0$ with full faithfulness of ι^0 gives

$$\omega^\infty \mathrm{kgl}_S(X) = \mathrm{Map}_{\mathrm{SH}(S)}(M_S(X), \iota^0 r^0 \mathrm{KGL}_S) \simeq \mathrm{Map}_{\mathrm{SH}(S)}(M_S(X), \mathrm{KGL}_S) = \mathrm{KH}(X),$$

so $\omega^\infty \mathrm{kgl}_S \simeq \mathrm{KH}|_{\mathrm{Sm}_S}$. Since smooth schemes over a regular Noetherian base remain regular, $\mathrm{KH}(X) \simeq \mathrm{K}(X)$ on Sm_S , which is connective by Proposition 1.3.11.

For the second claim, we have $\mathrm{Fil}_{\mathrm{slice}}^j \mathrm{KGL}_S \simeq \mathbb{T}_S^{\otimes j} \otimes \mathrm{kgl}_S$ by the identification established above. Since $\mathrm{kgl}_S \in \mathrm{SH}^{\mathrm{veff}}(S)$, Proposition 3.2.15(i) applied to the Tate level of index j gives that $\omega^\infty(\mathbb{T}_S^{\otimes j} \otimes \mathrm{kgl}_S)$ is $(j - \mathrm{Kr.dim}(S))$ -connective. $\square$

3.2.4. Orientation

A recurring theme in algebraic topology is that a cohomology theory becomes effectively computable on vector bundles — and on the projective bundles built from them — precisely when it is equipped with *orientation* data. At the heart of this theory lies a single, strikingly simple idea: orientation is a *retraction*. Concretely, the basepoint inclusion $\mathbb{T}_S \rightarrow \mathcal{P}ic$ (or its topological shadow $P\mathbb{C}P^1 \hookrightarrow P\mathbb{C}P^\infty \simeq BU(1)$, after stabilization $\mathbb{S}^2 \rightarrow \mathcal{P}ic(\mathbb{S})$) becomes split after tensoring with a ring spectrum E exactly when E carries an orientation. From this retraction, the Thom isomorphism, Chern classes, the projective bundle formula, and formal group laws all unfold as formal consequences.

We first develop the picture in the topological setting (Section 3.2.4), referring to [MS74; BT82] for classical proofs and to [Lin25] for the modern functorial treatment. We then transcribe the whole story motivically (Section 3.2.4), following [BH21; AI25]. Throughout we work with homotopy commutative ring spectra $E \in \mathrm{CAlg}(\mathrm{hSp})$ (resp. $\mathrm{CAlg}(\mathrm{hSH}(S))$); every ring spectrum relevant to these notes carries a genuine $\mathbb{E}_\infty$ -enhancement in $\mathrm{CAlg}(\mathrm{Sp})$ (resp. $\mathrm{CAlg}(\mathrm{SH}(S))$). For $E \in \mathrm{SH}(S)$ and $X \in \mathrm{SH}(S)$ we continue to write $E(j)(X) = \mathrm{map}_{\mathrm{SH}(S)}(X, E \otimes \mathbb{T}_S^{\otimes j})$ for the weight- j E -cohomology spectrum, in line with Section 3.1.2.

The topological case

Write $\mathcal{P}ic(\mathbb{S}) \subset \mathrm{Sp}^\simeq$ for the **Picard anima** of the sphere spectrum, the grouplike $\mathbb{E}_\infty$ -monoid of invertible objects of Sp ; its connected components are indexed by $\mathbb{Z}$ via $\Sigma^n \mathbb{S} \mapsto n$. The universal complex line bundle $\gamma_1 \rightarrow BU(1) \simeq P\mathbb{C}P^\infty$ fits into the diagram

$$P\mathbb{C}P^1 \hookrightarrow P\mathbb{C}P^\infty \xrightarrow{\gamma_1} \mathcal{P}ic(\mathbb{S}),$$

and its Thom spectrum identifies with $MU(1) \simeq \Sigma^\infty P\mathbb{C}P^\infty[-2]$ [Lin25, Corollary 4.3]; under this identification, the inclusion $P\mathbb{C}P^1 \hookrightarrow P\mathbb{C}P^\infty$ becomes a map of spectra $\mathbb{S} \rightarrow MU(1)$ realising the inclusion of a fiber [Lin25, Lemma 4.1].

Definition 3.2.18 (Complex orientation as retraction [Lin25, Definition 1.5]). A **complex orientation** of $E \in \mathrm{CAlg}(\mathrm{hSp})$ is a retraction

$$r_E: MU(1) \otimes E \longrightarrow \mathbb{S} \otimes E \simeq E$$

of the unit map $\mathbb{S} \otimes E \rightarrow MU(1) \otimes E$ induced by $P\mathbb{C}P^1 \hookrightarrow P\mathbb{C}P^\infty$. Equivalently, via the $(-) \otimes E$ adjunction, it is a map $x_E: MU(1) \rightarrow E$ whose restriction along $\mathbb{S} \rightarrow MU(1)$ is the unit of E ; in cohomological language, $x_E \in \widetilde{E}^2(P\mathbb{C}P^\infty)$ restricting to $1 \in \pi_0 E \simeq \widetilde{E}^2(P\mathbb{C}P^1)$. We say E is *complex-orientable* when such a retraction exists.

Remark 3.2.19 (Equivalent formulations of complex orientation). The retraction r_E of Definition 3.2.18 is one of several equivalent encodings of complex orientability. Specialising to the universal complex line bundle γ_1 , r_E is exactly the data of a Thom class for γ_1 in the sense of [Lin25, Definition 3.1, Proposition 4.4]; the same notion of *E-orientation* extends to any complex vector bundle $\zeta: X \rightarrow BU$ via either of the following:

- (i) (**Thom class.**) A class $\tau \in E^0(M\zeta)$ whose pullback along every fiber inclusion $\mathbb{S}^{2n} \hookrightarrow M\zeta$ (n the local rank) is, after the canonical $2n$ -shift, a unit in $\pi_0 E$ [Lin25, Definition 3.1];
- (ii) (**Trivialization.**) A natural equivalence $\zeta \otimes E \simeq \text{const}_E$ of the X -parametrized family of E -module spectra [Lin25, Proposition 3.3].

The equivalence (i) $\Leftrightarrow$ (ii) is [Lin25, Proposition 3.3 & Exercise 3.8]; specialising to $\zeta = \gamma_1$, both reduce to the retraction r_E via [Lin25, Proposition 4.4]. The retraction formulation is the most economical, the Thom-class formulation the most classical, and the trivialization formulation the most conceptual.

Theorem 3.2.20 (Thom isomorphism, topological [Lin25, Proposition 3.4]). *Let $\zeta: X \rightarrow BU$ be E -orientable and $M\zeta$ its Thom spectrum. There is a natural equivalence*

$$M\zeta \otimes E \simeq \Sigma_+^\infty X \otimes E.$$

For a rank- n complex vector bundle $V \rightarrow B$ this recovers the classical Thom isomorphism $E^q(B) \xrightarrow{\sim} \widetilde{E}^{q+2n}(\text{Th}(V))$; pulling the Thom class along the zero section defines the **Euler class** $e(V) \in E^{2n}(B)$, and the cofiber sequence of the unit sphere bundle $\mathbb{S}(V) \rightarrow B \rightarrow \text{Th}(V)$ produces the **Gysin sequence**

$$\dots \rightarrow E^q(B) \xrightarrow{p^*} E^q(\mathbb{S}(V)) \xrightarrow{\int} E^{q-2n+1}(B) \xrightarrow{\smile e(V)} E^{q+1}(B) \rightarrow \dots$$

Proof sketch. Taking colimits over X of the trivialization Remark 3.2.19(ii) identifies $M\zeta \otimes E$ with $\text{colim}_X \text{const}_E = \Sigma_+^\infty X \otimes E$. $\square$

The entire theory of complex characteristic classes specializes from this functorial picture by iterating the retraction.

Theorem 3.2.21 (Chern classes and projective bundle formula, topological [Lin25, Theorem 1.9]). *Let (E, r_E) be complex-oriented.*

- (i) *For a line bundle $\mathcal{L}$ on B classified by $[\mathcal{L}]: B \rightarrow \mathbb{P}\mathbb{C}\mathbb{P}^\infty$, the pullback $c_1(\mathcal{L}) := [\mathcal{L}]^* x_E \in E^2(B)$ is the **first Chern class**.*
- (ii) *There is a unique family of classes $c_i(V) \in E^{2i}(B)$ for rank- n complex vector bundles V , extending c_1 and satisfying the **projective bundle formula***

$$E^*(\mathbb{P}(V)) \simeq E^*(B)[[t]] / (t^n - c_1(V)t^{n-1} + \dots + (-1)^n c_n(V)),$$

where $t = c_1(\mathcal{O}_{\mathbb{P}(V)}(1))$.

- (iii) *The universal computation reads*

$$E^*(BU(n)) \simeq E^*[c_1, \dots, c_n], \quad |c_i| = 2i.$$

Remark 3.2.22 (Top Chern class and splitting principle, topological). The top Chern class coincides with the Euler class, $c_n(V) = e(V) \in E^{2n}(B)$. The **splitting principle** holds: the map $BU(1)^{\times n} \rightarrow BU(n)$ induces an injection on E -cohomology, and pulling a rank- n bundle V back along the flag bundle $\text{Fl}(V) \rightarrow B$ realizes it as an iterated extension of line bundles, so any polynomial identity among Chern classes valid for sums of line bundles is universally valid.

Remark 3.2.23 (Formal group law and Quillen universality, topological). The tensor product of line bundles is classified by a map $\otimes: P\mathbb{C}P^\infty \times P\mathbb{C}P^\infty \rightarrow P\mathbb{C}P^\infty$; pulling x_E back along $\otimes$ yields a power series

$$F_E(x, y) := \otimes^* x_E \in E^*[[x, y]]$$

computing $c_1(\mathcal{L}_1 \otimes \mathcal{L}_2) = F_E(c_1(\mathcal{L}_1), c_1(\mathcal{L}_2))$. The unital, commutative, associative structure of $\otimes$ makes F_E a graded **formal group law** over E^* . Singular cohomology $H\mathbb{Z}$ yields the additive law $F = x + y$. For complex K -theory, let $\beta_{\text{top}} \in KU^{-2}(*)^{\times}$ be the Bott element and choose the orientation $x = \beta_{\text{top}}^{-1}([\mathcal{L}] - 1)$; then the genuinely graded multiplicative law is

$$F_{KU}(x, y) = x + y + \beta_{\text{top}}xy.$$

Only after passing to the degree-zero periodic coordinate $u = \beta_{\text{top}}x$ does this become $u + v + uv$. Complex cobordism MU is itself complex-orientable and carries the *universal* formal group law. **Quillen's theorem** [Lin25, Theorem 1.14] asserts that the map

$$\text{Hom}_{\text{CAlg}(\text{hSp})}(MU, E) \longrightarrow \{\text{complex orientations of } E\}, \quad f \mapsto f^*(x_{MU}),$$

is a bijection, and the composite $L \rightarrow \pi_* MU$ from Lazard's ring is an isomorphism. This universality is the organizing principle of chromatic stable homotopy theory [Rav86; Lur10; Lin25].

The motivic case

For $A \in \text{SH}(S)$ we use the cohomological convention

$$E^{p,q}(A) := \pi_0 \text{Map}_{\text{SH}(S)}(A, E \otimes S^{p,q}).$$

We write $E^{p,q}(X)$ when $A = M_S(X)$ and use the reduced motive for pointed spaces. In particular, $E^{2,1}(X) = \pi_0 E(1)(M_S(X))$ because $S^{2,1} = \mathbb{T}_S$; this fixes the signs used below.

In the motivic setting, the sphere spectrum $\mathbb{S}$ is replaced by the motivic sphere $\mathbb{1}_S$ and Sp by $\text{SH}(S)$; cohomology classes live in the bigraded theory E^{**} of a motivic ring spectrum, and the Tate weight records $\mathbb{G}_m$ -twists invisible under Betti realization. The object replacing $P\mathbb{C}P^\infty \simeq BU(1)$ is the Picard stack of line bundles $B\mathbb{G}_m$. We package it as the connective **line-bundle Picard spectrum** $\mathcal{P}ic$ in $\text{Shv}_{\text{Nis}, \mathbb{A}^1}(\text{Sms}_S, \text{Sp})$, with $\Omega^\infty \mathcal{P}ic \simeq B\mathbb{G}_m$. By a slight abuse of notation we also write $\mathcal{P}ic$ for its image in $\text{SH}(S)$ under σ^∞ .

This object should not be confused with $\text{Sph}(S) := \text{Pic}(\text{SH}(S))$, the Picard anima of invertible motivic spectra and the actual analogue of the topological $\mathcal{P}ic(\mathbb{S}) = \text{Pic}(\text{Sp})$. The motivic J -homomorphism from line bundles to $\text{Sph}(S)$ appears in Section 3.5.1; the orientation construction here instead uses the source $B\mathbb{G}_m$. The role of $P\mathbb{C}P^1 \hookrightarrow P\mathbb{C}P^\infty$ is played by the composite

$$\mathbb{T}_S \xrightarrow{[O(1)]} \widetilde{M}_S(\Omega^\infty \mathcal{P}ic) \xrightarrow{\varepsilon} \mathcal{P}ic$$

in $\text{SH}(S)$, where the first arrow is $\widetilde{M}_S$ applied to the classifying map $\mathbb{P}_S^1 \rightarrow \Omega^\infty \mathcal{P}ic$ of the Serre twist (using $\widetilde{M}_S(\mathbb{P}_S^1, \infty) \simeq \mathbb{T}_S$ from Notation 3.1.2) and the second is the counit of $\Sigma^\infty \dashv \Omega^\infty$. With the reduced motivic spectrum functor $\widetilde{M}_S$ of Notation 3.1.2 in hand, the orientation architecture of Section 3.2.4 has its motivic incarnation in the universal first Chern class of Definition 3.2.24.

Definition 3.2.24 (Motivic orientation [BEM25, §3.4]). An **orientation** of $E \in \text{CAlg}(\text{hSH}(S))$ is a homotopy class of maps

$$c_E: \widetilde{M}_S(\Omega^\infty \mathcal{P}ic) \longrightarrow E \otimes \mathbb{T}_S$$

fitting into the commutative triangle

$$\begin{array}{ccc} \tilde{M}_S(\Omega^\infty \mathcal{P}ic) & \xrightarrow{c_E} & E \otimes \mathbb{T}_S \\ \uparrow [O(1)] & \nearrow \eta_E \otimes \mathbb{T}_S & \\ \mathbb{T}_S & & \end{array}$$

where the left vertical arrow $[O(1)]: \mathbb{T}_S \rightarrow \tilde{M}_S(\Omega^\infty \mathcal{P}ic)$ is $\tilde{M}_S$ applied to the classifying map $\mathbb{P}_S^1 \rightarrow \Omega^\infty \mathcal{P}ic$ of the Serre twist (using $\tilde{M}_S(\mathbb{P}_S^1, \infty) \simeq \mathbb{T}_S$ from Notation 3.1.2), and $\eta_E: \mathbb{1}_S \rightarrow E$ is the unit. We say E is *orientable* if such a c_E exists, and call (E, c_E) an *oriented* motivic ring spectrum.

Remark 3.2.25 (Equivalent formulations of orientation). The data of an orientation on E admits several equivalent presentations:

- (i) **(Universal first Chern class.)** The morphism $c_E: \tilde{M}_S(\Omega^\infty \mathcal{P}ic) \rightarrow E \otimes \mathbb{T}_S$ together with the basepoint triangle of Definition 3.2.24.
- (ii) **(Retraction.)** By the $\otimes$ -adjunction, equivalently a retraction

$$r_E: \tilde{M}_S(\Omega^\infty \mathcal{P}ic) \otimes E \longrightarrow \mathbb{T}_S \otimes E$$

of $[O(1)] \otimes \text{id}_E: \mathbb{T}_S \otimes E \rightarrow \tilde{M}_S(\Omega^\infty \mathcal{P}ic) \otimes E$; explicitly, r_E is the composite of $c_E \otimes \text{id}_E$ with the multiplication of E , while $c_E = r_E \circ (\text{id} \otimes \eta_E)$.

- (iii) **(Thom class.)** A class $\tau \in \tilde{E}^{2,1}(\tilde{M}_S(\Omega^\infty \mathcal{P}ic))$ whose restriction to $\mathbb{T}_S$ along $[O(1)]$ is the unit $1 \in \pi_0 E$; here $\tilde{M}_S(\Omega^\infty \mathcal{P}ic)$ is the (reduced motivic) Thom spectrum of the tautological line bundle on $\Omega^\infty \mathcal{P}ic$, after a $\mathbb{T}_S$ -shift.
- (iv) **(Cohomological section.)** A section of the restriction map $E^{\tilde{M}_S(\Omega^\infty \mathcal{P}ic)} \rightarrow E^{\mathbb{T}_S}$ in $\text{SH}(S)$.

The Chern-class formulation ((i)) is the one we take as the definition; the retraction formulation ((ii)) parallels the topological retraction of Definition 3.2.18 ($\text{MU}(1) \otimes E \rightarrow \mathbb{S} \otimes E$).

Remark 3.2.26 (Additive descent along the Picard counit). An orientation c_E may or may not descend along the counit

$$\varepsilon: \tilde{M}_S(\Omega^\infty \mathcal{P}ic) \longrightarrow \mathcal{P}ic$$

of $\Sigma^\infty + \Omega^\infty$ (Notation 3.1.2) to a morphism $\tilde{c}_E: \mathcal{P}ic \rightarrow E \otimes \mathbb{T}_S$ on the Picard spectrum itself. When such a factorization exists, $c_E = \tilde{c}_E \circ \varepsilon$, and it records the extra assertion that the resulting first Chern class is additive under tensor product of line bundles.

This extra assertion does *not* hold for the KGL-valued orientation itself: its class $1 - [\mathcal{L}^\vee]$ obeys the multiplicative formal group law $x + y = xy$, rather than the additive law. What does descend is the orientation induced on the weight-one slice $s^1 \text{KGL}$ (equivalently on $s^0 \text{KGL} = \mathbb{H}\mathbb{Z}$ after the Bott shift). The factorization through $\mathcal{P}ic$ is precisely the content of Lemma 3.2.39, and produces the additive class $c_1^{\mathbb{A}}$ of Section 3.2.5; compare [BEM25, §4.2.1].

Remark 3.2.27 (Three commentaries on the definition). First, the target $\mathbb{T}_S \otimes E$ rather than E encodes the *motivic weight shift*: since $\mathbb{T}_S \simeq S^{2,1}$, an orientation produces classes of bidegree $(2, 1)$, refining the classical fact that c_1 lives in H^2 . Second, the basepoint condition — c_E restricted to $\mathbb{T}_S$ is $\eta_E \otimes \mathbb{T}_S$ — is the motivic avatar of the classical requirement that $c_1(\mathcal{O}_{\mathbb{P}^1}(1))$ generate $H^2(\mathbb{P}^1; \mathbb{Z})$ (compare the topological case of Definition 3.2.18). Third, although $\tilde{M}_S(\Omega^\infty \mathcal{P}ic)$ looks much larger than $\mathbb{T}_S$, the underlying motivic space $\Omega^\infty \mathcal{P}ic \simeq \text{B}\mathbb{G}_m$ admits a $\mathbb{T}_S$ -cellular filtration whose reduced motive is presented as a filtered colimit

$$\text{colim}_m \tilde{M}_S(\mathbb{P}_S^m) \xrightarrow{\sim} \tilde{M}_S(\Omega^\infty \mathcal{P}ic)$$

— with transition maps $\mathbb{P}_S^m \hookrightarrow \mathbb{P}_S^{m+1}$ the linear inclusions and each map $\mathbb{P}_S^m \rightarrow \Omega^\infty \mathcal{P}ic$ classifying $\mathcal{O}(1)$ [MV99, §4, Proposition 3.7]. Reading a chosen c_E along this colimit gives compatible Chern classes on all the finite projective spaces, while the basepoint condition fixes their normalization on the first cell $\mathbb{T}_S \simeq \widetilde{M}_S(\mathbb{P}_S^1, \infty)$. Each $\widetilde{M}_S(\mathbb{P}_S^m)$ is 1-effective (the cofiber sequences $\widetilde{M}_S(\mathbb{P}_S^{m-1}) \rightarrow \widetilde{M}_S(\mathbb{P}_S^m) \rightarrow \mathbb{T}_S^{\otimes m}$ build it as an iterated extension of Tate twists $\mathbb{T}_S^{\otimes i}$ for $1 \leq i \leq m$), hence so is the colimit $\widetilde{M}_S(\Omega^\infty \mathcal{P}ic)$, so c_E factors through the 1-effective cover

$$\mathrm{Fil}_{\mathrm{slice}}^1(E \otimes \mathbb{T}_S) \simeq \mathrm{Fil}_{\mathrm{slice}}^0(E) \otimes \mathbb{T}_S$$

of the slice filtration (Construction 3.2.8); composing with the canonical $\mathrm{Fil}_{\mathrm{slice}}^0(E) \rightarrow s^0 E$ yields induced orientations of $\mathrm{Fil}_{\mathrm{slice}}^0(E)$ and of $s^0 E$ [BEM25, Remark 3.46]. This chain is what Section 3.2.5 specializes to $E = \mathrm{KGL}$ (Example 3.2.31) and to $s^0 \mathrm{KGL} \simeq \mathbb{H}\mathbb{Z}$ to produce the first Chern class map $\widetilde{M}_S(\Omega^\infty \mathcal{P}ic) \rightarrow \mathbb{Z}(1)^\mathbb{A}$. The cellular filtration fixes the normalization on the first $\mathbb{T}_S$ -cell, but its higher cells retain genuine orientation data; it does not make an orientation unique. This is conceptually parallel to, but distinct from, the higher-rank *splitting principle* for vector bundles via flag bundles discussed in Remark 3.2.32.

For a line bundle $\mathcal{L}$ on a smooth S -scheme Y , the classifying map $[\mathcal{L}]: Y \rightarrow \Omega^\infty \mathcal{P}ic$ composes with the Chern class map c_E of Remark 3.2.25(i) to give the **first Chern class**

$$c_1(\mathcal{L}) \in \pi_0 E(1)(M_S(Y)).$$

Multiplication in E combined with the diagonal $\Delta: Y \rightarrow Y \times_S Y$ defines a cup-product map

$$c_1(\mathcal{L}) \cdot - : \underline{\mathrm{Map}}(M_S(Y), E) \longrightarrow \underline{\mathrm{Map}}(M_S(Y), E \otimes \mathbb{T}_S)$$

in $\mathrm{SH}(S)$ (note that source and target differ by a Tate twist, so this is not literally an endomorphism — each cup product with c_1 raises weight by one), iterating to powers $c_1(\mathcal{L})^i \in \pi_0 E(i)(M_S(Y))$ living in weight- i cohomology. Applied to $Y = \mathbb{P}_S^d$ with $\mathcal{L} = \mathcal{O}(1)$, these powers are the building blocks of the projective bundle formula.

Definition 3.2.28 (Projective bundle formula). An oriented motivic commutative ring spectrum $(E, c_E) \in \mathrm{CAlg}(\mathrm{hSH}(S))$ **satisfies the projective bundle formula** if, for every $d \geq 1$, the map

$$\sum_{i=0}^d c_1(\mathcal{O}(1))^i \cdot \pi^*: \bigoplus_{i=0}^d \mathbb{T}_S^{\otimes(-i)} \otimes E \longrightarrow \underline{\mathrm{Map}}(M_S(\mathbb{P}_S^d), E)$$

is an equivalence in $\mathrm{SH}(S)$, where $\pi: \mathbb{P}_S^d \rightarrow S$ is the structure map.

Theorem 3.2.29 (Projective bundle formula for oriented spectra). *Every oriented $(E, c_E) \in \mathrm{CAlg}(\mathrm{hSH}(S))$ satisfies the projective bundle formula.*

Proof sketch. This is the standard projective bundle theorem for an oriented motivic ring spectrum; in the present setting it is recalled in [BEM25, §3.4]. The orientation supplies the normalized class $t = c_1(\mathcal{O}(1))$, and the theorem says that $1, t, \dots, t^d$ form an $E(S)$ -basis of $E(\mathbb{P}_S^d)$. Translating the cohomological shifts into tensor powers of $\mathbb{T}_S$ gives exactly the displayed equivalence in Definition 3.2.28. This argument does not require a vanishing assertion obtained merely by comparing slice-effectivity degrees. $\square$

Remark 3.2.30 (Relation to the pbf condition of Proposition 3.1.8). The $\mathbb{P}^1$ -bundle formula attached to a Q_S -module in Construction 3.1.4 is precisely the $d = 1$ case of Definition 3.2.28; Theorem 3.2.29 then says that the higher-degree formulas are automatic consequences of orientation.

Example 3.2.31 (KGL is oriented; the Bott element as restricted first Chern class). The motivic K-theory spectrum KGL_S carries a canonical orientation

$$c^{\mathrm{KGL}}: \widetilde{M}_S(\Omega^\infty \mathcal{P}ic) \longrightarrow \mathrm{KGL}_S \otimes \mathbb{T}_S$$

obtained from the orientation map $1 - (-)^\vee: \Omega^\infty \mathcal{P}ic \rightarrow \Omega^\infty k$ of [BEM25, Definition 4.4] (written there as K^{cn} in our notation k) by applying $\widetilde{M}_S$ (using the $\sigma^\infty \dashv \omega^\infty$ adjunction) and Bott periodization. For each smooth S -scheme Y and line bundle $\mathcal{L}$ on Y this classifies the first Chern class

$$c_1^{\mathrm{KGL}}(\mathcal{L}) = 1 - [\mathcal{L}^\vee] \in \mathrm{KH}_0(Y).$$

Specializing to $\mathcal{L} = \mathcal{O}(1)$ on $\mathbb{P}_S^1$ gives

$$c_1^{\mathrm{KGL}}(\mathcal{O}(1)) = 1 - [\mathcal{O}(1)^\vee] = 1 - [\mathcal{O}(-1)] = \beta,$$

recovering the Bott element of Section 3.1.6 and verifying the basepoint condition of Definition 3.2.24. The canonical orientation is defined by $1 - (-)^\vee$ with this Bott normalization; the K-theoretic projective bundle formula describes its compatible restrictions to $\widetilde{M}_S(\mathbb{P}_S^m)$ through

$$\mathrm{KH}(\mathbb{P}_S^\infty) \simeq \lim_m \mathrm{KH}(\mathbb{P}_S^m) \simeq \mathrm{KH}(S)[[t]]$$

with $t = 1 - [\mathcal{O}(-1)]$ (cf. Remark 3.2.27). This KGL-valued class is not additive under tensor product and therefore does not descend through the Picard count; its image in the weight-one slice does descend by Lemma 3.2.39. Combined with Theorem 3.2.29, the orientation nevertheless yields the projective bundle formula for KGL_S in all degrees.

Remark 3.2.32 (Higher Chern classes, Euler class, and formal group law). Once Theorem 3.2.29 and Example 3.2.31 are in hand, the classical template promotes the first Chern class to higher characteristic classes. Let $\mathcal{E}$ be a rank- r vector bundle on a smooth S -scheme Y , and write $\pi: \mathbb{P}(\mathcal{E}) \rightarrow Y$ for its projective bundle. It carries the universal quotient line bundle $\mathcal{O}_{\mathbb{P}(\mathcal{E})}(1)$. Applying Theorem 3.2.29 relatively — a formal consequence of the absolute statement, cf. [AI25, Lemma 3.2.3] — produces a unique identity

$$c_1(\mathcal{O}_{\mathbb{P}(\mathcal{E})}(1))^r = \sum_{i=1}^r (-1)^{i-1} c_i(\mathcal{E}) \cdot c_1(\mathcal{O}_{\mathbb{P}(\mathcal{E})}(1))^{r-i}$$

in $\pi_0 E(r)(M_S(\mathbb{P}(\mathcal{E})))$, with each $c_i(\mathcal{E})$ pulled back along $\pi: \mathbb{P}(\mathcal{E}) \rightarrow Y$. This identity uniquely defines the **higher Chern classes** $c_i(\mathcal{E}) \in \pi_0 E(i)(M_S(Y))$ for $1 \leq i \leq r$. The top Chern class $c_r(\mathcal{E})$ is the **Euler class** $e(\mathcal{E}) \in \pi_0 E(r)(M_S(Y))$, paralleling the topological identity $c_n(V) = e(V)$ of Remark 3.2.22. The **splitting principle** for vector bundles holds: pulling back $\mathcal{E}$ to the flag bundle $\mathrm{Fl}(\mathcal{E}) \rightarrow Y$ reduces computations to the case where $\mathcal{E}$ is a direct sum of line bundles, exactly as in topology. This is the rank- r analogue of the $\mathbb{T}_S$ -cellular splitting for $\widetilde{M}_S(\Omega^\infty \mathcal{P}ic)$ in Remark 3.2.27: there the orientation propagates from $\mathbb{T}_S$ to all of $\widetilde{M}_S(\Omega^\infty \mathcal{P}ic)$; here it propagates from line bundles to all vector bundles.

The tensor product of line bundles, classified by the multiplication $\mu: \Omega^\infty \mathcal{P}ic \times \Omega^\infty \mathcal{P}ic \rightarrow \Omega^\infty \mathcal{P}ic$ on the motivic Picard stack $\mathrm{B}\mathbb{G}_m$, pulled back along c_1 produces a power series

$$F_E(x, y) = \mu^* c_1 \in E^{**}(S)[[x, y]]$$

— the **formal group law** of E — computing $c_1(\mathcal{L}_1 \otimes \mathcal{L}_2) = F_E(c_1(\mathcal{L}_1), c_1(\mathcal{L}_2))$. Distinguished examples:

- $H\mathbb{Z}$, with the additive law $F(x, y) = x + y$;
- KGL , with the graded multiplicative law

$$F_{KGL}(x, y) = x + y - \beta xy, \quad \beta \in KGL^{-2, -1}(S)^\times.$$

Indeed, multiplication by the invertible class β identifies the graded coordinate $x = c_1^{KGL}(\mathcal{L})$ with the degree-zero class $t = \beta x = 1 - [\mathcal{L}^\vee] \in KH_0$, and

$$t(\mathcal{L}_1 \otimes \mathcal{L}_2) = t(\mathcal{L}_1) + t(\mathcal{L}_2) - t(\mathcal{L}_1)t(\mathcal{L}_2).$$

Thus Bott periodicity identifies graded pieces; it does not set $\beta = 1$. This is the formula $\otimes^*(c) = x_1 + x_2 - \beta x_1 x_2$ of [AHI24a, Lemma 8.3]. Its sign differs from the topological coordinate chosen in Remark 3.2.23 because here c_1 is defined using dual line bundles; the change of coordinate $x \mapsto -x$ interchanges the two signs.

- MGL , whose formal group law will be shown in Section 3.5 to be universal in the Lazard sense — over a field, integrally in characteristic zero, and after inverting the characteristic exponent in general (Theorem 3.5.18).

MGL provides a rich intermediate layer between $H\mathbb{Z}$ and KGL , the basis of the motivic chromatic picture.

Remark 3.2.33 (Motivic Thom isomorphism, Euler class, and Gysin sequence). The motivic Thom isomorphism, Euler class, and Gysin sequence transcribe *verbatim* from their topological avatars in Theorem 3.2.20, with each rank- n complex vector bundle replaced by a motivic vector bundle of rank n and each topological shift by $2n$ replaced by a Tate twist $\mathbb{T}_S^{\otimes n}$:

- For an oriented E and a rank- n motivic vector bundle $\zeta: X \rightarrow BGL_n$ with Thom space $\mathrm{Th}(\zeta)$, there is a natural equivalence $\widetilde{M}_S(\mathrm{Th}(\zeta)) \otimes E \simeq \mathbb{T}_S^{\otimes n} \otimes M_S(X) \otimes E$, recovering the Thom isomorphism $E^{p,q}(X) \xrightarrow{\sim} \widetilde{E}^{p+2n, q+n}(\mathrm{Th}(\zeta))$;
- Pulling back the Thom class along the zero section gives the **motivic Euler class** $e(V) \in E^{2n,n}(B)$;
- Writing $z: B \rightarrow V$ for the zero section and $V^\times := V \setminus z(B)$, the localization cofiber sequence produces a **motivic Gysin sequence** relating $E^{*,*}(B) \simeq E^{*,*}(V)$, $E^{*,*}(V^\times)$, and the Euler class.

See [BH21, Proposition 16.28] for the motivic Thom isomorphism in the framework of motivic Thom spectra (recalled as Proposition 3.5.12), and [AHI24b, §2] for the modern Gysin-map theory underlying the motivic Gysin sequence.

The universal example is the motivic Thom spectrum MGL of Section 3.5. Over a field, Theorem 3.5.13 says that every orientation of $E \in \mathrm{CAlg}(\mathrm{hSH}(k))$ factors uniquely through a morphism $MGL \rightarrow E$, the motivic analogue of Quillen’s theorem of Remark 3.2.23. Over a general base we use only the tautological orientation of MGL and the Thom isomorphisms it induces for MGL -algebras.

3.2.5. First Chern class map

An orientation of a motivic ring spectrum $E \in \mathrm{CAlg}(\mathrm{hSH}(S))$, in the sense of Definition 3.2.24, is a universal first Chern class morphism

$$c_E: \widetilde{M}_S(\Omega^\infty \mathcal{P}ic) \longrightarrow E \otimes \mathbb{T}_S$$

out of the reduced motive of the motivic Picard stack (Notation 3.1.2), fitting into a basepoint triangle with $\mathbb{T}_S$. Specialized to $E = KGL_S$, this records the K-theoretic Chern class $\mathcal{L} \mapsto$

$1 - [\mathcal{L}^\vee]$ of Example 3.2.31. This subsection refines c_{KGL_S} , via the slice filtration in weight 1, to a Chern class map landing in $\mathbb{A}^1$ -motivic cohomology:

$$c: \widetilde{M}_S(\Omega^\infty \mathcal{P}ic) \longrightarrow \mathbb{Z}(1)^\mathbb{A}, \quad (3.2.3)$$

constructed on smooth schemes over a regular noetherian base from the first two unstable slices of K-theory. The resulting map is the Atiyah–Hirzebruch edge of the $\mathbb{A}^1$ -motivic filtration on KH in weight 1.

Factoring through $\mathcal{P}ic$. The source of (3.2.3) is, by the $\Sigma^\infty \dashv \Omega^\infty$ adjunction at the level of presheaves of spectra, the free $\mathbb{E}_\infty$ -group object on the underlying pointed motivic space of $\mathcal{P}ic$:

$$\widetilde{M}_S(\Omega^\infty \mathcal{P}ic) \simeq \sigma^\infty \Sigma^\infty \Omega^\infty \mathcal{P}ic.$$

It therefore remembers only the pointed motivic space $\Omega^\infty \mathcal{P}ic \simeq \mathrm{BG}_m$ classifying line bundles, but *not* the fact that line bundles can be tensored together; in particular c as constructed in (3.2.3) is not *a priori* additive under $\otimes$. The Picard spectrum $\mathcal{P}ic \in \mathrm{Shv}_{\mathrm{Nis}, \mathbb{A}^1}(\mathrm{Sm}_S, \mathrm{Sp})$ does record that additive structure (its $\mathbb{E}_\infty$ -monoid structure is exactly the tensor product of line bundles), and the two are linked by the counit

$$\varepsilon: \widetilde{M}_S(\Omega^\infty \mathcal{P}ic) \longrightarrow \mathcal{P}ic$$

of the same adjunction. The compatibility Lemma 3.2.39 below establishes that c factors through ε to a map

$$c_1^\mathbb{A}: \mathcal{P}ic \longrightarrow \mathbb{Z}(1)^\mathbb{A},$$

which is then the genuine first Chern class: $c_1^\mathbb{A}(\mathcal{L})$ depends only on the isomorphism class of $\mathcal{L} \in \mathrm{Pic}(Y)$ and is additive, $c_1^\mathbb{A}(\mathcal{L}_1 \otimes \mathcal{L}_2) = c_1^\mathbb{A}(\mathcal{L}_1) + c_1^\mathbb{A}(\mathcal{L}_2)$.

Notation. $\mathbb{Z}(1)^\mathbb{A} = \omega^{\infty s^1} \mathrm{KGL}_S$ denotes the weight-1 $\mathbb{A}^1$ -motivic cohomology spectrum, formally defined in Definition 3.2.42 below. Following the T-spectra convention of Remark 3.2.43, $\mathbb{Z}(j)^\mathbb{A}$ carries no $[-2j]$ desuspension, so c and $c_1^\mathbb{A}$ land in $\mathbb{Z}(1)^\mathbb{A}$ itself rather than $\mathbb{Z}(1)^\mathbb{A}[2]$ as in [BEM25, §4].

The construction below adapts [BEM25, §4.2.1]. We state every auxiliary lemma used and give the structural proofs; detailed geometric and topological proofs of those lemmas are collected in the author’s companion Oberseminar notes *Low Weights*, §§1–3. Two further structural consequences are taken up later: the cdh-extension to qcqs schemes — parts (3) and (4) of [BEM25, Theorem 4.39] deferred from Theorem 3.2.48 — in Section 3.2.7, and the description of $\mathrm{KH}/\mathrm{Fil}_{\mathbb{A}, \mathrm{cdh}}^2 \mathrm{KH}$ via the derived Picard anima in Section 3.2.8.

Geometric and connectivity preliminaries

Throughout this subsection let S be a regular noetherian scheme. To compress notation in the unstable slice category we write

$$\mathrm{Shv}_S(j) := \mathrm{Shv}_{\mathrm{Nis}, \mathbb{A}^1}(\mathrm{Sm}_S, \mathrm{Sp})(j)$$

for the j -effective subcategory of Section 3.2.2, and $\mathrm{Shv}_S(j)^\perp$ for its right orthogonal complement in $\mathrm{Shv}_{\mathrm{Nis}, \mathbb{A}^1}(\mathrm{Sm}_S, \mathrm{Sp})$ (the full subcategory of G with $\mathrm{map}(H, G) \simeq 0$ for every $H \in \mathrm{Shv}_S(j)$).

The first lemma collects the regularity-dependent facts about $\Gamma_\tau(-, \mathbb{Z})$ and $\Gamma_\tau(-, \mathbb{G}_m)$ that drive the whole computation: cohomological vanishing, change of topology from Zariski to Nisnevich, and $\mathbb{A}^1$ -invariance. (Recall our standing derived convention: Γ_τ denotes derived τ -global sections, and H_τ^n is the cohomology of those derived sections.)

Lemma 3.2.34. *Let S be a regular noetherian scheme. Then:*

(i) *There are natural isomorphisms*

$$H_{\text{Zar}}^n(S, \mathbb{Z}) = \begin{cases} \mathbb{Z}^{\pi_0(S)} & n = 0, \\ 0 & n > 0, \end{cases} \quad H_{\text{Zar}}^n(S, \mathbb{G}_m) = \begin{cases} \mathcal{O}(S)^\times & n = 0, \\ \text{Pic}(S) & n = 1, \\ 0 & n > 1. \end{cases}$$

(ii) *The change-of-topology maps*

$$\Gamma_{\text{Zar}}(S, \mathbb{Z}) \rightarrow \Gamma_{\text{Nis}}(S, \mathbb{Z}), \quad \Gamma_{\text{Zar}}(S, \mathbb{G}_m) \rightarrow \Gamma_{\text{Nis}}(S, \mathbb{G}_m)$$

are equivalences.

(iii) *The canonical maps $\Gamma_{\text{Nis}}(\mathbb{A}_S^1, \mathbb{Z}) \rightarrow \Gamma_{\text{Nis}}(S, \mathbb{Z})$ and $\Gamma_{\text{Nis}}(\mathbb{A}_S^1, \mathbb{G}_m) \rightarrow \Gamma_{\text{Nis}}(S, \mathbb{G}_m)$ are equivalences.*

Proof sketch. Part (1) is classical: vanishing of higher cohomology of constant sheaves on irreducible schemes for $\mathbb{Z}$, and the length-1 Weil-divisor resolution

$$0 \rightarrow \mathbb{G}_m \rightarrow \bigoplus_{\eta \in S^{(0)}} i_{\eta,*} \mathcal{O}_\eta^\times \rightarrow \bigoplus_{x \in S^{(1)}} i_{x,*} \mathbb{Z} \rightarrow 0$$

on the regular (hence normal) scheme S for $\mathbb{G}_m$.

Part (2) reduces by sheafification to the stalk case — a regular noetherian local ring R of dimension d — and proceeds by induction on d . The Nisnevich-distinguished henselization square

$$\begin{array}{ccc} \text{Spec}(R^h) \setminus \{\mathfrak{m}R^h\} & \longrightarrow & \text{Spec}(R^h) \\ \downarrow & & \downarrow \\ \text{Spec}(R) \setminus \{\mathfrak{m}\} & \longrightarrow & \text{Spec}(R) \end{array}$$

is sent by $\Gamma_{\text{Nis}}(-, F)$ to a cartesian square (Nisnevich descent); henselianity of R^h kills its higher Nisnevich cohomology so the right-hand vertex agrees with Zariski. The inductive step shows that $\Gamma_{\text{Zar}}(-, F)$ also carries the square to a cartesian one. For $d = 0$ trivially; for $d = 1$ direct computation using the discrete valuation; for $d \geq 2$ one reduces to checking that the restriction maps $\Gamma_{\text{Zar}}(\text{Spec}(R), F) \rightarrow \Gamma_{\text{Zar}}(\text{Spec}(R) \setminus \{\mathfrak{m}\}, F)$ are equivalences, which holds by connectedness for $F = \mathbb{Z}$ and by Grothendieck's $\text{Pic}(\text{Spec}(R) \setminus \{\mathfrak{m}\}) = 0$ ([Stacks, Tag 0F2H](#)) plus normality $R^\times \simeq H^0(\text{Spec}(R) \setminus \{\mathfrak{m}\}, \mathbb{G}_m)$ for $F = \mathbb{G}_m$.

Part (3) follows from (2) by replacing Nis with Zar and applying (1): for $\mathbb{Z}$ note that $\pi_0(S) = \pi_0(\mathbb{A}_S^1)$, and for $\mathbb{G}_m$ use $A^\times = A[T]^\times$ on reduced rings together with Traverso–Swan invariance of Pic under $A \rightarrow A[T]$ on seminormal (in particular regular) rings. $\square$

Next, a clean way to identify the truncations of the unstable filtration: split a fiber sequence into an effective part and a right-orthogonal part.

Observation 3.2.35. Let $j \geq 0$ and suppose

$$A \longrightarrow E \longrightarrow B$$

is a fiber sequence in $\text{Shv}_{\text{Nis}, \mathbb{A}^1}(\text{Sm}_S, \text{Sp})$ with $A \in \text{Shv}_S(j)$ and $B \in \text{Shv}_S(j)^\perp$. Then there are canonical equivalences

$$A \simeq \text{Fil}_{\text{un-slice}}^j(E), \quad B \simeq E / \text{Fil}_{\text{un-slice}}^j(E).$$

Proof sketch. Since $\mathrm{Fil}_{\mathrm{un-slice}}^j(E) \rightarrow E$ is universal for maps from j -effective objects, the inclusion $A \hookrightarrow E$ factors as $A \xrightarrow{\phi} \mathrm{Fil}_{\mathrm{un-slice}}^j(E) \rightarrow E$. Let $C := \mathrm{cofib}(\phi)$; closure of $\mathrm{Shv}_S(j)$ under colimits gives $C \in \mathrm{Shv}_S(j)$. The octahedral axiom applied to $A \rightarrow \mathrm{Fil}_{\mathrm{un-slice}}^j(E) \rightarrow E$ produces a cofiber sequence

$$C \longrightarrow B \longrightarrow E/\mathrm{Fil}_{\mathrm{un-slice}}^j(E),$$

whose first map is zero because $C \in \mathrm{Shv}_S(j)$ and $B \in \mathrm{Shv}_S(j)^\perp$. Hence the sequence splits, exhibiting $C[1] \in \mathrm{Shv}_S(j)^\perp$ as a retract of $E/\mathrm{Fil}_{\mathrm{un-slice}}^j(E)$; combined with $C[1] \in \mathrm{Shv}_S(j)$ this forces $C = 0$, so ϕ is an equivalence. $\square$

Finally, a connectivity criterion that reduces effectivity of a Nisnevich sheaf of spectra to effectivity of its underlying $\Sigma^\infty \Omega^\infty$ -resolution — the engine that converts the Grassmannian computation below into a statement about $k^{\mathrm{rk}=0}$ and its fiber.

Observation 3.2.36. Let $E \in C_S$ be Nisnevich-locally connective. If

$$L_{\mathbb{A}^1, \mathrm{Nis}} \Sigma^\infty \Omega^\infty E \in \mathrm{Shv}_S(j),$$

then $E \in \mathrm{Shv}_S(j)$.

Proof sketch. Any spectrum A admits the comonadic resolution $A \simeq \mathrm{colim}_n (\Sigma^\infty \Omega^\infty(A[n]))[-n]$. For Nisnevich-locally connective E , each $\Omega^\infty(E[n])$ is Nisnevich-locally the iterated bar construction $B^n(\Omega^\infty E)$, built from products and colimits of $\Omega^\infty E$; the formula $\Sigma^\infty(A_1 \times A_2) \simeq \Sigma^\infty A_1 \oplus \Sigma^\infty A_2 \oplus (\Sigma^\infty A_1 \otimes \Sigma^\infty A_2)$ then expresses $\Sigma^\infty \Omega^\infty(E[n])$ as built from $\Sigma^\infty \Omega^\infty E$ by colimits, sums, tensors, and desuspensions. Closure of $\mathrm{Shv}_S(j)$ under these operations finishes the argument. $\square$

Unstable slices of K -theory

Recall from Section 3.2.2 the unstable slice filtration $\mathrm{Fil}_{\mathrm{un-slice}}^\star$ on $\mathrm{Shv}_{\mathrm{Nis}, \mathbb{A}^1}(\mathrm{Sm}_S, \mathrm{Sp})$, together with the Beck–Chevalley comparisons

$$\mathrm{Fil}_{\mathrm{un-slice}}^j \omega^\infty E \longrightarrow \omega^\infty \mathrm{Fil}_{\mathrm{slice}}^j(E), \quad s_u^j \omega^\infty E \longrightarrow \omega^\infty s^j E$$

attached to any $E \in \mathrm{SH}(S)$. Applied to $E = \mathrm{KGL}_S$ they yield maps

$$s_u^0 \omega^\infty \mathrm{KGL}_S \longrightarrow \omega^\infty s^0 \mathrm{KGL}_S \simeq \mathbb{Z}(0)^\mathbb{A}|_{\mathrm{Sm}_S}, \quad s_u^1 \omega^\infty \mathrm{KGL}_S \longrightarrow \omega^\infty s^1 \mathrm{KGL}_S \simeq \mathbb{Z}(1)^\mathbb{A}|_{\mathrm{Sm}_S}$$

in the conventions of Definition 3.2.42 (no $[-2j]$ desuspension; cf. Remark 3.2.43). Over the regular noetherian base S , every $Y \in \mathrm{Sm}_S$ is regular, so $K(Y) \simeq k(Y)$ (Proposition 1.3.11) and K is $\mathbb{A}^1$ -invariant on Sm_S by Quillen’s theorem; hence $K \simeq KH$ on Sm_S and $\omega^\infty \mathrm{KGL}_S \simeq k$ as Nisnevich sheaves of spectra on Sm_S . Identifying the two unstable slices of k is therefore the central computation.

Lemma 3.2.37. Let S be a regular noetherian scheme. In $\mathrm{Shv}_{\mathrm{Nis}, \mathbb{A}^1}(\mathrm{Sm}_S, \mathrm{Sp})$:

(i) The rank map $\mathrm{rk}: k \rightarrow \Gamma_{\mathrm{Nis}}(-, \mathbb{Z})$ induces an equivalence $s_u^0 k \xrightarrow{\sim} \Gamma_{\mathrm{Nis}}(-, \mathbb{Z})$, and hence

$$\mathrm{Fil}_{\mathrm{un-slice}}^1 k \simeq k^{\mathrm{rk}=0} := \mathrm{fib}(\mathrm{rk}).$$

(ii) The determinant $\det: k^{\mathrm{rk}=0} \rightarrow \Gamma_{\mathrm{Nis}}(-, \mathbb{G}_m)[1]$ induces an equivalence $s_u^1 k \xrightarrow{\sim} \Gamma_{\mathrm{Nis}}(-, \mathbb{G}_m)[1]$.

Proof sketch. Both assertions are instances of Observation 3.2.35. For (1), apply it to the fiber sequence $k^{\mathrm{rk}=0} \rightarrow k \rightarrow \Gamma_{\mathrm{Nis}}(-, \mathbb{Z})$ with $j = 1$. For (2), first set

$$F := \mathrm{fib}(\det: k^{\mathrm{rk}=0} \rightarrow \Gamma_{\mathrm{Nis}}(-, \mathbb{G}_m)[1]).$$

We will verify that F is 2-effective and that the target is right orthogonal to $\mathrm{Shv}_S(2)$; only after those verifications does Observation 3.2.35 identify F with $\mathrm{Fil}_{\mathrm{un-slice}}^2 k^{\mathrm{rk}=0}$. Two verifications are required in each case.

Right orthogonality. The category $\mathrm{Shv}_S(j)$ is generated under colimits and desuspensions by $\Sigma_+^\infty \mathbb{P}_X^j / \Sigma_+^\infty \mathbb{P}_X^{j-1} \simeq (\Sigma^\infty \mathbb{T}_S)^{\otimes j} \otimes \Sigma_+^\infty X$ for $X \in \mathrm{Sm}_S$, so it suffices to check that $\Gamma_{\mathrm{Nis}}(-, \mathbb{Z})$ is insensitive to passing from X to $\mathbb{P}_X^1$, and likewise for $\Gamma_{\mathrm{Nis}}(-, \mathbb{G}_m)$ from $\mathbb{P}_X^1$ to $\mathbb{P}_X^2$. Both reduce by Lemma 3.2.34(1)(2) to the corresponding Zariski computations (the projective-bundle vanishing in cohomology of $\mathbb{Z}$ and $\mathbb{G}_m$ on smooth S -schemes; cf. [Stacks, Tag 0BXJ](#)).

Effectivity. It remains to show $k^{\mathrm{rk}=0} \in \mathrm{Shv}_S(1)$ and $F \in \mathrm{Shv}_S(2)$. Morel–Voevodsky’s equivalence $\Omega^\infty k \simeq L_{\mathrm{Nis}, \mathbb{A}^1}(\mathbb{Z} \times \mathrm{BGL})$ [MV99, §4] gives $\Omega^\infty k^{\mathrm{rk}=0} \simeq L_{\mathrm{Nis}, \mathbb{A}^1} \mathrm{BGL}$. Pulling back the fiber sequence $L_{\mathrm{Nis}} \mathrm{BSL} \rightarrow L_{\mathrm{Nis}} \mathrm{BGL} \rightarrow L_{\mathrm{Nis}} \mathrm{B}\mathbb{G}_m$ along $L_{\mathbb{A}^1}$ — the base is pointed connected (π_0 -Nisnevich-stalk = Pic of a regular henselian local ring = 0), so by [Lur17, Lemma 5.5.6.17] taking the fiber commutes with sifted colimits, in particular with $L_{\mathbb{A}^1}$ — and using $\mathbb{A}^1$ -invariance of $L_{\mathrm{Nis}} \mathrm{BGL}$ (by Lindel’s theorem [Lin82], applied on Nisnevich stalks, which are regular local rings) and of $L_{\mathrm{Nis}} \mathrm{B}\mathbb{G}_m$ (Lemma 3.2.34(3)), one obtains a fiber sequence

$$L_{\mathrm{Nis}, \mathbb{A}^1} \mathrm{BSL} \longrightarrow L_{\mathrm{Nis}} \mathrm{BGL} \longrightarrow L_{\mathrm{Nis}} \mathrm{B}\mathbb{G}_m$$

of $\mathbb{A}^1$ -invariant Nisnevich sheaves, and hence $\Omega^\infty F \simeq L_{\mathrm{Nis}, \mathbb{A}^1} \mathrm{BSL}$. Both $k^{\mathrm{rk}=0}$ and F are Nisnevich-locally connective (long exact sequences plus connectivity of k on regular Sm_S), so by Observation 3.2.36 it suffices to show $L_{\mathrm{Nis}, \mathbb{A}^1} \Sigma^\infty \mathrm{BGL} \in \mathrm{Shv}_S(1)$ and $L_{\mathrm{Nis}, \mathbb{A}^1} \Sigma^\infty \mathrm{BSL} \in \mathrm{Shv}_S(2)$.

Effectivity input. The remaining standard calculation is

$$L_{\mathrm{Nis}, \mathbb{A}^1} \Sigma^\infty \mathrm{BGL} \in \mathrm{Shv}_S(1), \quad L_{\mathrm{Nis}, \mathbb{A}^1} \Sigma^\infty \mathrm{BSL} \in \mathrm{Shv}_S(2).$$

This is the Grassmannian/special-linear-Grassmannian effectivity computation used in [BEM25, §4.2.1]. The ordinary Grassmannian case follows from its Schubert cells; the special-linear refinement requires keeping track of the determinant trivialization and should not be replaced by the assertion that its finite model is simply the same projective homogeneous space G/P . Taking this computation as the geometric input, Observation 3.2.36 gives $k^{\mathrm{rk}=0} \in \mathrm{Shv}_S(1)$ and $F \in \mathrm{Shv}_S(2)$. Applying Observation 3.2.35 now gives $F \simeq \mathrm{Fil}_{\mathrm{un-slice}}^2 k^{\mathrm{rk}=0}$ and proves the second assertion. $\square$

The first Chern class on regular bases

We assemble the construction of $c_1^{\mathbb{A}}$ promised in (3.2.3) and verify the two compatibilities required of it: descent along the counit $\varepsilon: \widetilde{M}_S(\Omega^\infty \mathcal{P}ic) \rightarrow \mathcal{P}ic$, and compatibility with the Atiyah–Hirzebruch edge map of the $\mathbb{A}^1$ -motivic filtration.

The weight-0 piece of Lemma 3.2.37 together with the Beck–Chevalley comparison endows $\mathbb{Z}(\star)^{\mathbb{A}}$ with a $\mathbb{Z}$ -linear structure: the composite

$$\Gamma_{\mathrm{Nis}}(-, \mathbb{Z}) \simeq s_u^0 k \longrightarrow \omega^\infty s^0 \mathrm{KGL}_S = \mathbb{Z}(0)^{\mathbb{A}}|_{\mathrm{Sm}_S}$$

is a map of $\mathbb{E}_\infty$ -algebras (since $s^0 \mathrm{KGL}_S \simeq \mathrm{H}\mathbb{Z}_S$ is an $\mathbb{E}_\infty$ -ring spectrum and rk respects the multiplicative structure), and over this base ring $\mathbb{Z}(1)^{\mathbb{A}}$ is a module. The weight-1 piece then promotes to the universal first Chern class.

Construction 3.2.38. Let S be a regular noetherian scheme. On smooth S -schemes, the composite

$$\mathcal{P}ic \xleftarrow{\sim} \Gamma_{\text{Nis}}(-, \mathbb{G}_m)[1] \xleftarrow{\sim} s_u^1 k \longrightarrow \omega^\infty s^1 \text{KGL}_S = \mathbb{Z}(1)^\mathbb{A}|_{\text{Sm}_S}$$

— where the first equivalence identifies $\mathcal{P}ic$ on Sm_S via Lemma 3.2.34 (which removes both the truncation $\tau^{\leq 1}$ and the change of topology), the second is Lemma 3.2.37(2), and the third is the Beck–Chevalley comparison — defines the **first Chern class**

$$c_1^\mathbb{A}: \mathcal{P}ic \longrightarrow \mathbb{Z}(1)^\mathbb{A}.$$

For a line bundle $\mathcal{L}$ on $Y \in \text{Sm}_S$, the class $[\mathcal{L}] \in \text{Pic}(Y) = \pi_0 \Gamma_{\text{Nis}}(Y, \mathbb{G}_m)[1]$ maps to

$$c_1^\mathbb{A}(\mathcal{L}) \in \pi_0 \mathbb{Z}(1)^\mathbb{A}(Y),$$

which (by construction) depends only on the isomorphism class of $\mathcal{L}$ and is additive in $\otimes$.

The Chern class so constructed agrees with the orientation-induced map $\widetilde{M}_S(\Omega^\infty \mathcal{P}ic) \rightarrow \mathbb{Z}(1)^\mathbb{A}$ of (3.2.3) via the counit ε , as we now make precise.

Lemma 3.2.39. *Let S be a regular noetherian scheme.*

- (i) *The mapping anima $\text{Map}(\widetilde{M}_S(\Omega^\infty \mathcal{P}ic), \Gamma_{\text{Nis}}(-, \mathbb{Z}))$ in $\text{Shv}_{\text{Nis}, \mathbb{A}^1}(\text{Sm}_S, \text{Sp})$ is contractible. In particular, the universal orientation map $1 - (-)^\vee: \widetilde{M}_S(\Omega^\infty \mathcal{P}ic) \rightarrow k^{\text{rk}=0}$ of Example 3.2.31 factors uniquely through $k^{\text{rk}=0}$.*
- (ii) *The resulting diagram*

$$\begin{array}{ccc} \widetilde{M}_S(\Omega^\infty \mathcal{P}ic) & \xrightarrow{1 - (-)^\vee} & k^{\text{rk}=0} \\ \varepsilon \downarrow & & \downarrow \det \\ \mathcal{P}ic & \xrightarrow{\sim} & \Gamma_{\text{Nis}}(-, \mathbb{G}_m)[1] \end{array}$$

commutes up to homotopy in $\text{Shv}_{\text{Nis}, \mathbb{A}^1}(\text{Sm}_S, \text{Sp})$, where the left vertical arrow is the counit $\varepsilon: \widetilde{M}_S(\Omega^\infty \mathcal{P}ic) \simeq \sigma^\infty \Sigma^\infty \Omega^\infty \mathcal{P}ic \rightarrow \mathcal{P}ic$ and the bottom identification is the equivalence of Lemma 3.2.34 (after removing the implicit $\tau^{\leq 1}$ on smooth schemes over a regular base).

Proof sketch. (1) $\widetilde{M}_S(\Omega^\infty \mathcal{P}ic) \simeq \sigma^\infty \Sigma^\infty \Omega^\infty \mathcal{P}ic$ is Nisnevich-locally 1-connective: it is the reduced suspension spectrum of the Nisnevich-locally *connected* pointed space $B\mathbb{G}_m$, so even π_0 vanishes locally. On the other hand $\Gamma_{\text{Nis}}(-, \mathbb{Z})$ is concentrated in degree 0 on regular bases (Lemma 3.2.34(1)). The mapping anima between them is therefore contractible, so $1 - (-)^\vee$ has zero composite with the rank map.

(2) By the $\Sigma^\infty \dashv \Omega^\infty$ adjunction, commutativity of the diagram amounts to showing that the unstable composite

$$\begin{array}{c} \Omega^\infty \mathcal{P}ic \xrightarrow{1 - (-)^\vee} \Omega^\infty k^{\text{rk}=0} \xrightarrow{\det} \Omega^\infty \Gamma_{\text{Zar}}(-, \mathbb{G}_m)[1] \\ \downarrow \sim \\ \Omega^\infty \mathcal{P}ic \end{array}$$

is the identity. By uniqueness of the orientation [BEM25, Definition 4.4] it suffices to verify this composite on the universal classes $[O(1)]: \mathbb{P}_S^m \rightarrow \Omega^\infty \mathcal{P}ic$, i.e., to check that $\det: K_0^{\text{rk}=0}(\mathbb{P}_S^m) \rightarrow$

$\mathrm{Pic}(\mathbb{P}_S^m)$ sends $1 - [\mathcal{O}(-1)]$ to $\mathcal{O}(1)$. This in turn reduces to the general fact $\det([\mathcal{V}] - d) = \det(\mathcal{V})$ for a rank- d bundle $\mathcal{V}$, applied to $\mathcal{V} = \mathcal{L}$ a line bundle: Morel–Voevodsky’s representability [MV99, §4, Proposition 3.7] identifies the classifying map $S \xrightarrow{[\mathcal{L}]} L_{\mathrm{Nis}} B\mathbb{G}_m \rightarrow L_{\mathrm{Nis}, \mathbb{A}^1} BGL \simeq \Omega^\infty k^{\mathrm{rk}=0}$ with $[\mathcal{L}] - 1 \in K_0^{\mathrm{rk}=0}(S)$; since the determinant of $GL_1 \hookrightarrow GL$ is the identity on $\mathbb{G}_m$, the composite is the identity on $L_{\mathrm{Nis}} B\mathbb{G}_m$ and we conclude $\det([\mathcal{V}] - 1) = \mathcal{V}$ for every line bundle $\mathcal{V}$. Applying this with $\mathcal{V} = \mathcal{L}^\vee$ and using that $\det$ is a map of group-like $\mathbb{E}_\infty$ -objects, $\det(1 - [\mathcal{L}^\vee]) = \det([\mathcal{L}^\vee] - 1)^{-1} = (\mathcal{L}^\vee)^{-1} = \mathcal{L}$. $\square$

Naturality of the Beck–Chevalley transformation $\mathrm{Fil}_{\mathrm{un-slice}}^j \omega^\infty \rightarrow \omega^\infty \mathrm{Fil}_{\mathrm{slice}}^j$ across the cofiber projection $\mathrm{Fil}_{\mathrm{slice}}^j \rightarrow s^j$, applied to KGL_S at $j = 1$, identifies $c_1^\mathbb{A}$ as the edge map of the Atiyah–Hirzebruch filtration in weight 1. (Recall from Theorem 3.2.48(ii) that $\mathrm{gr}_\mathbb{A}^1 KH \simeq \mathbb{Z}(1)^\mathbb{A}$ in our T-spectra convention, with no [2] shift.)

Corollary 3.2.40. *Let S be a regular noetherian scheme. The diagram of presheaves of spectra on Sm_S*

$$\begin{array}{ccc} k^{\mathrm{rk}=0} & \xrightarrow{\quad} & \mathrm{Fil}_\mathbb{A}^1 KH \\ \det \downarrow & & \downarrow \text{edge} \\ \Gamma_{\mathrm{Nis}}(-, \mathbb{G}_m)[1] & \xrightarrow{c_1^\mathbb{A}} & \mathbb{Z}(1)^\mathbb{A} \end{array}$$

commutes, where the right vertical arrow is the edge map of the $\mathbb{A}^1$ -motivic filtration (Definition 3.2.42) and the bottom horizontal arrow is the first Chern class of Construction 3.2.38 (composed with $\Gamma_{\mathrm{Nis}}(-, \mathbb{G}_m)[1] \xrightarrow{\sim} \mathcal{P}ic$ on Sm_S).

Proof sketch. Compare $\mathrm{Fil}_{\mathrm{un-slice}}^j k \rightarrow \omega^\infty \mathrm{Fil}_{\mathrm{slice}}^j KGL_S$ with the cofiber projection to $s_u^j k \rightarrow \omega^\infty s^j KGL_S$ at $j = 1$; the two together with $\mathrm{Fil}_{\mathrm{un-slice}}^1 k \simeq k^{\mathrm{rk}=0}$ (Lemma 3.2.37(1)) assemble the displayed square. $\square$

Remark 3.2.41. Concretely, $c_1^\mathbb{A}(\mathcal{L}) \in \pi_0 \mathbb{Z}(1)^\mathbb{A}(Y)$ depends only on the isomorphism class of $\mathcal{L} \in \mathrm{Pic}(Y)$ (since ε is precisely the universal map collapsing $\tilde{M}_S(\Omega^\infty \mathcal{P}ic)$ to its tensor-product-aware completion $\mathcal{P}ic$), and is additive in $\otimes$: $c_1^\mathbb{A}(\mathcal{L}_1 \otimes \mathcal{L}_2) = c_1^\mathbb{A}(\mathcal{L}_1) + c_1^\mathbb{A}(\mathcal{L}_2)$. By Example 3.2.31 the KGL-valued Chern class is $c_1^{\mathrm{KGL}}(\mathcal{L}) = 1 - [\mathcal{L}^\vee]$; the slice projection then gives the $\mathbb{A}^1$ -motivic formula. The formal group law underlying $c_1^\mathbb{A}$ on $\mathbb{H}\mathbb{Z}_S \simeq s^0 KGL_S$ is the additive law (Remark 3.2.32), as expected for an ordinary motivic cohomology theory.

3.2.6. $\mathbb{A}^1$ -motivic cohomology

Let S be a qcqs scheme. Applying the slice filtration of Construction 3.2.8 to the motivic K-theory spectrum KGL_S simultaneously assembles a motivic cohomology theory as its graded pieces and endows $KH(S)$ with a natural multiplicative filtration.

Definition 3.2.42 ($\mathbb{A}^1$ -motivic cohomology). For each qcqs scheme S and $j \in \mathbb{Z}$, the $\mathbb{A}^1$ -motivic cohomology of weight j is

$$\mathbb{Z}(j)^\mathbb{A}(S) := \mathrm{map}_{\mathrm{SH}(S)}(\mathbb{1}_S, s^j KGL_S),$$

and the $\mathbb{A}^1$ -motivic filtration on KH is

$$\mathrm{Fil}_\mathbb{A}^\star KH(S) := \mathrm{map}_{\mathrm{SH}(S)}(\mathbb{1}_S, \mathrm{Fil}_{\mathrm{slice}}^\star KGL_S).$$

These assemble into presheaves of (respectively, filtered) spectra on $\mathrm{Sch}^{\mathrm{qcqs}}$.

Remark 3.2.43 (Convention vs. [BEM25]). Our $\mathbb{Z}(j)^\mathbb{A}$ differs from [BEM25, Definition 4.15] by a $[2j]$ shift: their definition is $\mathrm{map}_{\mathrm{SH}(S)}(\mathbb{1}_S, s^j \mathrm{KGL}_S)[-2j]$, whereas ours is the mapping spectrum without desuspension. This is the same convention difference as in our $\mathbb{T}_S$ -graded $E(j) = \omega^\infty(E \otimes \mathbb{T}_S^{\otimes j})$ (Warning 3.1.5). In particular, BEM’s motivic cohomology groups translate to ours via $H^p(X, \mathbb{Z}(j))_{\mathrm{BEM}} = \pi_{-p} \mathbb{Z}(j)_{\mathrm{BEM}}^\mathbb{A}(X) = \pi_{2j-p} \mathbb{Z}(j)^\mathbb{A}(X)$. The $\mathbb{A}^1$ -motivic filtration $\mathrm{Fil}_\mathbb{A}^\star \mathrm{KH}$, by contrast, has the same definition in both conventions.

Remark 3.2.44 (Levels of the motivic Eilenberg–MacLane spectrum). Write $\mathrm{HZ}_S := s^0 \mathrm{KGL}_S \in \mathrm{CAlg}(\mathrm{SH}(S))$. The Bott identification $s^j \mathrm{KGL}_S \simeq \mathbb{T}_S^{\otimes j} \otimes s^0 \mathrm{KGL}_S$ established in the slice filtration for KGL above presents $\mathbb{Z}(j)^\mathbb{A}$ as the j -th level of HZ_S in the $\mathbb{T}$ -convention of Section 3.1.2:

$$\mathbb{Z}(j)^\mathbb{A}(S) \simeq \mathrm{map}_{\mathrm{SH}(S)}(\mathbb{1}_S, \mathrm{HZ}_S \otimes \mathbb{T}_S^{\otimes j}) = \mathrm{HZ}_S(j)(S).$$

Fixing S and letting $Y \in \mathrm{Sm}_S$ vary, the restriction of $\mathbb{Z}(j)^\mathbb{A}$ to Sm_S is the $\mathbb{A}^1$ -invariant Nisnevich sheaf of spectra $\omega^\infty s^j \mathrm{KGL}_S$; in particular, using $\mathrm{KGL}_Y \simeq f_Y^* \mathrm{KGL}_S$ (Proposition 3.1.12) together with the fact that the slice filtration commutes with smooth pullback (Remark 3.2.10),

$$\mathbb{Z}(j)^\mathbb{A}(Y) \simeq \mathrm{map}_{\mathrm{SH}(S)}(M_S(Y), s^j \mathrm{KGL}_S), \quad \mathrm{Fil}_\mathbb{A}^j \mathrm{KH}(Y) \simeq \mathrm{map}_{\mathrm{SH}(S)}(M_S(Y), \mathrm{Fil}_{\mathrm{slice}}^j \mathrm{KGL}_S)$$

for every $Y \in \mathrm{Sm}_S$.

Remark 3.2.45 (Multiplicativity and exhaustiveness). Since $\mathrm{Fil}_{\mathrm{slice}}^\star$ is lax symmetric monoidal (Construction 3.2.8) and $\mathrm{map}_{\mathrm{SH}(S)}(\mathbb{1}_S, -)$ is lax symmetric monoidal, $\mathrm{Fil}_\mathbb{A}^\star \mathrm{KH}$ is an $\mathbb{E}_\infty$ -algebra in filtered presheaves of spectra, and $\mathbb{Z}(\star)^\mathbb{A}$ inherits the structure of a graded $\mathbb{E}_\infty$ -algebra. Exhaustiveness of the slice filtration implies $\mathrm{colim}_{j \rightarrow -\infty} \mathrm{Fil}_\mathbb{A}^j \mathrm{KH} \simeq \mathrm{KH}$.

For applications over non-regular schemes and for cdh descent, we also need a variant of $\mathbb{Z}(\star)^\mathbb{A}$ built by pulling back the slice filtration from the absolute base. The need for such a variant comes from the fact that the slice filtration does not commute with arbitrary pullbacks (Remark 3.2.10); the KGL-spectra themselves, however, are stable under arbitrary pullback (Proposition 3.1.12).

Definition 3.2.46 ($\mathbb{A}^1$ -cdh-motivic cohomology). For any qcqs scheme S with structure map $f_S: S \rightarrow \mathrm{Spec}(\mathbb{Z})$ and $j \in \mathbb{Z}$, set

$$\mathbb{Z}(j)^{\mathbb{A}, \mathrm{cdh}}(S) := \mathrm{map}_{\mathrm{SH}(S)}(\mathbb{1}_S, f_S^* s^j \mathrm{KGL}_\mathbb{Z}),$$

and

$$\mathrm{Fil}_{\mathbb{A}, \mathrm{cdh}}^\star \mathrm{KH}(S) := \mathrm{map}_{\mathrm{SH}(S)}(\mathbb{1}_S, f_S^* \mathrm{Fil}_{\mathrm{slice}}^\star \mathrm{KGL}_\mathbb{Z}).$$

These are presheaves of (respectively, filtered) spectra on $\mathrm{Sch}^{\mathrm{qcqs}}$.

Remark 3.2.47 (Comparison map and cdh descent). By Proposition 3.1.12, $f_S^* \mathrm{KGL}_\mathbb{Z} \simeq \mathrm{KGL}_S$, so the base change transformation of Remark 3.2.10 yields a canonical multiplicative map

$$f_S^* \mathrm{Fil}_{\mathrm{slice}}^\star \mathrm{KGL}_\mathbb{Z} \longrightarrow \mathrm{Fil}_{\mathrm{slice}}^\star (f_S^* \mathrm{KGL}_\mathbb{Z}) \simeq \mathrm{Fil}_{\mathrm{slice}}^\star \mathrm{KGL}_S,$$

which is not an equivalence in general. Applying $\mathrm{map}_{\mathrm{SH}(S)}(\mathbb{1}_S, -)$ and passing to graded pieces, it induces natural comparison maps

$$\mathrm{Fil}_{\mathbb{A}, \mathrm{cdh}}^\star \mathrm{KH} \longrightarrow \mathrm{Fil}_\mathbb{A}^\star \mathrm{KH}, \quad \mathbb{Z}(\star)^{\mathbb{A}, \mathrm{cdh}} \longrightarrow \mathbb{Z}(\star)^\mathbb{A}.$$

Since SH is a cdh sheaf on $\mathrm{Sch}^{\mathrm{qcqs}}$ (Corollary 2.3.12) and the targets $s^j \mathrm{KGL}_\mathbb{Z}$ and $\mathrm{Fil}_{\mathrm{slice}}^\star \mathrm{KGL}_\mathbb{Z}$ are fixed objects of $\mathrm{SH}(\mathrm{Spec}(\mathbb{Z}))$, the presheaves $\mathbb{Z}(j)^{\mathbb{A}, \mathrm{cdh}}$ and $\mathrm{Fil}_{\mathbb{A}, \mathrm{cdh}}^\star \mathrm{KH}$ on $\mathrm{Sch}^{\mathrm{qcqs}}$ are cdh sheaves, justifying the “cdh” in the notation.

The following theorem summarizes the basic structural properties of both theories; we defer its proof to [BEM25]. The refinements concerning $\mathbb{Z}(0)^\mathbb{A}$ and $\mathbb{Z}(1)^\mathbb{A}$ — namely the $\mathbb{Z}$ -linear structure and the motivic first Chern class of [BEM25, Theorem 4.39(3)&(4)] — are taken up in Section 3.2.5.

Theorem 3.2.48 ([BEM25, Theorem 4.39]). *Let S be a qcqs scheme. Then:*

- (i) $\mathrm{Fil}_{\mathbb{A}, \mathrm{cdh}}^\star \mathrm{KH}(S)$ and $\mathrm{Fil}_{\mathbb{A}}^\star \mathrm{KH}(S)$ are functorial, multiplicative, exhaustive $\mathbb{Z}$ -indexed filtrations on $\mathrm{KH}(S)$, with graded pieces naturally and multiplicatively given³ by

$$\mathrm{gr}_{\mathbb{A}, \mathrm{cdh}}^j \mathrm{KH}(S) \simeq \mathbb{Z}(j)^{\mathbb{A}, \mathrm{cdh}}(S), \quad \mathrm{gr}_{\mathbb{A}}^j \mathrm{KH}(S) \simeq \mathbb{Z}(j)^\mathbb{A}(S) \quad \text{for } j \in \mathbb{Z}.$$

The filtrations are related by a natural multiplicative map $\mathrm{Fil}_{\mathbb{A}, \mathrm{cdh}}^\star \mathrm{KH}(S) \rightarrow \mathrm{Fil}_{\mathbb{A}}^\star \mathrm{KH}(S)$, which on graded pieces recovers the comparison map of Remark 3.2.47.

- (ii) For each $j \in \mathbb{Z}$, the presheaves $\mathbb{Z}(j)^{\mathbb{A}, \mathrm{cdh}}$ and $\mathrm{Fil}_{\mathbb{A}, \mathrm{cdh}}^\star \mathrm{KH}$ are finitary, $\mathbb{A}^1$ -invariant cdh sheaves on $\mathrm{Sch}^{\mathrm{qcqs}}$; meanwhile $\mathbb{Z}(j)^\mathbb{A}$ and $\mathrm{Fil}_{\mathbb{A}}^\star \mathrm{KH}$ are $\mathbb{A}^1$ -invariant Nisnevich sheaves which commute with filtered colimits along smooth affine transition maps.

Corollary 3.2.49 (Projective bundle formula). *For each $j \in \mathbb{Z}$ and $d \geq 0$, the map*

$$\sum_{i=0}^d c_1(\mathcal{O}(1))^i \cdot \pi^*: \bigoplus_{i=0}^d \mathbb{Z}(j-i)^{\mathbb{A}, \mathrm{cdh}}(S) \xrightarrow{\sim} \mathbb{Z}(j)^{\mathbb{A}, \mathrm{cdh}}(\mathbb{P}_S^d)$$

is an equivalence, and the analogous statement holds for $\mathbb{Z}(\star)^\mathbb{A}$.

Proof sketch. By Example 3.2.31, KGL_S carries a canonical orientation, and $f_S^* \mathrm{KGL}_{\mathbb{Z}} \simeq \mathrm{KGL}_S$ (Proposition 3.1.12). The slice filtration $\mathrm{Fil}_{\mathrm{slice}}^\star$ is lax symmetric monoidal (Construction 3.2.8), so the 0-th slice $\mathrm{HZ}_S := s^0 \mathrm{KGL}_S$ inherits an orientation from KGL_S , as does $f_S^* s^0 \mathrm{KGL}_{\mathbb{Z}}$. Applying Theorem 3.2.29 to HZ_S (resp. $f_S^* s^0 \mathrm{KGL}_{\mathbb{Z}}$) and taking the j -th Tate level via Remark 3.2.44 yields the claim. $\square$

Lisse and cdh motivic cohomology

Following [BEM25, §7], we record a second construction of motivic cohomology on qcqs schemes, obtained by left Kan extension from smooth schemes over a base and cdh sheafification. This variant will be the target of the main comparison theorem stated below.

Definition 3.2.50 (Lisse motivic cohomology [BEM25, Definition 7.1]). Let B be either a field or a mixed-characteristic Dedekind domain, and $j \in \mathbb{Z}$. The **weight- j lisse motivic cohomology** of qcqs B -schemes is the left Kan extension to $\mathrm{Sch}_B^{\mathrm{qcqs}}$ of the restriction of $\mathbb{A}^1$ -motivic cohomology to smooth B -schemes:

$$\mathbb{Z}(j)_B^{\mathrm{lse}} := L_B^{\mathrm{sm}}(\mathbb{Z}(j)^\mathbb{A}|_{\mathrm{Sm}_B}): (\mathrm{Sch}_B^{\mathrm{qcqs}})^{\mathrm{op}} \longrightarrow \mathrm{D}(\mathbb{Z}).$$

When $B = \mathrm{Spec}(\mathbb{Z})$ we drop the subscript and write $\mathbb{Z}(j)^{\mathrm{lse}}$.

Definition 3.2.51 (cdh-motivic cohomology [BEM25, Definition 7.8]). For $j \in \mathbb{Z}$, the **weight- j cdh-motivic cohomology** is the cdh sheafification of $\mathbb{Z}(j)^{\mathrm{lse}}$:

$$\mathbb{Z}(j)^{\mathrm{cdh}} := L_{\mathrm{cdh}} \mathbb{Z}(j)^{\mathrm{lse}}: (\mathrm{Sch}^{\mathrm{qcqs}})^{\mathrm{op}} \longrightarrow \mathrm{D}(\mathbb{Z}).$$

³This statement appears in [BEM25, Theorem 4.39(1)] as $\mathrm{gr}_{\mathbb{A}}^j \mathrm{KH}(S) \simeq \mathbb{Z}(j)^\mathbb{A}(S)[2j]$ (and likewise for the cdh-variant); the $[2j]$ shift is absent here because our $\mathbb{Z}(j)^\mathbb{A}$ already incorporates it (Remark 3.2.43).

By [BEM25, Remark 4.20], $\mathbb{Z}(j)^{\mathbb{A}} = 0$ for $j < 0$; consequently $\mathbb{Z}(j)^{\text{lse}} = 0$ and $\mathbb{Z}(j)^{\text{cdh}} = 0$ in negative weights.

Remark 3.2.52 (Comparison map). By the universal property of $\mathbb{Z}(\star)^{\text{cdh}}$ as the cdh sheafification of a left Kan extension, and since $\mathbb{Z}(\star)^{\mathbb{A}, \text{cdh}}$ is a cdh sheaf (Theorem 3.2.48(ii)), to produce a canonical multiplicative map of graded $\mathbb{E}_{\infty}$ -algebras

$$\mathbb{Z}(\star)^{\text{cdh}} \longrightarrow \mathbb{Z}(\star)^{\mathbb{A}, \text{cdh}}$$

it suffices to produce one on $\text{Sm}_{\mathbb{Z}}$. On smooth $\mathbb{Z}$ -schemes the slice-base-change comparison map $\mathbb{Z}(\star)^{\mathbb{A}, \text{cdh}} \rightarrow \mathbb{Z}(\star)^{\mathbb{A}}$ of Remark 3.2.47 is an equivalence (cf. Remark 3.2.10 for smooth f), and composing with the identity of $\mathbb{Z}(\star)^{\mathbb{A}}|_{\text{Sm}_{\mathbb{Z}}}$ and taking the inverse yields the desired map.

The central comparison theorem of [BEM25] identifies, after $\mathbb{A}^1$ -localization, the cdh-motivic cohomology of Definition 3.2.51 with the $\mathbb{A}^1$ -cdh-motivic cohomology of Definition 3.2.46.

Theorem 3.2.53 ([BEM25, Theorem 9.1]). *The canonical comparison map*

$$L_{\mathbb{A}^1} \mathbb{Z}(\star)^{\text{cdh}} \xrightarrow{\sim} \mathbb{Z}(\star)^{\mathbb{A}, \text{cdh}}$$

is an equivalence of $\mathbb{E}_{\infty}$ -algebras of graded cdh sheaves of complexes on qcqs schemes.

3.2.7. Extension to qcqs schemes

The same left-Kan-extend-then-cdh-sheafify procedure that organized the lisse and cdh cohomology theories of Section 3.2.6 — culminating in the comparison Theorem 3.2.53 — applies, mutatis mutandis, to the construction of $c_1^{\mathbb{A}}$ in Section 3.2.5, extending it from regular noetherian bases to all qcqs schemes. This establishes parts (3) and (4) of [BEM25, Theorem 4.39] deferred from Theorem 3.2.48. The two key inputs are the rigidity / LKE framework of [BEM25, §2.3] and the Kerz–Strunk–Tamme identification $L_{\text{cdh}} k \simeq \text{KH}$ [KST18, Theorem 6.3], applied piecewise to k , $k^{rk=0}$, $\Gamma_{\text{Nis}}(-, \mathbb{Z})$, and $\Gamma_{\text{Nis}}(-, \mathbb{G}_m)$.

The mechanism is encapsulated by the following commutative square of LKE/restriction operations, which we cite from [BEM25] without proof.

Proposition 3.2.54 ([BEM25, Proposition 2.23]). *Let $\mathcal{C}$ be a presentable category. The following diagram of $\mathcal{C}$ -valued functor categories commutes up to canonical equivalence:*

$$\begin{array}{ccccccc} \text{Fun}(\text{Sch}_{\mathbb{Z}}^{\text{qcqs}, \text{op}}, \mathcal{C}) & \longrightarrow & \text{Fun}(\text{CAlg}_{\mathbb{Z}}, \mathcal{C}) & \xrightarrow{=} & \text{Fun}(\text{CAlg}_{\mathbb{Z}}, \mathcal{C}) & \longrightarrow & \text{Fun}(\text{CAlg}_{\mathbb{Z}}^{? \text{loc}}, \mathcal{C}) \\ \uparrow L_{\mathbb{Z}}^{\text{sm}} & & \uparrow & & \uparrow & & \uparrow \\ \text{Fun}(\text{Sm}_{\mathbb{Z}}^{\text{op}}, \mathcal{C}) & \longrightarrow & \text{Fun}(\text{SmAlg}_{\mathbb{Z}}, \mathcal{C}) & \xrightarrow{\sim} & \text{Fun}^{\text{fin}}(\text{IndSmAlg}_{\mathbb{Z}}, \mathcal{C}) & \rightarrow & \text{Fun}(\text{EssSmAlg}_{\mathbb{Z}}^{? \text{loc}}, \mathcal{C}) \end{array}$$

where $? \text{loc} \in \{\text{loc}, \text{hloc}, \text{shloc}\}$ ranges over local, henselian local, or strict henselian local algebras, the vertical arrows are left Kan extensions along the fully faithful inclusions, the horizontal arrow labelled $\sim$ is again a left Kan extension, and the remaining horizontal arrows are restrictions.

The left Kan extension functor is denoted $L_{\mathbb{Z}}^{\text{sm}}$ as in Definition 3.2.50. The point of Proposition 3.2.54 for our purposes is the following: if a presheaf F on $\text{Sch}_{\mathbb{Z}}^{\text{qcqs}}$ is left Kan extended from $\text{Sm}_{\mathbb{Z}}$, then F on all of $\text{Sch}_{\mathbb{Z}}^{\text{qcqs}}$ is determined by its restriction to local $\mathbb{Z}$ -algebras (and in fact already by the restriction to essentially smooth local $\mathbb{Z}$ -algebras). This is precisely the

situation for k , $k^{\mathrm{rk}=0}$, $\Gamma_{\mathrm{Nis}}(-, \mathbb{Z})$, and $\Gamma_{\mathrm{Nis}}(-, \mathbb{G}_m)$, which lets us extend the regular-noetherian constructions of Construction 3.2.38 and Corollary 3.2.40 to all qcqs schemes by cdh sheafification.

Theorem 3.2.55 ([BEM25, Theorem 4.39(3)(4)]). *For any qcqs scheme S :*

(i) (Weight zero and $\mathbb{Z}$ -linear structure.) *There is a natural map of presheaves of $\mathbb{E}_\infty$ -algebras*

$$\Gamma_{\mathrm{cdh}}(-, \mathbb{Z}) \longrightarrow \mathbb{Z}(0)^{\mathbb{A}, \mathrm{cdh}}$$

on qcqs schemes, fitting into a commutative triangle

$$\begin{array}{ccc} \Gamma_{\mathrm{cdh}}(-, \mathbb{Z}) & \longrightarrow & \mathbb{Z}(0)^{\mathbb{A}, \mathrm{cdh}} \\ & \nwarrow \mathrm{rk} & \uparrow \text{edge} \\ & & \mathrm{KH} \end{array}$$

of presheaves of spectra.

(ii) (Weight one and first Chern class.) *There is a natural map*

$$c_1^{\mathbb{A}, \mathrm{cdh}}: \Gamma_{\mathrm{cdh}}(-, \mathbb{G}_m)[1] \longrightarrow \mathbb{Z}(1)^{\mathbb{A}, \mathrm{cdh}}$$

of presheaves of spectra on qcqs schemes, fitting into a commutative diagram

$$\begin{array}{ccc} \mathrm{KH}^{\mathrm{rk}=0} & \longrightarrow & \mathrm{Fil}_{\mathbb{A}, \mathrm{cdh}}^1 \mathrm{KH} \\ \det \downarrow & & \downarrow \text{edge} \\ \Gamma_{\mathrm{cdh}}(-, \mathbb{G}_m)[1] & \xrightarrow{c_1^{\mathbb{A}, \mathrm{cdh}}} & \mathbb{Z}(1)^{\mathbb{A}, \mathrm{cdh}} \end{array}$$

where $\mathrm{KH}^{\mathrm{rk}=0} := \mathrm{fib}(\mathrm{rk}: \mathrm{KH} \rightarrow \Gamma_{\mathrm{cdh}}(-, \mathbb{Z}))$.

Remark 3.2.56 (Convention vs. [BEM25]). Following the T-spectra convention of Remark 3.2.43, $\mathbb{Z}(1)^{\mathbb{A}, \mathrm{cdh}}$ already absorbs the $[2]$ shift carried explicitly by [BEM25, Theorem 4.39(4)]: their statement reads $c_1^{\mathbb{A}, \mathrm{cdh}}: \Gamma_{\mathrm{cdh}}(-, \mathbb{G}_m)[-1] \rightarrow \mathbb{Z}(1)^{\mathbb{A}, \mathrm{cdh}}$ with right-hand edge map landing in $\mathbb{Z}(1)^{\mathbb{A}, \mathrm{cdh}}[2]$. The two statements agree after a global $[2]$ -shift; we display the unshifted version to match our $\mathbb{Z}(1)^{\mathbb{A}, \mathrm{cdh}}$.

Proof sketch. Step 1: each piece is LKE from $\mathrm{Sm}_{\mathbb{Z}}$. The three functors $\mathrm{CAlg}_{\mathbb{Z}}^{\mathrm{loc}} \rightarrow \mathrm{D}(\mathbb{Z})$, sending $A \mapsto \mathbb{Z}$, $A \mapsto A^\times$, and $A \mapsto k(A)$, are each LKE from $\mathrm{EssSmAlg}_{\mathbb{Z}}^{\mathrm{loc}}$: the constant $\mathbb{Z}$ trivially; $A^\times$ via corepresentability by the smooth affine $\mathbb{Z}$ -scheme $\mathbb{G}_{m, \mathbb{Z}}$ (cofinality argument); and k by Bhatt–Lurie, which we established as Proposition 3.1.13 in Section 3.1.6 (cf. [Elm+20, Example A.0.6]). By Proposition 3.2.54, the corresponding presheaves $\Gamma_{\mathrm{Nis}}(-, \mathbb{Z})$, $\Gamma_{\mathrm{Nis}}(-, \mathbb{G}_m)$, and k on $\mathrm{Sch}_{\mathbb{Z}}^{\mathrm{qcqs}}$ are LKE from $\mathrm{Sm}_{\mathbb{Z}}$, as is the fiber $k^{\mathrm{rk}=0} = \mathrm{fib}(k \rightarrow \Gamma_{\mathrm{Nis}}(-, \mathbb{Z}))$: pointwise left Kan extension into the stable target is exact, hence preserves fiber sequences. No filteredness of the comma categories is required.

Step 2: extend the natural transformations. On the smooth $\mathbb{Z}$ -base $S = \mathrm{Spec}(\mathbb{Z})$ (regular noetherian), Construction 3.2.38 and Corollary 3.2.40 supply on $\mathrm{Sm}_{\mathbb{Z}}$:

$$\mathrm{rk}: k \rightarrow \Gamma_{\mathrm{Nis}}(-, \mathbb{Z}), \quad \det: k^{\mathrm{rk}=0} \rightarrow \Gamma_{\mathrm{Nis}}(-, \mathbb{G}_m)[1],$$

the maps $\Gamma_{\text{Nis}}(-, \mathbb{Z}) \rightarrow \mathbb{Z}(0)^{\mathbb{A}}$ and $c_1^{\mathbb{A}}: \Gamma_{\text{Nis}}(-, \mathbb{G}_m)[1] \rightarrow \mathbb{Z}(1)^{\mathbb{A}}$, and the compatibility diagram of Corollary 3.2.40. The universal property of $L_{\mathbb{Z}}^{\text{sm}}$ states

$$\text{Map}(L_{\mathbb{Z}}^{\text{sm}} F, G) \simeq \text{Map}(F|_{\text{Sm}_{\mathbb{Z}}}, G|_{\text{Sm}_{\mathbb{Z}}}).$$

Since every source presheaf is LKE from $\text{Sm}_{\mathbb{Z}}$ by Step 1, every map and commutative diagram extends uniquely to $\text{Sch}_{\mathbb{Z}}^{\text{qcqs}}$.

Step 3: cdh sheafify. Apply L_{cdh} . By Kerz–Strunk–Tamme [KST18, Theorem 6.3] (cf. [BEM25, Theorem 7.12(1)]) we have $L_{\text{cdh}} k \simeq \text{KH}$; by definition $L_{\text{cdh}} \Gamma_{\text{Nis}}(-, F) = \Gamma_{\text{cdh}}(-, F)$ for $F = \mathbb{Z}$ or $\mathbb{G}_m$. The targets $\mathbb{Z}(j)^{\mathbb{A}, \text{cdh}}$ are cdh sheaves by Theorem 3.2.48(ii), and their restriction to $\text{Sm}_{\mathbb{Z}}$ recovers $\mathbb{Z}(j)^{\mathbb{A}}|_{\text{Sm}_{\mathbb{Z}}}$ by smooth base change of the slice filtration (Remark 3.2.10; cf. Remark 3.2.52). Cdh sheafification of the extended maps therefore produces the diagrams (1) and (2). $\square$

Remark 3.2.57 (Rational comparison). [BEM25, Corollary 4.60(1)] records that the maps of Theorem 3.2.55 are rational equivalences — a consequence of the fact that $\mathbb{Q}(j)^{\mathbb{A}, \text{cdh}}$ is a direct summand of $\text{KH}_{\mathbb{Q}}$ [BEM25, Theorem 4.47(2)] and that $\text{KH}_{\mathbb{Q}}$ is itself the cdh sheafification of an LKE from $\text{Sm}_{\mathbb{Z}}$ [BEM25, proof of Theorem 7.12(1)], so that one may LKE- and-cdh-sheafify the rational equivalences over regular noetherian bases ([BEM25, Proposition 4.56]). The promotion from rational to integral equivalences is deeper and uses inputs from [BEM25, §6] (notably [BEM25, Corollary 6.26], asserting that $c_1^{\mathbb{A}}$ is an equivalence for smooth schemes over a field or mixed-characteristic Dedekind domain) together with the comparison $\mathbb{Z}(j)^{\text{cdh}} \simeq \mathbb{Z}(j)^{\mathbb{A}, \text{cdh}}$ for $j \leq 1$ implicit in [BEM25, Theorem 7.12(5)(6)]. We import the resulting integral identifications directly in Section 3.2.8.

3.2.8. The second filtration quotient

Combining the $c_1^{\mathbb{A}, \text{cdh}}$ of Section 3.2.7 with the integral low-weight identifications

$$\mathbb{Z}(0)^{\mathbb{A}, \text{cdh}} \simeq \Gamma_{\text{cdh}}(-, \mathbb{Z}), \quad \mathbb{Z}(1)^{\mathbb{A}, \text{cdh}} \simeq \Gamma_{\text{cdh}}(-, \mathbb{G}_m)[1]$$

of [BEM25, Corollary 9.12(1)] (translated to the T-spectra convention of Remark 3.2.43) describes the first two filtration quotients of $\text{Fil}_{\mathbb{A}, \text{cdh}}^{\star} \text{KH}$ in terms of the *derived Picard anima* $\mathcal{P}ic^{\dagger}(S) := \mathcal{P}ic(\text{D}(S))$: the Knudsen–Mumford determinant lifts the rank map $\text{rk}: \text{KH} \rightarrow \Gamma_{\text{cdh}}(-, \mathbb{Z})$ to $\det: \text{KH} \rightarrow L_{\text{cdh}} \mathcal{P}ic^{\dagger}$, and the fibers of rk and $\det$ recover Fil^1 and Fil^2 respectively (after cdh sheafification).

Definition 3.2.58 (Derived Picard anima). For a qcqs scheme S , the **derived Picard anima** is

$$\mathcal{P}ic^{\dagger}(S) := \mathcal{P}ic(\text{D}(S)),$$

the Picard anima of the symmetric monoidal derived category $\text{D}(S)$ of quasi-coherent complexes on S , i.e., the anima of $\otimes$ -invertible objects. Zariski-locally on S , every $\otimes$ -invertible object of $\text{D}(S)$ has the form $\mathcal{L}[n]$ for a line bundle $\mathcal{L}$ and an integer n ; the cohomological grading n is locally constant on S , yielding a **rank map**

$$\text{rk}: \mathcal{P}ic^{\dagger} \longrightarrow \Gamma(-, \mathbb{Z})$$

of presheaves of spectra on Sch^{qcqs} whose fiber over 0 is the ordinary Picard spectrum $\mathcal{P}ic$:

$$\mathcal{P}ic \longrightarrow \mathcal{P}ic^{\dagger} \xrightarrow{\text{rk}} \Gamma(-, \mathbb{Z}).$$

This fiber sequence exhibits $\mathcal{P}ic^\dagger$ as a $\mathbb{Z}$ -graded extension of $\mathcal{P}ic$ in presheaves of spectra; the rank map records the integer shift on $\otimes$ -invertibles, while $\mathcal{P}ic$ records the underlying line bundle.

Construction 3.2.59 (Knudsen–Mumford determinant). The **Knudsen–Mumford determinant** sends a perfect complex to its top exterior power on connective resolutions, assembling into a map

$$\det: K \longrightarrow \mathcal{P}ic^\dagger$$

of presheaves of spectra on Sch^{qcqs} (equivalently, of grouplike $\mathbb{E}_\infty$ -anima). It is additive in the sense that direct sum of perfect complexes is carried to tensor product of graded lines. The rank map $\text{rk}: K \rightarrow \Gamma(-, \mathbb{Z})$ factors as

$$K \xrightarrow{\det} \mathcal{P}ic^\dagger \xrightarrow{\text{rk}} \Gamma(-, \mathbb{Z}),$$

and the restriction $\det|_{K^{\text{rk}=0}}: K^{\text{rk}=0} \rightarrow \mathcal{P}ic$ recovers the classical determinant on rank-0 K-theory used in Construction 3.2.38.

On smooth schemes over a regular noetherian base, where $K \simeq KH$, the first two quotients may already be written before cdh sheafification as

$$K/\text{Fil}_{\mathbb{A}}^1 K \simeq \Gamma(-, \mathbb{Z}) \quad \text{and} \quad K/\text{Fil}_{\mathbb{A}}^2 K \simeq \mathcal{P}ic^\dagger;$$

the corollary below records the cdh-sheafified version, which is the form actually needed for qcqs schemes.

Corollary 3.2.60. *For every qcqs scheme S , there are natural equivalences of presheaves of spectra on Sch^{qcqs} :*

$$\text{Fil}_{\mathbb{A}, \text{cdh}}^1 KH \simeq L_{\text{cdh}} \text{fib}(\text{rk}: K \rightarrow \mathbb{Z}[0]), \quad \text{Fil}_{\mathbb{A}, \text{cdh}}^2 KH \simeq L_{\text{cdh}} \text{fib}(\det: K \rightarrow \mathcal{P}ic^\dagger).$$

Equivalently,

$$KH/\text{Fil}_{\mathbb{A}, \text{cdh}}^1 KH \simeq \Gamma_{\text{cdh}}(-, \mathbb{Z}), \quad KH/\text{Fil}_{\mathbb{A}, \text{cdh}}^2 KH \simeq L_{\text{cdh}} \mathcal{P}ic^\dagger.$$

Proof sketch. Write $\text{Fil}^j := \text{Fil}_{\mathbb{A}, \text{cdh}}^j KH$. By Theorem 3.2.48(ii) (in the T-spectra convention with no $[2j]$ shift),

$$\text{Fil}^0/\text{Fil}^1 \simeq \mathbb{Z}(0)^{\mathbb{A}, \text{cdh}}, \quad \text{Fil}^1/\text{Fil}^2 \simeq \mathbb{Z}(1)^{\mathbb{A}, \text{cdh}}.$$

The integral identifications cited above ([BEM25, Corollary 9.12(1)], translated convention) rewrite these as $\Gamma_{\text{cdh}}(-, \mathbb{Z})$ and $\Gamma_{\text{cdh}}(-, \mathbb{G}_m)[1] \simeq L_{\text{cdh}} \mathcal{P}ic$ respectively.

The Fil^1 formula. The first identification gives $\text{Fil}^1 \simeq \text{fib}(KH \rightarrow \Gamma_{\text{cdh}}(-, \mathbb{Z}))$. Left exactness of L_{cdh} , together with $L_{\text{cdh}} K \simeq KH$ [KST18, Theorem 6.3] and $L_{\text{cdh}} \mathbb{Z}[0] \simeq \Gamma_{\text{cdh}}(-, \mathbb{Z})$, then yields

$$L_{\text{cdh}} \text{fib}(\text{rk}: K \rightarrow \mathbb{Z}[0]) \simeq \text{fib}(KH \rightarrow \Gamma_{\text{cdh}}(-, \mathbb{Z})) \simeq \text{Fil}^1.$$

The Fil^2 formula. By Construction 3.2.59, the rank map factors as $KH \xrightarrow{\det} L_{\text{cdh}} \mathcal{P}ic^\dagger \xrightarrow{\text{rk}} \Gamma_{\text{cdh}}(-, \mathbb{Z})$ (apply L_{cdh} to Definition 3.2.58), and the restriction $\det|_{\text{Fil}^1}: \text{Fil}^1 \rightarrow L_{\text{cdh}} \mathcal{P}ic$ recovers (after L_{cdh}) the determinant on rank-0 K-theory used to build $c_1^{\mathbb{A}, \text{cdh}}$. Consider the

diagram of presheaves of spectra

$$\begin{array}{ccccc}
 \mathrm{Fil}^2 & \longrightarrow & \mathrm{Fil}^1 & \longrightarrow & \mathrm{Fil}^0 = \mathrm{KH} \\
 \downarrow & & \downarrow \scriptstyle \mathrm{det} & & \downarrow \scriptstyle \mathrm{det} \\
 0 & \longrightarrow & L_{\mathrm{cdh}} \mathcal{P}ic & \longrightarrow & L_{\mathrm{cdh}} \mathcal{P}ic^+ \\
 \downarrow & & \downarrow & & \downarrow \scriptstyle \mathrm{rk} \\
 0 & \longrightarrow & 0 & \longrightarrow & \Gamma_{\mathrm{cdh}}(-, \mathbb{Z})
 \end{array}$$

whose commutativity follows from $\mathrm{rk} = \mathrm{rk} \circ \mathrm{det}$ together with the factorization $\mathrm{det}|_{\mathrm{Fil}^1}$ landing in $L_{\mathrm{cdh}} \mathcal{P}ic$. We chase cartesianness in four steps:

- (i) The *bottom-right square* is cartesian by Definition 3.2.58 (applied after L_{cdh}).
- (ii) The *right rectangle* (columns 2–3, all rows) is cartesian by the Fil^1 formula above.
- (iii) Pasting (1) and (2), the *top-right square* is cartesian: $\mathrm{Fil}^1 \simeq \mathrm{KH} \times_{L_{\mathrm{cdh}} \mathcal{P}ic^+} L_{\mathrm{cdh}} \mathcal{P}ic$.
- (iv) By the commutative edge-map square of Theorem 3.2.55(ii) and the integral equivalence $c_1^{\mathbb{A}, \mathrm{cdh}}: L_{\mathrm{cdh}} \mathcal{P}ic \xrightarrow{\sim} \mathbb{Z}(1)^{\mathbb{A}, \mathrm{cdh}}$, the map $\mathrm{det}: \mathrm{Fil}^1 \rightarrow L_{\mathrm{cdh}} \mathcal{P}ic$ identifies with the quotient map $\mathrm{Fil}^1 \rightarrow \mathrm{Fil}^1/\mathrm{Fil}^2$. Hence the *top-left square* is cartesian. Pasting with (3) makes the entire top rectangle cartesian.

Cartesianness of the top rectangle gives $\mathrm{Fil}^2 \simeq \mathrm{fib}(\mathrm{det}: \mathrm{KH} \rightarrow L_{\mathrm{cdh}} \mathcal{P}ic^+) \simeq L_{\mathrm{cdh}} \mathrm{fib}(\mathrm{det}: \mathrm{K} \rightarrow \mathcal{P}ic^+)$ by left exactness of L_{cdh} , and equivalently $\mathrm{KH}/\mathrm{Fil}^2 \simeq L_{\mathrm{cdh}} \mathcal{P}ic^+$. $\square$

Remark 3.2.61. Presenting $\mathrm{Fil}_{\mathbb{A}, \mathrm{cdh}}^j \mathrm{KH}$ as $L_{\mathrm{cdh}} \mathrm{fib}(\mathrm{K} \rightarrow \cdots)$ rather than $\mathrm{fib}(\mathrm{KH}^{\mathrm{rk}=0} \rightarrow L_{\mathrm{cdh}} \mathcal{P}ic)$ has the advantage that the filtration *quotients* $\Gamma_{\mathrm{cdh}}(-, \mathbb{Z})$ and $L_{\mathrm{cdh}} \mathcal{P}ic^+$ become immediately visible. The two formulations agree because L_{cdh} is left exact and in particular preserves fibers.

Remark 3.2.62 (Comparison with the orientation-theoretic Chern class). The cdh -extended diagram of Theorem 3.2.55(ii) expresses $c_1^{\mathbb{A}, \mathrm{cdh}}$ as the edge map of the $\mathrm{Fil}^1/\mathrm{Fil}^2$ truncation; Corollary 3.2.60 packages the same data structurally, replacing the edge map by the fiber of the Knudsen–Mumford determinant. The two viewpoints are formally equivalent (and dual): the first emphasizes weight-by-weight Chern classes, the second emphasizes the global geometric content of the filtration. The interplay between $c_1^{\mathbb{A}, \mathrm{cdh}}$ and $\mathcal{P}ic^+$ is the motivic shadow of the K-theoretic decomposition $\mathrm{det}: \mathrm{KH} \rightarrow L_{\mathrm{cdh}} \mathcal{P}ic^+$, recovering the classical statement that the first Chern class detects the determinant of a perfect complex.

3.2.9. The slice conjecture and consequences

The comparison theorem Theorem 3.2.53, together with the vanishing of $\mathbb{Z}(\star)^{\mathrm{cdh}}$ in negative weights, has a striking consequence first conjectured by Voevodsky: the slice filtration on KGL is stable under arbitrary base change. The key intermediate input is a vanishing for the slice filtration on the 0-th slice $\mathrm{H}\mathbb{Z}_{\mathbb{Z}}$.

Corollary 3.2.63 ([BEM25, Corollary 9.2]). *For every qcqs scheme S with structure map $f_S: S \rightarrow \mathrm{Spec}(\mathbb{Z})$,*

$$\mathrm{Fil}_{\mathrm{slice}}^1(f_S^* s^0 \mathrm{KGL}_{\mathbb{Z}}) \simeq 0.$$

Proof. Bott periodicity $\mathbb{T}_{\mathbb{Z}} \otimes \mathrm{KGL}_{\mathbb{Z}} \simeq \mathrm{KGL}_{\mathbb{Z}}$ together with the slice covariance $s^j(\mathbb{T} \otimes -) \simeq \mathbb{T} \otimes s^{j-1}(-)$ established in Section 3.2.3 gives $s^0 \mathrm{KGL}_{\mathbb{Z}} \otimes \mathbb{T}_{\mathbb{Z}}^{-1} \simeq s^{-1} \mathrm{KGL}_{\mathbb{Z}}$. By definition of $\mathrm{Fil}_{\mathrm{slice}}^1 = \iota^1 r^1$ and the generators $\mathbb{T}_S \otimes M_S(Y)[i]$ of $\mathrm{SH}^{\mathrm{eff}}(S)(1)$, the vanishing $\mathrm{Fil}_{\mathrm{slice}}^1(f_S^* s^0 \mathrm{KGL}_{\mathbb{Z}}) \simeq 0$ amounts to

$$\mathrm{Map}_{\mathrm{SH}(S)}(\mathbb{T}_S \otimes M_S(Y)[i], f_S^* s^0 \mathrm{KGL}_{\mathbb{Z}}) \simeq 0 \quad \text{for all } Y \in \mathrm{Sm}_S, i \in \mathbb{Z}.$$

Using the adjunction $\mathbb{T}_S \otimes - \dashv (-) \otimes \mathbb{T}_S^{-1}$, base change of $\otimes$, and the smooth-pullback identification $\mathrm{Map}_{\mathrm{SH}(S)}(M_S(Y), -) \simeq \mathrm{Map}_{\mathrm{SH}(Y)}(\mathbb{1}_Y, f_Y^*(-))$, the displayed mapping spectrum rewrites as

$$\mathrm{Map}_{\mathrm{SH}(Y)}(\mathbb{1}_Y, f_Y^* s^{-1} \mathrm{KGL}_{\mathbb{Z}})[-i] = \mathbb{Z}(-1)^{\mathbb{A}, \mathrm{cdh}}(Y)[-i].$$

By Theorem 3.2.53, $\mathbb{Z}(-1)^{\mathbb{A}, \mathrm{cdh}} \simeq L_{\mathbb{A}^1} \mathbb{Z}(-1)^{\mathrm{cdh}}$, and the latter is zero since cdh -motivic cohomology vanishes in negative weights (see the discussion preceding Remark 3.2.52). The claim follows. $\square$

Theorem 3.2.64 (Slice conjecture, [BEM25, Theorem 9.3]; Voevodsky’s Conjecture 17 in [Voe02]).

For every morphism $f: S' \rightarrow S$ of qcqs schemes and every $j \in \mathbb{Z}$, the comparison map of Remark 3.2.10 is an equivalence:

$$f^* \mathrm{Fil}_{\mathrm{slice}}^j \mathrm{KGL}_S \xrightarrow{\sim} \mathrm{Fil}_{\mathrm{slice}}^j \mathrm{KGL}_{S'}, \quad f^* s^j \mathrm{KGL}_S \xrightarrow{\sim} s^j \mathrm{KGL}_{S'}.$$

Consequently, $\mathbb{Z}(j)^{\mathbb{A}, \mathrm{cdh}} \simeq \mathbb{Z}(j)^{\mathbb{A}}$ on $\mathrm{Sch}^{\mathrm{qcqs}}$ for every $j \in \mathbb{Z}$.

Proof. By Proposition 3.1.12 we may reduce to $S = \mathrm{Spec}(\mathbb{Z})$ and write $E = \mathrm{KGL}_{\mathbb{Z}}$, $f = f_{S'}$. The natural map $f^* \mathrm{Fil}_{\mathrm{slice}}^0 E \rightarrow \mathrm{Fil}_{\mathrm{slice}}^0 f^* E$ fits in a cofiber sequence whose cofiber is $\mathrm{Fil}_{\mathrm{slice}}^0(f^* \mathrm{Fil}_{\mathrm{slice}}^{<0} E)$, where $\mathrm{Fil}_{\mathrm{slice}}^{<0} E := \mathrm{cofib}(\mathrm{Fil}_{\mathrm{slice}}^0 E \rightarrow E)$. Since $\mathrm{Fil}_{\mathrm{slice}}^0 = \iota^0 r^0$ is exact (r^0 is a right adjoint, hence preserves fiber sequences, and ι^0 preserves them by closure of $\mathrm{SH}^{\mathrm{eff}}$ under extensions), induction along the cofiber sequences $\mathrm{Fil}_{\mathrm{slice}}^{j+1} E \rightarrow \mathrm{Fil}_{\mathrm{slice}}^j E \rightarrow s^j E$ for $j < 0$ reduces the vanishing to

$$\mathrm{Fil}_{\mathrm{slice}}^0(f^* s^j \mathrm{KGL}_{\mathbb{Z}}) \simeq 0 \quad \text{for every } j < 0.$$

By Bott periodicity and the formula $\mathrm{Fil}_{\mathrm{slice}}^0(\mathbb{T}_{S'}^{\otimes j} \otimes -) \simeq \mathbb{T}_{S'}^{\otimes j} \otimes \mathrm{Fil}_{\mathrm{slice}}^{-j}(-)$ from Section 3.2.3,

$$\mathrm{Fil}_{\mathrm{slice}}^0(f^* s^j \mathrm{KGL}_{\mathbb{Z}}) \simeq \mathbb{T}_{S'}^{\otimes j} \otimes \mathrm{Fil}_{\mathrm{slice}}^{-j}(f^* s^0 \mathrm{KGL}_{\mathbb{Z}}).$$

For $j < 0$ we have $-j \geq 1$. The inclusion $\mathrm{SH}^{\mathrm{eff}}(S')(-j) \subset \mathrm{SH}^{\mathrm{eff}}(S')(1)$ factors ι^{-j} through ι^1 , so by passage to right adjoints r^{-j} factors through r^1 . Hence $\mathrm{Fil}_{\mathrm{slice}}^1(f^* s^0 \mathrm{KGL}_{\mathbb{Z}}) \simeq 0$ from Corollary 3.2.63 forces $r^1(f^* s^0 \mathrm{KGL}_{\mathbb{Z}}) \simeq 0$, hence $\mathrm{Fil}_{\mathrm{slice}}^{-j}(f^* s^0 \mathrm{KGL}_{\mathbb{Z}}) \simeq 0$.

The same Bott-shift argument transports the equivalence at $j = 0$ to every $j \in \mathbb{Z}$, both for $\mathrm{Fil}_{\mathrm{slice}}^j$ and (passing to graded pieces) for s^j . Applying $\mathrm{map}_{\mathrm{SH}(-)}(\mathbb{1}, -)$ yields $\mathbb{Z}(j)^{\mathbb{A}, \mathrm{cdh}} \simeq \mathbb{Z}(j)^{\mathbb{A}}$ on $\mathrm{Sch}^{\mathrm{qcqs}}$. $\square$

Definition 3.2.65 (cdh-filtered K-theory). The **cdh-filtered K-theory** on qcqs $\mathbb{Z}$ -schemes is

$$\mathrm{Fil}_{\mathrm{cdh}}^{\star} \mathrm{KH} := L_{\mathrm{cdh}} L^{\mathrm{sm}}(\mathrm{Fil}_{\mathbb{A}}^{\star} \mathrm{KH}|_{\mathrm{Sm}_{\mathbb{Z}}}),$$

where L^{sm} denotes left Kan extension along $\mathrm{Sm}_{\mathbb{Z}}^{\mathrm{op}} \hookrightarrow \mathrm{Sch}^{\mathrm{qcqs}, \mathrm{op}}$.

Corollary 3.2.66. On qcqs schemes,

$$L_{\mathbb{A}^1} \mathrm{Fil}_{\mathrm{cdh}}^{\star} \mathrm{KH} \xrightarrow{\sim} \mathrm{Fil}_{\mathbb{A}, \mathrm{cdh}}^{\star} \mathrm{KH}.$$

Proof. Evaluate the comparison on a qcqs scheme X . Its stage 0 is an equivalence: both sides identify with $\mathrm{KH}(X)$, using $L_{\mathrm{cdh}}L^{\mathrm{sm}}k \simeq \mathrm{KH}$. Its map on every associated graded piece is an equivalence by Theorem 3.2.53. The cofiber sequences

$$F^{j+1} \longrightarrow F^j \longrightarrow \mathrm{gr}^j F$$

now give the result stage by stage: induction upward from $j = 0$ treats $j \geq 0$, and induction downward treats $j < 0$. Thus no completeness assertion for either unbounded filtration is needed. This is the filtered induction used in [BEM25, proof of Corollary 9.5]. $\square$

Corollary 3.2.67. *Let X be a qcqs scheme of valutive dimension $\leq d$. Then for every $i \in \mathbb{Z}$,*

$$\mathrm{Fil}_{\mathbb{A}}^i \mathrm{KH}(X) \in \mathrm{Sp}_{\geq i-d}.$$

Proof. By Corollary 3.2.66 and Theorem 3.2.64, it is enough to estimate

$$(L_{\mathbb{A}^1} \mathrm{Fil}_{\mathrm{cdh}}^i \mathrm{KH})(X) \simeq \left| [m] \mapsto \mathrm{Fil}_{\mathrm{cdh}}^i \mathrm{KH}(X \times \Delta_{\mathbb{Z}}^m) \right|.$$

The valutive dimension of $X \times \Delta_{\mathbb{Z}}^m$ is at most $d + m$. The cdh-filtration bound [BEM25, Theorem 7.12(3)] therefore makes its m -th simplicial term $(i - d - m)$ -connective. The skeletal filtration of geometric realization shows that a simplicial spectrum whose m -th term is $(r - m)$ -connective has r -connective realization: at the m -th skeletal step the new summand is a retract of the m -th term shifted by $[m]$. Taking $r = i - d$ gives the claimed bound. This is the argument of [BEM25, proof of Corollary 9.5]; in particular, it does not infer connectivity of derived global sections directly from stalkwise connectivity. $\square$

Definition 3.2.68 (Derived category of motives). The **derived category of motives** over a qcqs scheme X is

$$\mathrm{DM}(X) := \mathrm{Mod}_{\mathrm{HZ}_X}(\mathrm{SH}(X)) = \mathrm{Mod}_{s^0 \mathrm{KGL}_X}(\mathrm{SH}(X)),$$

where $\mathrm{HZ}_X = s^0 \mathrm{KGL}_X$ is the motivic Eilenberg–MacLane spectrum of Remark 3.2.44.

Proposition 3.2.69 ([BEM25, Corollary 9.10]). *The assignment $X \mapsto \mathrm{DM}(X)$ upgrades to a six-functor formalism on qcqs schemes, compatible with the six-functor formalism on $\mathrm{SH}(-)$ of Theorem 2.3.10.*

Proof sketch. The traditional reference, via $\mathrm{HZ}^{\mathbb{A}}$ being an absolute motivic spectrum, is [CD19, Proposition 11.4.7]; we record here the modern argument.

By Theorem 2.3.10 the motivic stable homotopy category SH is itself a Pr^L -valued six-functor formalism on qcqs schemes, with !-able morphisms those locally of finite presentation. Base-changing along the symmetric monoidal functor

$$- \otimes \mathrm{HZ}_{\mathrm{Spec}(\mathbb{Z})} : \mathrm{SH}(\mathbb{Z}) \longrightarrow \mathrm{DM}(\mathbb{Z})$$

defines a candidate six-functor formalism $X \mapsto \mathrm{SH}(X) \otimes_{\mathrm{SH}(\mathbb{Z})} \mathrm{DM}(\mathbb{Z})$. By Theorem 3.2.64, the natural map $f^* \mathrm{HZ}_{\mathrm{Spec}(\mathbb{Z})} \xrightarrow{\sim} \mathrm{HZ}_X$ is an equivalence, so the canonical comparison

$$\mathrm{SH}(X) \otimes_{\mathrm{SH}(\mathbb{Z})} \mathrm{DM}(\mathbb{Z}) \xrightarrow{\sim} \mathrm{DM}(X)$$

is an equivalence in Pr^L [Lur17, Theorem 4.8.4.6], transporting the six-functor formalism from the left-hand side to DM . $\square$

Remark 3.2.70 (Explicit description of DM [BEM25, Remark 9.11]). The category $\mathrm{DM}(X)$ is more accessible than it might initially appear. Under our convention (Definition 3.2.42, no degree shift), the motivic Eilenberg–MacLane spectrum is $H\mathbb{Z}_X = H(\mathbb{Z}(\star)^\mathbb{A})$ via the functor of Definition 3.1.9, and Proposition 3.1.8 applied with Sp replaced by $D(\mathbb{Z})$ yields

$$\mathrm{DM}(X) \simeq \mathrm{Mod}_{\mathbb{Z}(\star)^\mathbb{A}} \left(\mathrm{GrShv}_{\mathrm{Nis}, \mathbb{A}^1}(\mathrm{Sm}_X, D(\mathbb{Z})) \right)_{c, \mathrm{pbf}}.$$

Concretely, $\mathrm{DM}(X)$ is the category of $\mathbb{Z}(\star)^\mathbb{A}$ -modules $M^\star$ in $\mathrm{GrShv}_{\mathrm{Nis}, \mathbb{A}^1}(\mathrm{Sm}_X, D(\mathbb{Z}))$ satisfying the $\mathbb{P}^1$ -bundle formula: multiplication by the first Chern class $c_1^\mathbb{A}(\mathcal{O}(1)) \in \pi_0\mathbb{Z}(1)^\mathbb{A}(\mathbb{P}_X^1)$ induces an equivalence

$$M^j \xrightarrow{\sim} \mathrm{fib}(M^{j+1}(\mathbb{P}_-^1) \xrightarrow{\infty^*} M^{j+1})$$

for every $j \in \mathbb{Z}$, in line with the bonding-map formulation of the pbf in Construction 3.1.4. In particular, if $\mathbb{A}^1$ -motivic cohomology is accepted as a black box, $\mathrm{DM}(X)$ can be defined without ever mentioning $\mathrm{SH}(X)$.

3.3. FRAMED CORRESPONDENCES AND THE MOTIVIC RECOGNITION

PRINCIPLE

Remark 3.2.16 announced an unstable description of $\mathrm{SH}^{\mathrm{veff}}(k)$ over a perfect field k , in terms of grouplike sheaves on the category $\mathrm{Corr}^{\mathrm{fr}}(\mathrm{Sm}_k)$ of *framed correspondences*. This section develops the framework. After motivating the appearance of framed finite syntomic transfers via the six-functor formalism (Section 3.3.1), we introduce framed correspondences in their normally-framed presentation and assemble the resulting categories of framed motivic objects (Sections 3.3.2 and 3.3.3); state and outline the proof of the motivic recognition principle of [Elm+21] (cited as EHKS $\mathbb{Y}$) in Section 3.3.4; review Voevodsky’s lemma, the geometric input on which the proof rests (Section 3.3.5); and discuss the extent to which the various pieces extend beyond perfect fields (Section 3.3.6). The closing subsection records consequences: an explicit description of the motivic sphere spectrum, a framed presentation of $H\mathbb{Z}$, and a framed model for connective K -theory complementing the Bott-periodic construction of Section 3.1.6.

3.3.1. Six-functor heuristics for transfers

The starting point is the adjunction

$$\sigma^\infty : \mathrm{Shv}_{\mathrm{Nis}, \mathbb{A}^1}(\mathrm{Sm}_S, \mathrm{Sp}) \rightleftarrows \mathrm{SH}^{\mathrm{eff}}(S) : \omega^\infty$$

established in Section 3.1. The right-hand side is described intrinsically in terms of Sm_S ; the question now is what additional structure on $\omega^\infty(E)$, viewed as a sheaf on Sm_S , recovers the effective motivic spectrum E . Following Hoyo $\mathbb{S}$ we organize this into two questions:

- Q1.** What additional structure does $\omega^\infty(E)$ carry, beyond being Nisnevich-local and $\mathbb{A}^1$ -invariant?
- Q2.** Is this structure enough to recover E from $\omega^\infty(E)$?

The answers, due over a perfect field to EHKS $\mathbb{Y}$ [Elm+21] after groundwork of Voevodsky and Garkusha–Panin, are

- A1.** $\omega^\infty(E)$ carries transfers along finite syntomic morphisms equipped with a K -theoretic trivialization of the cotangent complex.

A2. Yes, when S is the spectrum of a perfect field.

We sketch in this subsection where the transfers come from, by way of the six-functor formalism developed in Section 2.2. The reader should bear in mind from the outset (Remark 3.3.2) that this argument is heuristic only: it produces a transfer for each individual framed map but does *not* package them into a coherent functor. The proof of the recognition principle in Sections 3.3.4 and 3.3.5 proceeds along entirely different lines.

Finite étale transfers from purity

For any morphism f that is $!$ -able in our six-functor formalism on SH (i.e. locally of finite presentation, the class fixed in Theorem 2.3.10), the kernel-level adjunction $\mathbb{L}_X \dashv \omega_f$ of Theorem 2.3.20 gives, after Ψ , the basic identity

$$f^! \simeq f^* \otimes \omega_f \quad \text{when } f \text{ is cohomologically smooth.}$$

For f cohomologically étale, Proposition 2.3.35 sharpens this to $f^! \simeq f^*$, and for f cohomologically proper to $f_! \simeq f_*$. A finite étale morphism $f: Y \rightarrow X$ is both proper and étale, so combining the two identities and applying the counit of $f_! \dashv f^!$ produces a natural map

$$f_* f^* E \simeq f_! f^! E \xrightarrow{\text{counit}} E,$$

which on cohomology yields a wrong-way transfer $\omega^\infty(E)(Y) \rightarrow \omega^\infty(E)(X)$. This is the algebro-geometric counterpart of the manifold-level transfer for finite covers $\Sigma_+^\infty X \rightarrow \Sigma_+^\infty Y$ in classical homotopy theory.

Extension to lci morphisms via Déglise–Jin–Khan

Beyond the étale range, the construction extends thanks to a result of Déglise–Jin–Khan in their theory of motivic fundamental classes.

Theorem 3.3.1 (Déglise–Jin–Khan [DJK21, Theorem 3.3.2], purity transformation [DJK21, §2.5]). *For every smoothable lci morphism $f: Y \rightarrow X$ of schemes, the fundamental class η_f in twisted bivariant cohomology induces a canonical natural transformation*

$$\mathrm{pur}(\eta_f): \Sigma^{\mathbb{L}_f} f^* \longrightarrow f^!,$$

where $\mathbb{L}_f$ is the cotangent complex of f , viewed as an element of $K(Y)$ via its perfect-complex class. Equivalently, twisting by $\Sigma^{-\mathbb{L}_f}$ yields a natural transformation $f^* \rightarrow \Sigma^{-\mathbb{L}_f} f^!$. The transformation is compatible with composition and with tor-independent base change, and reduces to the purity equivalence $f^! \simeq f^* \otimes \omega_f$ whenever f is smooth.

For a finite syntomic (i.e. flat lci) morphism $f: Y \rightarrow X$, properness gives $f_! \simeq f_*$, and a framing — a path $\mathbb{L}_f \simeq 0$ in $\Omega^\infty K(Y)$ — trivializes the Thom twist via the construction recalled below. After this trivialization, the purity transformation gives a map $f^* E \rightarrow f^! E$; it need not be an equivalence for a general lci morphism. The transfer is the composite

$$f_* f^* E \simeq f_! f^* E \xrightarrow{f_! \mathrm{pur}(\eta_f)} f_! f^! E \xrightarrow{\text{counit}} E$$

where the middle arrow includes the chosen identification $\Sigma^{\mathbb{L}_f} f^* E \simeq f^* E$. Finite syntomic morphisms with K -theoretic framings are thus the algebro-geometric avatar of finite étale maps, with the framing supplying the twist-trivialization needed to construct the wrong-way map.

Thom twists by K -theoretic classes

The Thom-twist symbol $\Sigma^{-\mathbb{L}_f}$ above is a slight extension of the Thom isomorphism machinery of Remark 3.2.33. Let $\mathcal{P}ic(\mathrm{SH}(S))$ denote the Picard spectrum of $\mathrm{SH}(S)$, whose underlying $\mathbb{E}_\infty$ -monoid is the moduli of $\otimes$ -invertible motivic spectra; this is *not* the line-bundle Picard spectrum $\mathcal{P}ic$ of Section 3.2.4 but the strictly larger object containing all Tate twists $\mathbb{T}_S^{\otimes j}$, sphere shifts, and line-bundle Thom spaces. For a finite locally free sheaf $\mathcal{E}$ on X , the classical Thom space

$$\mathrm{Th}_X(\mathcal{E}) := \mathbb{V}(\mathcal{E}) / (\mathbb{V}(\mathcal{E}) \setminus 0)$$

is $\otimes$ -invertible after passage to $\mathrm{SH}(X)$. The assignment $\mathcal{E} \mapsto \mathrm{Th}_X(\mathcal{E})$ is symmetric monoidal in $\mathcal{E}$ and hence factors through the group-completion of $\mathrm{Vect}(X)$, defining a map of $\mathbb{E}_\infty$ -spaces

$$\mathrm{Th}_X : \Omega^\infty K(X) \longrightarrow \Omega^\infty \mathcal{P}ic(\mathrm{SH}(X)).$$

For $\xi \in \Omega^\infty K(X)$, the **Thom twist** by ξ is the auto-equivalence

$$\Sigma^\xi := \mathrm{Th}_X(\xi) \otimes - : \mathrm{SH}(X) \longrightarrow \mathrm{SH}(X).$$

A framing of f is by definition a path $\mathbb{L}_f \simeq 0$ in $\Omega^\infty K(Y)$; under Th_Y it produces a path $\Sigma^{\mathbb{L}_f} \simeq \mathrm{id}$ in the auto-equivalences of $\mathrm{SH}(Y)$, trivializing the twist.

Remark 3.3.2 (The heuristic does not produce a coherent theory). The above construction yields a transfer for each individual framed finite syntomic morphism, but not a coherent functor on a category of framed correspondences. The transfer depends on the framing (a continuous datum), not just on f , and even the homotopy-class-level coherence is non-trivial. The actual construction of the category $\mathrm{Corr}^{\mathrm{fr}}(\mathrm{Sm}_S)$ in [Elm+21, §4] proceeds through a labelled span construction; the six-functor argument is heuristic only.

3.3.2. Framed correspondences

A *framed correspondence* from a smooth S -scheme X to a smooth S -scheme Y is, informally, a span

$$\begin{array}{ccc} & Z & \\ f \swarrow & & \searrow g \\ X & & Y \end{array}$$

in which f is finite syntomic and Z comes equipped with a K -theoretic trivialization $\mathbb{L}_f \simeq 0$. Three inequivalent presentations of this data appear in the literature; they coincide only after motivic localization. We adopt the *normally framed* presentation for explicit work with framed presheaves and reserve the *tangentially framed* presentation for the construction of $\mathrm{Corr}^{\mathrm{fr}}(\mathrm{Sm}_S)$ as a category (Remark 3.3.6).

Normally framed correspondences

Definition 3.3.3 (Normally framed correspondence). Following [Elm+21, Definition 2.2.2], let $X, Y \in \mathrm{Sch}_S$ and $n \geq 0$. A **normally framed correspondence of level n** from X to Y over S consists of:

- (i) a span $X \xleftarrow{f} Z \xrightarrow{g} Y$ over S in which f is finite syntomic;
- (ii) a closed immersion $i : Z \hookrightarrow \mathbb{A}_X^n$ over X ;
- (iii) a trivialization $\tau : \mathcal{O}_Z^{\oplus n} \xrightarrow{\sim} \mathcal{N}_i$ of the conormal sheaf of i .

Denote the resulting discrete groupoid by $\text{Corr}_S^{\text{nfr},n}(X, Y)$. Closed immersion into $\mathbb{A}_X^{n+1}$ via $(x, 0)$ defines a stabilization

$$\begin{aligned} \sigma_{X,Y}: \text{Corr}_S^{\text{nfr},n}(X, Y) &\hookrightarrow \text{Corr}_S^{\text{nfr},n+1}(X, Y), \\ (Z, i, \tau) &\longmapsto (Z, i \times \{0\}, \tau \oplus \text{id}), \end{aligned}$$

and we set $\text{Corr}_S^{\text{nfr}}(X, Y) := \text{colim}_n \text{Corr}_S^{\text{nfr},n}(X, Y)$.

Remark 3.3.4 (*K-theoretic content*). The conormal trivialization τ is the categorical avatar of a stable framing of $\mathbb{L}_f$ in $K(Z)$. Indeed, factoring f as the closed immersion $i: Z \hookrightarrow \mathbb{A}_X^n$ followed by the smooth projection $p: \mathbb{A}_X^n \rightarrow X$ produces a cofibre sequence of cotangent complexes

$$i^*\mathbb{L}_p \longrightarrow \mathbb{L}_f \longrightarrow \mathbb{L}_i,$$

with $i^*\mathbb{L}_p = \mathcal{O}_Z^{\oplus n}$ in degree zero and $\mathbb{L}_i \simeq \mathcal{N}_i[1]$. Passing to $K(Z)$ gives the identity

$$[\mathbb{L}_f] = [\mathcal{O}_Z^{\oplus n}] - [\mathcal{N}_i] \in K(Z),$$

so a trivialization $\tau: \mathcal{O}_Z^{\oplus n} \simeq \mathcal{N}_i$ produces a path $[\mathbb{L}_f] \simeq 0$ in $\Omega^\infty K(Z)$, that is, a stable framing of $\mathbb{L}_f$ in the sense of Section 3.3.1.

Three presentations and their comparison

Remark 3.3.5 (The three presentations). Three presentations of framed correspondences are useful in different contexts.

- (i) *Equationally framed* correspondences (originally introduced by Voevodsky and developed by Garkusha–Panin, see [Elm+21, §2.1] for a recollection) record a closed subscheme $Z \subseteq \mathbb{A}_X^n$ together with a map $\phi: (\mathbb{A}_X^n)_Z^h \rightarrow \mathbb{A}^n$ from the *henselization* of $\mathbb{A}_X^n$ along Z such that $\phi^{-1}(0) = Z$. By Voevodsky’s lemma (Lemma 3.3.13), these data compute the mapping anima between $\mathbb{P}^1$ -suspension spectra and Thom spectra of trivial bundles.
- (ii) *Normally framed* correspondences (Definition 3.3.3) record only the closed embedding $Z \hookrightarrow \mathbb{A}_X^n$ together with a trivialization of its conormal sheaf. This presentation is best suited to geometric descriptions of underlying spaces of motivic spectra: when Y is quasi-projective over S , the functor $\text{Corr}_S^{\text{nfr}}(-, Y)$ is ind-representable by quasi-projective S -schemes [Elm+21, §5.1].
- (iii) *Tangentially framed* correspondences [Elm+21, §2.3] record only the homotopy-theoretic data: a finite syntomic span as above, together with a trivialization of $\mathbb{L}_f$ in $K(Z)$. Unlike the previous two, tangentially framed correspondences are the mapping anima of a category whose objects are smooth S -schemes; this is the category $\text{Corr}^{\text{fr}}(\text{Sm}_S)$ of Definition 3.3.7.

The presheaves of equationally framed, normally framed, and tangentially framed correspondences are pairwise motivically equivalent [Elm+21, Corollary 2.3.27], so for the purpose of computing mapping anima in $H(S)$ they are interchangeable.

Remark 3.3.6 (Why tangential framing is needed for a category). The closed immersion $Z \hookrightarrow \mathbb{A}_X^n$ in the data of a normally framed correspondence prevents direct composition: given a second normally framed span $Y \leftarrow W \rightarrow T$ with $W \hookrightarrow \mathbb{A}_Y^m$, the pullback $Z \times_Y W$ embeds into $\mathbb{A}_X^{n+m}$ only after stabilization, and the choice of this embedding is not canonical at the level of the strict definition. The tangential presentation eliminates this issue by remembering only the K -theoretic class $[\mathbb{L}_f]$ together with a path to zero, which is composable

in the obvious way: the cofibre sequence $g^*\mathbb{L}_f \rightarrow \mathbb{L}_{fg} \rightarrow \mathbb{L}_g$ in $K(W)$ yields a composite framing $[\mathbb{L}_{fg}] \approx 0$ from framings of f and g separately.

Definition 3.3.7 (Category of framed correspondences). Following [Elm+21, §4], the symmetric monoidal category $\text{Corr}^{\text{fr}}(\text{Sm}_S)$ has:

- objects: smooth S -schemes;
- mapping anima: $\text{Map}_{\text{Corr}^{\text{fr}}}(X, Y) = \text{Corr}_S^{\text{fr}}(X, Y)$ (tangentially framed);
- composition: pullback of spans together with the cofibre-sequence framing of Remark 3.3.6;
- symmetric monoidal structure: induced by the Cartesian product $\times_S$ on Sm_S and direct sum of framings.

The unit is S itself. There is a canonical symmetric monoidal functor

$$\gamma: \text{Sm}_S \longrightarrow \text{Corr}^{\text{fr}}(\text{Sm}_S)$$

sending a morphism $f: X \rightarrow Y$ to the trivially framed graph $\text{span } X \xleftarrow{\text{id}} X \xrightarrow{f} Y$ at level $n = 0$.

Sheaf properties

Notation 3.3.8 (Framed presheaves). For each $Y \in \text{Sm}_S$ we write

$$h_S^{\text{fr}}(Y) := \text{Corr}_S^{\text{fr}}(-, Y) \in \text{PShv}(\text{Sm}_S),$$

the presheaf of framed correspondences into Y , and similarly $h_S^{\text{nfr}}(Y) = \text{Corr}_S^{\text{nfr}}(-, Y)$. By construction both take values in $\mathbb{E}_\infty$ -anima: the empty correspondence is the unit and disjoint union of framings is the product.

Proposition 3.3.9 (Sheaf properties [Elm+21, Proposition 2.2.4]). For every $Y \in \text{Sm}_S$:

- Each finite-level presheaf $h_S^{\text{nfr}, n}(Y)$ is an fpqc sheaf, and the colimit $h_S^{\text{nfr}}(Y)$ is a sheaf for the quasi-compact fpqc topology.
- $h_S^{\text{nfr}}(Y)$ satisfies closed gluing.
- For every Nisnevich covering sieve $R \hookrightarrow Y$ generated by a single map, the induced map $h_S^{\text{nfr}}(R) \rightarrow h_S^{\text{nfr}}(Y)$ is a Nisnevich equivalence (the source being defined by extending h_S^{nfr} to presheaves via colimits).

For the tangentially framed presheaf $h_S^{\text{fr}}(Y)$ itself, the situation is subtler: it is still a Nisnevich sheaf on qcqs schemes and satisfies the analogue of ((iii)) [Elm+21, Proposition 2.3.7], but closed gluing genuinely fails for it [Elm+21, Remark 2.3.26]. For motivic purposes this is immaterial, since $h_S^{\text{nfr}}(Y) \rightarrow h_S^{\text{fr}}(Y)$ is a motivic equivalence [Elm+21, Corollary 2.3.27].

3.3.3. Categories of framed motivic objects

The promotion of h_S^{fr} to a sheaf valued in spectra requires care because $\text{Corr}^{\text{fr}}(\text{Sm}_S)$ is naturally only *semiadditive*; spectrum-valued sheaves are recovered by group-completing and stabilizing.

Definition 3.3.10 (Categories of framed motivic objects [Elm+21, §3.2–3.3]). Let $H^{\text{fr}}(S) \subseteq \text{PShv}_\Sigma(\text{Corr}^{\text{fr}}(\text{Sm}_S))$ denote the full subcategory of presheaves of anima on $\text{Corr}^{\text{fr}}(\text{Sm}_S)$ that take finite coproducts in Sm_S to products of anima, are Nisnevich-local, and are $\mathbb{A}^1$ -invariant.

Its grouplike subcategory is denoted $H^{\text{fr}}(S)^{\text{gp}}$. The corresponding category of **framed motivic S^1 -spectra** is

$$\text{SH}^{S^1, \text{fr}}(S) := \text{PShv}_{\Sigma}(\text{Corr}^{\text{fr}}(\text{Sm}_S), \text{Sp})_{\text{Nis}, \mathbb{A}^1},$$

the full subcategory of Sp -valued presheaves on $\text{Corr}^{\text{fr}}(\text{Sm}_S)$ which take finite coproducts to products, are Nisnevich-local, and are $\mathbb{A}^1$ -invariant. The full $\mathbb{T}_S$ -stabilization is

$$\text{SH}^{\text{fr}}(S) := \text{Sp}_{\mathbb{T}_S}(H^{\text{fr}}(S)),$$

the universal presentably symmetric monoidal category under $H^{\text{fr}}(S)$ in which $\gamma^*\mathbb{T}_S$ becomes $\otimes$ -invertible (Definition 2.1.5).

The functor $\gamma: \text{Sm}_S \rightarrow \text{Corr}^{\text{fr}}(\text{Sm}_S)$ induces an adjunction

$$\gamma^*: H_*(S) \rightleftarrows H^{\text{fr}}(S) : \gamma_*,$$

together with S^1 - and $\mathbb{T}_S$ -stabilized variants. The right adjoint γ_* sends a framed sheaf to its underlying Nisnevich sheaf with $\mathbb{E}_{\infty}$ -structure; the left adjoint γ^* freely adjoins framed transfers. Conservativity and colimit-preservation of the stabilized γ_* [Elm+21, Proposition 3.5.2] (at the unstable level, γ_* is conservative and preserves sifted colimits) make these adjunctions amenable to comparison with the unframed side.

3.3.4. The motivic recognition principle

Theorem 3.3.11 (Motivic recognition principle [Elm+21, Theorem 3.5.14]; see also the lecture notes [BE25]). *Let k be a perfect field. The functors $\gamma_*\Sigma_{\mathbb{T}, \text{fr}}^{\infty}$ and $\gamma_*\Sigma_{\mathbb{G}_m, \text{fr}}^{\infty}$ induce equivalences of symmetric monoidal categories:*

$$(i) \text{ (Unstable form.) } \gamma_*\Sigma_{\mathbb{T}, \text{fr}}^{\infty}: H^{\text{fr}}(k)^{\text{gp}} \xrightarrow{\sim} \text{SH}^{\text{veff}}(k);$$

$$(ii) \text{ (Stable form.) } \gamma_*\Sigma_{\mathbb{G}_m, \text{fr}}^{\infty}: \text{SH}^{S^1, \text{fr}}(k) \xrightarrow{\sim} \text{SH}^{\text{eff}}(k).$$

After $\mathbb{T}_S$ -stabilization, the pullback functor $\gamma^*: \text{SH}(k) \rightarrow \text{SH}^{\text{fr}}(k)$ is itself an equivalence (the reconstruction theorem, [Elm+21, Theorem 3.5.12]). The transfers produced one map at a time by the heuristic of Section 3.3.1 do not suffice to construct these equivalences (Remark 3.3.2); assembling the framed transfers coherently requires the genuinely different argument outlined below.

Remark 3.3.12 (The dictionary). Theorem 3.3.11 is the precise motivic counterpart of $\text{Sp}_{\geq 0} \simeq \text{CMon}^{\text{gp}}(\text{An})$:

Classical		Motivic
$\text{Sp}_{\geq 0}$	$\longleftrightarrow$	$\text{SH}^{\text{veff}}(k)$
$\text{CMon}^{\text{gp}}(\text{An})$	$\longleftrightarrow$	$H^{\text{fr}}(k)^{\text{gp}}$
finite sets Fin_*	$\longleftrightarrow$	$\text{Corr}^{\text{fr}}(\text{Sm}_k)$
Σ_+^{∞}	$\longleftrightarrow$	$\Sigma_{\mathbb{T}, \text{fr}}^{\infty}$

The role of the collapse maps $\langle n \rangle \rightarrow \langle 1 \rangle$ in Fin_* — the wrong-way maps, when one regards a pointed map of finite sets as a span whose backward leg is a subset inclusion — is played by the framed transfers in Corr^{fr} : both encode the abstract addition law that promotes a product-preserving presheaf of anima to a presheaf of $\mathbb{E}_{\infty}$ -anima.

The proof of Theorem 3.3.11 factors into three steps.

Step 1: Cancellation. The first input is a motivic cancellation theorem in the framed setting: for every $M \in \mathrm{SH}^{S^1, \mathrm{fr}}(k)$, the unit map

$$M \longrightarrow \Omega_{\mathbb{G}_m}(\mathbb{G}_m^{\mathrm{fr}} \otimes M)$$

is an equivalence [Elm+21, Theorem 3.5.8]. This is the framed analogue of Voevodsky’s cancellation theorem in $\mathrm{DM}^{\mathrm{eff}}(k)$: the underlying cancellation for *equationally* framed correspondences is due to Ananyevskiy–Garkusha–Panin (with the characteristic-2 case completed by Druzhinin–Panin), and *loc. cit.* deduces the tangentially framed statement above from it via the comparison of Remark 3.3.5. As a consequence, $\Sigma_{\mathbb{G}_m, \mathrm{fr}}^\infty: \mathrm{SH}^{S^1, \mathrm{fr}}(k) \rightarrow \mathrm{SH}^{\mathrm{fr}}(k)$ is fully faithful; combining this with the S^1 -recognition principle for grouplike framed motivic spaces — which rests on the strict $\mathbb{A}^1$ -invariance theorem of Garkusha–Panin — the composite $\Sigma_{\mathbb{T}, \mathrm{fr}}^\infty: \mathrm{H}^{\mathrm{fr}}(k)^{\mathrm{gp}} \rightarrow \mathrm{SH}^{\mathrm{fr}}(k)$ is fully faithful as well. Geometrically, the $\mathbb{G}_m$ -suspension obstruction in the unframed setting — which prevents the analogous statement there — is repaired by framed transfers.

Step 2: Reconstruction. The reconstruction theorem [Elm+21, Theorem 3.5.12] establishes that

$$\gamma^*: \mathrm{SH}(k) \longrightarrow \mathrm{SH}^{\mathrm{fr}}(k)$$

is itself a symmetric monoidal equivalence. Conservativity and colimit-preservation of γ_* reduce the question to checking that the unit map

$$\Sigma_{\mathbb{T}}^\infty X_+ \longrightarrow \gamma_* \gamma^* \Sigma_{\mathbb{T}}^\infty X_+$$

is a motivic equivalence for every $X \in \mathrm{Sm}_k$. The right-hand side is computed level-wise by the $\mathbb{G}_m$ -prespectrum

$$(B_{\mathrm{mot}}^\infty L_{\mathrm{mot}} h^{\mathrm{fr}}(X_+), B_{\mathrm{mot}}^\infty L_{\mathrm{mot}} h^{\mathrm{fr}}(\mathbb{G}_m \wedge X_+), \dots),$$

where $B_{\mathrm{mot}}^\infty: \mathrm{CMon}(\mathrm{H}(k)) \rightarrow \mathrm{SH}^{S^1}(k)$ denotes the motivic infinite delooping, left adjoint to Ω^∞ . The comparison with the standard $\mathbb{T}_S$ -suspension prespectrum is then reduced, via the equational-to-tangential comparison [Elm+21, Corollary 2.3.27], to the framed motives theorem of Garkusha–Panin for infinite perfect fields, recalled in [Elm+21, §3.4].

Step 3: Image identification. Once γ^* is known to be an equivalence, the two restricted statements of Theorem 3.3.11 follow by identifying the images of the various stabilization functors:

- For ((i)), the image of $\gamma_* \Sigma_{\mathbb{T}, \mathrm{fr}}^\infty$ is the colimit closure of $\{\Sigma_{\mathbb{T}}^\infty X_+\}_{X \in \mathrm{Sm}_k}$ in $\mathrm{SH}(k)$, which coincides with $\mathrm{SH}^{\mathrm{veff}}(k)$: although Definition 3.2.13 takes the closure of these generators under colimits *and* extensions, over a perfect field k this colimit closure is already closed under extensions [Bac17, remark following Proposition 4], so the two agree.
- For ((ii)), the image is the colimit closure of $\{\Sigma_{\mathbb{T}}^{-n} \Sigma_{\mathbb{T}}^\infty X_+\}_{X \in \mathrm{Sm}_k, n \geq 0}$, which is $\mathrm{SH}^{\mathrm{eff}}(k)$ by Definition 3.2.5.

3.3.5. Voevodsky’s lemma and the comparison $\mathrm{Corr}^{\mathrm{eff}} \rightarrow \mathrm{Corr}^{\mathrm{nfr}} \rightarrow \mathrm{Corr}^{\mathrm{fr}}$

The cancellation and reconstruction theorems of Steps 1 and 2 ultimately rest on a geometric input due to Voevodsky, identifying framed correspondences as the mapping anima for $\mathbb{P}^1$ -suspension spectra at the unstable level.

Lemma 3.3.13 (Voevodsky’s lemma [Elm+21, Proposition A.1.4]). *Let X be an S -scheme and $U \subseteq X$ an open subscheme. The Nisnevich sheafification of the pointed cofibre X/U in $\mathrm{PShv}(\mathrm{Sm}_S)_*$ is given on $Y \in \mathrm{Sm}_S$ by the discrete groupoid*

$$L_{\mathrm{Nis}}(X/U)(Y) = \{ (Z, \phi) \mid Z \subseteq Y \text{ a closed subset, } \phi: Y_Z^h \rightarrow X \text{ over } S, \phi^{-1}(X \setminus U) = Z \},$$

where Y_Z^h denotes the henselization of Y along Z , viewed as a pro-object (of algebraic spaces étale over Y). Moreover, the sheafification is automatically étale-local.

The relevance to framed correspondences is the following specialization: the discrete groupoid of *equationally framed correspondences* of level n from $X \in \mathrm{Sm}_S$ to $Y \in \mathrm{Sm}_S$ (Remark 3.3.5(1)) is

$$\mathrm{Corr}_S^{\mathrm{efr}, n}(X, Y) := \{ (Z, \phi, g) \mid Z \hookrightarrow \mathbb{A}_X^n \text{ is closed and finite over } X, \\ \phi: (\mathbb{A}_X^n)_Z^h \rightarrow \mathbb{A}^n, \quad \phi^{-1}(0) = Z, \quad g: (\mathbb{A}_X^n)_Z^h \rightarrow Y \}.$$

Here $f: Z \rightarrow X$ is the (finite) projection and the map to Y is the restriction $g|_Z$, so that (Z, ϕ, g) presents the span $X \xleftarrow{f} Z \xrightarrow{g|_Z} Y$. For $X, Y \in \mathrm{Sm}_S$, Lemma 3.3.13 specializes to a natural equivalence

$$\mathrm{Map}_{\mathrm{PShv}(\mathrm{Sm}_S)_*} \left(X_+ \wedge (\mathbb{P}^1, \infty)^{\wedge n}, L_{\mathrm{Nis}}(Y_+ \wedge \mathbb{A}^n / (\mathbb{A}^n \setminus 0)) \right) \\ \simeq \mathrm{Corr}_S^{\mathrm{efr}, n}(X, Y),$$

see [Elm+21, Corollary A.1.7]. This is the geometric bridge between equationally framed data and mapping anima in stabilized motivic categories.

The proof of the recognition principle then proceeds by interposing the normally framed correspondences of Definition 3.3.3 between the equationally framed and the tangentially framed ones. One constructs natural maps of presheaves on Sm_S :

$$\begin{array}{ccc} \mathrm{Corr}_S^{\mathrm{efr}}(X, Y) & \longrightarrow & \mathrm{Map}_{\mathrm{SH}(S)}(\Sigma_+^\infty X, \Sigma_+^\infty Y) \\ \downarrow & & \\ \mathrm{Corr}_S^{\mathrm{nfr}}(X, Y) & & \\ \downarrow & & \\ \mathrm{Corr}_S^{\mathrm{fr}}(X, Y) & & \end{array}$$

and shows that both vertical maps are motivic equivalences in X [Elm+21, Corollaries 2.2.21 and 2.3.25]. The horizontal map is the geometric input from Lemma 3.3.13, and the chain of comparisons reduces calculations of $\mathbb{P}^1$ -loop spaces to computations on the framed side.

3.3.6. Extension to arbitrary base schemes

The recognition principle of Theorem 3.3.11 is a statement over a perfect field, since both the cancellation theorem and the reduction to Garkusha–Panin make essential use of strict $\mathbb{A}^1$ -invariance, which is known only over perfect fields. The reconstruction theorem alone, however, generalizes:

Theorem 3.3.14 (Hoyois [Hoy21, Theorem 18]). *For every scheme S , the functor*

$$\gamma^*: \mathrm{SH}(S) \longrightarrow \mathrm{SH}^{\mathrm{fr}}(S)$$

is a symmetric monoidal equivalence.

The proof in [Hoy21] proceeds via a framed analogue of the Morel–Voevodsky localization theorem: for a closed immersion $i: Z \hookrightarrow S$ with open complement j , the null-sequence $j_{\#}j^* \rightarrow \text{id} \rightarrow i_*i^*$ of endofunctors on $H^{\text{fr}}(S)$ is a cofibre sequence [Hoy21, Theorem 10]. This implies that framed motivic spectra satisfy Ayoub’s axioms [Ayo07] and hence enjoy the full six-functor formalism of Section 2.2 on the framed side. Theorem 3.3.14 then follows by Zariski descent, proper pushforwards (made available by the six operations), and conservativity of pullback to points, ultimately invoking the perfect-field case of Theorem 3.3.11 over the prime fields $\mathbb{Q}$ and $\mathbb{F}_p$. As a consequence, the conclusion that γ_* encodes framed transfers on $\Omega_{\mathbb{T}_S}^{\infty} E$ holds over arbitrary base schemes, even though the recognition principle as an equivalence with $H^{\text{fr}}(S)^{\text{gp}}$ does not.

3.3.7. Consequences

The motivic sphere spectrum

Example 3.3.15 (Explicit description of $\mathbb{1}_k$). Under Theorem 3.3.11(i), the motivic sphere spectrum $\mathbb{1}_k \in \text{SH}^{\text{veff}}(k)$ is identified with the group completion of $h_k^{\text{fr}}(\text{Spec } k)$, the Nisnevich-localized $\mathbb{A}^1$ -invariant presheaf of finite syntomic schemes equipped with stable framings. Specializing to the unit object, this is the framed motivic Barratt–Priddy–Quillen theorem of [Elm+21, Theorem 3.5.18], and computing π_0 recovers Morel’s classical identification $\pi_0(\mathbb{1}_k) \simeq \text{GW}(k)$ [Mor12]. The underlying anima of $\mathbb{1}_k$ at the unstable level admits a model in terms of Hilbert schemes of framed points in affine space [Elm+21, §5.1].

Motivic cohomology

Theorem 3.3.16 ([Hoy21, Theorem 21]). *For every scheme S there is a canonical equivalence of motivic $\mathbb{E}_{\infty}$ -ring spectra*

$$\Sigma_{\text{fr}}^{\infty} \mathbb{Z} \simeq H\mathbb{Z}_S \quad \text{in } \text{SH}(S),$$

where $H\mathbb{Z}_S$ is Spitzweck’s motivic cohomology spectrum and $\mathbb{Z}$ on the left denotes the constant framed sheaf of abelian groups on $\text{Corr}^{\text{fr}}(\text{Sm}_S)$.

After collapsing framings, the underlying transfer structure on $H\mathbb{Z}$ recovers Voevodsky’s category of finite Nisnevich correspondences. This is the natural starting point for the motivic Dold–Kan correspondence and the Eilenberg–Zilber theorem.

Connective K-theory: a framed model

We close with the framed construction of connective K-theory due to Hoyois–Jelisiejew–Nardin–Totaro–Yakerson [Hoy+25], used as a key technical tool in [BEM25, §4.1.3]. It complements the Bott-periodic construction of KGL_S in Section 3.1.6.

The presheaf Vect_S of finite locally free sheaves on Sm_S is an $\mathbb{E}_{\infty}$ -monoid under direct sum, and its group completion carries finite locally free transfers via tensor product and pushforward of vector bundles. Pulling back along the symmetric monoidal forgetful $\text{Corr}^{\text{fr}}(\text{Sm}_S) \rightarrow \text{Corr}^{\text{ff}}(\text{Sm}_S)$ promotes $\text{Vect}_S^{\text{gp}}$ to

$$\text{Vect}_S^{\text{gp}} \in \text{CAlg}(\text{PShv}_{\Sigma}(\text{Corr}^{\text{fr}}(\text{Sm}_S))).$$

Crucially, the group completion $\text{Vect}_S^{\text{gp}}$ — algebraic K-theory — is left Kan extended from smooth algebras [Elm+20, Example A.0.6(1)], hence compatible with arbitrary base change in S ; so the framed suspension spectrum

$$\mathcal{V}_S := \Sigma_{\text{fr}}^{\infty}(\text{Vect}_S^{\text{gp}}) \in \text{CAlg}(\text{SH}(S))$$

defines an absolute motivic spectrum. The Bott class of Section 3.1.6 lifts to a canonical map $\mathbb{T}_S \rightarrow \mathcal{V}_S$ in $\mathrm{SH}(S)$ classifying the virtual bundle $\mathcal{O} - \mathcal{O}(-1)$ on $\mathbb{P}_S^1$, whose composite with the multiplicative comparison map $\mathcal{V}_S \rightarrow \mathrm{KGL}_S$ recovers the Bott element $\beta = 1 - [\mathcal{O}(-1)]$.

Proposition 3.3.17 ([BEM25, Corollary 9.6]). *For every qcqs scheme S , the canonical map*

$$\mathcal{V}_S \xrightarrow{\sim} \mathrm{kg}l_S$$

is an equivalence of $\mathbb{E}_\infty$ -algebras in $\mathrm{SH}(S)$, where $\mathrm{kg}l_S$ is the effective cover of KGL_S from Construction 3.2.8. Inverting the Bott element gives a canonical equivalence

$$\mathcal{V}_S[\beta^{-1}] \xrightarrow{\sim} \mathrm{KGL}_S$$

in $\mathrm{CAlg}(\mathrm{SH}(S))$.

The case $S = \mathrm{Spec}(k)$ with k a field is [Hoy+25, Corollary 5.2], and is where the recognition principle enters decisively: by Theorem 3.3.11(i) the very effective cover of $\mathcal{V}_k[\beta^{-1}] \simeq \mathrm{KGL}_k$ is computed as $\Sigma_{\mathrm{fr}}^\infty \Omega_{\mathrm{fr}}^\infty(\mathcal{V}_k[\beta^{-1}])$, and one identifies this with $\mathcal{V}_k$ itself by exhibiting $\Omega_{\mathbb{T}_k}^\infty \mathcal{V}_k \simeq \Omega^\infty K$. The case of arbitrary qcqs S , established in [BEM25], reduces by base change to $S = \mathrm{Spec}(\mathbb{Z})$ — using stability under pullback of both $\mathcal{V}$ (an absolute spectrum) and $\mathrm{kg}l$ (the hard input, part of the main theorem of *op. cit.*) — and over the Dedekind base $\mathbb{Z}$ the statement is checked on residue and fraction fields, where the field-level result applies.

Proposition 3.3.17 thus gives a framed-theoretic construction of KGL_S , parallel to the Bott-periodic construction of Definition 3.1.11. The two constructions are compatible by design: the Bott element $\beta = 1 - [\mathcal{O}(-1)]$ of Section 3.1.6 is the image of the framed Bott class above under the linearization $\mathrm{Vect}^{\mathrm{gp}} \rightarrow \Omega^\infty K \rightarrow \Omega^\infty \mathrm{KH}$.

3.4. THE ZEROth SLICE OF THE SPHERE

The comparison Theorem 3.2.53 identified the slices of KGL with $\mathbb{A}^1$ -motivic cohomology, and Definition 3.2.68 took the resulting equivalence $s^0 \mathrm{KGL}_X \simeq H\mathbb{Z}_X$ as the definition of the motivic Eilenberg–MacLane spectrum. This closing section revisits the zeroth slice from the classical, geometric direction, following Bachmann–Elmanto [BE21] and Levine [Lev08], and proves two complementary statements:

- (i) over a perfect field k , the unit map induces an equivalence $s^0(\mathbb{1}_k) \xrightarrow{\sim} s^0(\mathrm{KGL}_k)$ (Theorem 3.4.8), via a birational analysis of the framed Hilbert scheme resting on the recognition principle (Theorem 3.3.11);
- (ii) Levine’s homotopy coniveau tower computes the slice filtration explicitly, and through it Bloch’s higher Chow groups re-identify $s^0 \mathrm{KGL} \simeq H\mathbb{Z}$ (Theorems 3.4.12 and 3.4.14).

Together they give the slice theorem of Levine and Voevodsky: $s^0(\mathbb{1}_k) \simeq H\mathbb{Z}_k$ (Corollary 3.4.15).

Throughout, k is a field. We write $\mathrm{Shv}_{\mathrm{Nis}, \mathbb{A}^1}(\mathrm{Sm}_k, \mathrm{Sp})$ for the ∞ -category of $\mathbb{A}^1$ -invariant Nisnevich sheaves of spectra on Sm_k — the home of the unstable slice filtration of Section 3.2.2 and the target of $\omega^\infty: \mathrm{SH}(k) \rightarrow \mathrm{Shv}_{\mathrm{Nis}, \mathbb{A}^1}(\mathrm{Sm}_k, \mathrm{Sp})$.

3.4.1. Birational description of the slice filtration

The slice filtration admits a description in terms of birational geometry, due to Bachmann–Elmanto.

Definition 3.4.1. An open immersion $U \hookrightarrow X$ in Sm_k is d -**birational** if U contains every point of X of codimension $\leq d$; equivalently, $X \setminus U$ has codimension $\geq d + 1$. A presheaf F on Sm_k is d -**rationally invariant** if $F(X) \xrightarrow{\sim} F(U)$ for every d -birational open immersion $U \hookrightarrow X$.

For $d = 0$ an open immersion is 0-birational precisely when U is dense, so 0-birational invariance is the classical notion of a birational sheaf. We write L_{bir}^d for the Bousfield localization of $\mathrm{Shv}_{\mathrm{Nis}, \mathbb{A}^1}(\mathrm{Sm}_k, \mathrm{Sp})$ onto the d -rationally invariant objects.

Proposition 3.4.2 ([BE21, §3]). *Let k be a perfect field and $d \geq 0$. The localizing subcategory $\mathrm{Shv}_{\mathrm{Nis}, \mathbb{A}^1}(\mathrm{Sm}_k, \mathrm{Sp})(d + 1)$ of $(d + 1)$ -effective objects (Section 3.2.2) coincides with the subcategory of L_{bir}^d -acyclic objects, yielding a localization sequence*

$$\mathrm{Shv}_{\mathrm{Nis}, \mathbb{A}^1}(\mathrm{Sm}_k, \mathrm{Sp})(d + 1) \hookrightarrow \mathrm{Shv}_{\mathrm{Nis}, \mathbb{A}^1}(\mathrm{Sm}_k, \mathrm{Sp}) \xrightarrow{L_{\mathrm{bir}}^d} L_{\mathrm{bir}}^d \mathrm{Shv}_{\mathrm{Nis}, \mathbb{A}^1}(\mathrm{Sm}_k, \mathrm{Sp}).$$

Consequently, for every $E \in \mathrm{Shv}_{\mathrm{Nis}, \mathbb{A}^1}(\mathrm{Sm}_k, \mathrm{Sp})$ there is a cofibre sequence

$$\mathrm{Fil}_{\mathrm{un-slice}}^{d+1}(E) \longrightarrow E \longrightarrow L_{\mathrm{bir}}^d(E).$$

Remark 3.4.3. Proposition 3.4.2 is the slice-filtration analogue of the Postnikov cofibre sequence $\tau_{\geq d+1} E \rightarrow E \rightarrow \tau_{\leq d} E$: the $(d + 1)$ -effective cover plays the role of the connective cover, and d -birational localization that of the Postnikov truncation.

Proof sketch. The generator $\Sigma^\infty(\mathbb{T}^{\wedge(d+1)} \wedge X_+)$ of $\mathrm{Shv}_{\mathrm{Nis}, \mathbb{A}^1}(\mathrm{Sm}_k, \mathrm{Sp})(d + 1)$ is the cofibre of (the suspension spectrum of) the open immersion

$$X \times (\mathbb{A}^{d+1} \setminus 0) \hookrightarrow X \times \mathbb{A}^{d+1},$$

which is d -birational since its complement $X \times \{0\}$ has codimension $d + 1$; hence every $(d + 1)$ -effective object is L_{bir}^d -acyclic. Conversely, the cofibre X/U of any d -birational open immersion is $(d + 1)$ -effective: when $Z = X \setminus U$ is smooth of codimension $c \geq d + 1$, the purity equivalence $X/U \simeq \mathrm{Th}_Z(N_{Z/X})$ exhibits it as c -effective, and for general Z the perfectness of k supplies a filtration $\emptyset = Z_m \subseteq \cdots \subseteq Z_0 = Z$ with each $Z_i \setminus Z_{i+1}$ smooth, reducing to the smooth case. The two inclusions identify $\mathrm{Shv}_{\mathrm{Nis}, \mathbb{A}^1}(\mathrm{Sm}_k, \mathrm{Sp})(d + 1)$ with the L_{bir}^d -acyclics, and the cofibre sequence is the recollement triangle evaluated on E . $\square$

Corollary 3.4.4. *Define L_{bir}^d on $\mathrm{SH}(k)$ analogously, by inverting all shifts $\Sigma_{\mathbb{T}}^\infty f_+[n]$ ($n \in \mathbb{Z}$) of d -birational open immersions f in Sm_k . The argument of Proposition 3.4.2 applies verbatim — the generators $\Sigma_{\mathbb{T}}^\infty(\mathbb{T}^{\wedge(d+1)} \wedge X_+)[n]$ of $\mathrm{SH}^{\mathrm{eff}}(k)(d + 1)$ are cofibres of d -birational immersions, and conversely X/U is $(d + 1)$ -effective by purity and stratification — and identifies the L_{bir}^d -acyclics with $\mathrm{SH}^{\mathrm{eff}}(k)(d + 1)$. In particular, for effective $E \in \mathrm{SH}^{\mathrm{eff}}(k)$ there is a cofibre sequence*

$$\mathrm{Fil}_{\mathrm{slice}}^{d+1}(E) \longrightarrow E \longrightarrow L_{\mathrm{bir}}^d(E),$$

so that $L_{\mathrm{bir}}^0(E) \simeq s^0(E)$.

3.4.2. The framed Hilbert scheme and the zeroth slice of the sphere

The recognition principle (Theorem 3.3.11) identifies the underlying space of an effective motivic spectrum with a group-completed framed presheaf: for $X \in \mathrm{Sm}_k$,

$$\Omega_{\mathbb{T}}^\infty \Sigma_{\mathbb{T}}^\infty X_+ \simeq L_{\mathrm{mot}} h_k^{\mathrm{fr}}(X)^{\mathrm{gp}}.$$

To apply the codimension bookkeeping of Definition 3.4.1 we need the right-hand side to be representable by honest schemes. This is supplied by the normally framed presentation (Definition 3.3.3) and the framed Hilbert scheme.

Construction 3.4.5 (Framed Hilbert scheme [Elm+21, §5.1]). Let $\mathrm{Hilb}^{\mathrm{lci}}(\mathbb{A}^n) \subseteq \mathrm{Hilb}(\mathbb{A}^n)$ be the open subscheme of the Hilbert scheme of points parametrizing finite lci subschemes, and let $Z_{\mathrm{lci}} \rightarrow \mathrm{Hilb}^{\mathrm{lci}}(\mathbb{A}^n)$ be the restriction of the universal subscheme $Z_{\mathrm{univ}} \subseteq \mathbb{A}^n \times \mathrm{Hilb}(\mathbb{A}^n)$; it is finite syntomic. Let $p: Z_{\mathrm{lci}} \rightarrow \mathrm{Hilb}^{\mathrm{lci}}(\mathbb{A}^n)$ denote the projection and $T \rightarrow Z_{\mathrm{lci}}$ the GL_n -torsor of trivializations $\mathcal{O}_{Z_{\mathrm{lci}}}^{\oplus n} \simeq \mathcal{N}_{Z_{\mathrm{lci}}/\mathbb{A}^n}$ of the conormal sheaf. For $X \in \mathrm{Sm}_k$ quasi-projective, set

$$\mathrm{Hilb}^{\mathrm{fr}}(\mathbb{A}^n, X) := p_*(T \times_k X),$$

the Weil restriction along the finite syntomic p of the smooth Z_{lci} -scheme $T \times_k X$. Since $T \times_k X$ is smooth and quasi-projective over Z_{lci} and p is finite syntomic, the Weil restriction exists and $\mathrm{Hilb}^{\mathrm{fr}}(\mathbb{A}^n, X)$ is smooth. There is a natural identification

$$\mathrm{Corr}_k^{\mathrm{fr}}(Y, X) \simeq \mathrm{colim}_n \mathrm{Hilb}^{\mathrm{fr}}(\mathbb{A}^n, X)(Y),$$

so $h_k^{\mathrm{fr}}(X)$ is ind-represented by the ind-smooth scheme $\mathrm{Hilb}^{\mathrm{fr}}(\mathbb{A}^\infty, X) = \mathrm{colim}_n \mathrm{Hilb}^{\mathrm{fr}}(\mathbb{A}^n, X)$.

The geometry of this ind-scheme is the central input: smoothness comes for free with the Weil-restriction presentation above, while the degreewise *rationality* of $\mathrm{Hilb}^{\mathrm{fr}}(\mathbb{A}^n, \mathrm{Spec} k)$ — via the birational identification of $\mathrm{Hilb}_d^{\mathrm{lci}}(\mathbb{A}^n)$ with $\mathrm{Sym}^d(\mathbb{A}^n)$ — is established by Bachmann–Elmanto [BE21, §2]. The complementary K -theoretic identification $\mathcal{V}_k \simeq \mathrm{kgl}_k$ of Proposition 3.3.17 has its origin in the study of the Hilbert scheme of infinite affine space by Hoyois–Jelisiejew–Nardin–Totaro–Yakerson [Hoy+25].

Lemma 3.4.6 ([BE21, §§2–3]). *If $U \hookrightarrow X$ is a d -birational open immersion of smooth quasi-projective k -schemes, then $\mathrm{Hilb}^{\mathrm{fr}}(\mathbb{A}^n, U) \hookrightarrow \mathrm{Hilb}^{\mathrm{fr}}(\mathbb{A}^n, X)$ is again d -birational. Consequently $h_k^{\mathrm{fr}}(U) \rightarrow h_k^{\mathrm{fr}}(X)$ is an L_{bir}^d -equivalence, and L_{bir}^d commutes with the forgetful functor $\gamma_*: \mathrm{H}^{\mathrm{fr}}(k) \rightarrow \mathrm{H}(k)$.*

Proposition 3.4.7 ([BE21, §§2 and 4]). *The forgetful and rank maps*

$$\mathrm{Hilb}^{\mathrm{fr}}(\mathbb{A}^\bullet, \mathrm{Spec} k) \xrightarrow{\mathrm{fgt}} \mathrm{Vect} \xrightarrow{\mathrm{rk}} \mathbb{N}$$

are L_{bir}^0 -equivalences; the composite is the degree.

Proof sketch. The torsor T is Zariski-locally $\mathrm{GL}_n \simeq \mathbb{A}^{n^2} \setminus \{\det = 0\}$, so $\mathrm{Hilb}^{\mathrm{fr}}(\mathbb{A}^n, \mathrm{Spec} k) = p_* T$ is an affine-bundle-like scheme over $\mathrm{Hilb}^{\mathrm{lci}}(\mathbb{A}^n)$, which decomposes by degree into rational, $\mathbb{A}^1$ -connected components (on the locus of reduced length- d subschemes, $\mathrm{Hilb}^{\mathrm{lci}}(\mathbb{A}^n)$ agrees with $\mathrm{Sym}^d(\mathbb{A}^n)$). After 0-birational localization only the discrete set of degrees survives, identifying $L_{\mathrm{bir}}^0 \mathrm{Hilb}^{\mathrm{fr}}(\mathbb{A}^\bullet, \mathrm{Spec} k)$ with $\mathbb{N}$; the rank map $\mathrm{Vect} \rightarrow \mathbb{N}$ is an L_{bir}^0 -equivalence for the same reason (the Grassmannians are rational). See [BE21, §§2 and 4] for the details. $\square$

Theorem 3.4.8 ([BE21, §4]). *Let k be a perfect field. The unit map $\mathbb{1}_k \rightarrow \mathrm{kgl}_k$ is an L_{bir}^0 -equivalence in $\mathrm{SH}(k)$ (in the sense of Corollary 3.4.4); equivalently, since both source and target are effective, it induces an equivalence on zeroth slices*

$$s^0(\mathbb{1}_k) \xrightarrow{\sim} s^0(\mathrm{KGL}_k).$$

Proof sketch. By Proposition 3.4.7 the comparison $h_k^{\text{fr}}(\text{Spec } k) \rightarrow \text{Vect}$ is an L_{bir}^0 -equivalence in $H(k)$, and by Lemma 3.4.6 it remains one in $H^{\text{fr}}(k)$. The framed suspension $\Sigma_{\text{fr}}^\infty$ preserves L_{bir}^0 -equivalences (it is built from left adjoints sending 0-birational immersions to the maps inverted in Corollary 3.4.4; cf. [BE21, §3]), so using $\mathcal{V}_k = \Sigma_{\text{fr}}^\infty \text{Vect}^{\text{gp}} \simeq \text{kg}l_k$ (Proposition 3.3.17), the unit $\mathbb{1}_k = \Sigma_{\text{fr}}^\infty(h_k^{\text{fr}}(\text{Spec } k)^{\text{gp}}) \rightarrow \text{kg}l_k$ becomes an L_{bir}^0 -equivalence in $\text{SH}(k)$. For an effective $E \in \text{SH}^{\text{eff}}(k)$, Corollary 3.4.4 with $d = 0$ gives $L_{\text{bir}}^0(E) \simeq \text{cofib}(\text{Fil}_{\text{slice}}^1 E \rightarrow E) \simeq s^0(E)$; applied to $\mathbb{1}_k$ and $\text{kg}l_k$ this turns the L_{bir}^0 -equivalence into $s^0(\mathbb{1}_k) \simeq s^0(\text{kg}l_k) = s^0(\text{KGL}_k)$, the last equality because $\text{kg}l_k = \text{Fil}_{\text{slice}}^0 \text{KGL}_k$. $\square$

3.4.3. Levine’s homotopy coniveau tower

The slice filtration of Construction 3.2.8 is defined by an abstract universal property. Levine [Lev08] gave an explicit geometric model for it, the *homotopy coniveau tower*, a homotopy-invariant refinement of the classical coniveau filtration.

Recall the classical coniveau filtration of a presheaf of spectra E on a Noetherian scheme: writing $E_Z(U) := \text{fib}(E(U) \rightarrow E(U \setminus Z))$ for sections supported on Z ,

$$E^{(p)}(U) := \text{colim}_{Z \subseteq U, \text{codim } Z \geq p} E_Z(U), \quad \text{gr}^{(p)} E(U) \simeq \bigoplus_{x \in U^{(p)}} E_x(\mathcal{O}_{U,x}).$$

This filtration is functorial only for flat maps — the wrong variance for $\mathbb{A}^1$ -localization, whose transition maps are closed immersions. The homotopy coniveau tower repairs this by imposing support conditions transverse to all faces of the cosimplicial scheme $\mathbb{A}^\bullet$.

Definition 3.4.9 ([Lev08]). For $E: \text{Sm}_k^{\text{op}} \rightarrow \text{Sp}$ and $p \geq 0$, set

$$E_n^{(p)}(X) := \text{colim}_Z E_Z(X \times \Delta^n),$$

the colimit over closed $Z \subseteq X \times \Delta^n$ meeting every face $X \times \Delta^i$ in codimension $\geq p$ (here $\Delta^\bullet$ is the cosimplicial scheme of Definition 1.2.10; recall that its faces are *not* coordinate inclusions under $\Delta^n \simeq \mathbb{A}^n$), and

$$E^{(p)}(X) := \text{colim}_{[n] \in \Delta^{\text{op}}} E_n^{(p)}(X).$$

These assemble into the **homotopy coniveau tower**

$$\dots \rightarrow E^{(p)} \rightarrow E^{(p-1)} \rightarrow \dots \rightarrow E^{(0)} = L_{\mathbb{A}^1} E,$$

whose p -th graded piece is supported on the codimension- p points of $X \times \Delta^\bullet$ transverse to the faces.

Theorem 3.4.10 (Levine [Lev08]). Let k be a perfect field and $E \in \text{Shv}_{\text{Nis}, \mathbb{A}^1}(\text{Sm}_k, \text{Sp})$. If k is finite, impose Levine’s axiom A3: for every finite Galois extension k'/k with group G and every $X \in \text{Sm}_k$, the canonical map

$$E(X) \longrightarrow E(X_{k'})^{hG}$$

is an equivalence after inverting $|G|$. This condition is automatic when E is the zeroth level of a fibrant $\mathbb{P}^1$ - Ω -spectrum [Lev08, Remark 2.1.1]. Then:

- (i) (Functoriality.) The tower $E^{(\star)}$, a priori functorial only for smooth morphisms, extends to a functor $\text{Sm}_k^{\text{op}} \rightarrow \text{Fil}(\text{Sp})$; the extension rests on Chow’s moving lemma.
- (ii) Each $E^{(p)}$ is $\mathbb{A}^1$ -invariant.

- (iii) ($\mathbb{P}^1$ -delooping.) For $p \geq 1$ there is a natural equivalence $(\Omega_{\mathbb{P}^1} E)^{(p-1)} \simeq \Omega_{\mathbb{P}^1} (E^{(p)})$.
- (iv) (Localization.) For a closed immersion $Z \hookrightarrow X$ with open complement U , there is a fibre sequence $E_Z^{(p)}(X) \rightarrow E^{(p)}(X) \rightarrow E^{(p)}(U)$; in particular $E^{(p)}$ is a Nisnevich sheaf.
- (v) For $q \geq p$, the canonical map $(E^{(q)})^{(p)} \xrightarrow{\sim} E^{(q)}$ is an equivalence.

Proof sketch. For (i), a morphism $f: Y \rightarrow X$ induces f^* only on the sub-presheaf $E_f^{(p)}(X)$ of supports in good position relative to f ; Chow’s moving lemma asserts $E_f^{(p)}(X) \xrightarrow{\sim} E^{(p)}(X)$, and functoriality is obtained by passing to an auxiliary category of schemes-with-good-position-data on which the assignment is strictly functorial and whose fibres over Sm_k are weakly contractible. Statement (iii) is the “horn moving lemma”; (iv) rests on the étale-neighbourhood invariance of $E_Z^{(p)}$; and (v) is a further moving-lemma argument. We refer to [Lev08] for the (substantial) details. $\square$

Corollary 3.4.11 ([Lev08]). *Under the hypotheses of Theorem 3.4.10, the homotopy coniveau tower computes the unstable slice filtration:*

$$\mathrm{Fil}_{\mathrm{un-slice}}^p(E) \simeq E^{(p)}, \quad p \geq 0.$$

Via the $\mathbb{P}^1$ -delooping of Theorem 3.4.10(iii), the slices of a $\mathbb{T}$ -spectrum E' are computed levelwise by the homotopy coniveau tower of the S^1 -spectra $\omega^\infty(\mathbb{T}^{\wedge j} \otimes E')$.

3.4.4. Bloch’s cycle complex and the identification $s^0\mathrm{KGL} \simeq H\mathbb{Z}$

Applied to algebraic K -theory, the homotopy coniveau tower recovers Bloch’s higher Chow groups, by a theorem of Bloch and Levine.

Theorem 3.4.12 ([Blo86; Lev08]). *Let k be a perfect field and let $K: \mathrm{Sm}_k^{\mathrm{op}} \rightarrow \mathrm{Sp}$ be the algebraic K -theory presheaf. The associated graded of its homotopy coniveau tower is Bloch’s cycle complex,*

$$\mathrm{gr}^{(p)} K \simeq |z^p(-, \bullet)|,$$

where $z^p(X, n)$ is the group of codimension- p cycles on $X \times \Delta^n$ meeting each face in the expected codimension.

Definition 3.4.13. The **higher Chow groups** of $X \in \mathrm{Sm}_k$ are $\mathrm{CH}^p(X, m) := H_m(z^p(X, \bullet))$, and the **motivic cohomology** is $H^n(X, \mathbb{Z}(p)) := \mathrm{CH}^p(X, 2p - n)$. With the weight convention of Definition 3.2.42 (no topological shift), Theorem 3.4.12 reads $\mathrm{gr}^{(p)} K \simeq \mathbb{Z}(p)$, the weight- p motivic cohomology spectrum.

Theorem 3.4.14 ([Lev08]). *Over a perfect field k , the zeroth slice of K -theory is the integral motivic cohomology spectrum:*

$$s^0(\mathrm{KGL}_k) \simeq H\mathbb{Z}_k.$$

Proof sketch. Algebraic K -theory is the zeroth level of the $\mathbb{P}^1$ - Ω -spectrum KGL , so over a finite field it satisfies axiom A3 by [Lev08, Remark 2.1.1]; hence Corollary 3.4.11 applies over every perfect field. Together with Theorem 3.4.12, it reduces the claim to $\mathrm{gr}^{(0)} K(F) \simeq \mathbb{Z}$ for every field F . By definition $\mathrm{gr}^{(0)} K(F) = \mathrm{cofib}(K^{(1)}(F) \rightarrow K(F))$, and $K^{(1)}(F)$ is the fibre of $K(F) \simeq K(\mathbb{A}_F^\bullet) \rightarrow K(\Delta_0^\bullet)$, where Δ_0^n is the semilocalization of $\mathbb{A}_F^n$ at its $n + 1$ vertices. One must show that the simplicial diagram $\cdots \rightrightarrows K(\Delta_0^1) \rightrightarrows K(F) \rightarrow \mathbb{Z}$ is a colimit, i.e. that each $K(\Delta_0^n) \rightarrow \mathrm{Map}(\partial\Delta^n, K(\Delta_0^n))$ is surjective on π_0 . Comparing with the analogous diagram for homotopy K -theory and using that $\partial\Delta^n$ is K_1 -regular [Vor79], the map $K \rightarrow \mathrm{KH}$ is an

isomorphism on $\pi_{\leq 1}$ over these semilocal rings, reducing the surjectivity to the elementary case of KH. Hence $\mathrm{gr}^{(0)}K(F) \simeq \mathbb{Z}$, and $s^0(\mathrm{KGL}_k) \simeq H\mathbb{Z}_k$. $\square$

Corollary 3.4.15 (Slice theorem). *For a perfect field k , the zeroth slice of the motivic sphere spectrum is motivic cohomology:*

$$s^0(\mathbb{1}_k) \simeq H\mathbb{Z}_k.$$

Proof. Combine Theorem 3.4.8 and Theorem 3.4.14: $s^0(\mathbb{1}_k) \simeq s^0(\mathrm{KGL}_k) \simeq H\mathbb{Z}_k$. $\square$

Remark 3.4.16. The identification $s^0\mathrm{KGL} \simeq H\mathbb{Z}$ obtained here over a field by Levine’s tower is the classical counterpart of the integral, base-independent statement built into Definition 3.2.68, where $H\mathbb{Z}_X = s^0\mathrm{KGL}_X$ was extracted from the comparison Theorem 3.2.53 between the slices of KGL and $\mathbb{A}^1$ -motivic cohomology. The framed route additionally yields $s^0(\mathbb{1}_k) \simeq s^0(\mathrm{KGL}_k)$, a statement about the sphere that is invisible to the purely KGL-theoretic argument of Section 3.2.9. For the relation of both routes to algebraic cobordism — and why the Hopkins–Morel–Hoyois equivalence of Section 3.5.3 does *not* constitute a third one — see Remark 3.5.23.

3.5. THE MOTIVIC THOM SPECTRUM MGL

This closing section constructs the protagonist that the orientation theory of Section 3.2.4 has been pointing at all along: the *algebraic cobordism spectrum* MGL. It has a tautological orientation over every base, and over a field it is the universal oriented motivic ring spectrum in the homotopy-category sense of Theorem 3.5.13. The chapter so far has been organized around KGL: its slice filtration produced $H\mathbb{Z} = s^0\mathrm{KGL}$ and $\mathbb{A}^1$ -motivic cohomology (Section 3.2), and the framed technology of Sections 3.3 and 3.4 identified $s^0(\mathbb{1}_k) \simeq s^0(\mathrm{KGL}_k) \simeq H\mathbb{Z}_k$. We now show that both protagonists are governed by a single object. From MGL one reaches $H\mathbb{Z}$ by *killing* the Lazard generators — the Hopkins–Morel–Hoyois equivalence $\mathrm{MGL}/(a_1, a_2, \dots)[1/c] \simeq H\mathbb{Z}[1/c]$ of Section 3.5.3 — and one reaches KGL by *inverting* the first of them — the motivic Conner–Floyd theorem of Section 3.5.4. Everything in between is the motivic avatar of the chromatic picture, interpolated precisely by the slices $s^t\mathrm{MGL} \simeq H\mathbb{Z} \otimes \mathbb{T}^{\otimes t} \otimes \mathbb{L}_t$.

The topological template is the complex cobordism spectrum MU, whose role as universal complex-oriented theory and entry point to chromatic homotopy theory was recalled in Remark 3.2.23. But MU is not merely a universal gadget: by Pontryagin–Thom theory its homotopy groups are *geometric*, namely cobordism classes of stably almost-complex manifolds, and Quillen described $\mathrm{MU}^*(X)$ directly in terms of proper maps of manifolds. The motivic story repeats both aspects. The universal side is the content of Sections 3.5.1 and 3.5.2; the geometric side appears twice, once through Levine–Morel’s algebraic cobordism Ω^* — generated by projective morphisms $Z \rightarrow X$ from smooth varieties, with $\Omega^*(X) \cong \mathrm{MGL}^{2*,*}(X)$ in characteristic zero (Corollary 3.5.26) — and once through the framed model $\mathrm{MGL} \simeq \Sigma_{\mathbb{T}, \mathrm{fr}}^\infty \mathrm{FSyn}$ of Section 3.5.5, which continues the Hilbert-scheme thread of Section 3.4: where K-theory counts vector bundles (Proposition 3.3.17), cobordism counts finite syntomic schemes.

Throughout this section we follow the modern construction of Bachmann–Hoyois [BH21, §16], which produces MGL as the Thom spectrum of the *motivic J-homomorphism* and yields its $\mathbb{E}_\infty$ - and normed structures for free; the original definition by Grassmannian colimits, due to Voevodsky [Voe98], is recovered as a computation.

3.5.1. Thom spectra and the motivic J-homomorphism

In topology, MU is the Thom spectrum of the J-homomorphism $j: BU \rightarrow \mathcal{P}ic(\mathbb{S}) \subset \mathcal{S}p$, i.e. the colimit of the functor sending a point of BU to the invertible spectrum determined by the corresponding virtual vector bundle. From this perspective the $\mathbb{E}_\infty$ -structure of MU is inherited from a single fact: j is a map of $\mathbb{E}_\infty$ -monoids. The motivic transcription requires two preliminary constructions: an invertible motivic spectrum $\mathbb{S}^\xi$ for every vector bundle ξ , and a colimit-formation turning a family of such data into one spectrum.

Warning 3.5.1 (Two Picard objects). Two different “Picard” objects now share the stage. The motivic Picard spectrum $\mathcal{P}ic$ of Section 3.2.4, with $\Omega^\infty \mathcal{P}ic \simeq B\mathbb{G}_m$, classifies *line bundles*; it is an object of $SH(S)$. The **Picard anima of the stable category**,

$$\mathrm{Sph}(S) := \mathcal{P}ic(SH(S)) \subset SH(S)^\simeq,$$

is the grouplike $\mathbb{E}_\infty$ -anima of $\otimes$ -invertible *motivic spectra*; it is the motivic analogue of the Picard anima $\mathcal{P}ic(\mathbb{S})$ of Section 3.2.4 and is *not* an object of $SH(S)$ but a presheaf of anima $X \mapsto \mathrm{Sph}(X)$ on Sm_S . We reserve the notation Sph for the latter, following [BH21, §16.1].

Construction 3.5.2 (Thom spectra of vector bundles [BH21, §16.2]). Let X be a scheme and ξ a vector bundle on X , with $\mathrm{Vect}(X)$ the groupoid of vector bundles under direct sum. The **Thom space** of ξ is the quotient $\mathrm{Th}_X(\xi) := \xi/(\xi \setminus 0)$ of motivic spaces, pointed by the collapsed complement of the zero section; for the trivial bundle, $\mathrm{Th}_X(\mathcal{O}^n) \simeq \mathbb{T}_X^{\otimes n}$ by Corollary 1.4.9. Since open complements of zero sections multiply under direct sum, the assignment $\xi \mapsto \mathrm{Th}_X(\xi)$ refines to a symmetric monoidal functor to pointed Nisnevich sheaves, and composing with motivic localization and stabilization yields a symmetric monoidal functor

$$\mathbb{S}^{(-)}: \mathrm{Vect}(X) \longrightarrow SH(X), \quad \xi \longmapsto \mathbb{S}^\xi := \Sigma^\infty \mathrm{Th}_X(\xi).$$

Each $\mathbb{S}^\xi$ is $\otimes$ -invertible: Zariski-locally ξ is trivial, so $\mathbb{S}^\xi$ is locally a Tate twist $\mathbb{T}^{\otimes n}$, and invertibility glues. Hence $\mathbb{S}^{(-)}$ lands in $\mathrm{Sph}(X)$.

Definition 3.5.3 (Motivic J-homomorphism [BH21, §16.2]). Since $\mathrm{Sph}(X)$ is a *grouplike* $\mathbb{E}_\infty$ -anima, the symmetric monoidal functor $\mathbb{S}^{(-)}$ of Construction 3.5.2 factors uniquely through the group completion $\mathrm{Vect}(X)^{\mathrm{gp}}$. As X varies, Sph is a Zariski sheaf, hence right Kan extended from affine schemes, while the right Kan extension of $X \mapsto \mathrm{Vect}(X)^{\mathrm{gp}}$ from affines is the Thomason–Trobaugh K -theory anima $\Omega^\infty K$. The resulting natural transformation of presheaves of grouplike $\mathbb{E}_\infty$ -anima

$$j: \Omega^\infty K \longrightarrow \mathrm{Sph}, \quad \xi \longmapsto \mathbb{S}^\xi,$$

is the **motivic J-homomorphism**. It is natural for pullbacks and even for finite étale norms [BH21, §16.2].

Remark 3.5.4 (The J-homomorphism factors through KH). The presheaf Sph is in fact a *cdh* sheaf, and the *cdh* sheafification of $\Omega^\infty K$ is $\Omega^\infty KH$ by the theorem of Kerz–Strunk–Tamme [KST18] (see [BH21, Remark 16.11]). The J-homomorphism therefore factors through homotopy K -theory — the same spectrum whose slice filtration defined $\mathbb{A}^1$ -motivic cohomology in Section 3.2.6. One consequence [BH21, Remark 16.11]: the Thom spectrum functor below inverts motivic equivalences of presheaves over $\Omega^\infty K$, over an arbitrary base.

Definition 3.5.5 (Motivic Thom spectrum functor [BH21, Definition 16.1, Remark 16.5]). Let S be a scheme. The **motivic Thom spectrum functor**

$$\mathbb{M}_S: \mathrm{PShv}(\mathrm{Sm}_S)_{/\mathrm{Sph}} \longrightarrow SH(S)$$

sends $\phi: A \rightarrow \mathbf{Sph}$ to the colimit

$$\mathbb{M}_S(\phi) \simeq \operatorname{colim}_{\substack{f: U \rightarrow S \text{ smooth} \\ \alpha \in A(U)}} f_{\#} \phi(\alpha),$$

indexed by the category of elements of A ; here $f_{\#}: \mathbf{SH}(U) \rightarrow \mathbf{SH}(S)$ is the left adjoint to f^* from the six-functor formalism of Chapter 2. (Bachmann–Hoyois define $\mathbb{M}_S$ on the larger right-lax slice $\mathbf{PShv}((\mathbf{Sm}_S)_{//\mathbf{SH}})$, but the displayed restriction suffices for MGL and its relatives [BH21, Remark 16.3]. We write $\mathbb{M}_S$ rather than the M_S of *op. cit.* to avoid a collision with our notation for motivic spectra of smooth S -schemes, Notation 3.1.2.)

Remark 3.5.6 (First properties). Three properties of $\mathbb{M}_S$ will be used repeatedly.

- (i) **(Suspension spectra.)** For the constant family $A \rightarrow \mathbf{Sph}$, $\alpha \mapsto \mathbb{1}$, one recovers $\mathbb{M}_S(A, \mathbb{1}) \simeq \Sigma_+^{\infty} L_{\text{mot}} A$ [BH21, Example 16.6]: the Thom spectrum functor extends the (motivically localized) suspension spectrum functor, twisting it by the prescribed family of invertible spectra.
- (ii) **(Multiplicativity.)** $\mathbb{M}_S$ is symmetric monoidal; in particular it sends $\mathbb{E}_{\infty}$ -maps $A \rightarrow \mathbf{Sph}$ to $\mathbb{E}_{\infty}$ -rings in $\mathbf{SH}(S)$. On the span category

$$\operatorname{Span}(\mathbf{Sm}_S, \text{all}, \text{fst}),$$

it sends natural transformations to normed spectra [BH21, §16.3].

- (iii) **(Base change.)** For $f: T \rightarrow S$ one has $f^* \mathbb{M}_S \simeq \mathbb{M}_T f^*$ [BH21, Lemma 16.7], by smooth base change.

With the machine in place, algebraic cobordism is one line. Write $e: \Omega^{\infty} K^{\text{rk}=0} \hookrightarrow \Omega^{\infty} K$ for the inclusion of the rank-0 part, as in the proof of Lemma 3.2.39.

Definition 3.5.7 (Algebraic cobordism spectrum). The **algebraic cobordism spectrum** of a scheme S is the motivic Thom spectrum of the rank-0 motivic J-homomorphism:

$$\text{MGL}_S := \mathbb{M}_S(j \circ e: \Omega^{\infty} K^{\text{rk}=0} \rightarrow \mathbf{Sph}).$$

Theorem 3.5.8 (Comparison with Voevodsky’s definition [BH21, Theorem 16.13]). Let $\text{Gr}_n = \operatorname{colim}_k \text{Gr}_n(\mathbb{A}_S^k)$ be the infinite Grassmannian of n -planes with tautological bundle γ_n , and let $\gamma: \text{Gr}_{\infty} = \operatorname{colim}_n \text{Gr}_n \rightarrow \Omega^{\infty} K^{\text{rk}=0}$ classify $\gamma_n - \mathcal{O}^n$ on the n -th stage. For every scheme S , the map γ induces an equivalence

$$\operatorname{colim}_n \mathbb{T}_S^{\otimes(-n)} \otimes \Sigma^{\infty} \text{Th}(\gamma_n) \xrightarrow{\sim} \text{MGL}_S,$$

where the transition maps are induced by $\mathbb{T} \otimes \text{Th}(\gamma_n) \simeq \text{Th}(\gamma_{n+1}|_{\text{Gr}_n}) \rightarrow \text{Th}(\gamma_{n+1})$. The left-hand side is Voevodsky’s original definition [Voe98].

Proof sketch. The essential point is that γ is a motivic equivalence: Zariski-locally, $\Omega^{\infty} K^{\text{rk}=0}$ is the group completion of $\coprod_n \text{BGL}_n$, and $\text{Gr}_{\infty} \simeq L_{\text{Nis}, \mathbb{A}^1} \text{BGL}$ by Morel–Voevodsky’s representability theorem (compare the proof of Lemma 3.2.39). When S is regular, $\Omega^{\infty} K$ is an $\mathbb{A}^1$ -invariant Nisnevich sheaf on $\mathbf{Sm}_S$, so $\mathbb{M}_S(j \circ -)$ inverts motivic equivalences (Remark 3.5.4) and the claim follows. The general case reduces to $S = \operatorname{Spec} \mathbb{Z}$ by the base-change property of Remark 3.5.6, since $\Omega^{\infty} K^{\text{rk}=0}|_{\mathbf{Sm}_S}$ is, up to Zariski equivalence, pulled back from $\mathbf{Sm}_{\mathbb{Z}}$ [BH21, Lemma 16.12]. $\square$

Remark 3.5.9 ($\mathbb{E}_\infty$ -structure, norms, and periodization). Since $j \circ e$ is a map of $\mathbb{E}_\infty$ -objects, MGL_S is an $\mathbb{E}_\infty$ -ring spectrum for free, and since j is natural on the span category, MGL_S is even a *normed* spectrum: this is [BH21, Theorem 16.19], the original motivation for the Thom-spectrum formalism of *op. cit.* (earlier constructions of the $\mathbb{E}_\infty$ -structure relied on point-set models; see the references there). We refer to Chapter A for the norm formalism. Applying M_S to j itself, rather than to its rank-0 part, produces the $(2, 1)$ -periodization

$$\mathrm{M}_S(j) \simeq \bigvee_{n \in \mathbb{Z}} \mathrm{MGL}_S \otimes \mathbb{T}^{\otimes n}$$

[BH21, Remark 16.14], the MGL-analogue of the Bott-periodic KGL of Sections 3.1.2 and 3.1.6; it is again a normed spectrum.

Remark 3.5.10 (Other structure groups). Replacing K-theory by the K -theory of forms — equivalently, restricting the tautological bundles to the families SL_n , Sp_{2n} , O_n , SO_n of structure groups — the same machine produces normed motivic Thom spectra MSL , MSp , MO , MSO [BH21, Examples 16.22–16.24]. They are universal homes for the corresponding weakened notions of orientation; in particular MSL is universal for SL -oriented theories, while Chow–Witt and Milnor–Witt theories provide related examples rather than being represented by MSL itself. We will not pursue them here, except to note in Section 3.5.5 that their framed models run entirely parallel to that of MGL.

3.5.2. The universal oriented ring spectrum

We now equip MGL with its tautological orientation over an arbitrary base. Over a field, Theorem 3.5.13 proves that it is the initial oriented theory in the homotopy-category sense, the motivic counterpart of Quillen’s theorem for MU recalled in Remark 3.2.23.

Construction 3.5.11 (The tautological orientation). By Remark 3.2.25((iii)), the reduced motive of the Picard stack is the $\mathbb{T}$ -shifted Thom spectrum of the tautological line bundle: under the identification $\Omega^\infty \mathcal{P}ic \simeq \mathrm{colim}_m \mathbb{P}_S^m$ of Remark 3.2.27, with $\mathrm{Gr}_1 = \mathbb{P}_S^\infty$ and $\gamma_1 = \mathcal{O}(-1)$,

$$\widetilde{\mathrm{M}}_S(\Omega^\infty \mathcal{P}ic) \simeq \Sigma^\infty \mathrm{Th}(\gamma_1) = \mathrm{MGL}(1) \otimes \mathbb{T}_S, \quad \mathrm{MGL}(1) := \mathbb{T}_S^{\otimes(-1)} \otimes \Sigma^\infty \mathrm{Th}(\gamma_1).$$

Let $u_1: \mathrm{MGL}(1) \rightarrow \mathrm{MGL}$ be the first structure map of the colimit of Theorem 3.5.8. The **tautological orientation** of MGL is the composite

$$c^{\mathrm{MGL}}: \widetilde{\mathrm{M}}_S(\Omega^\infty \mathcal{P}ic) \simeq \mathrm{MGL}(1) \otimes \mathbb{T}_S \xrightarrow{u_1 \otimes \mathbb{T}_S} \mathrm{MGL} \otimes \mathbb{T}_S.$$

The basepoint condition of Definition 3.2.24 holds: restricting along $[O(1)]: \mathbb{T}_S \rightarrow \widetilde{\mathrm{M}}_S(\Omega^\infty \mathcal{P}ic)$ corresponds to restricting $\mathrm{Th}(\gamma_1)$ along $\mathbb{P}^0 \hookrightarrow \mathbb{P}^\infty$, where γ_1 trivializes; the resulting map $\mathbb{T}_S \rightarrow \mathrm{MGL} \otimes \mathbb{T}_S$ is the unit cell $\mathbb{1} \simeq \mathrm{MGL}(0) \rightarrow \mathrm{MGL}$ tensored with $\mathbb{T}_S$, i.e. $\eta_{\mathrm{MGL}} \otimes \mathbb{T}_S$. This is the orientation denoted th^{MGL} in [PPR08]. By Theorem 3.2.29, MGL satisfies the projective bundle formula, and Remark 3.2.32 equips it with Chern classes and a formal group law.

Proposition 3.5.12 (Thom isomorphism [BH21, Proposition 16.28, Examples 16.30 and 16.31]). Let S be a scheme and $\xi \in K_0^{\mathrm{rk}=0}(X)$ a virtual bundle of rank 0 on $X \in \mathrm{Sm}_S$.

- (i) There is a canonical equivalence of MGL-modules $\mathbb{S}^\xi \otimes \mathrm{MGL}_X \simeq \mathrm{MGL}_X$; equivalently, for a rank- n bundle ξ , $\Sigma^\infty \mathrm{Th}(\xi) \otimes \mathrm{MGL} \simeq \mathbb{T}^{\otimes n} \otimes \mathrm{M}_S(X) \otimes \mathrm{MGL}$.
- (ii) Taking ξ universal, the shearing equivalence identifies the co-operations: $\mathrm{MGL} \otimes \mathrm{MGL} \simeq \mathrm{MGL} \otimes \Sigma_+^\infty \mathrm{L}_{\mathrm{mot}} \Omega^\infty K^{\mathrm{rk}=0}$.

More generally, for any homotopy MGL-algebra E , base change along $\mathrm{MGL} \rightarrow E$ yields $\Sigma^\infty \mathrm{Th}(\xi) \otimes E \simeq \mathbb{T}^{\otimes n} \otimes M_S(X) \otimes E$. When $S = \mathrm{Spec}(k)$ is a field, Theorem 3.5.13 applies this statement to every oriented E and recovers the Thom isomorphism of Remark 3.2.33.

Proof sketch. This is an instance of the Mahowald-style Thom isomorphism for Thom spectra: for a grouplike A_∞ -object A with an A_∞ -map $\psi: A \rightarrow \mathrm{Sph}$ and any $\phi: B \rightarrow A$, the *shearing automorphism* $(a, b) \mapsto (a + \phi(b), b)$ of $A \times B$ over A induces $\mathbb{M}_S(\psi) \otimes \mathbb{M}_S(\psi\phi) \simeq \mathbb{M}_S(\psi) \otimes \Sigma_+^\infty L_{\mathrm{mot}} B$, using only that $\mathbb{M}_S$ is symmetric monoidal. Apply this with $\psi = j \circ e$ and $\phi = \xi$, resp. $\phi = \mathrm{id}$. $\square$

Theorem 3.5.13 (Universality; Panin–Pimenov–Röndigs [PPR08]). *Let k be a field and $E \in \mathrm{CAlg}(\mathrm{hSH}(k))$ a homotopy commutative motivic ring spectrum. The assignment $\varphi \mapsto \varphi_*(c^{\mathrm{MGL}})$ defines a bijection*

$$\mathrm{Hom}_{\mathrm{CAlg}(\mathrm{hSH}(k))}(\mathrm{MGL}, E) \xrightarrow{\sim} \{\text{orientations of } E\}.$$

Proof sketch. Given an orientation c_E , Remark 3.2.32 produces Chern classes and, via the splitting principle, a Thom class $th(\xi) \in \tilde{E}^{2n,n}(\mathrm{Th}(\xi))$ for every rank- n bundle ξ , multiplicative in ξ and natural; in particular classes $th(\gamma_n) \in \tilde{E}^{2n,n}(\mathrm{Th}(\gamma_n))$ compatible with the transition maps of Theorem 3.5.8. The projective bundle formula computes $E^{*,*}(\mathrm{Gr}_n)$ as a power series ring over $E^{*,*}(k)$ on Chern classes, and the Thom isomorphism determined by $th(\gamma_n)$ then computes $\tilde{E}^{*,*}(\mathrm{Th}(\gamma_n))$ as a free module; the transition maps in the resulting inverse system are surjective, so $\lim^1$ vanishes by Mittag-Leffler and the compatible system $\{th(\gamma_n)\}$ assembles uniquely into a class $\varphi \in E^{0,0}(\mathrm{MGL})$, i.e. a map $\varphi: \mathrm{MGL} \rightarrow E$ with $\varphi_*(c^{\mathrm{MGL}}) = c_E$. Multiplicativity of φ is checked on the products $\mathrm{Gr}_m \times \mathrm{Gr}_n$ by the Whitney formula, using again the $\lim^1$ -vanishing. We refer to [PPR08] for the details. $\square$

Remark 3.5.14 (Refinements and history). A version of Theorem 3.5.13 was first formulated by Vezzosi [Vez01]. The theorem is an *unstructured* universality: it classifies maps in the homotopy category. Promoting it to a classification by coherent $\mathbb{E}_\infty$ -mapping spaces inside SH is subtler. In the *non- $\mathbb{A}^1$ -invariant* world of Warning 2.1.30, Annala–Hoyois–Iwasa construct a canonical $\mathbb{E}_\infty$ -structure on $\mathrm{MGL} \in \mathrm{MotSp}(S)$ over qcqs derived schemes and prove that it is initial among oriented objects of $\mathrm{CAlg}(\mathrm{hMotSp}(S))$ [AHI24a, Theorem 1.1]. This remains a homotopy-category universal property; it should not be read as a theorem about the full coherent mapping space of $\mathbb{E}_\infty$ -rings. Their Conner–Floyd theorem for genuine K -theory (recalled in Chapter 2) is its first application. The $\mathbb{A}^1$ -invariant MGL studied here is the image of theirs under the localization $\mathrm{MotSp}(S) \rightarrow \mathrm{SH}(S)$.

Construction 3.5.15 (The Lazard classifying map). By Construction 3.5.11 and Remark 3.2.32, MGL carries a formal group law F_{MGL} over the graded coefficient ring

$$\mathrm{MGL}^{-2*,-*}(S) := \bigoplus_{n \in \mathbb{Z}} \mathrm{MGL}^{-2n,-n}(S), \quad \mathrm{MGL}^{-2n,-n}(S) = \pi_{2n,n} \mathrm{MGL}(S).$$

By Lazard’s theorem (Remark 3.2.23), formal group laws are classified by the Lazard ring $L \cong \mathbb{Z}[a_1, a_2, \dots]$, $|a_i| = i$; we write

$$\theta_S: L \longrightarrow \mathrm{MGL}^{-2*,-*}(S)$$

for the classifying homomorphism of F_{MGL} , natural in S . Theorem 3.5.13 now expresses the *weak* universality of F_{MGL} : for every oriented (E, c_E) over a field, the classifying map of F_E factors as $L \xrightarrow{\theta} \mathrm{MGL}^{-2*,-*} \xrightarrow{\varphi_*} E^{-2*,-*}$. Whether θ itself is an isomorphism — the *strong*

universality, Quillen’s theorem for MU — is a much deeper question: at the time of [PPR08] it was known only as an unpublished announcement of Hopkins and Morel in characteristic zero. Its resolution is the subject of the next subsection.

3.5.3. The Hopkins–Morel–Hoyois equivalence

Quillen’s computation $\pi_*\mathrm{MU} \cong \mathbb{L}$ admits a spectrum-level reformulation: the quotient of MU by a regular sequence of polynomial generators $a_1, a_2, \dots \in \pi_*\mathrm{MU}$ is an Eilenberg–Mac Lane spectrum,

$$\mathrm{MU}/(a_1, a_2, \dots) \simeq H\mathbb{Z}.$$

Read from right to left, this presents ordinary cohomology as the “additive shadow” of cobordism. The motivic analogue was announced by Hopkins and Morel in 2004 (a proof was sketched in a Harvard seminar, recounted in [Hop04]; see also Ayoub’s report [Ayo05]) and proved, in a strengthened form covering positive characteristic, by Hoyois [Hoy15], using the motivic Steenrod algebra of Voevodsky and its positive-characteristic extension by Hoyois–Kelly–Østvær.

Definition 3.5.16 (Chromatic quotients). Fix generators $L = \mathbb{Z}[a_1, a_2, \dots]$ and regard $a_i \in \pi_{2i,i} \mathrm{MGL}(S)$ via the classifying map θ_S of Construction 3.5.15, i.e. as maps $a_i: \mathbb{T}^{\otimes i} \otimes \mathrm{MGL} \rightarrow \mathrm{MGL}$ of MGL-modules. Define inductively

$$\mathrm{MGL}/(a_1, \dots, a_n) := \mathrm{cofib}\left(\mathbb{T}^{\otimes n} \otimes \mathrm{MGL}/(a_1, \dots, a_{n-1}) \xrightarrow{a_n} \mathrm{MGL}/(a_1, \dots, a_{n-1})\right)$$

in MGL-modules, and $\mathrm{MGL}/(a_1, a_2, \dots) := \mathrm{colim}_n \mathrm{MGL}/(a_1, \dots, a_n)$. (Hoyois imposes a mild *adequacy* condition on the chosen generators, removed a posteriori; see [Hoy15, §7].)

Recall from Remark 3.2.32 that $H\mathbb{Z}$ carries an orientation with *additive* formal group law, constructed in Lemma 3.2.39. By Theorem 3.5.13 it classifies a ring map $\vartheta: \mathrm{MGL} \rightarrow H\mathbb{Z}$. Since the formal group law of $H\mathbb{Z}$ is additive, $\vartheta_*: L \rightarrow H\mathbb{Z}^{-2*,-*}$ kills every a_i with $i \geq 1$, so ϑ factors canonically through a map of MGL-modules

$$\Theta: \mathrm{MGL}/(a_1, a_2, \dots) \longrightarrow H\mathbb{Z}.$$

Theorem 3.5.17 (Hopkins–Morel–Hoyois [Hoy15, Theorem 7.12]). *Let k be a perfect field of characteristic exponent c (i.e. $c = 1$ in characteristic 0 and $c = p$ in characteristic p), and let S be essentially smooth over k . Then*

$$\Theta[1/c]: \mathrm{MGL}/(a_1, a_2, \dots)[1/c] \xrightarrow{\sim} H\mathbb{Z}[1/c]$$

is an equivalence in $\mathrm{SH}(S)$. In particular, in characteristic zero $\mathrm{MGL}/(a_1, a_2, \dots) \simeq H\mathbb{Z}$ integrally.

Outline of the proof. Write $f := \Theta[1/c]$ and assume for simplicity $\mathrm{char} k = 0$. Hoyois’s proof [Hoy15] has the structure

$$\begin{aligned} (1) \text{ and } (2) &\implies (3) \ H\mathbb{Z} \otimes f \text{ is an equivalence,} \\ (3) \text{ and } (4) \ \mathrm{MGL}_{\leq 0} &\simeq H\mathbb{Z}_{\leq 0} \implies (5) \ f \text{ is an equivalence,} \end{aligned}$$

where $(-)_{\leq 0}$ denotes truncation in the homotopy t -structure of Section 3.1.7.

- (i) Rationally, $H\mathbb{Q}$ is the motivic Landweber-exact spectrum of the additive formal group over $\mathbb{Q}$ (Theorem 3.5.24 below), and the statement follows from the Landweber-exactness machinery of Naumann–Spitzweck–Østvær.

- (ii) Both sides of $H\mathbb{Z}/\ell \otimes f$ are cellular $H\mathbb{Z}/\ell$ -modules, so it suffices to compare homotopy groups. The left side is computed by elementary cell-counting from Definition 3.5.16; the right side, $\pi_{*,*}(H\mathbb{Z}/\ell \otimes H\mathbb{Z})$, is the heart of the matter: by Voevodsky’s computation of the motivic Steenrod algebra (extended to characteristic $p > 0$, $\ell \neq p$, by Hoyois–Kelly–Østvær) it is the kernel of the Bockstein on the dual Steenrod algebra. The comparison also uses Röndigs–Østvær’s identification of $H\mathbb{Z}/\ell$ -modules with motivic spectra with $\mathbb{Z}/\ell$ -transfers.
- (iii) is elementary arithmetic (an integral map that is a rational and mod- ℓ equivalence for all ℓ , between spectra with bounded $H\mathbb{Z}$ -homological cell structure, is an $H\mathbb{Z}$ -homology equivalence).
- (iv) By a theorem of Morel, the homotopy sheaves of the truncation $\mathbb{1}_{\leq 0}$ are computed by Milnor–Witt K -theory; the homotopy sheaves of $MGL_{\leq 0}$ are identified with the cokernel of the Hopf map η acting on those of $\mathbb{1}_{\leq 0}$, which recovers Milnor K -theory — the same sheaves as for $H\mathbb{Z}_{\leq 0}$ by the classical comparison $H^{n,n}(L, \mathbb{Z}) \cong K_n^M(L)$.
- (v) Finally, $MGL/(a_1, a_2, \dots)$ is $H\mathbb{Z}$ -local: the homotopy t -structure is left complete, so it suffices to see that each truncation is $H\mathbb{Z}_{\leq 0}$ -local; by a theorem of Gutiérrez–Röndigs–Spitzweck–Østvær [GRSØ12] these truncations are $MGL_{\leq 0}$ -modules, and (4) converts $MGL_{\leq 0}$ -locality into $H\mathbb{Z}_{\leq 0}$ -locality. Combined with (3), this forces $\text{fib}(f) = 0$.

□

Theorem 3.5.18 (Strong universality). *Let k be a field of characteristic exponent c . The Lazard classifying map of Construction 3.5.15 induces an isomorphism [Hoy15, Proposition 8.2]*

$$\begin{aligned} \theta_k[1/c]: L[1/c] &\xrightarrow{\cong} MGL^{-2*,-*}(k)[1/c], \\ MGL^{-2*,-*}(k)[1/c] &= \bigoplus_n \pi_{2n,n} MGL(k)[1/c]. \end{aligned}$$

In particular, in characteristic zero the formal group law of MGL is the universal one, integrally. This fulfils the promise of Remark 3.2.32.

Proof sketch. Injectivity follows from Lazard’s injective homomorphism and the commutative motivic Hurewicz square

$$L \hookrightarrow \mathbb{Z}[b_1, b_2, \dots],$$

which detects the classifying map $L \rightarrow MGL_{**}$, as in the proof of [Hoy15, Proposition 8.2]; complex realization gives an additional check only when k embeds into $\mathbb{C}$. For surjectivity, one proves by induction that, for every m ,

$$L/(a_1, \dots, a_m) \longrightarrow \pi_{2*,*}(MGL/(a_1, \dots, a_m))$$

is surjective. The induction uses the connectivity of these quotients in the homotopy t -structure and Theorem 3.5.17 in the colimit; the case $m = 0$ is the claim. See [Hoy15, §8.2]. □

Corollary 3.5.19 (Cellularity [Hoy15, Proposition 8.1]). *Over a base essentially smooth over a field, $HA[1/c]$ is cellular — an iterated colimit of bigraded spheres $S^{p,q}$ — for every $A \in \text{Sp}(\text{sMod}_{\mathbb{Z}[1/c]})$. Indeed, MGL is cellular by Theorem 3.5.8 (Grassmannians have algebraic cell structures), hence so is the quotient $MGL/(a_1, a_2, \dots)[1/c] \simeq H\mathbb{Z}[1/c]$.*

Remark 3.5.20 (Beyond fields). Over more general bases the situation is the subject of Spitzweck’s monograph [Spi18]: he constructs an $\mathbb{E}_\infty$ -motivic cohomology spectrum over

any Dedekind-domain base, compatible with base change, representing Bloch–Levine motivic cohomology on smooth schemes, and proves a generalization of the Hopkins–Morel isomorphism in that setting. Over a perfect field all constructions in sight — Voevodsky’s $H\mathbb{Z}$, Spitzweck’s spectrum, and our $s^0\text{KGL}$ of Definition 3.2.68 — agree by Theorem 3.4.14. We emphasize that Theorem 3.5.17 is proved by Steenrod-algebra and t -structure techniques and consumes *no* slice-theoretic input; in particular it is logically independent of the zeroth-slice theorem of Section 3.4. How the two interlock is the subject of the next subsection.

3.5.4. Slices, Conner–Floyd, and the chromatic picture

In topology, the chromatic organization of stable homotopy rests on $\pi_*\text{MU} \cong \mathbb{L}$: localizations and quotients of MU realize the strata of the moduli of formal groups, with $H\mathbb{Z}$ (additive stratum) and KU (multiplicative stratum) as the first two layers. This subsection assembles the motivic counterpart. The slice filtration of Section 3.2.2 plays the role of the Postnikov filtration, and the two theorems already in our hands — the zeroth-slice theorem of Section 3.4 and the Hopkins–Morel–Hoyois equivalence — combine to compute *all* slices of MGL , of every Landweber-exact spectrum, and of the sphere itself.

The starting point is a purely geometric observation of Spitzweck, valid over any Noetherian base of finite Krull dimension.

Proposition 3.5.21 ([Spi10, Theorem 3.1, Corollary 3.3]). *The cofiber of the unit $\mathbb{1}_S \rightarrow \text{MGL}_S$ lies in $\mathbb{T} \otimes \text{SH}(S)^{\text{eff}}$; it is in fact built from the cells $\mathbb{T}^{\otimes n} \otimes \Sigma^\infty(\text{Gr}_\bullet)_+$ with $n \geq 1$ arising from Theorem 3.5.8. Consequently MGL_S is effective and the unit induces an equivalence*

$$s^0(\mathbb{1}_S) \xrightarrow{\sim} s^0(\text{MGL}_S).$$

The sphere and algebraic cobordism are thus indistinguishable at the level of zeroth slices — a first, weight-zero shadow of the chromatic philosophy that the sphere is assembled from cobordism data. Combining this with the zeroth-slice theorem $s^0(\mathbb{1}_k) \simeq H\mathbb{Z}_k$ of Corollary 3.4.15, the orientation map $\vartheta: \text{MGL} \rightarrow H\mathbb{Z}$ is identified with the projection of MGL onto its zeroth slice [Spi10, Corollary 4.7], and the Hopkins–Morel–Hoyois theorem then computes the entire slice tower:

Theorem 3.5.22 (Slices of MGL ; Voevodsky’s conjecture [Voe02], proved by Spitzweck + Hopkins–Morel–Hoyois [Spi10, Theorem 4.6], [Hoy15, §8.3]). *Let k be a perfect field of characteristic exponent c . For every $t \geq 0$ there are canonical equivalences of $H\mathbb{Z}$ -modules*

$$s^t(\text{MGL}_k)[1/c] \simeq H\mathbb{Z}[1/c] \otimes \mathbb{T}^{\otimes t} \otimes_{\mathbb{Z}} L_t,$$

where L_t is the degree- t part of the Lazard ring (a finitely generated free abelian group), and $s^t\text{MGL} = 0$ for $t < 0$.

Proof sketch. By Proposition 3.5.21 and Corollary 3.4.15, $s^0\text{MGL}_k \simeq H\mathbb{Z}_k$ and the zeroth-slice projection is ϑ . Spitzweck’s filtration argument [Spi10, Theorem 4.6] presents $\text{Fil}_{\text{slice}}^t\text{MGL}$ as a colimit of the MGL -cells $\mathbb{T}^{\otimes n} \otimes \text{MGL}$ indexed by monomials of degree $n \geq t$ in the a_i , provided the map $\text{MGL} \rightarrow \text{MGL}/(a_1, a_2, \dots)$ is the zeroth-slice projection; Theorem 3.5.17 supplies exactly this (after inverting c), and the cofibers of the resulting tower are visibly $H\mathbb{Z} \otimes \mathbb{T}^{\otimes t} \otimes L_t$. The lift of the equivalence to $H\mathbb{Z}$ -modules uses [GRSØ12]. $\square$

Remark 3.5.23 (The logical interlock). It is worth isolating the logical structure. Two independent inputs entered: the zeroth-slice theorem (proved in Section 3.4 by Levine’s coniveau

tower, or by the framed Hilbert-scheme geometry) and the Hopkins–Morel–Hoyois equivalence (proved by Steenrod-algebra techniques, Remark 3.5.20). Neither implies the other by the arguments presented here: the slice computation *consumes* both. In particular, contrary to a tempting misreading, Theorem 3.5.17 does not by itself furnish a third proof of $s^0(\mathbb{1}) \simeq H\mathbb{Z}$ — what it furnishes, given Proposition 3.5.21, is the equivalence of the two statements “ $s^0(\mathbb{1}_k) \simeq H\mathbb{Z}_k$ ” and “ $\mathrm{MGL} \rightarrow \mathrm{MGL}/(a_1, a_2, \dots)$ is the zeroth-slice projection”. Compare Remark 3.4.16.

Theorem 3.5.24 (Motivic Landweber exactness; Naumann–Spitzweck–Østvær [NSØ09, Theorem 7.3]). *Let S be a Noetherian scheme of finite Krull dimension and A_* a Landweber-exact graded L -algebra (e.g. any localization or regular quotient of L satisfying Landweber’s criterion). Then*

$$X \mapsto \mathrm{MGL}_{*,*}(X) \otimes_L A_*$$

is a bigraded homology theory on $\mathrm{SH}(S)$, represented by a motivic spectrum E_A which is stable under base change in S [NSØ09, §§8–9]. The displayed tensor product is the ordinary graded tensor product; multiplicative structures are inherited when the input carries the requisite diagonal.

The slices and the convergence of the slice tower for these spectra are then controlled by the Hopkins–Morel–Hoyois theorem:

Theorem 3.5.25 ([Hoy15, Theorem 8.5, Theorem 8.12]; see also [Spi10; Spi12]). *Let S be essentially smooth over a field of characteristic exponent c , let M_* be a Landweber-exact $L[1/c]$ -module, and let $E \in \mathrm{SH}(S)$ be the associated motivic spectrum.*

- (i) *There are canonical equivalences of $H\mathbb{Z}$ -modules $s^t E \simeq HM_t \otimes \mathbb{T}^{\otimes t}$.*
- (ii) *The slice filtration of E is complete and convergent; the resulting strongly convergent spectral sequence reads*

$$H^{p+2t, q+t}(X, M_t) \implies E^{p,q}(X), \quad X \in \mathrm{Sm}_S.$$

For $M_ = L[1/c]$ this is the slice spectral sequence $H^{p+2t, q+t}(X, \mathbb{Z}) \otimes L_t[1/c] \implies \mathrm{MGL}^{p,q}(X)[1/c]$, the motivic analogue of the topological Atiyah–Hirzebruch spectral sequence for MU .*

Corollary 3.5.26 (Comparison with algebraic cobordism [Hoy15, Corollary 8.15], after Levine [Lev09]). *Let k be a field of characteristic zero and $X \in \mathrm{Sm}_k$. There is a natural isomorphism of graded rings*

$$\Omega^*(X) \cong \mathrm{MGL}^{2*,*}(X),$$

where Ω^ is the algebraic cobordism of Levine–Morel [LM07].*

Remark 3.5.27 (Geometric cobordism cycles). Corollary 3.5.26 is the precise sense in which MGL -cohomology is *geometric* in characteristic zero, resolving a conjecture of Levine–Morel. The theory Ω^* is built from *cobordism cycles*: $\Omega^n(X)$ is generated by classes $[f: Z \rightarrow X]$ of projective morphisms from smooth varieties with $\mathrm{codim} = n$, modulo relations forced by the axiomatics of oriented Borel–Moore theories — it is, by construction, the universal such theory, with formal group law governing c_1 of tensor products exactly as in Remark 3.2.32. Levine’s proof of the comparison [Lev09] presupposed the spectral sequence of Theorem 3.5.25, which the Hopkins–Morel–Hoyois theorem then supplied; the corollary is thus a genuine joint achievement of the geometric and homotopical methods. In the Pontryagin–Thom dictionary it is the motivic analogue of Quillen’s geometric description of MU^* , and Section 3.5.5 will exhibit a second, framed incarnation of the same principle, valid over any base.

Theorem 3.5.28 (Slices of the sphere [Hoy15, Theorem 8.7], after Levine [Lev13]). *Let S be essentially smooth over a field of characteristic exponent c , and let $\mathbb{L}^\bullet$ be the graded cosimplicial ring associated with the Hopf algebroid (L, LB) classifying formal group laws and strict isomorphisms. Then for every t ,*

$$s^t(\mathbb{1})[1/c] \simeq H\mathbb{L}_t^\bullet \otimes \mathbb{T}^{\otimes t}[1/c],$$

where the degree- t part $\mathbb{L}_t^\bullet$ is viewed as a chain complex via dual Dold–Kan. For $t = 0$ this recovers $s^0(\mathbb{1})[1/c] \simeq H\mathbb{Z}[1/c]$, consistently with Corollary 3.4.15.

Remark 3.5.29 (Slice tower as motivic Adams–Novikov). The cosimplicial object (L, LB) is precisely the cobar construction computing the E_2 -page of the Adams–Novikov spectral sequence in topology. Theorem 3.5.28 thus says: *the slice spectral sequence of the motivic sphere is an Adams–Novikov-type spectral sequence whose input is motivic cohomology with coefficients in the algebraic cobordism cooperations.* This perspective, due to Levine [Lev13], is the engine behind the computations of the first stable motivic stems by Röndigs–Spitzweck–Østvær, and it identifies the slice filtration as the correct motivic home of chromatic phenomena.

Having reached $H\mathbb{Z}$ from MGL by killing all the generators, we now reach the chapter’s other protagonist by inverting one of them.

Theorem 3.5.30 (Motivic Conner–Floyd; Spitzweck–Østvær [SØ09], Spitzweck [Spi10, Theorem 5.2]). *Let S be a Noetherian scheme of finite Krull dimension. Choose the generators $x_i := a_i$ so that $x_1 \in L_1$ is the coefficient of the multiplicative deformation (the Bott class). Then:*

(i) *The canonical map of MGL-modules*

$$x_1^{-1} \text{MGL}/(x_2, x_3, \dots) \xrightarrow{\sim} \text{KGL}$$

is an equivalence; equivalently, KGL is the motivic Landweber-exact spectrum of the multiplicative formal group law over $\mathbb{Z}[\beta^{\pm 1}] \cong x_1^{-1}L/(x_2, x_3, \dots)$.

(ii) *For every $X \in \text{SH}(S)$ there is a natural isomorphism of bigraded coefficient modules*

$$\text{MGL}^{*,*}(X) \otimes_L \mathbb{Z}[\beta^{\pm 1}] \xrightarrow{\cong} \text{KGL}^{*,*}(X).$$

Here $\otimes_L$ is the ordinary graded tensor product. If X carries a compatible diagonal — for example $X = \Sigma_{\mathbb{T}}^\infty Y_+$ for a scheme or motivic space Y — the comparison is multiplicative and hence is an isomorphism of bigraded rings.

Remark 3.5.31 (Closing the circle). Theorem 3.5.30 closes the loop that this chapter opened. The spectrum KGL, constructed in Section 3.1.6 from the Bott-periodic algebra machinery, is recovered from MGL by a one-line chromatic operation, and its orientation (Example 3.2.31), formal group law (Remark 3.2.32), and slices are all visible from the cobordism side: applying Theorem 3.5.25 to $M_* = \mathbb{Z}[\beta^{\pm 1}]$ gives $s^t \text{KGL}[1/c] \simeq H\mathbb{Z}[1/c] \otimes \mathbb{T}^{\otimes t}$ for all $t \in \mathbb{Z}$, recovering — over fields, after inverting the characteristic exponent — the slice description that Theorem 3.2.64 established integrally over qcqs bases by entirely different (BEM) methods. Conversely, the non- $\mathbb{A}^1$ -invariant upgrade of Theorem 3.5.30(ii), where KGL is replaced by genuine (non-KH) algebraic K -theory, is the Conner–Floyd theorem of Annala–Hoyois–Iwasa [AHI24a, Theorem 1.2] in MotSp, already previewed in Chapter 2 (see Warning 2.1.30).

Remark 3.5.32 (Snaith presentation). There is yet a third route between the protagonists, bypassing MGL: the motivic Snaith theorem of Gepner–Snaith [GS09] and Spitzweck–Østvær [SØ09],

$$\text{KGL} \simeq \Sigma_+^\infty(\Omega^\infty \mathcal{P}ic)[1/\beta] = \Sigma_+^\infty \mathbb{P}^\infty[1/\beta],$$

inverting the Bott class $\beta \in \pi_{2,1}\Sigma_+^\infty\mathbb{P}^\infty$ of Section 3.1.6 (compare the lift $\mathbb{T}_S \rightarrow \mathcal{V}_S$ of Proposition 3.3.17). A cautionary subtlety: the periodized cobordism spectrum $\bigvee_n \mathrm{MGL} \otimes \mathbb{T}^{\otimes n}$ carries both the normed $\mathbb{E}_\infty$ -structure of Remark 3.5.9 and a Gepner–Snaith $\mathbb{E}_\infty$ -structure via the analogous presentation $\Sigma_+^\infty \mathrm{L}_{\mathrm{mot}}(\Omega^\infty K^{\mathrm{rk}=0})[1/\beta]$, and these do *not* coincide [BH21, Remark 16.20].

Remark 3.5.33 (The motivic chromatic picture). Theorem 3.5.24 opens the door to the full chromatic apparatus over a field: Landweber-exact L-algebras produce motivic Brown–Peterson spectra BP (first constructed by Vezzosi [Vez01] via the Quillen idempotent), Johnson–Wilson theories $E(n)$, and connective covers thereof, with slices computed by Theorem 3.5.25 [Spi10; Spi12]. The tower

$$\begin{aligned} H\mathbb{Z} = \text{“height } \infty \text{ shadow”} & \longleftarrow \mathrm{BP}, E(n), \dots \\ & \longleftarrow \mathrm{KGL} = \text{“height 1”} \end{aligned}$$

interpolates between the chapter’s two protagonists exactly as $H\mathbb{Z}$ and KU are interpolated in topology [Rav86; Lur10; Lin25], with the caveat that motivic “chromatic” phenomena are richer: the sphere also sees the weight direction (η is not nilpotent, Section 3.1.7), so the MGL -based picture captures only the part of $\mathrm{SH}(k)$ visible to oriented theories.

3.5.5. The framed model: cobordism and finite syntomic schemes

The results so far portray MGL through its universal property. We close the chapter — and the main text — with the complementary *geometric* portrait, in the style of Pontryagin–Thom theory: just as $\Omega^\infty \mathrm{MU}$ is the cobordism space of stably almost-complex manifolds, the infinite $\mathbb{P}^1$ -loop space of MGL is a moduli space of *finite syntomic schemes*. This is the theorem of Elmanto–Hoyois–Khan–Sosnilo–Yakerson [Elm+20], and it is the natural culmination of the framed technology of Sections 3.3 and 3.4: the motivic recognition principle (Theorem 3.3.11) reduced infinite $\mathbb{P}^1$ -loop spaces to framed transfers, the framed Hilbert scheme (Construction 3.4.5) computed the sphere, and the moduli of vector bundles computed kgl (Proposition 3.3.17). Cobordism slots into the same pattern.

Let FSyn_S denote the presheaf of groupoids on Sm_S sending X to the groupoid of **finite syntomic X -schemes** (finite, flat, and local complete intersection morphisms $Z \rightarrow X$), an $\mathbb{E}_\infty$ -object under disjoint union. It is naturally a presheaf with finite-syntomic transfers. Restricting those transfers along the symmetric monoidal forgetful functor from tangentially framed correspondences to finite-syntomic correspondences makes FSyn_S an object of $H^{\mathrm{fr}}(S)$ in the sense of Definition 3.3.10 [Elm+20, §§2–3]. This does not assert that an individual finite syntomic map has a canonical framing: virtual rank 0 of its cotangent complex does not supply a trivialization of its K -theory class.

Theorem 3.5.34 ([Elm+20, Theorem 3.4.1]). *For every scheme S there is an equivalence of $\mathbb{E}_\infty$ -ring spectra*

$$\mathrm{MGL}_S \simeq \Sigma_{\mathbb{T}, \mathrm{fr}}^\infty \mathrm{FSyn}_S \quad \text{in } \mathrm{SH}(S) \simeq \mathrm{SH}^{\mathrm{fr}}(S)$$

(Section 3.3.6 for the identification of the two sides’ homes). More generally, for $n \geq 1$ the twists $\mathbb{T}^{\otimes n} \otimes \mathrm{MGL}_S$ are the framed suspension spectra of the moduli of finite quasi-smooth derived schemes of virtual dimension $-n$.

The juxtaposition with Proposition 3.3.17 is the conceptual payoff of the framed point of view — the two protagonists of this chapter count the two most basic classes of geometric objects over a variety. The first equivalence says that K -theory counts vector bundles, while

the second says that cobordism counts finite syntomic schemes:

$$\begin{aligned} \mathrm{kgl}_S &\simeq \Sigma_{\mathrm{fr}}^\infty (\mathrm{Vect}_S^{\mathrm{gp}}), \\ \mathrm{MGL}_S &\simeq \Sigma_{\mathbb{T}, \mathrm{fr}}^\infty \mathrm{FSyn}_S. \end{aligned}$$

Theorem 3.5.35 (Hilbert-scheme model [Elm+20, Corollary 3.4.2, Theorem 3.5.2]). *Let S be pro-smooth over a field. There are equivalences of spaces*

$$\Omega_{\mathbb{T}}^\infty \mathrm{MGL}_S \simeq \mathrm{L}_{\mathrm{zar}} \mathrm{L}_{\mathbb{A}^1} (\mathrm{FSyn}_S)^{\mathrm{gp}} \simeq \mathrm{L}_{\mathrm{zar}} (\mathrm{L}_{\mathbb{A}^1} \mathrm{Hilb}^{\mathrm{lci}}(\mathbb{A}_S^\infty))^{\mathrm{gp}},$$

where $\mathrm{Hilb}^{\mathrm{lci}}(\mathbb{A}_S^\infty) = \mathrm{colim}_n \mathrm{Hilb}^{\mathrm{lci}}(\mathbb{A}_S^n)$ is the Hilbert scheme of finite local complete intersections in infinite affine space, a smooth ind-scheme and a commutative monoid up to $\mathbb{A}^1$ -homotopy. The cited Hilbert-scheme comparison does not identify the multiplicative structures, so no $\mathbb{E}_\infty$ qualifier is imposed on the second equivalence.

This is the exact MGL-counterpart of the framed Hilbert scheme that computed $s^0(\mathbb{1})$ in Section 3.4.2, and of the Hilbert scheme of $\mathbb{A}^\infty$ computing K -theory behind Proposition 3.3.17: “cobordism classes” of finite syntomic schemes are parametrized by an honest (ind-)scheme, and stabilization in $\mathbb{A}^\infty$ plus group completion is all that separates the moduli from the cohomology theory.

Theorem 3.5.36 (Modules over algebraic cobordism [Elm+20, Theorems 4.1.3 and 4.1.4]). *Let $\mathrm{SH}^{\mathrm{fsyn}}(S)$ and $\mathrm{H}^{\mathrm{fsyn}}(S)$ be defined as in Definition 3.3.10, with finite syntomic correspondences in place of framed ones.*

(i) *For every scheme S there is an equivalence of symmetric monoidal categories*

$$\mathrm{Mod}_{\mathrm{MGL}}(\mathrm{SH}(S)) \simeq \mathrm{SH}^{\mathrm{fsyn}}(S).$$

(ii) *For k a perfect field this restricts to $\mathrm{Mod}_{\mathrm{MGL}}(\mathrm{SH}^{\mathrm{veff}}(k)) \simeq \mathrm{H}^{\mathrm{fsyn}}(k)^{\mathrm{gp}}$ under the recognition equivalence $\mathrm{SH}^{\mathrm{veff}}(k) \simeq \mathrm{H}^{\mathrm{fr}}(k)^{\mathrm{gp}}$ of Theorem 3.3.11.*

Analogous statements hold for MSL with oriented finite syntomic correspondences [Elm+20, §4.2].

Remark 3.5.37 (Taking an MGL-module is taking transfers). Theorem 3.5.36 is the cobordism analogue of the equivalence $\mathrm{Mod}_{H\mathbb{Z}} \simeq \mathrm{DM}$ between motivic-cohomology modules and Voevodsky motives (compare Definition 3.2.68): a module structure over a cohomology theory is the same as a system of wrong-way maps of the corresponding geometric type — finite syntomic transfers for MGL, finite *flat* transfers conjecturally for the sphere itself, transfers along framed correspondences universally. Within this dictionary motivic cohomology acquires a third description: by [Elm+20, Theorem 4.3.1], for every scheme S there is an equivalence of $\mathbb{E}_\infty$ -algebras

$$H\mathbb{Z}_S \simeq \Sigma_{\mathbb{T}, \mathrm{fsyn}}^\infty \mathbb{Z}_S \quad \text{in } \mathrm{Mod}_{\mathrm{MGL}}(\mathrm{SH}(S)) \simeq \mathrm{SH}^{\mathrm{fsyn}}(S),$$

where $H\mathbb{Z}_S$ is Spitzweck’s spectrum (Remark 3.5.20) and $\mathbb{Z}_S$ the constant sheaf: *motivic cohomology is the MGL-linearization of the constant sheaf $\mathbb{Z}$* . This refines, over general bases, the perfect-field statement $H\mathbb{Z} \simeq s^0 \mathrm{MGL}$ of Proposition 3.5.21 and Corollary 3.4.15.

We can now survey the notes from their endpoint. The stable motivic category $\mathrm{SH}(S)$ was built in Chapter 2 as the universal $\mathbb{A}^1$ -invariant six-functor formalism; inside it, this chapter located the K -theory spectrum KGL and extracted from its slice filtration the motivic cohomology $H\mathbb{Z} = s^0 \mathrm{KGL}$ and the $\mathbb{A}^1$ -motivic cohomology $\mathbb{Z}(j)^\mathbb{A}$. The present section revealed the organizing object behind all of it: the canonically oriented theory MGL, universal

over a field in the sense of Theorem 3.5.13 and geometric on both of its faces (cobordism cycles in characteristic zero, finite syntomic schemes over any base). It has $H\mathbb{Z}$ as its additive quotient under the Hopkins–Morel–Hoyois hypotheses, KGL as its multiplicative Landweber specialization, and the sphere sharing its zeroth slice; the slice spectral sequence binds these theories together. In the non- $\mathbb{A}^1$ -invariant world of MotSp (Warning 2.1.30) this organizing role persists: MGL has a canonical $\mathbb{E}_\infty$ -structure, is initial among oriented objects of $\text{CAlg}(\text{hMotSp}(S))$, and governs genuine K -theory [AHI24a]. This is the natural boundary of the present account: motivic cohomology is retained as a structure internal to motivic homotopy theory, while its separate syntomic, étale, and cdh comparison theory is not pursued here.

NORMS ON MOTIVIC SPECTRA

This appendix sketches the construction of **norm functors** due to Bachmann–Hoyois [BH21], which were previewed in Remark 1.0.2 and underlie the multiplicative structure used at several points in Chapters 2 and 3. The construction proceeds in three steps—sifted cocompletion, motivic localization, and $\mathbb{T}$ -stabilization—each a universal cocompletion. We therefore begin by recalling the cocompletion machinery in Section A.1, then turn to the norm functors themselves in Section A.2.

A.1. SIFTED COCOMPLETION AND ANIMATION

A.1.1. Algebraic structures as product-preserving presheaves

A classical observation is that the category $\mathbf{Grp}$ of groups can be reconstructed from its full subcategory $\mathbf{FinFree} \subseteq \mathbf{Grp}$ of finitely generated free groups. Write $F_n \in \mathbf{FinFree}$ for the free group on n generators; note that $\mathbf{FinFree}$ has finite coproducts, with $F_m \sqcup F_n \simeq F_{m+n}$.

Given a finite-product-preserving presheaf $\underline{G}: \mathbf{FinFree}^{\mathrm{op}} \rightarrow \mathbf{Set}$, set $G := \underline{G}(F_1)$. Since $F_n \simeq F_1^{\sqcup n}$ in $\mathbf{FinFree}$, one has $\underline{G}(F_n) \simeq G^n$. The codiagonal-and-multiplication morphism $F_1 \rightarrow F_2$ in $\mathbf{FinFree}$ (sending the generator to ab) induces, after $\underline{G}$, the map

$$G \times G \simeq \underline{G}(F_2) \longrightarrow \underline{G}(F_1) \simeq G,$$

which is the group multiplication. The unit and inverse morphisms of $\mathbf{FinFree}$ likewise produce the unit and inverse of G , and one obtains an equivalence of 1-categories

$$\mathrm{Fun}^{\Pi}(\mathbf{FinFree}^{\mathrm{op}}, \mathbf{Set}) \simeq \mathbf{Grp},$$

where Fun^{Π} denotes the full subcategory of finite-product-preserving functors. The same pattern recovers $\mathbf{CRing}$, $\mathbf{Mod}_R$, and similar algebraic categories from their respective subcategories of free objects of finite rank. Sifted cocompletion is the universal-property reformulation of this observation.

A.1.2. Sifted colimits

Definition A.1.1 (Sifted category). A nonempty category $\mathcal{I}$ is **sifted** if the diagonal $\Delta: \mathcal{I} \rightarrow \mathcal{I} \times \mathcal{I}$ is cofinal. A colimit indexed by a sifted $\mathcal{I}$ is called a **sifted colimit**.

The defining feature of sifted colimits is that they commute with finite products in $\mathbf{An}$: for $X, Y: \mathcal{I} \rightarrow \mathbf{An}$, cofinality of Δ gives

$$\mathrm{colim}_{i \in \mathcal{I}} (X_i \times Y_i) \simeq \mathrm{colim}_{(i,j) \in \mathcal{I} \times \mathcal{I}} (X_i \times Y_j) \simeq (\mathrm{colim}_i X_i) \times (\mathrm{colim}_j Y_j).$$

Two paradigmatic examples are filtered categories and Δ^{op} , and these in fact span the full notion:

Proposition A.1.2 ([Lur09, Corollary 5.5.8.17]). *A functor between categories with all small colimits preserves all sifted colimits if and only if it preserves filtered colimits and geometric realizations (i.e., Δ^{op} -indexed colimits).*

In slogan form: *sifted colimits = filtered colimits + reflexive coequalizers*, the latter being the truncated form of geometric realizations.

A.1.3. Sifted cocompletion

Definition A.1.3 (Sifted cocompletion). Let C be a small category with finite coproducts. The **sifted cocompletion** of C is the smallest full subcategory

$$\mathcal{P}_{\Sigma}(C) \subseteq \text{PShv}(C) = \text{Fun}(C^{\text{op}}, \text{An})$$

containing the representables and closed under sifted colimits.

The same sifted-vs.-products commutation argument from Section A.1.1 applies: every $F \in \mathcal{P}_{\Sigma}(C)$, written as a sifted colimit $F \simeq \text{colim}_{i \in I} y(X_i)$ of representables, satisfies $F(A \sqcup B) \simeq F(A) \times F(B)$. Conversely, every product-preserving presheaf arises this way:

Proposition A.1.4 ([Lur09, Definition 5.5.8.8, Proposition 5.5.8.10, Lemma 5.5.8.14]). *Let C be a small category with finite coproducts. The Yoneda embedding identifies*

$$\mathcal{P}_{\Sigma}(C) \simeq \text{Fun}^{\Pi}(C^{\text{op}}, \text{An}).$$

The defining universal property is then:

Proposition A.1.5 ([Lur09, Proposition 5.5.8.15]). *Let C be a small category with finite coproducts and $\mathcal{D}$ a category with all sifted colimits. Restriction along the Yoneda embedding induces an equivalence*

$$\text{Fun}^{\text{sift}}(\mathcal{P}_{\Sigma}(C), \mathcal{D}) \xrightarrow{\sim} \text{Fun}(C, \mathcal{D}).$$

In words: $\mathcal{P}_{\Sigma}(C)$ is freely generated under sifted colimits by C . By the dual identification of Proposition A.1.4, it is equivalently *the universal way to add sifted colimits while preserving the existing finite coproducts*—which is the perspective taken in Remark 1.0.2.

Remark A.1.6 (Symmetric monoidal refinement). The sifted cocompletion is presentable and compactly generated. The Yoneda image consists of compact projective objects and generates under sifted colimits (not, in general, under filtered colimits alone) [Lur09, Proposition 5.5.8.10 and Lemma 5.5.8.14]. It inherits a symmetric monoidal structure from any symmetric monoidal structure on C via Day convolution; with this enhancement, its universal property refines to symmetric monoidal functors that preserve sifted colimits [Lur17, §4.8.1]. This is the form in which we shall apply it.

A.1.4. Animation

The classical equivalence $\text{Grp} \simeq \text{Fun}^{\Pi}(\text{FinFree}^{\text{op}}, \text{Set})$ lifts by replacing Set with An . Recall that an object P of a 1-category C with sifted colimits is *compact projective* if $\text{Hom}_C(P, -)$ preserves sifted colimits; equivalently, it preserves filtered colimits and reflexive coequalizers, the 1-categorical shadow of geometric realizations. Write $C^{\text{cp}} \subseteq C$ for the full subcategory of compact projective objects. A 1-category C is *compactly projectively generated* when C^{cp} is essentially small and the canonical map $\mathcal{P}_{\Sigma}(C^{\text{cp}}) \rightarrow C$ (using the 1-categorical sifted cocompletion) is an equivalence.

Definition A.1.7 (Animation). Let $\mathcal{C}$ be a compactly projectively generated 1-category. The **animation** of $\mathcal{C}$ is

$$\mathrm{Ani}(\mathcal{C}) := \mathcal{P}_{\Sigma}(\mathcal{C}^{\mathrm{cp}}) \simeq \mathrm{Fun}^{\Pi}((\mathcal{C}^{\mathrm{cp}})^{\mathrm{op}}, \mathbf{An}),$$

where $\mathcal{P}_{\Sigma}$ is the (default) sifted cocompletion of Definition A.1.3, applied to $\mathcal{C}^{\mathrm{cp}}$ viewed as a category.

Examples: $\mathrm{Ani}(\mathbf{Set}) = \mathbf{An}$; $\mathrm{Ani}(\mathbf{CRing})$ is the category of *animated commutative rings*, the home of derived/animated algebraic geometry; similarly for $\mathrm{Ani}(\mathbf{Mod}_R)$ and $\mathrm{Ani}(\mathbf{Grp})$.

Remark A.1.8 (Animation vs. derived category). Lurie [Lur09, §5.5.8] calls the sifted cocompletion the *nonabelian derived category*. For a compactly projectively generated abelian category $\mathcal{A}$, simplicial resolutions and the Dold–Kan correspondence identify its animation with the connective derived category,

$$\mathrm{Ani}(\mathcal{A}) \simeq \mathbf{D}_{\geq 0}(\mathcal{A})$$

[Lur17, §1.3.3]. Thus animation is the connective derived enhancement of $\mathcal{A}$; passing to the unbounded derived category requires a further stabilization, not merely animation.

A.1.5. The general cocompletion framework

Sifted cocompletion is one instance of a general procedure. For the standard classes of colimit shapes used below, let $\mathcal{K} \subseteq \mathcal{K}'$ and write $\mathrm{Cat}(\mathcal{K})$ for the (very large) category of categories admitting all $\mathcal{K}$ -indexed colimits, with morphisms preserving them. The corresponding forgetful functor $\mathrm{Cat}(\mathcal{K}') \rightarrow \mathrm{Cat}(\mathcal{K})$ admits a left adjoint:

Definition A.1.9 ($\mathcal{K}'$ -cocompletion). For $\mathcal{C} \in \mathrm{Cat}(\mathcal{K})$, the $\mathcal{K}'$ -**cocompletion** $\mathcal{P}_{\mathcal{K}}^{\mathcal{K}'}(\mathcal{C}) \in \mathrm{Cat}(\mathcal{K}')$ is characterized by the universal property

$$\mathrm{Fun}^{\mathcal{K}'}(\mathcal{P}_{\mathcal{K}}^{\mathcal{K}'}(\mathcal{C}), \mathcal{D}) \simeq \mathrm{Fun}^{\mathcal{K}}(\mathcal{C}, \mathcal{D})$$

for every $\mathcal{D} \in \mathrm{Cat}(\mathcal{K}')$, where $\mathrm{Fun}^{\mathcal{K}}$ denotes functors preserving all $\mathcal{K}$ -colimits.

Familiar instances populate the framework:

Example A.1.10.

- (i) $\mathcal{K} = \emptyset, \mathcal{K}' = \mathbf{all}$: $\mathcal{P}^{\mathbf{all}}(\mathcal{C}) = \mathbf{PShv}(\mathcal{C})$.
- (ii) $\mathcal{K} = \emptyset, \mathcal{K}' = \{\text{filtered}\}$: $\mathbf{Ind}(\mathcal{C})$.
- (iii) $\mathcal{K} = \emptyset, \mathcal{K}' = \{\text{sifted}\}$: the free sifted cocompletion $\mathcal{P}^{\text{sift}}(\mathcal{C})$. When $\mathcal{C}$ has finite coproducts, this is the category denoted $\mathcal{P}_{\Sigma}(\mathcal{C})$ in Definition A.1.3 [Lur09, Remark 5.5.8.16].
- (iv) $\mathcal{K} = \{\text{finite coproducts}\}, \mathcal{K}' = \mathbf{all}$: again $\mathcal{P}_{\Sigma}(\mathcal{C})$, by the dual perspective of Proposition A.1.4.
- (v) $\mathcal{K} = \emptyset, \mathcal{K}' = \{\text{Idem}\}$: the idempotent completion $\mathcal{C}^{\text{idem}}$.

The fact that sifted cocompletion appears under two complementary descriptions (items 3 and 4 of Example A.1.10) reflects a general phenomenon: $\{\text{sifted}\}$ and $\{\text{finite coproducts}\}$ are *complementary* classes of colimits, in the sense that adjoining one to a category preserving the other gives the same answer. This is why $\mathcal{P}_{\Sigma}(\mathcal{C})$ may equally be thought of as “free sifted cocompletion of $\mathcal{C}$ ” or as “free cocompletion preserving finite coproducts of $\mathcal{C}$ ”.

The norm-functor construction will combine Definition A.1.9 with two further universal constructions: each stage in the chain

$$\mathrm{Sm}_{T,+}^{\mathrm{qp}} \xrightarrow{(1)} \mathcal{P}_{\Sigma}(\mathrm{Sm}_T^{\mathrm{qp}})_* \xrightarrow{(2)} \mathrm{H}_*(T) \xrightarrow{(3)} \mathrm{SH}(T)$$

is a universal construction on the previous one — the sifted cocompletion, then the motivic localization, then the $\otimes$ -inversion of $\mathbb{T}$. A symmetric monoidal functor on the source then extends whenever its image satisfies the corresponding preservation condition, though the precise universal property invoked differs from stage to stage (see Section A.2.3).

A.2. NORM FUNCTORS

A.2.1. Weil restriction

Write $\mathrm{Sm}_S^{\mathrm{qp}} \subseteq \mathrm{Sm}_S$ for the full subcategory of smooth quasi-projective S -schemes and $\mathrm{Sm}_{S,+}^{\mathrm{qp}}$ for the corresponding category of objects $X_+ = X \sqcup S$.

For a finite locally free morphism $p: T \rightarrow S$ in classical algebraic geometry, the **Weil restriction** $R_p X$ of a quasi-projective T -scheme X is the S -scheme characterized by

$$\mathrm{Hom}_S(Y, R_p X) \simeq \mathrm{Hom}_T(Y \times_S T, X);$$

it exists and is quasi-projective, and is smooth when X is [BH21, §2.3], so R_p is right adjoint to the pullback $p^*: \mathrm{Sm}_S^{\mathrm{qp}} \rightarrow \mathrm{Sm}_T^{\mathrm{qp}}$ on smooth quasi-projective schemes. (Without quasi-projectivity the Weil restriction still exists as an algebraic space.) In the pointed setting, Weil restriction extends to a symmetric monoidal functor $p_{\otimes}: \mathrm{Sm}_{T,+}^{\mathrm{qp}} \rightarrow \mathrm{Sm}_{S,+}^{\mathrm{qp}}$ with respect to the smash product. The quasi-projective restriction is harmless after Nisnevich localization, since every smooth scheme is Nisnevich locally quasi-projective [BH21, §4.1]. The core question of [BH21] is: can $p_{\otimes}$ be extended to motivic homotopy theory?

A.2.2. The norm functor theorem

Theorem A.2.1 (Bachmann–Hoyois [BH21, Theorem 3.3, Proposition 4.5]). *Let $p: T \rightarrow S$ be a finite locally free morphism. Then there exists a symmetric monoidal functor*

$$p_{\otimes}: \mathrm{H}_*(T) \longrightarrow \mathrm{H}_*(S)$$

satisfying:

- (i) $p_{\otimes}$ preserves sifted colimits;
- (ii) $p_{\otimes}(X_+) \simeq (R_p X)_+$ for $X \in \mathrm{Sm}_T$ quasi-projective;
- (iii) if $T = S^{\sqcup n}$ and p is the fold map, then $p_{\otimes}$ is the n -fold smash product;
- (iv) if p is moreover finite étale, $p_{\otimes}$ extends further to a symmetric monoidal functor $p_{\otimes}: \mathrm{SH}(T) \rightarrow \mathrm{SH}(S)$.

(Bachmann–Hoyois construct the functor on $\mathcal{P}_{\Sigma}(\mathrm{Sm}_T)_*$ in the greater generality of universally clopen p , with descent to H_* for integral p .)

A.2.3. Construction via universal cocompletions

By Proposition A.1.5 and the symmetric monoidal refinement of Remark A.1.6, the symmetric monoidal Weil restriction $p_{\otimes}: \mathrm{Sm}_{T,+}^{\mathrm{qp}} \rightarrow \mathrm{Sm}_{S,+}^{\mathrm{qp}}$ extends along the universal sequence

$$\mathrm{Sm}_{T,+}^{\mathrm{qp}} \xrightarrow{(1)} \mathcal{P}_{\Sigma}(\mathrm{Sm}_T^{\mathrm{qp}})_* \xrightarrow{(2)} \mathrm{H}_*(T) \xrightarrow{(3)} \mathrm{SH}(T)$$

in stages, provided the relevant preservation property holds at each stage:

- (i) **Sifted cocompletion (unconditional).** Since $\mathrm{Sm}_T^{\mathrm{qp}}$ has finite coproducts, $p_{\otimes}$ extends along (1) uniquely to a symmetric monoidal sifted-cocontinuous functor on $\mathcal{P}_{\Sigma}(\mathrm{Sm}_T^{\mathrm{qp}})_*$ by Proposition A.1.5.
- (ii) **Motivic localization.** Since $H_*(T) = \mathcal{P}_{\Sigma}(\mathrm{Sm}_T)_*[(\mathrm{Nis} \cup \mathbb{A}^1)^{-1}]$ is a localization, the extension of step (1) descends to $H_*(T)$ if and only if $p_{\otimes}$ sends Nisnevich and $\mathbb{A}^1$ -equivalences to motivic equivalences. This holds whenever p is integral [BH21, Theorem 3.3(4)], in particular for finite locally free p .
- (iii) **$\mathbb{T}$ -stabilization.** Since $\mathrm{SH}(T) = H_*(T)[\mathbb{T}_T^{-1}]$ is the $\otimes$ -inversion of $\mathbb{T}_T$ (Definition 2.1.5), one asks whether the composite $\Sigma^{\infty}p_{\otimes}: H_*(T) \rightarrow \mathrm{SH}(S)$ extends across this inversion. The required criterion is that $\Sigma^{\infty}p_{\otimes}(\mathbb{T}_T)$ be $\otimes$ -invertible in $\mathrm{SH}(S)$. A subtlety intervenes: $p_{\otimes}$ does *not* preserve colimits, so the Pr^L -universal property underlying Definition 2.1.5 does not apply. Bachmann–Hoyois prove a refined universal property of the $\otimes$ -inversion relative to symmetric monoidal functors preserving only *filtered* colimits [BH21, Lemma 4.1], for which this stable invertibility criterion suffices; and $\Sigma^{\infty}p_{\otimes}(\mathbb{T}_T) \simeq \mathbb{S}^{\mathrm{R}_p\mathbb{A}^1}$ is indeed invertible in $\mathrm{SH}(S)$ for finite étale p [BH21, Lemma 4.4].

This is a paradigmatic example of how the universal-property approach pays off: a hard “can we extend?” question reduces to checking elementary preservation properties at each stage of the tower.

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